II

VOLUME II · EXAMINATION PRACTICE

Clinical SBA Questions Only

1,500 clinical single-best-answer questions

No answer keys · Designed for independent practice

Foundation 270 · Intermediate 730 · Advanced 500

Syllabus-organised · Five options per question

Independent revision material · Specialist review is still required

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Using this question book

What this file includes. Every question (Q1-Q1500) is presented without a correct-answer indication. Use the separate Answers & Justifications PDF to mark your responses.

Navigation. Topics are arranged by difficulty, then clinical syllabus block. Use the PDF bookmarks to jump between Foundation, Intermediate and Advanced sections.

Timed practice. A Diploma-style written paper contains 150 SBAs over 180 minutes according to the supplied examination material; verify current RCSEd candidate instructions for your sitting.

Scoring. Award 1 mark for the single best answer; award zero for an incorrect or unanswered response. The matching Answers PDF uses the same Q-number IDs, even when questions are grouped by level rather than original number.

Practice tip. Read all five plausible choices. Identify the decisive patient-specific finding before selecting the most appropriate management action.

Printed companion to the GitHub Pages question bank

- 1,500 question IDs, five answer options each.
- 270 Foundation / 730 Intermediate / 500 Advanced.
- Author: Dr. Kam O Man, Owen.
- Editorial edition: November 2026.

PART I / EXAMINATION QUESTIONS

Answer keys and rationales are separated into Part II. The original question IDs are preserved so your website notes and exported progress can still be cross-referenced.

Foundation questions · 270 items

BLOCK A · Applied anatomy (35)

Q2 FOUNDATION · Block A

A 34-year-old with tension-type headaches and TMD has a muscle tender to palpation above the zygomatic arch, in the temple.
Which muscle is this most likely to be?

A	Masseter
B	Medial pterygoid
C	Lateral pterygoid
D	Temporalis
E	Anterior belly of digastric

See answer and justification for Q2 →

Q3 FOUNDATION · Block A

After a fall, a 22-year-old has a fracture of the left condylar neck. The radiograph shows the condylar head displaced anteromedially.
Which muscle has displaced it?

A	Lateral pterygoid (inferior head)
B	Masseter
C	Temporalis
D	Medial pterygoid
E	Posterior belly of digastric

See answer and justification for Q3 →

Q5 FOUNDATION · Block A

A patient has weakness during mandibular retrusion, but elevation remains powerful. Examination suggests selective damage to the horizontal fibres of one first-arch masticatory muscle. Which muscle is most directly implicated?

A	Anterior temporalis
B	Posterior temporalis
C	Superficial masseter
D	Medial pterygoid
E	Inferior head of lateral pterygoid

See answer and justification for Q5 →

Q11 FOUNDATION · Block A

A patient gags strongly during upper impressions.
Which nerves form the afferent and efferent limbs of the gag reflex?

A	Trigeminal afferent, facial efferent
B	Glossopharyngeal afferent, vagus efferent
C	Vagus afferent, glossopharyngeal efferent
D	Hypoglossal afferent, vagus efferent
E	Facial afferent, hypoglossal efferent

See answer and justification for Q11 →

Q14 FOUNDATION · Block A

A newly delivered mandibular complete denture retains well at rest. On forceful clenching the distobuccal flange causes a reproducible sore spot and the posterior saddle dislodges. Pressure-indicator paste shows a localised interference and there is no obvious working-side occlusal premature contact. What is the most appropriate first correction?

A	Increase the vertical dimension to reduce masseter recruitment
B	Relieve the distobuccal flange after confirming the functional masseteric notch
C	Remove the indirect retainer and increase distal clasp retention
D	Increase the lingual flange extension to stabilise the whole denture
E	Prescribe adhesive without modifying the contour

See answer and justification for Q14 →

Q15 FOUNDATION · Block A

An upper denture is dislodged when the patient opens wide or moves the jaw sideways. The distobuccal flange is very thick.
Which structure is interfering with it?

A	Zygomatic buttress
B	Hamular notch
C	Masseter
D	Coronoid process
E	Maxillary tuberosity

See answer and justification for Q15 →

Q16 FOUNDATION · Block A

You are marking the post-dam on a master cast.
Which landmark is a useful guide to its position?

A	Incisive papilla
B	Palatal rugae
C	Fovea palatini, near the vibrating line
D	Midline raphe
E	Zygomatic buttress

See answer and justification for Q16 →

Q20 FOUNDATION · Block A

A patient reports swelling of the cheek on eating, with an over-extended upper buccal flange opposite the upper second molar.
Which structure could be compressed?

A	Submandibular duct
B	Sublingual ducts
C	Parotid duct
D	Buccal fat pad
E	Facial artery

See answer and justification for Q20 →

Q24 FOUNDATION · Block A

A patient has an upper canine abscess with swelling near the inner corner of the eye.
What is the main danger of this spread?

A	Ludwig's angina
B	Cavernous sinus thrombosis via the angular and ophthalmic veins
C	Mediastinitis
D	Parotitis
E	Trismus

See answer and justification for Q24 →

Q26 FOUNDATION · Block A

An inferior alveolar nerve block is given.
Into which space is the solution deposited?

A	Submandibular space
B	Buccal space
C	Sublingual space
D	Pterygomandibular space
E	Canine space

See answer and justification for Q26 →

Q28 FOUNDATION · Block A

After a successful inferior alveolar block, the buccal gingiva of 46 is still sensitive during a crown preparation.
Which nerve needs anaesthetising?

A	Lingual nerve
B	Nerve to mylohyoid
C	Mental nerve
D	Auriculotemporal nerve
E	Long buccal nerve, the sensory branch of the anterior division

See answer and justification for Q28 →

Q29 FOUNDATION · Block A

You are describing the nerve to mylohyoid to a student.
Which muscles does it supply?

A	Mylohyoid and posterior belly of digastric
B	Mylohyoid and anterior belly of digastric
C	Geniohyoid and mylohyoid
D	Genioglossus and mylohyoid
E	Mylohyoid alone

See answer and justification for Q29 →

Q30 FOUNDATION · Block A

After middle-ear surgery, a patient loses taste in the anterior tongue and experiences dry mouth, but tongue motor function is preserved. Which affected fibre pathway most directly explains the paired symptoms?

A	Glossopharyngeal sensory fibres only
B	Hypoglossal motor fibres
C	Chorda tympani carrying taste and parasympathetic fibres via the lingual nerve
D	Long buccal nerve sensory fibres
E	Greater petrosal fibres to the pterygopalatine ganglion

See answer and justification for Q30 →

Q31 FOUNDATION · Block A

After parotid surgery, a patient sweats and flushes over the cheek when eating.
Which nerve is involved?

A	Great auricular nerve
B	Facial nerve
C	Auriculotemporal nerve
D	Lingual nerve
E	Buccal nerve

See answer and justification for Q31 →

Q32 FOUNDATION · Block A

A patient with a skull-base injury is noted to have damaged the artery passing through the foramen spinosum.
Which branch of the maxillary artery is this?

A	Middle meningeal artery
B	Inferior alveolar artery
C	Sphenopalatine artery
D	Infraorbital artery
E	Deep temporal artery

See answer and justification for Q32 →

Q34 FOUNDATION · Block A

You are explaining the blood supply to a hypertrophied masseter.
Which part of the maxillary artery gives the masseteric artery?

A	First (mandibular) part
B	Second (pterygoid) part
C	Third (pterygopalatine) part
D	The facial artery directly
E	The superficial temporal artery alone

See answer and justification for Q34 →

Q35 FOUNDATION · Block A

Which statement about the nerve to medial pterygoid is correct?

A	It arises from the anterior division of V3
B	It arises from the posterior division of V3
C	It is a branch of the facial nerve
D	It is a branch of the buccal nerve
E	It arises from the main trunk of V3, before it divides

See answer and justification for Q35 →

Q37 FOUNDATION · Block A

A coronoid fracture fragment is pulled upwards towards the temporal fossa.
Which muscle inserts into the coronoid process?

A	Masseter
B	Lateral pterygoid
C	Temporalis
D	Medial pterygoid
E	Buccinator

See answer and justification for Q37 →

Q38 FOUNDATION · Block A

A patient protrudes during a jaw-relation record and the operator guides the mandible back.
Which muscle is the main retractor of the mandible?

A	Lateral pterygoid
B	Masseter (superficial)
C	Medial pterygoid
D	Anterior belly of digastric
E	Temporalis (posterior, horizontal fibres)

See answer and justification for Q38 →

Q39 FOUNDATION · Block A

You ask a patient to slide the mandible to the left during an excursive check.
Which muscles on which side mainly produce this movement?

A	Left lateral and medial pterygoids
B	Left masseter
C	Right temporalis
D	Right lateral and medial pterygoids
E	Left digastric

See answer and justification for Q39 →

Q40 FOUNDATION · Block A

A 28-year-old has a reciprocal click in the right TMJ.
Which muscle attaches to the articular disc and is most associated with disc displacement?

A	Medial pterygoid
B	Lateral pterygoid (superior head)
C	Masseter (deep)
D	Temporalis
E	Posterior belly of digastric

See answer and justification for Q40 →

Q41 FOUNDATION · Block A

A patient has weakness of the palatoglossal arch after a brainstem lesion, while tongue protrusion is normal. Which nerve supplies palatoglossus?

A	Hypoglossal nerve
B	Glossopharyngeal nerve alone
C	Facial nerve
D	Mandibular nerve
E	Vagus via the pharyngeal plexus

See answer and justification for Q41 →

Q45 FOUNDATION · Block A

A student asks which nerve supplies geniohyoid.
What is the correct answer?

A	Nerve to mylohyoid
B	C1 fibres travelling with the hypoglossal nerve
C	Facial nerve
D	Lingual nerve
E	Glossopharyngeal nerve

See answer and justification for Q45 →

Q47 FOUNDATION · Block A

You are checking the posterior extension of a new lower complete denture.
How much of the retromolar pad should the base cover?

A	None of it
B	A narrow strip at its front edge
C	At least the anterior half to two-thirds
D	All of it, onto the pterygomandibular raphe
E	It depends only on the OVD

See answer and justification for Q47 →

Q49 FOUNDATION · Block A

A new maxillary complete denture shows focal blanching and pain on pressure over a large midline bony prominence while the remainder of the palate bears the appliance comfortably. Which adjustment best addresses this focal cause?

A	Create selective relief over the torus while preserving support elsewhere
B	Relieve the entire posterior palatal seal broadly
C	Reduce both buccal flanges equally
D	Increase the occlusal vertical dimension
E	Use a thick resilient liner over the entire palate without local assessment

See answer and justification for Q49 →

Q51 FOUNDATION · Block A

An elderly edentulous patient has a severely resorbed mandibular ridge and new numbness of the lower lip that occurs when wearing the denture. A palpable opening lies close to the ridge crest. What requires priority assessment?

A	Posterior palatal seal position
B	Labial frenum extension only
C	Masseteric notch contour
D	Pressure over the superficial mental foramen, with referral if symptoms persist
E	Maxillary tuberosity undercut

See answer and justification for Q51 →

Q52 FOUNDATION · Block A

A patient on anticholinergic medication complains of a dry mouth at rest.
Which gland produces most of the unstimulated saliva?

A	Parotid
B	Sublingual
C	Submandibular (about 65–70%)
D	Minor palatal glands
E	Von Ebner's glands

See answer and justification for Q52 →

Q53 FOUNDATION · Block A

A patient has had radiotherapy to both parotid glands.
Which saliva is reduced most?

A	Stimulated saliva
B	Unstimulated saliva only
C	Mucous saliva only
D	Minor gland saliva only
E	None, because the submandibular glands compensate fully

See answer and justification for Q53 →

Q54 FOUNDATION · Block A

You see a small salivary stone at the duct opening beside the lingual frenum.
Which duct opens here?

A	Parotid duct
B	Sublingual ducts of Rivinus
C	Duct of Bartholin only
D	Submandibular (Wharton's) duct, at the sublingual papilla
E	Nasolacrimal duct

See answer and justification for Q54 →

Q55 FOUNDATION · Block A

You explain parotid secretomotor supply to a student.
What is the correct parasympathetic pathway?

A	Facial nerve, chorda tympani, lingual nerve
B	Glossopharyngeal nerve, lesser petrosal nerve, otic ganglion, auriculotemporal nerve
C	Vagus nerve, pharyngeal plexus
D	Superior cervical ganglion, external carotid plexus
E	Facial nerve, greater petrosal nerve, pterygopalatine ganglion

See answer and justification for Q55 →

Q58 FOUNDATION · Block A

A patient has TMJ inflammation and the vascular supply is discussed.
Which arteries supply the TMJ?

A	Superficial temporal and maxillary arteries
B	Facial and lingual arteries
C	Inferior alveolar artery only
D	Occipital artery
E	Ascending pharyngeal artery only

See answer and justification for Q58 →

Q59 FOUNDATION · Block A

A student asks how to classify the TMJ.
Which description is correct?

A	Fibrous joint
B	Primary cartilaginous joint
C	Synovial hinge joint only
D	Synovial ginglymoarthrodial (hinge and gliding) joint
E	Secondary cartilaginous joint

See answer and justification for Q59 →

Q60 FOUNDATION · Block A

During an inferior alveolar block, the solution may be deflected by a ligament running from the spine of the sphenoid to the lingula.
Which ligament is this?

A	Lateral ligament
B	Stylomandibular ligament
C	Sphenomandibular ligament
D	Pterygomandibular raphe
E	Stylohyoid ligament

See answer and justification for Q60 →

Q67 FOUNDATION · Block A

An edentulous patient has large undercuts on both maxillary tuberosities.
What is the usual approach?

A	Choose a path or block out one side, reducing surgically only if both still prevent insertion
B	Ignore them
C	Fill both undercuts with acrylic
D	Use a soft liner alone
E	Extract any remaining molars

See answer and justification for Q67 →

Q68 FOUNDATION · Block A

A partially dentate patient needs a lower Kennedy Class I RPD. Large bilateral mandibular tori leave no room for a lingual bar.
What is appropriate?

A	A lingual bar over the tori
B	A lingual plate over the tori
C	No major connector
D	A labial bar, or surgical removal of the tori if a sound design needs it
E	A sublingual bar

See answer and justification for Q68 →

BLOCK B · Physiology and pathology (25)

Q72 FOUNDATION · Block B

Two upper complete dentures have similar tissue adaptation. One repeatedly drops when a posterior border seal is broken by speech, whereas it is retentive at rest. Which physical change best explains the loss of retention?

A	Reduced adhesive contact after a large internal gap develops
B	Failure of the peripheral seal compromises pressure-related retention
C	A sudden increase in acrylic elastic modulus
D	Increased porcelain-to-acrylic tooth bonding
E	Excessive clasp friction at a natural abutment

See answer and justification for Q72 →

Q76 FOUNDATION · Block B

A patient sips an acidic drink and stimulated saliva begins to neutralise it.
What is the main buffer in stimulated saliva?

A	Bicarbonate
B	Phosphate
C	Proteins
D	Urea
E	Ammonia

See answer and justification for Q76 →

Q77 FOUNDATION · Block B

A hyposalivating older denture wearer has recurrent mucosal candidosis. Which component of normal saliva contributes antigen-specific mucosal immune exclusion and is reduced when whole salivary delivery is poor?

A	Salivary bicarbonate
B	Amylase
C	Secretory immunoglobulin A
D	Mucin alone
E	Calcium phosphate

See answer and justification for Q77 →

Q82 FOUNDATION · Block B

A public-health lecture quotes the WHO goal for tooth retention.
How many functional natural teeth should be retained throughout life?

A	10
B	20
C	28
D	24
E	32

See answer and justification for Q82 →

Q85 FOUNDATION · Block B

You are explaining long-term ridge resorption to a patient after extractions.
In the long term, how much more height does the anterior mandible lose than the anterior maxilla?

A	The same
B	About twice as much
C	About four times as much
D	About ten times as much
E	Less

See answer and justification for Q85 →

Q86 FOUNDATION · Block B

A patient asks when her ridges will shrink most after extractions.
When is resorption fastest?

A	After 5 years
B	In the first 6–12 months
C	After 10 years
D	At a constant rate for life
E	It does not resorb

See answer and justification for Q86 →

Q87 FOUNDATION · Block B

An edentulous mandibular arch is planned for an overdenture; two vital, well-supported canines remain and have favourable root morphology. The patient wants to preserve proprioceptive support and avoid surgery if possible. What option best meets that goal?

A	Extract both canines before constructing a conventional complete denture
B	Retain and prepare suitable canine roots as overdenture abutments
C	Rely on continuous denture adhesive rather than abutments
D	Place implants immediately without discussing root retention
E	Remove all teeth and increase OVD

See answer and justification for Q87 →

Q89 FOUNDATION · Block B

A 56-year-old lost 36 ten years ago and wants it replaced. 26 has over-erupted by 3 mm.
What is the main consequence for planning?

A	None
B	Increased OVD
C	Posterior crossbite
D	Reduced interocclusal space, so the over-erupted tooth must be managed first
E	TMJ ankylosis

See answer and justification for Q89 →

Q91 FOUNDATION · Block B

Patient X has sugar in ten cups of tea a day. Patient Y has the same total sugar in one dessert.
What is the most important caries advice for patient X?

A	Only total sugar intake matters
B	Brush after every cup
C	Use a mouthwash after every cup
D	Reduce the frequency of sugar intake
E	Rinse with water after each cup

See answer and justification for Q91 →

Q92 FOUNDATION · Block B

A partially dentate patient with recession has multiple root caries lesions.
Which organism is particularly associated with root caries?

A	Actinomyces
B	Lactobacillus only
C	Candida
D	Streptococcus sanguinis
E	Porphyromonas gingivalis

See answer and justification for Q92 →

Q93 FOUNDATION · Block B

A histology slide shows an early enamel caries lesion.
Which zone shows the greatest demineralisation?

A	Surface zone
B	Body of the lesion
C	Dark zone
D	Translucent zone
E	Sound enamel

See answer and justification for Q93 →

Q96 FOUNDATION · Block B

An adult with partial dentures is at high caries risk.
Which professional fluoride application is recommended?

A	Fluoride varnish 22,600 ppm at least twice a year
B	5000 ppm varnish monthly
C	1450 ppm varnish once a year
D	2800 ppm gel weekly
E	226 ppm rinse daily in the surgery

See answer and justification for Q96 →

Q97 FOUNDATION · Block B

A partially dentate patient with exposed roots, high plaque retention around an RPD and recently developed xerostomia has several new root caries lesions. What modifiable factor should be addressed directly alongside local plaque control?

A	Ridge width
B	Border-moulding material
C	Salivary flow and its medication-related causes
D	Torus palatinus height
E	Temporomandibular translation

See answer and justification for Q97 →

Q99 FOUNDATION · Block B

A 38-year-old has smooth, silky enamel on the upper incisors, cupped cusps, and amalgam restorations standing proud of the surrounding tooth.
What is the most likely diagnosis?

A	Attrition
B	Abrasion
C	Abfraction
D	Caries
E	Erosion

See answer and justification for Q99 →

Q100 FOUNDATION · Block B

A 50-year-old has flat, shiny wear facets that match on opposing teeth, fractured cusps and hypertrophic masseters.
What is the most likely diagnosis?

A	Erosion
B	Abrasion
C	Attrition
D	Abfraction
E	Caries

See answer and justification for Q100 →

Q101 FOUNDATION · Block B

A 45-year-old has sharp-edged, V-shaped cervical notches on the upper canines and premolars, worse on the left. She is right-handed and brushes hard.
What is the most likely diagnosis?

A	Erosion
B	Abrasion
C	Attrition
D	Caries
E	Perimylolysis

See answer and justification for Q101 →

Q104 FOUNDATION · Block B

You screen a new patient's tooth wear with the BEWE.
How is the BEWE scored?

A	0–4 for each tooth surface
B	As a DMFT count
C	0–5 for each tooth
D	Using the Smith and Knight index
E	0–3 for the worst surface in each sextant, with the sum guiding management

See answer and justification for Q104 →

Q105 FOUNDATION · Block B

You need to record wear on each surface for a research audit.
Which index scores 0–4 on every tooth surface?

A	BEWE
B	BPE
C	Smith and Knight Tooth Wear Index
D	ICDAS
E	DMFS

See answer and justification for Q105 →

Q110 FOUNDATION · Block B

A 70-year-old denture wearer has painful fissures at both corners of the mouth.
Which investigations are indicated?

A	Biopsy of each corner
B	A swab and no other tests
C	Full blood count, haematinics and blood glucose
D	A radiograph of the mandible
E	No investigations

See answer and justification for Q110 →

Q117 FOUNDATION · Block B

A patient drinks heavily and smokes daily. They ask whether stopping one habit matters if the other continues. Which statement best supports a balanced oral cancer prevention discussion?

A	Combined alcohol and tobacco exposure can act synergistically and reducing both lowers preventable risk
B	Alcohol matters only after oral cancer develops
C	Smoking is relevant only when denture stomatitis is present
D	Areca nut alone determines risk independently of other exposures
E	Cessation has no value once mucosal screening is established

See answer and justification for Q117 →

Q118 FOUNDATION · Block B

At a denture review you find a red and white patch on the lateral tongue that the patient had not noticed.
What does NICE NG12 advise?

A	Urgent referral on the suspected cancer pathway
B	Review in 6 months
C	Antifungal treatment and review
D	Routine referral
E	Photograph only

See answer and justification for Q118 →

Q121 FOUNDATION · Block B

A 58-year-old with poorly controlled type 2 diabetes has periodontitis.
How are the two conditions related?

A	They are unrelated
B	Diabetes affects only caries
C	Periodontitis improves diabetes, but not the reverse
D	Diabetes affects only implants
E	The relationship is bidirectional

See answer and justification for Q121 →

Q123 FOUNDATION · Block B

A patient asks how to clean her cobalt–chromium partial denture.
Which advice is correct?

A	Soak it overnight in household bleach
B	Do not soak it in hypochlorite, because it damages Co-Cr
C	Boil it weekly
D	Scrub it with abrasive toothpaste
E	Never remove it

See answer and justification for Q123 →

Q125 FOUNDATION · Block B

A 50-year-old with generalised periodontitis wants a bridge.
What is the usual sequence of periodontal care before definitive prosthodontics?

A	Surgery, then hygiene instruction
B	Bridge first, then periodontal care
C	Hygiene and risk-factor control, subgingival instrumentation, surgery where needed, then supportive care
D	Antibiotics alone
E	Extraction of all mobile teeth first

See answer and justification for Q125 →

Q127 FOUNDATION · Block B

A patient with marked xerostomia from anticholinergic medication has active root caries despite using 1,450-ppm fluoride toothpaste. They seek a realistic prevention plan. What is the most appropriate immediate approach?

A	Offer definitive crowns for all exposed roots before altering risk factors
B	Ask the patient to stop the medication without consulting the prescriber
C	Review medication contributions with the prescriber and intensify fluoride, diet and salivary support
D	Use a citric-acid saliva substitute in a dentate patient
E	Rely on RPD replacement to restore salivary flow

See answer and justification for Q127 →

BLOCK C · Dental materials science (20)

Q133 FOUNDATION · Block C

A Kennedy Class II RPD is planned on a premolar abutment. The surveyor finds a 0.5-mm undercut in a position suitable for a long gingivally approaching clasp, whereas an adjacent site offers only 0.25 mm for a short cast clasp. What is the best design principle?

A	All 0.5-mm undercuts must be occupied by short cast Co–Cr clasp arms
B	Clasp material can be selected solely from the depth without considering arm dimensions
C	A 0.75-mm undercut is always safe for every metal clasp
D	Match the undercut to clasp material, length, cross-section, approach and abutment support so flexure remains within the elastic limit
E	Deeper retention is always preferable even when the periodontium is compromised

See answer and justification for Q133 →

Q136 FOUNDATION · Block C

A sports mouthguard and a soft denture liner both need to absorb energy without permanent deformation. Which property describes this?

A	Toughness
B	Resilience
C	Hardness
D	Ductility
E	Creep

See answer and justification for Q136 →

Q148 FOUNDATION · Block C

An alginate impression for study casts is left in a sealed bag over the weekend before pouring. What is the main problem?

A	It becomes too elastic
B	It releases hydrogen
C	It polymerises further
D	It bonds to the tray
E	Syneresis and imbibition distort it

See answer and justification for Q148 →

Q159 FOUNDATION · Block C

A two-stage putty–wash impression gives a tight crown.
What is the recognised risk of this technique?

A	Putty displacing the wash
B	Hydrogen release
C	Too little working time
D	Latex inhibition
E	Distortion if the set putty is reseated without adequate relief

See answer and justification for Q159 →

Q160 FOUNDATION · Block C

A nurse asks how to disinfect an alginate impression before sending it to the laboratory.
What is correct?

A	Autoclave it
B	Soak it overnight in hypochlorite
C	Rinse it, then use a material-specific disinfectant for the time on the label, avoiding prolonged immersion
D	Do nothing
E	Dry it with hot air

See answer and justification for Q160 →

Q161 FOUNDATION · Block C

A technician adds extra water to stone so it pours easily.
What is the effect on the cast?

A	Stronger, with greater expansion
B	Weaker, with less expansion and a slower set
C	No change
D	Faster set
E	Harder surface

See answer and justification for Q161 →

Q162 FOUNDATION · Block C

A technician spatulates stone for twice the recommended time.
What is the effect?

A	Slower set and less expansion
B	Weaker cast
C	No change
D	The stone will not set
E	A faster set and more expansion

See answer and justification for Q162 →

Q172 FOUNDATION · Block C

A wax pattern is finished at 5 p.m. and the technician plans to invest it the next morning.
What is the best advice?

A	Invest promptly
B	Store it in the fridge overnight
C	Leave it in hot water
D	Re-wax the margins in the morning
E	It makes no difference

See answer and justification for Q172 →

Q173 FOUNDATION · Block C

A student asks how heat-cured PMMA polymerisation starts.
What is the initiator?

A	Hydroquinone
B	Camphorquinone
C	Tertiary amine
D	Benzoyl peroxide
E	Ethylene glycol dimethacrylate

See answer and justification for Q173 →

Q182 FOUNDATION · Block C

A practice switches to digitally milled, pre-polymerised PMMA denture bases.
What is the main material advantage?

A	Higher thermal conductivity
B	Very low residual monomer and porosity, with no polymerisation shrinkage in the clinic
C	Radiopacity
D	Self-repairing surface
E	Chemical bond to Co-Cr

See answer and justification for Q182 →

Q184 FOUNDATION · Block C

A complete denture with porcelain posterior teeth opposes natural lower premolars and molars.
What is the main risk?

A	Fracture of the base
B	Discolouration
C	Debonding of the porcelain teeth
D	Abrasion of the opposing natural teeth
E	Candida

See answer and justification for Q184 →

Q185 FOUNDATION · Block C

A tissue conditioner is used on inflamed mucosa before new impressions.
What is it made from?

A	Heat-cured PMMA
B	Silicone rubber
C	PEMA powder with an aromatic ester plasticiser in ethanol
D	Zinc oxide and eugenol
E	Polyether

See answer and justification for Q185 →

Q190 FOUNDATION · Block C

A Co-Cr clasp fractured on first insertion. The laboratory reports a high carbon content in the alloy.
What does excess carbon do?

A	Improves castability
B	Increases ductility
C	Reduces stiffness
D	Forms excess carbides, making the alloy brittle
E	Increases corrosion resistance

See answer and justification for Q190 →

Q192 FOUNDATION · Block C

A clinician compares implant abutment metals.
What is the elastic modulus of titanium, and how is it protected from corrosion?

A	About 220 GPa; chromium oxide
B	About 20 GPa; tin oxide
C	About 2 GPa; no oxide
D	About 110 GPa
E	About 90 GPa; nickel oxide

See answer and justification for Q192 →

Q203 FOUNDATION · Block C

A student asks why zirconia resists crack propagation.
What is transformation toughening?

A	Cubic to tetragonal change on heating
B	Stress at a crack tip converts tetragonal to monoclinic zirconia, with about 3–5% volume expansion that closes the crack
C	Glass infiltration of cracks
D	Heat treatment after milling
E	Ion exchange at the surface

See answer and justification for Q203 →

Q204 FOUNDATION · Block C

A zirconia hip implant failed in the 1990s after roughening in body fluids.
What is this phenomenon?

A	Low-temperature degradation (ageing): spontaneous tetragonal to monoclinic change in moisture
B	Galvanic corrosion
C	Creep
D	Fatigue only
E	Hydrolysis of silane

See answer and justification for Q204 →

Q210 FOUNDATION · Block C

A glass ionomer cement is chosen for a metal–ceramic crown on a patient with high caries risk.
What are its main properties?

A	No fluoride release; insoluble immediately
B	High bond to zirconia via MDP
C	Eugenol content and sedation
D	Fluoride release, CTE close to tooth, and sensitivity to moisture early on
E	The highest bond strength of any cement

See answer and justification for Q210 →

Q213 FOUNDATION · Block C

A crown does not seat fully and the cement layer is thick.
What is the ISO maximum film thickness for luting cements?

A	5 µm
B	10 µm
C	25 µm
D	100 µm
E	250 µm

See answer and justification for Q213 →

Q214 FOUNDATION · Block C

You are bonding a resin-bonded bridge to enamel.
What is the standard enamel etch?

A	10% polyacrylic acid for 10 s
B	5% hydrofluoric acid for 60 s
C	2% chlorhexidine for 30 s
D	30–37% phosphoric acid for 15–30 s
E	Air abrasion only

See answer and justification for Q214 →

Q217 FOUNDATION · Block C

A resin-bonded bridge is made with a cobalt–chromium wing.
What is the current standard for bonding the wing?

A	Perforated wings retained by resin (Rochette)
B	Electrolytically etched metal (Maryland)
C	Lost-salt crystals (Virginia)
D	Air-abraded base metal bonded with an MDP-containing resin cement
E	Zinc phosphate cement

See answer and justification for Q217 →

BLOCK D · Pharmacology, pain control, emergencies and the medically compromised patient (22)

Q224 FOUNDATION · Block D

Minutes after several cartridges, a patient reports tingling around the lips and ringing in the ears, then becomes drowsy.
What is happening?

A	A vasovagal faint
B	Anaphylaxis
C	Hypoglycaemia
D	Early local anaesthetic toxicity
E	An adrenaline reaction only

See answer and justification for Q224 →

Q226 FOUNDATION · Block D

An inferior alveolar block repeatedly fails on a lower molar with an acute periapical abscess.
What is the main pharmacological reason?

A	Low tissue pH reduces the non-ionised fraction of the anaesthetic
B	The anaesthetic has expired
C	Adrenaline is inactivated
D	Accessory supply from the facial nerve
E	The patient is resistant to amides

See answer and justification for Q226 →

Q228 FOUNDATION · Block D

An inhalation sedation machine is being checked before use.
What is the minimum oxygen concentration it must deliver?

A	10%
B	15%
C	21%
D	25%
E	30% (a maximum of 70% nitrous oxide)

See answer and justification for Q228 →

Q234 FOUNDATION · Block D

The patient in the previous scenario is allergic to penicillin.
What is an appropriate alternative?

A	Clarithromycin 500 mg twice daily (or metronidazole 400 mg three times daily for anaerobes)
B	Amoxicillin at a lower dose
C	Co-amoxiclav
D	Phenoxymethylpenicillin with antihistamine cover
E	Flucloxacillin

See answer and justification for Q234 →

Q238 FOUNDATION · Block D

A patient with a prosthetic aortic valve needs crown preparation with retraction cord.
What is the current guidance on antibiotic prophylaxis?

A	Amoxicillin 3 g before every appointment
B	Clindamycin for all invasive procedures
C	Not routinely recommended
D	Never consider it
E	Give it only for impressions

See answer and justification for Q238 →

Q241 FOUNDATION · Block D

A 48 kg, 82-year-old woman with mild hepatic impairment needs paracetamol.
What is correct?

A	Give 2 g four times daily
B	No limit applies
C	The usual maximum is 4 g in 24 hours, reduced for low body weight and hepatic impairment
D	Paracetamol is contraindicated
E	Give 1 g hourly as needed

See answer and justification for Q241 →

Q243 FOUNDATION · Block D

A patient on warfarin needs an extraction before a partial denture. Her INR this morning is 3.2.
What is appropriate?

A	Stop warfarin for 3 days
B	Proceed in primary care with local haemostatic measures
C	Refer to hospital
D	Give vitamin K
E	Defer until the INR is below 2

See answer and justification for Q243 →

Q245 FOUNDATION · Block D

A patient taking apixaban twice daily needs several extractions (a higher bleeding-risk procedure).
What does SDCEP advise?

A	Miss the morning dose
B	Stop apixaban for a week
C	Switch to warfarin
D	Take a double dose
E	No change at all

See answer and justification for Q245 →

Q246 FOUNDATION · Block D

A patient taking rivaroxaban once each morning needs a higher-risk extraction.
What does SDCEP advise?

A	Stop for 5 days
B	Delay the morning dose until about 4 hours after haemostasis
C	Miss two days
D	Give vitamin K
E	Bridge with heparin

See answer and justification for Q246 →

Q247 FOUNDATION · Block D

A patient on dual antiplatelet therapy (aspirin and clopidogrel) after a coronary stent needs a tooth extracted before a denture.
What is correct?

A	Stop clopidogrel for 7 days
B	Stop aspirin only
C	Stop both
D	Refer for hospital admission
E	Do not stop either

See answer and justification for Q247 →

Q248 FOUNDATION · Block D

A patient with moderate haemophilia A needs a lower molar extracted.
What is the safe approach?

A	Liaise with the haemophilia centre
B	Proceed with an inferior alveolar block and sutures
C	Give aspirin for pain
D	Prescribe tranexamic acid and proceed with no liaison
E	Give vitamin K

See answer and justification for Q248 →

Q252 FOUNDATION · Block D

You are assessing MRONJ risk in several patients.
Who is at highest risk?

A	Osteoporosis patient on oral alendronate for 2 years
B	Osteoporosis patient with no drug treatment
C	Cancer patient on high-dose IV zoledronic acid
D	Patient on vitamin D
E	Patient who stopped alendronate 10 years ago after 1 year

See answer and justification for Q252 →

Q254 FOUNDATION · Block D

A patient on oral bisphosphonates for 6 years needs a simple extraction.
What does SDCEP advise about a drug holiday?

A	Stop for 3 months beforehand
B	Drug holidays are not recommended for routine dental care
C	Stop for 6 months afterwards
D	Switch to denosumab
E	Stop permanently

See answer and justification for Q254 →

Q260 FOUNDATION · Block D

A patient's GP letter gives an HbA1c result.
At or above which value is diabetes diagnosed?

A	30 mmol/mol
B	42 mmol/mol
C	48 mmol/mol (6.5%)
D	55 mmol/mol
E	70 mmol/mol

See answer and justification for Q260 →

Q261 FOUNDATION · Block D

An undiagnosed 54-year-old with periodontitis and recurrent candidosis has an HbA1c of 45 mmol/mol.
What does this indicate?

A	Normal glycaemia
B	Type 1 diabetes
C	Well-controlled diabetes
D	Prediabetes (42–47 mmol/mol)
E	Hypoglycaemia

See answer and justification for Q261 →

Q263 FOUNDATION · Block D

Minutes after an impression, a patient develops wheeze, facial swelling, urticaria and a falling blood pressure.
What is the first drug and dose?

A	Chlorphenamine 10 mg IV
B	Hydrocortisone 200 mg IM
C	Adrenaline 1:1000, 500 micrograms IM into the anterolateral thigh, repeated after 5 minutes if needed
D	Salbutamol 100 micrograms
E	Adrenaline 1:10,000, 1 mg IV

See answer and justification for Q263 →

Q264 FOUNDATION · Block D

An 8-year-old develops anaphylaxis in the surgery.
What is the IM adrenaline dose?

A	150 micrograms
B	300 micrograms
C	500 micrograms
D	1 mg
E	50 micrograms

See answer and justification for Q264 →

Q265 FOUNDATION · Block D

A 65-year-old with known angina develops central chest tightness during a crown preparation.
What is the first drug to give?

A	Aspirin 75 mg
B	Oxygen only
C	Adrenaline 500 micrograms
D	Salbutamol
E	GTN spray 400 micrograms sublingually

See answer and justification for Q265 →

Q272 FOUNDATION · Block D

A patient on long-term corticosteroids collapses with profound hypotension after a stressful procedure.
What is appropriate?

A	Glucagon
B	GTN
C	Salbutamol
D	Call 999 and give hydrocortisone if prescribed and you are trained
E	Adrenaline 1 mg IV

See answer and justification for Q272 →

Q273 FOUNDATION · Block D

A patient collapses, is unresponsive and is not breathing normally.
What is correct adult CPR?

A	30 compressions to 2 breaths, at 100–120 a minute and 5–6 cm deep, with an AED as soon as available
B	15:2 at 60 a minute
C	5 breaths, then 15 compressions
D	Compressions 2–3 cm deep
E	Wait for the ambulance before compressions

See answer and justification for Q273 →

Q275 FOUNDATION · Block D

A patient becomes acutely unwell in the surgery.
How should oxygen be given?

A	2 L/min via nasal cannula
B	15 L/min via a non-rebreathe mask
C	5 L/min via a venturi mask
D	Only if the patient asks
E	Never before the ambulance arrives

See answer and justification for Q275 →

Q282 FOUNDATION · Block D

A colleague has told a patient to stop rivaroxaban for 3 days before impressions for a new denture.
What is the best response?

A	Continue the rivaroxaban
B	Agree, and stop it for a week instead
C	Switch to warfarin
D	Give vitamin K
E	Refer to hospital for the impressions

See answer and justification for Q282 →

BLOCK E · Assessment, diagnosis, imaging, classifications and planning (20)

Q294 FOUNDATION · Block E

A new patient has a BPE code 4 in one sextant and code 2 elsewhere.
What does the BSP recommend?

A	Oral hygiene instruction only
B	A full six-point periodontal chart of the whole dentition
C	A chart of that sextant only
D	A repeat BPE in a year
E	Immediate referral for surgery

See answer and justification for Q294 →

Q298 FOUNDATION · Block E

An upper arch is missing 14 and 15 only.
What is the Kennedy classification?

A	Class III
B	Class I
C	Class II
D	Class IV
E	Class III modification 1

See answer and justification for Q298 →

Q299 FOUNDATION · Block E

An upper arch is missing 12, 11, 21 and 22 only.
What is the Kennedy classification?

A	Class III
B	Class III modification 1
C	Class IV
D	Class I
E	Class IV modification 1

See answer and justification for Q299 →

Q305 FOUNDATION · Block E

A mandibular ridge has adequate height but a narrow, sharp crest.
What is its Cawood and Howell class?

A	Class III
B	Class IV (knife-edge)
C	Class V
D	Class VI
E	Class II

See answer and justification for Q305 →

Q306 FOUNDATION · Block E

An edentulous mandible has a flat ridge of inadequate height and width.
What is its Cawood and Howell class?

A	Class II
B	Class III
C	Class IV
D	Class VI
E	Class V

See answer and justification for Q306 →

Q307 FOUNDATION · Block E

An edentulous mandible is depressed, with loss of basal bone.
What is its Cawood and Howell class?

A	Class V
B	Class VI
C	Class IV
D	Class III
E	Class I

See answer and justification for Q307 →

Q310 FOUNDATION · Block E

A 50-year-old's worst interproximal bone loss reaches the middle third of the root.
What is the BSP stage?

A	Stage I
B	Stage II
C	Stage IV
D	Stage III
E	It cannot be staged

See answer and justification for Q310 →

Q311 FOUNDATION · Block E

A 45-year-old with interproximal bone loss estimated at 35% of root length has a bone-loss-to-age ratio below 1.0. The patient smokes 15 cigarettes per day and has no longitudinal radiographs. How should the grade be approached?

A	Automatically assign Grade A from the patient’s age
B	Assign Stage I because the bone loss does not exceed 50%
C	Grade on tooth mobility alone without periodontal charting
D	Recognise that heavy smoking is a Grade C modifier despite a ratio compatible with Grade B, and document uncertainty and full case findings
E	Assign Grade B and ignore smoking because only radiographs influence grade

See answer and justification for Q311 →

Q312 FOUNDATION · Block E

A 30-year-old has 60% interproximal bone loss at the worst site.
What is the BSP grade?

A	Grade A
B	Grade B
C	Stage only
D	It cannot be graded
E	Grade C (60 ÷ 30 = 2.0)

See answer and justification for Q312 →

Q316 FOUNDATION · Block E

A treated patient still has several 5 mm pockets that bleed.
What is the periodontal status?

A	Currently stable
B	In remission
C	Currently unstable
D	Healthy
E	Stage I

See answer and justification for Q316 →

Q321 FOUNDATION · Block E

A lower incisor moves more than 1 mm horizontally and can be depressed.
What is its Miller mobility grade?

A	Grade 1
B	Grade 3
C	Grade 2
D	Grade 0
E	Grade 4

See answer and justification for Q321 →

Q324 FOUNDATION · Block E

A patient's BEWE scores across the six sextants sum to 15.
What does this indicate?

A	No risk
B	Low risk
C	High risk: oral hygiene and dietary assessment, find the cause, monitor, and consider restoration
D	Medium risk only
E	Immediate full-mouth crowns

See answer and justification for Q324 →

Q327 FOUNDATION · Block E

A lateral cephalogram shows an ANB angle of 3°.
What is the skeletal pattern?

A	Class II
B	Class III
C	Severe Class II
D	It cannot be interpreted
E	Class I (normal is about 2–4°)

See answer and justification for Q327 →

Q328 FOUNDATION · Block E

You assess a lower premolar as a bridge abutment.
What are the ideal and minimum acceptable crown–root ratios?

A	Ideal 1:1, minimum 2:1
B	Ideal 1:2, minimum 3:1
C	Ideal 3:1, minimum 1:1
D	Ideal 2:3, minimum 1:1
E	Any ratio is acceptable

See answer and justification for Q328 →

Q332 FOUNDATION · Block E

A colleague asks for CBCT on every new denture patient "as a baseline".
What is the correct principle?

A	CBCT is suitable for routine screening
B	CBCT replaces all intraoral radiographs
C	CBCT is justified only for a three-dimensional question that lower-dose imaging cannot answer
D	CBCT has a lower dose than a bitewing
E	CBCT needs no clinical evaluation

See answer and justification for Q332 →

Q334 FOUNDATION · Block E

An audit finds radiographs in the notes with no written findings.
What does IR(ME)R require of every exposure?

A	Justification only
B	Optimisation only
C	A second opinion
D	Justification, optimisation (ALARP) and a recorded clinical evaluation
E	Patient signature

See answer and justification for Q334 →

Q343 FOUNDATION · Block E

A patient with a sore tongue and recurrent ulcers has a high MCV.
What does this suggest?

A	Iron deficiency
B	Thalassaemia
C	Vitamin B12 or folate deficiency
D	Diabetes
E	Normal blood

See answer and justification for Q343 →

Q344 FOUNDATION · Block E

A root-filled premolar has 1.5 mm of sound coronal dentine on much of its circumference, no clinical evidence of root fracture and acceptable periodontal support. Before proposing a post-and-crown, what is the most important determinant of prognosis?

A	Post length alone
B	Choice of resin cement alone
C	A predictable ferrule combined with root/support prognosis
D	The maximum possible core diameter
E	The number of cementation appointments

See answer and justification for Q344 →

Q357 FOUNDATION · Block E

You measure the freeway space of a complete denture wearer.
What is the normal range?

A	0 mm
B	0.5 mm
C	10 mm
D	About 2–4 mm
E	8–12 mm

See answer and justification for Q357 →

Q359 FOUNDATION · Block E

A completely edentulous patient has bilateral large mandibular tori and a highly displaceable anterior ridge. Which clinical finding most affects the selection of an impression technique?

A	Patient eye colour
B	Anterior tissue displaceability and undercut geometry
C	Posterior tooth shade from old dentures
D	The size of the labial frenum alone
E	The type of articulator in the laboratory

See answer and justification for Q359 →

BLOCK F · Removable partial dentures (25)

Q371 FOUNDATION · Block F

You have surveyed a study cast for a lower RPD and are about to draw the design.
What is decided first?

A	The clasps
B	The major connector
C	The indirect retainers
D	The outline of the saddles (which spaces will be restored)
E	The tooth shade

See answer and justification for Q371 →

Q376 FOUNDATION · Block F

A student marks the survey line on a premolar.
What does the survey line represent?

A	The gingival margin
B	The occlusal plane
C	The greatest convexity of the tooth relative to the chosen path of insertion
D	The cementoenamel junction
E	The long axis

See answer and justification for Q376 →

Q377 FOUNDATION · Block F

A cast RPD design looks retentive on a static model but binds against a proximal tooth surface during trial insertion. What is the most appropriate next design assessment?

A	Increase the clasp undercut to overcome resistance
B	Re-survey insertion path and guide-plane geometry before adjusting retention
C	Alter only the acrylic tooth shade
D	Shorten every rest simultaneously
E	Add a thicker distal saddle before checking the framework

See answer and justification for Q377 →

Q389 FOUNDATION · Block F

A student lists the requirements of a clasp unit.
How much of the tooth's circumference should the clasp encircle?

A	More than 180°
B	90°
C	Less than 120°
D	Exactly 360°
E	Encirclement does not matter

See answer and justification for Q389 →

Q396 FOUNDATION · Block F

The retentive arm of a new clasp is pressing on the tooth when the denture is fully seated.
What is the correct position of the retentive tip?

A	In the occlusal third, under tension
B	On the survey line
C	Above the survey line
D	In the gingival third undercut, passive when fully seated
E	On the gingiva

See answer and justification for Q396 →

Q400 FOUNDATION · Block F

An RPI clasp is designed for 34, the abutment of a free-end saddle.
What are its three components?

A	Retentive arm, plate, I-bar
B	Mesial rest, distal proximal plate and buccal I-bar
C	Distal rest, mesial plate and lingual arm
D	Ring, plate, I-bar
E	Rest, plate and wrought wire

See answer and justification for Q400 →

Q405 FOUNDATION · Block F

A student asks what else indirect retainers achieve besides resisting rotation.
What is correct?

A	They give support and stop the major connector settling
B	They provide the main retention
C	They replace rests
D	They improve the shade
E	They have no other role

See answer and justification for Q405 →

Q407 FOUNDATION · Block F

At review, pressing on the lower free-end saddle lifts the canine rests from their seats.
What is needed?

A	Tighter clasps
B	New rests
C	Reline of the saddles
D	Removal of the indirect retainers
E	A new framework

See answer and justification for Q407 →

Q416 FOUNDATION · Block F

A dental bar (continuous clasp) on the cingula is suggested for a patient with recession.
What is its main drawback?

A	It traps food
B	It gives no indirect retention
C	It must cover the gingiva
D	It is invisible
E	It is very flexible

See answer and justification for Q416 →

Q420 FOUNDATION · Block F

A 60-year-old has worn a mucosa-borne acrylic RPD covering the gingival margins for 10 years.
What is the typical consequence?

A	Better periodontal health
B	Less caries
C	No effect
D	Bone gain
E	Gingival stripping, with more periodontal disease and caries

See answer and justification for Q420 →

Q425 FOUNDATION · Block F

A patient asks for a flexible nylon denture because she dislikes metal clasps.
What is its main disadvantage?

A	It is radiopaque
B	It is heavy
C	It is hard to adjust, reline or add to
D	It is always tooth-borne
E	It corrodes

See answer and justification for Q425 →

Q426 FOUNDATION · Block F

A patient asks about precision attachments for a lower RPD.
What is correct?

A	They need no crowns
B	They are simple Level 1 care
C	They need crowned abutments and usually lie beyond Level 2
D	They are cheaper than clasps
E	They cannot be used with Co-Cr

See answer and justification for Q426 →

Q427 FOUNDATION · Block F

26 has over-erupted into a lower edentulous space where an RPD is planned.
What is appropriate before designing?

A	Correct the over-eruption and occlusal plane (pre-prosthetic optimisation, a Level 2 task)
B	Ignore it
C	Extract 26
D	Increase the OVD
E	Use a soft liner

See answer and justification for Q427 →

Q430 FOUNDATION · Block F

A new Co-Cr RPD is fitted.
When should it first be reviewed?

A	At 6 months
B	About 1 week, then at regular recalls
C	Only if there are problems
D	At 2 years
E	The same day

See answer and justification for Q430 →

Q431 FOUNDATION · Block F

A patient has worn a Co-Cr RPD for 5 years.
What is the most important long-term maintenance?

A	Tightening clasps every year
B	Polishing the framework
C	Replacing the teeth
D	Monitoring caries and periodontal health at the abutments, and relining saddles as needed
E	Remaking every 2 years

See answer and justification for Q431 →

Q432 FOUNDATION · Block F

A patient is worried that an RPD will damage her remaining teeth.
What does the evidence (Bergman) show?

A	Any RPD destroys the abutments
B	Acrylic RPDs are harmless
C	A well-designed, tooth-supported Co-Cr RPD with good hygiene and recall does little harm
D	RPDs prevent caries
E	Only implants are safe

See answer and justification for Q432 →

Q437 FOUNDATION · Block F

A Kennedy Class IV RPD replaces several anterior teeth. During function, the prosthesis tends to lift anteriorly around a line through the anterior rests. Which design element most directly limits that rotation?

A	More labial flange bulk
B	Posteriorly positioned indirect retention at an appropriate distance from the fulcrum
C	Changing the denture teeth to porcelain
D	Making all connectors flexible
E	Increasing vertical dimension without checking occlusion

See answer and justification for Q437 →

Q440 FOUNDATION · Block F

A student asks when guide planes and rest seats should be prepared for a Co-Cr RPD.
What is the correct sequence?

A	Prepare freehand at the first visit
B	Prepare after the framework arrives
C	Prepare after the master impression
D	Survey the study cast and plan, prepare the teeth, then take the master impression
E	Prepare at the fit appointment

See answer and justification for Q440 →

Q441 FOUNDATION · Block F

An RPI assembly is planned on 34.
How long should the distal guide plane be?

A	Short, about 2–3 mm, so the plate can disengage under load
B	The full crown height
C	6 mm
D	There should be none
E	Angled away from the path

See answer and justification for Q441 →

Q446 FOUNDATION · Block F

A lower lingual plate is planned for a patient with mobile incisors.
What is required for it to succeed?

A	A deep sulcus
B	No rests
C	Gingival contact without relief
D	Avoiding the cingula
E	Excellent oral hygiene, as it covers the gingival margins

See answer and justification for Q446 →

Q448 FOUNDATION · Block F

A patient asks how to soak her Co-Cr RPD overnight.
What is correct?

A	Dilute hypochlorite
B	An alkaline peroxide cleanser
C	Boiling water
D	Vinegar
E	Bleach

See answer and justification for Q448 →

Q449 FOUNDATION · Block F

A patient asks whether to wear her new RPD at night.
What is the standard advice?

A	Remove it at night
B	Wear it day and night
C	Wear it only at night
D	Wear it only for meals
E	It makes no difference

See answer and justification for Q449 →

Q459 FOUNDATION · Block F

A student designs an RPD by first choosing an attractive clasp for each tooth.
What is wrong with this approach?

A	Nothing
B	Clasps should be chosen last of all
C	Clasps are optional
D	Rests are not needed
E	It reverses the design logic

See answer and justification for Q459 →

Q461 FOUNDATION · Block F

A lower Kennedy Class I Co–Cr RPD has sound abutment support, but the distal saddle compresses displaceable mucosa during chewing and the indirect retainer lifts. The metal framework is accurately seated. Which method is most likely to improve support before final processing?

A	Record a functional impression of the distal-extension saddle on the seated framework and produce an altered master cast
B	Place a longer distal rest and omit indirect retention
C	Increase clasp retention to resist vertical mucosal displacement
D	Use a mucostatic wash of the entire arch while the saddles are unloaded
E	Reduce saddle coverage to concentrate load directly beneath the occlusal table

See answer and justification for Q461 →

Q462 FOUNDATION · Block F

A colleague plans Akers clasps engaging the distobuccal undercuts of the terminal abutments of a lower free-end RPD.
What is the problem?

A	Too little retention
B	Poor aesthetics only
C	Clasps are too flexible
D	The clasps will torque the abutments distally as the saddle moves
E	There is no problem

See answer and justification for Q462 →

BLOCK G · Complete dentures (25)

Q471 FOUNDATION · Block G

A new upper complete denture stays up at rest but slides sideways when the patient chews.
Which property is deficient?

A	Retention
B	Support
C	Stability
D	Aesthetics
E	Vertical dimension

See answer and justification for Q471 →

Q472 FOUNDATION · Block G

An upper complete denture is stable at rest but repeatedly dislodges when the posterior border seal is broken during speech, despite adequate saliva and well-adapted fitting surfaces. What factor is the most likely primary driver?

A	Poor adhesion due to salivary evaporation
B	Loss of peripheral seal during functional movement
C	Loss of the elastomeric impression tray adhesive
D	Fracture toughness of the acrylic tooth
E	Excessive clasp reciprocation

See answer and justification for Q472 →

Q476 FOUNDATION · Block G

You take primary impressions of an edentulous patient in stock trays.
What is the main aim?

A	A mucostatic record of the ridge only
B	Recording the full sulcus depth so that special trays can be made
C	Recording the occlusion
D	Recording the post-dam
E	Choosing the shade

See answer and justification for Q476 →

Q480 FOUNDATION · Block G

You are at the jaw registration visit with wax rims.
What is recorded at this visit?

A	Only the shade
B	Only OVD
C	OVD, RCP, occlusal plane, centre and canine lines, lip line, tooth mould and shade
D	The post-dam only
E	Nothing; the laboratory decides

See answer and justification for Q480 →

Q485 FOUNDATION · Block G

You set the incisal level of an upper wax rim for a 45-year-old.
How much rim usually shows below the relaxed upper lip?

A	None
B	4–5 mm
C	6 mm
D	About 1–2 mm (less with age)
E	8 mm

See answer and justification for Q485 →

Q487 FOUNDATION · Block G

You set the lower posterior occlusal plane.
What is the usual landmark?

A	The lower lip
B	The mylohyoid ridge
C	About two-thirds up the retromolar pad, level with the lateral border of the tongue
D	The top of the retromolar pad
E	The external oblique ridge

See answer and justification for Q487 →

Q490 FOUNDATION · Block G

You use a Willis gauge during jaw registration.
What does it measure?

A	Pupil to lip
B	Base of the nose to the underside of the chin
C	Interpupillary distance
D	Inter-alar width
E	Ridge height

See answer and justification for Q490 →

Q491 FOUNDATION · Block G

You test the OVD of trial dentures phonetically.
What is the normal closest speaking space on "s"?

A	About 1–2 mm
B	0 mm
C	5 mm
D	8 mm
E	10 mm

See answer and justification for Q491 →

Q496 FOUNDATION · Block G

A student asks why complete dentures are made in the retruded contact position.
What is the reason?

A	It is the most comfortable
B	It gives the most protrusion
C	It increases OVD
D	It is the position patients habitually use
E	It is reproducible

See answer and justification for Q496 →

Q499 FOUNDATION · Block G

At try-in, the posterior teeth on one side do not meet and there is a slide on closure.
What is the correct action?

A	Grind the teeth until they meet
B	Process the dentures and adjust at fit
C	Take a new RCP registration and re-articulate
D	Add wax to the teeth
E	Accept it

See answer and justification for Q499 →

Q500 FOUNDATION · Block G

A technician uses an average-value articulator for complete dentures.
What are its fixed settings?

A	Condylar guidance about 30° and a Bonwill triangle with sides of about 10 cm
B	No lateral movement at all
C	Settings taken from a protrusive record
D	A fully adjustable Bennett angle
E	Condylar guidance of 0°

See answer and justification for Q500 →

Q504 FOUNDATION · Block G

A facebow record is taken for complete dentures on a semi-adjustable articulator.
What does it record?

A	The relationship of the maxillary arch to the hinge axis
B	The OVD
C	RCP
D	The lower occlusal plane
E	The shade

See answer and justification for Q504 →

Q506 FOUNDATION · Block G

A technician refers to Hanau's quint when setting balanced occlusion.
What are its five factors?

A	OVD, RCP, shade, mould, lip line
B	Freeway space, OVD, overjet, overbite, post-dam
C	Condylar guidance, incisal guidance, cusp angle, plane of orientation, compensating curve
D	Ridge height, width, shape, saliva, tongue
E	Retention, stability, support, aesthetics, speech

See answer and justification for Q506 →

Q507 FOUNDATION · Block G

A student asks why bilateral balanced occlusion is used in complete dentures.
What is its purpose?

A	Simultaneous contacts on working and non-working sides keep the dentures stable in excursions
B	It improves appearance
C	It increases bite force
D	It reduces OVD
E	It prevents Candida

See answer and justification for Q507 →

Q513 FOUNDATION · Block G

A technician sets the lower second molar on the ascending slope of the ramus.
What is the consequence?

A	The lower denture is pushed forward in function
B	Better support
C	Less cheek biting
D	Better aesthetics
E	No effect

See answer and justification for Q513 →

Q521 FOUNDATION · Block G

At try-in, a patient's "f" sounds like "v".
What does the "f/v" test check?

A	Posterior palatal thickness
B	The OVD alone
C	Lower incisor position
D	The post-dam
E	That the upper incisal edges touch the wet–dry line of the lower lip (upper incisor length and labial position)

See answer and justification for Q521 →

Q541 FOUNDATION · Block G

An immediate upper denture is fitted after extractions.
What is the correct aftercare?

A	Remove it at night from the first day
B	Review in 6 months
C	Leave it in for the first 24 hours, review the next day, and reline at about 3–6 months
D	Never reline it
E	Replace it after 1 month

See answer and justification for Q541 →

Q543 FOUNDATION · Block G

A patient with overdenture abutments asks how to protect them.
What is the key preventive measure?

A	Antibiotics
B	Fluoride gel placed in the fitting surface of the denture, with meticulous hygiene
C	Wearing the denture continuously
D	Avoiding brushing the roots
E	Chlorhexidine only

See answer and justification for Q543 →

Q548 FOUNDATION · Block G

A lab note says a denture was "rebased" rather than "relined".
What is the difference?

A	They are identical
B	A rebase adds a thin layer to the fitting surface
C	A reline replaces the teeth
D	A rebase replaces the whole base
E	A reline changes the occlusion

See answer and justification for Q548 →

Q553 FOUNDATION · Block G

A patient with failing complete dentures asks about implants at a Level 2 practice.
What is appropriate?

A	Place implants at once
B	Refuse to discuss implants
C	Tell the patient implants are never suitable
D	Offer adhesive only
E	Discuss implant-assisted dentures as an option and refer for implant surgery

See answer and justification for Q553 →

Q554 FOUNDATION · Block G

At the try-in, only the left posterior teeth contact and the mandible slides on closure.
What does this indicate?

A	Excessive OVD
B	An RCP registration error
C	A short post-dam
D	Porcelain teeth
E	Thin saliva

See answer and justification for Q554 →

Q556 FOUNDATION · Block G

You use a Fox's plane guide at the jaw registration visit.
What does it check?

A	The occlusal plane against the interpupillary and ala–tragal lines
B	The OVD
C	The shade
D	The centre line
E	The post-dam depth

See answer and justification for Q556 →

Q565 FOUNDATION · Block G

An edentulous patient asks whether she still needs dental check-ups.
What is the correct advice?

A	Yes: regular review of the oral mucosa and denture fit
B	No; edentulous patients need no recall
C	Only if in pain
D	Every 10 years
E	Only for denture repairs

See answer and justification for Q565 →

Q568 FOUNDATION · Block G

A laboratory asks which articulator to use for routine complete dentures.
What is usually adequate?

A	An average-value articulator
B	A fully adjustable articulator with a kinematic facebow
C	A simple hinge
D	No articulator
E	A fully programmed digital articulator

See answer and justification for Q568 →

Q569 FOUNDATION · Block G

A student asks which articulator uses a Bonwill triangle with sides of about 10 cm.
What is the answer?

A	A simple hinge
B	A fully adjustable articulator
C	A semi-adjustable arcon
D	An average-value articulator
E	A virtual articulator only

See answer and justification for Q569 →

BLOCK H · Fixed prosthodontics (24)

Q574 FOUNDATION · Block H

A short mandibular premolar is being prepared for a crown. The initial preparation has 22 degrees of total occlusal convergence, short walls and limited resistance to lateral dislodgement. Gingival health is satisfactory and some additional proximal tooth structure remains. What is the most defensible improvement?

A	Increase taper further to make cementation easier
B	Reduce the taper where feasible and consider auxiliary grooves or boxes while preserving healthy tooth tissue
C	Increase cement thickness to create an artificial axial wall
D	Deepen every margin subgingivally without reassessing periodontal support
E	Replace a full crown with a bridge solely to gain more retainers

See answer and justification for Q574 →

Q587 FOUNDATION · Block H

A student asks why provisional crowns are needed.
What are their purposes?

A	Only aesthetics
B	To protect the pulp, keep tooth position, protect the gingiva, restore function and appearance, and test the planned shape or OVD
C	Only to keep the patient happy
D	To retain the definitive cement
E	None

See answer and justification for Q587 →

Q599 FOUNDATION · Block H

A molar has just had root canal treatment and will be crowned in 3 months.
What is important meanwhile?

A	Leave the access open
B	A cotton pellet only
C	Nothing
D	A good coronal seal placed promptly
E	Antibiotics

See answer and justification for Q599 →

Q601 FOUNDATION · Block H

A student mixes up bridge terms.
What is a retainer?

A	The replacement tooth
B	The joint between units
C	The supporting tooth
D	The edentulous space
E	The restoration on the abutment tooth that holds the bridge

See answer and justification for Q601 →

Q603 FOUNDATION · Block H

A student quotes Ante's law.
What does it state?

A	The pontic should be wider than the abutment
B	Cantilevers are always contraindicated
C	Bridges should be no longer than three units
D	The total root surface area of the abutments should equal or exceed that of the teeth being replaced
E	Abutments must be vital

See answer and justification for Q603 →

Q604 FOUNDATION · Block H

An examiner asks about the status of Ante's law.
What is correct?

A	It is supported by strong trial evidence
B	It is historical and not well supported by evidence, though still examined
C	It is a legal requirement
D	It applies only to implants
E	It has never been taught

See answer and justification for Q604 →

Q609 FOUNDATION · Block H

A fixed-fixed conventional bridge is planned between two heavily restored abutments.
What is a recognised drawback of this design?

A	The least tooth destruction
B	No need for parallel preparations
C	The most tooth destruction, the need for parallel preparations, and a loose retainer may go unnoticed
D	No retainers
E	It cannot be used for bounded spans

See answer and justification for Q609 →

Q615 FOUNDATION · Block H

A 24-year-old has lost 22. 23 is unrestored with plenty of enamel, and the occlusion is favourable.
What is usually the first choice?

A	A conventional fixed-fixed bridge
B	An implant immediately
C	A cantilever metal–ceramic bridge
D	An acrylic RPD
E	A single-wing resin-bonded cantilever bridge

See answer and justification for Q615 →

Q616 FOUNDATION · Block H

A two-winged fixed-fixed resin-bonded bridge has been in place for 4 years. The patient reports a bad taste.
What is the likely problem?

A	Wing fracture
B	Pontic fracture
C	Differential movement has debonded one wing unnoticed, leading to caries
D	Allergy
E	Normal ageing

See answer and justification for Q616 →

Q618 FOUNDATION · Block H

A five-unit bridge from 13 to 17 has a pier abutment at 15. One terminal retainer keeps debonding.
Why, and what is the solution?

A	The span is too short; add a pontic
B	The pier acts as a fulcrum
C	A stiffer, thicker bridge
D	Temporary cement
E	Remove the pier abutment

See answer and justification for Q618 →

Q619 FOUNDATION · Block H

A minimally restored maxillary lateral incisor is missing; adjacent teeth have healthy enamel and the patient prefers to avoid implant surgery. What is the main rationale for considering a single-retainer resin-bonded bridge?

A	It always lasts longer than an implant crown
B	It requires no assessment of occlusal contacts
C	It preserves sound tooth structure and permits conservative replacement in a suitable case
D	It prevents caries regardless of hygiene
E	It is indicated even when enamel bonding is not feasible

See answer and justification for Q619 →

Q622 FOUNDATION · Block H

You are cementing a Co-Cr resin-bonded bridge wing.
What is the correct bonding protocol?

A	Zinc phosphate on dry enamel
B	Glass ionomer without isolation
C	Temporary cement first
D	Hydrofluoric acid on the metal
E	Air-abrade the metal (50 µm alumina), etch the enamel, and use an MDP-containing resin cement under rubber dam

See answer and justification for Q622 →

Q623 FOUNDATION · Block H

A 45-year-old bruxist with heavy anterior contacts asks for a resin-bonded bridge replacing 21.
What is the main concern?

A	Too much enamel
B	Bruxism and heavy occlusion make debonding more likely
C	Age
D	The lack of anaesthesia
E	The cost

See answer and justification for Q623 →

Q628 FOUNDATION · Block H

A bridge has a concave pontic covering the whole ridge, and the mucosa beneath is inflamed.
What is this pontic, and why has it failed?

A	A saddle (ridge lap) pontic; it is unhygienic
B	An ovate pontic; it is too small
C	A sanitary pontic; it traps food
D	A modified ridge lap; it is too narrow
E	A conical pontic; it is too short

See answer and justification for Q628 →

Q638 FOUNDATION · Block H

You are fitting feldspathic porcelain veneers.
What is the bonding protocol?

A	Zinc phosphate
B	Glass ionomer
C	Air abrasion and MDP
D	HF-etched, silanated ceramic and light-cured resin cement on etched enamel
E	Temporary cement

See answer and justification for Q638 →

Q645 FOUNDATION · Block H

A colleague insists every anterior restoration must follow the golden proportion.
What is correct?

A	It is mandatory
B	It always applies to natural smiles
C	It is a legal requirement
D	It is not a mandatory rule for a natural smile
E	It applies only to veneers

See answer and justification for Q645 →

Q647 FOUNDATION · Block H

A 19-year-old has white-spot lesions on the labial surfaces of the upper incisors after orthodontics.
What is the least invasive option?

A	Crowns
B	Veneers
C	Composite build-ups
D	Resin infiltration or microabrasion
E	No treatment possible

See answer and justification for Q647 →

Q648 FOUNDATION · Block H

A 17-year-old has peg-shaped upper lateral incisors after orthodontic space redistribution.
What is the most conservative restorative option?

A	Crowns
B	Porcelain veneers
C	Additive direct composite
D	Extraction and implants
E	An RBB

See answer and justification for Q648 →

Q659 FOUNDATION · Block H

A lower first molar has had root canal treatment and has an MOD cavity with thin cusps.
What is the appropriate restoration?

A	An MOD composite without coverage
B	Cuspal coverage with an onlay or crown
C	A post and amalgam
D	A veneer
E	No restoration

See answer and justification for Q659 →

Q663 FOUNDATION · Block H

A patient wants 46 replaced by a cantilever pontic from 45 alone. He has a heavy bite.
What is the best advice?

A	This is ideal
B	Use a larger pontic
C	Avoid it
D	Add a second pontic
E	Use temporary cement

See answer and justification for Q663 →

Q664 FOUNDATION · Block H

A rigid bridge from 13 to 17 (14 and 16 missing) has repeatedly debonded at 13.
What is the design problem and solution?

A	Too few pontics; add another
B	The bridge is too short
C	The shade is wrong
D	Temporary cement is needed
E	15 acts as a pier fulcrum

See answer and justification for Q664 →

Q665 FOUNDATION · Block H

A patient with a deep overbite and bruxism requests upper incisor veneers.
What is the best response?

A	Proceed with thin veneers
B	Explain that heavy occlusion contraindicates veneers, and discuss alternatives
C	Use temporary cement
D	Prepare deeply into dentine
E	Proceed without a mock-up

See answer and justification for Q665 →

Q668 FOUNDATION · Block H

You use a double-cord technique before an impression for 21.
What is its purpose?

A	The first cord controls fluid and the second displaces the tissue until just before the impression
B	To stop bleeding only
C	To push the margin deeper
D	To shrink the gingiva permanently
E	To replace the impression material

See answer and justification for Q668 →

Q669 FOUNDATION · Block H

A crown on 36 is overcontoured buccally over the furcation, and the gingiva there is chronically inflamed.
What is the definitive solution?

A	Chlorhexidine
B	Antibiotics
C	Gingival surgery only
D	Recontour or remake the crown with a flat emergence profile
E	Ignore it

See answer and justification for Q669 →

BLOCK I · Occlusion and temporomandibular disorders (16)

Q673 FOUNDATION · Block I

A colleague talks about recording centric relation.
What is centric relation?

A	Maximum intercuspation
B	A tooth-guided position
C	A jaw-to-jaw relation with the condyles in their most superior-anterior position against the eminence, discs correctly interposed, independent of the teeth
D	The rest position
E	The edge-to-edge position

See answer and justification for Q673 →

Q674 FOUNDATION · Block I

On guiding a patient into RCP, you find a slide into ICP.
What is normal?

A	A slide is always pathological
B	A slide occurs in fewer than 10% of people
C	A normal slide is 5 mm
D	A slide must always be eliminated
E	A slide of about 1 mm forwards occurs in about 90% of people

See answer and justification for Q674 →

Q675 FOUNDATION · Block I

You measure a dentate patient's rest position.
What is the normal freeway space?

A	0 mm
B	About 2–4 mm
C	6–8 mm
D	10 mm
E	0.5 mm

See answer and justification for Q675 →

Q676 FOUNDATION · Block I

In right lateral excursion, only the right canines contact.
What is this guidance called?

A	Group function
B	Balanced occlusion
C	A non-working interference
D	Mutually protected occlusion only
E	Canine guidance

See answer and justification for Q676 →

Q679 FOUNDATION · Block I

In right lateral excursion, the left second molars contact heavily.
What is this and when is it desirable?

A	A non-working (balancing) interference
B	A working-side contact; always desirable
C	Canine guidance
D	Group function; ideal
E	Anterior guidance

See answer and justification for Q679 →

Q680 FOUNDATION · Block I

In right lateral excursion, the right first molars contact before the right canine.
What is this?

A	Group function
B	Canine guidance
C	A working-side interference
D	A non-working interference
E	Balanced occlusion

See answer and justification for Q680 →

Q681 FOUNDATION · Block I

On closing, a contact on 17 diverts the mandible forward and sideways.
What is this?

A	A deflective contact
B	Canine guidance
C	Group function
D	Freeway space
E	The Bennett angle

See answer and justification for Q681 →

Q683 FOUNDATION · Block I

You inspect the mediolateral curvature of the posterior occlusal plane.
What is it called?

A	Curve of Spee
B	Compensating curve
C	Bonwill triangle
D	Balkwill angle
E	Curve of Wilson

See answer and justification for Q683 →

Q685 FOUNDATION · Block I

You record mandibular movements.
What are typical lateral excursion and protrusion ranges?

A	1–2 mm
B	20 mm each
C	4 mm lateral only
D	About 8–10 mm each
E	30 mm

See answer and justification for Q685 →

Q693 FOUNDATION · Block I

A patient had a closed lock 2 years ago. She now opens 44 mm without clicking.
What is the diagnosis?

A	Disc displacement with reduction
B	Disc displacement without reduction, without limited opening
C	Subluxation
D	Myalgia
E	Normal joint

See answer and justification for Q693 →

Q697 FOUNDATION · Block I

A patient asks whether her "bad bite" caused her TMD.
What is the current understanding of TMD aetiology?

A	Malocclusion is the main cause
B	It is purely genetic
C	It is multifactorial and biopsychosocial
D	It is always caused by trauma
E	It is infective

See answer and justification for Q697 →

Q702 FOUNDATION · Block I

A patient wants a soft splint for jaw pain because it is cheaper.
What is the concern?

A	Soft splints are illegal
B	They are for short-term use and may increase clenching in some patients
C	They cause caries
D	They are always more effective
E	They need monthly replacement

See answer and justification for Q702 →

Q711 FOUNDATION · Block I

A patient with intermittent clicking and myofascial pain has been advised online to undergo extensive occlusal grinding to eliminate a slide from RCP to ICP. There are no acute joint red flags. What is the most evidence-consistent initial management?

A	Reorganise the occlusion immediately to a canine-guided scheme
B	Provide comprehensive crowns to eliminate the reported click
C	Replace any dentures before completing a pain history
D	Prescribe prolonged antibiotics for the TMJ
E	Explain the uncertainty of a causal occlusal relationship and begin conservative, reversible self-care and pain management

See answer and justification for Q711 →

Q720 FOUNDATION · Block I

A closed lock of 4 months has not improved with self-care, a splint and exercises.
What is appropriate?

A	Continue indefinitely
B	Occlusal grinding
C	Extraction of molars
D	Full-mouth crowns
E	Referral

See answer and justification for Q720 →

Q723 FOUNDATION · Block I

A patient with acute TMJ arthralgia asks about pain relief.
What is appropriate?

A	Long-term opioids
B	Antibiotics
C	Simple analgesia, such as short-term NSAIDs
D	Steroid injections at the first visit
E	No medication

See answer and justification for Q723 →

Q726 FOUNDATION · Block I

A colleague prescribes a "Michigan splint".
What is it?

A	A soft lower splint
B	A full-coverage hard maxillary stabilisation splint
C	An anterior repositioning splint
D	A partial anterior bite plane
E	A night guard for snoring

See answer and justification for Q726 →

BLOCK J · Tooth wear, the failing dentition, interfaces and trauma (13)

Q735 FOUNDATION · Block J

A patient has completed stabilisation and reassessment.
What does the definitive phase include?

A	Only hygiene
B	Only extractions
C	Endodontics, periodontal surgery if needed, then fixed and removable prosthodontics
D	Only splints
E	Only maintenance

See answer and justification for Q735 →

Q741 FOUNDATION · Block J

A 45-year-old with localised anterior wear needs restoration.
What is the first-line restorative approach?

A	Conventional crowns
B	Extraction and implants
C	Porcelain veneers
D	Direct resin composite: additive, adhesive and repairable
E	Post crowns

See answer and justification for Q741 →

Q745 FOUNDATION · Block J

A patient asks how long the back teeth will be apart after Dahl composites.
What is typical?

A	A day
B	A week
C	3 years
D	They never meet
E	About 4–6 months, occasionally up to 18

See answer and justification for Q745 →

Q752 FOUNDATION · Block J

Surgical crown lengthening is proposed to gain height on worn upper incisors.
What is correct?

A	It is Level 1 care
B	It has no risks
C	It is Level 3, and risks root exposure and an unfavourable crown–root ratio
D	It always improves aesthetics
E	It preserves the crown–root ratio

See answer and justification for Q752 →

Q760 FOUNDATION · Block J

A bridge is planned between 14 and 16. 16 has a guarded prognosis.
What is the principle?

A	Use 16 as an abutment anyway
B	Splint 16 with the bridge to strengthen it
C	Extract 14
D	Teeth of doubtful prognosis should not be bridge abutments
E	Use a longer bridge

See answer and justification for Q760 →

Q762 FOUNDATION · Block J

You check readiness for definitive prosthodontics in primary care.
What are indicative full-mouth targets?

A	Plaque under 20% and bleeding under 30%
B	Plaque under 80%
C	Bleeding under 60%
D	No targets
E	Plaque 0% only

See answer and justification for Q762 →

Q766 FOUNDATION · Block J

A lower first molar with a Grade III furcation is proposed as a bridge abutment.
What is correct?

A	It is an ideal abutment
B	Root resection is routine Level 2 care
C	Furcation does not affect prognosis
D	Splint it with composite only
E	Furcation-involved molars are poor abutments

See answer and justification for Q766 →

Q779 FOUNDATION · Block J

A 4-year-old's primary upper central incisor has been avulsed.
What is correct?

A	Replant it immediately
B	Store it in milk and replant later
C	Splint for 6 months
D	Root-fill it extraorally and replant
E	Never replant an avulsed primary tooth

See answer and justification for Q779 →

Q782 FOUNDATION · Block J

A 13-year-old lost 21 in an accident and asks for an implant.
What is correct?

A	Place it immediately
B	Implants grow with the jaw
C	Age does not matter
D	An implant is the only option
E	A fixed implant is generally deferred until growth is complete

See answer and justification for Q782 →

Q789 FOUNDATION · Block J

An adult with active erosion and dentine sensitivity asks which toothpaste to use.
What is appropriate?

A	A whitening toothpaste
B	A charcoal toothpaste
C	Non-fluoride toothpaste
D	5000 ppm fluoride toothpaste
E	Baking soda only

See answer and justification for Q789 →

Q792 FOUNDATION · Block J

Dahl composites have just been placed on 13 to 23.
What is the immediate occlusal situation?

A	The posterior teeth are out of contact
B	All teeth contact evenly
C	Only molars contact
D	The OVD is reduced
E	The canines are out of contact

See answer and justification for Q792 →

Q798 FOUNDATION · Block J

A patient's full-mouth plaque score is 45% and bleeding score 50%. She wants a bridge now.
What is appropriate?

A	Proceed with the bridge
B	Use a resin-bonded bridge instead
C	Prescribe antibiotics and proceed
D	Improve hygiene and disease control first
E	Extract the abutments

See answer and justification for Q798 →

Q800 FOUNDATION · Block J

A 30-year-old missing 12 is offered orthodontic space closure.
What is the main disadvantage of this option?

A	It needs an implant
B	Time, appliances and Level 3 planning
C	It damages adjacent teeth
D	It is irreversible surgery
E	It is never possible

See answer and justification for Q800 →

BLOCK K · Implants and digital prosthodontics (13)

Q818 FOUNDATION · Block K

An edentulous patient wants an implant overdenture in the maxilla.
What is usual?

A	One implant
B	Usually 4 or more implants
C	Two implants
D	No implants
E	Ten implants

See answer and justification for Q818 →

Q823 FOUNDATION · Block K

A single implant crown has been in place for 3 years.
What are its commonest complications?

A	Implant fracture
B	Nerve injury
C	Sinus perforation
D	Bone graft failure
E	Abutment screw loosening and ceramic chipping

See answer and justification for Q823 →

Q832 FOUNDATION · Block K

A patient asks whether an implant crown will "feel" like a natural tooth.
What is correct?

A	It has a periodontal ligament
B	It has normal proprioception
C	It has no periodontal ligament or physiological movement
D	It moves more than a tooth
E	It has pulpal sensation

See answer and justification for Q832 →

Q835 FOUNDATION · Block K

A Level 2 dentist is asked to perform a sinus lift.
What is appropriate?

A	Proceed
B	Proceed under sedation
C	Proceed with a guide
D	Refer; sinus grafting and ridge augmentation are not routine Level 2 care
E	Proceed if the patient pays privately

See answer and justification for Q835 →

Q837 FOUNDATION · Block K

An intraoral scanner fails to obtain consistent margins on a deeply subgingival preparation because bleeding and soft tissue obscure the finish line. What is the most appropriate first step?

A	Proceed with the poorly defined digital margin and ask the lab to guess
B	Address isolation, haemostasis and tissue management before deciding whether scanning or a conventional impression is feasible
C	Increase the scan resolution without controlling the field
D	Move the margin apically without assessing periodontal implications
E	Use a larger shade tab as a scanner reference

See answer and justification for Q837 →

Q838 FOUNDATION · Block K

A laboratory asks for a scan in STL format.
What does an STL file contain?

A	Volumetric radiographic data
B	Colour and texture
C	Surface geometry as triangles, in an open format without colour
D	Jaw movement data
E	A patient's medical history

See answer and justification for Q838 →

Q844 FOUNDATION · Block K

A practice buys a 3D printer.
What are its typical prosthodontic uses?

A	Models, surgical guides, trays, splints and temporary crowns
B	Definitive implant bodies
C	Titanium abutments only
D	Zirconia blocks
E	Nothing clinical

See answer and justification for Q844 →

Q845 FOUNDATION · Block K

A patient asks about digital complete dentures.
What are their advantages and limitations?

A	They cannot be duplicated
B	They need more visits
C	Fewer visits, stored records and low-monomer milled bases with good fit
D	They always fit better than conventional dentures in all cases
E	They need no clinical records

See answer and justification for Q845 →

Q850 FOUNDATION · Block K

A multi-implant full-arch prosthesis needs a passive fit.
What is appropriate?

A	An intraoral scan alone without verification
B	A validated technique, with a verification jig or conventional splinted impression where indicated
C	Alginate
D	Any technique
E	Guessing

See answer and justification for Q850 →

Q853 FOUNDATION · Block K

A bruxist has a new implant crown.
What is appropriate?

A	No precautions
B	Remove the crown at night
C	A soft daytime splint
D	Layered porcelain
E	A protective night guard, as bruxism raises the risk of complications

See answer and justification for Q853 →

Q854 FOUNDATION · Block K

A patient had radiotherapy to the mandible and wants implants there.
What is correct?

A	Implants in irradiated bone have the same success
B	They are routine Level 2
C	They are contraindicated in all cases
D	Higher failure rates; Level 3 care
E	Only in the maxilla

See answer and justification for Q854 →

Q856 FOUNDATION · Block K

A patient with an HbA1c of 85 mmol/mol wants implants.
What is appropriate?

A	Proceed immediately
B	Ignore diabetes
C	Prescribe antibiotics and proceed
D	Refuse permanently
E	Recognise the raised risk and optimise glycaemic control first

See answer and justification for Q856 →

Q858 FOUNDATION · Block K

An implant has suppuration, a 7 mm pocket and radiographic bone loss.
What is appropriate?

A	Debride and review in a year
B	Refer as peri-implantitis (Level 3)
C	Ignore it
D	Retighten the screw
E	Replace the crown only

See answer and justification for Q858 →

BLOCK L · Complications, failures and maintenance (7)

Q867 FOUNDATION · Block L

A post-crowned premolar has a vertical root fracture with no ferrule.
What is the usual outcome?

A	Recementation
B	Extraction; prevention depends on ferrule and post design
C	A longer post
D	Composite repair
E	Splinting

See answer and justification for Q867 →

Q877 FOUNDATION · Block L

A patient returns three times with pain under a distal-extension RPD saddle. Repeated grinding has not helped; the framework seats accurately, but the mucosa under the saddle is mobile and the contralateral indirect retainer rises when pressure is applied. What is the most appropriate next investigation?

A	Increase retentive arm engagement on the terminal abutment
B	Assess tissue support, saddle adaptation, occlusal load and the need for a functional reline or altered-cast approach
C	Shorten the free-end saddle to minimise the area receiving load
D	Increase OVD until the saddle cannot sink
E	Remove every occlusal rest from the framework

See answer and justification for Q877 →

Q879 FOUNDATION · Block L

An edentulous patient asks how often to have her dentures checked.
What is the recommendation?

A	Every 10 years
B	Only when broken
C	At least annually, for mucosal examination, fit and occlusal wear
D	Never
E	Weekly

See answer and justification for Q879 →

Q882 FOUNDATION · Block L

A patient asks how to clean her acrylic complete dentures.
What is the correct advice?

A	Abrasive toothpaste
B	Boiling water
C	Brush with a soft brush and soap or non-abrasive cleanser over a basin of water, soak overnight, and leave them out at night
D	Bleach for metal dentures
E	Wear them day and night

See answer and justification for Q882 →

Q884 FOUNDATION · Block L

A practice wants fewer laboratory errors.
What are the best preventive measures?

A	Clear written prescriptions, inspection of impressions and casts at each stage, photographs and shade information, and agreed delivery times
B	Changing laboratory every month
C	Sending no prescription
D	Phoning the technician only
E	Accepting all work

See answer and justification for Q884 →

Q886 FOUNDATION · Block L

A patient asks how long a conventional bridge lasts.
What is the approximate benchmark survival?

A	About 50% at 10 years
B	About 99% at 20 years
C	About 89% at 10 years
D	About 20% at 5 years
E	It cannot be estimated

See answer and justification for Q886 →

Q888 FOUNDATION · Block L

A patient asks how long a modern resin-bonded bridge lasts.
What is the approximate benchmark?

A	About 20% at 5 years
B	About 80–90% at 5–10 years, with cantilevers performing well
C	About 99% at 20 years
D	About 50% at 1 year
E	They always fail

See answer and justification for Q888 →

BLOCK M · Governance, law and communication (15)

Q899 FOUNDATION · Block M

A 17-year-old wants an RBB and her parent disagrees.
What is the legal position?

A	Parents must consent
B	16–17-year-olds are presumed to have capacity to consent (Family Law Reform Act 1969)
C	Consent is impossible under 18
D	A court must decide
E	The dentist decides

See answer and justification for Q899 →

Q904 FOUNDATION · Block M

A dentist is planning CPD for the next cycle.
What are the enhanced CPD requirements?

A	At least 100 verifiable hours per 5-year cycle, at least 10 hours in every 2-year period, with a personal development plan
B	50 hours per year
C	No minimum
D	250 hours per 5 years
E	10 hours per 5 years

See answer and justification for Q904 →

Q905 FOUNDATION · Block M

A newly qualified dentist asks about indemnity.
What is correct?

A	It is optional
B	Appropriate indemnity is mandatory
C	Only needed for implants
D	The NHS always covers it
E	Only for private work

See answer and justification for Q905 →

Q907 FOUNDATION · Block M

After an error, a dentist apologises to the patient.
What is the legal status of an apology?

A	It admits negligence
B	It must be avoided
C	It voids indemnity
D	It must be in writing only
E	An apology is not an admission of liability

See answer and justification for Q907 →

Q918 FOUNDATION · Block M

A patient requests a copy of her records.
What does UK GDPR require?

A	A response within one month, normally free of charge
B	A response within 6 months with a fee
C	Refusal
D	Only summaries
E	A response within a year

See answer and justification for Q918 →

Q921 FOUNDATION · Block M

A new practice installs intraoral X-ray sets.
What does IRR17 require of the employer?

A	Nothing beyond buying the equipment
B	Registering with the GDC
C	Patient consent forms only
D	Registering with the HSE, appointing a radiation protection adviser and supervisor, and setting local rules and a controlled area
E	A CBCT unit

See answer and justification for Q921 →

Q922 FOUNDATION · Block M

You set up the controlled area for an intraoral X-ray set.
What is the typical extent?

A	The whole building
B	Within about 1.5 m of the tube and patient, and within the primary beam until it is attenuated
C	10 m
D	Only the patient's head
E	No area is needed

See answer and justification for Q922 →

Q925 FOUNDATION · Block M

You check the steam steriliser's display.
What is the standard sterilisation cycle?

A	134–137°C for at least 3 minutes
B	121°C for 1 minute
C	100°C for 10 minutes
D	160°C dry heat for 2 hours only
E	80°C for 30 minutes

See answer and justification for Q925 →

Q927 FOUNDATION · Block M

You review the practice's instrument cleaning.
What is best practice under HTM 01-05?

A	Manual scrubbing only
B	No cleaning before sterilising
C	A washer-disinfector
D	Soaking in bleach
E	Rinsing under the tap

See answer and justification for Q927 →

Q928 FOUNDATION · Block M

A nurse asks which steriliser tests are done daily.
What is correct?

A	None
B	An automatic control test, plus a steam penetration test for vacuum sterilisers
C	Annual validation only
D	A spore test every cycle
E	A weekly visual check

See answer and justification for Q928 →

Q932 FOUNDATION · Block M

A newly employed dentist will perform exposure-prone procedures.
What is required?

A	Nothing
B	An annual flu vaccine only
C	A chest radiograph
D	Hepatitis B vaccination with a documented response
E	A blood test for anaemia

See answer and justification for Q932 →

Q935 FOUNDATION · Block M

A wheelchair user cannot reach your upstairs surgery.
What does the Equality Act 2010 require?

A	Nothing
B	Refusal of care
C	A higher fee
D	Referral only to hospital
E	Reasonable adjustments for a person with a protected characteristic (disability)

See answer and justification for Q935 →

Q937 FOUNDATION · Block M

A dentist checks her RPD records against a published standard.
What is this activity?

A	Research
B	Audit, measuring practice against a standard
C	Service evaluation
D	A clinical trial
E	A case report

See answer and justification for Q937 →

Q944 FOUNDATION · Block M

A treatment reduces failure from 30% to 20%.
What is the number needed to treat?

A	1
B	3
C	5
D	10
E	20

See answer and justification for Q944 →

Q947 FOUNDATION · Block M

A dentist applies to provide Level 2 prosthodontics.
What is required?

A	Nothing beyond registration
B	A formal link with a consultant-led managed clinical network, a minimum caseload, outcome recording (PROMs and PREMs), audit and complication reporting
C	A specialist title
D	A PhD
E	Hospital employment

See answer and justification for Q947 →

BLOCK N · Integrated cases (10)

Q968 FOUNDATION · Block N

An NHS patient writes to complain about the cost and fit of her new denture.
What is the first required step?

A	Ignore it until she phones
B	Refund immediately without discussion
C	Acknowledge it within 3 working days and investigate locally
D	Refer it to the GDC
E	Refer it to the Ombudsman at once

See answer and justification for Q968 →

Q980 FOUNDATION · Block N

A trial reports that a new denture adhesive reduces denture failure from 40% to 30%.
What is the number needed to treat?

A	1
B	3
C	4
D	30
E	10

See answer and justification for Q980 →

Q981 FOUNDATION · Block N

A colleague requests CBCT at every annual implant review "to be safe".
What is correct?

A	Each exposure must be justified individually
B	CBCT is the routine standard
C	No radiographs are ever needed
D	A panoramic radiograph every 3 months
E	Patient preference decides

See answer and justification for Q981 →

Q982 FOUNDATION · Block N

A patient asks the practice to delete all her records because she is moving abroad.
What is the correct response?

A	Delete them immediately
B	Delete only the radiographs
C	Give her the original records
D	Explain that records must be retained (adults at least 11 years after last treatment), and offer copies
E	Shred them after a month

See answer and justification for Q982 →

Q983 FOUNDATION · Block N

A 17-year-old with capacity wants a resin-bonded bridge. Her mother refuses.
What is the correct position?

A	The mother's refusal stands
B	The 17-year-old is presumed to have capacity and can consent
C	A court order is needed
D	Wait until she is 18
E	The dentist decides

See answer and justification for Q983 →

Q986 FOUNDATION · Block N

You take an impression for a patient with known hepatitis B.
How should the impression be handled?

A	Autoclave it
B	Treat it as high risk and refuse to send it
C	Send it untreated with a warning label
D	Bag it without disinfection
E	Standard precautions: rinse, disinfect per protocol, label and send, as for every patient

See answer and justification for Q986 →

Q991 FOUNDATION · Block N

A patient with a severe gag reflex has never tolerated an upper complete denture.
Which option may help?

A	A thicker palate
B	An implant-retained palateless prosthesis (after referral), with desensitisation
C	A longer post-dam
D	A soft liner
E	Adhesive only

See answer and justification for Q991 →

Q992 FOUNDATION · Block N

A patient asks whether her resin-bonded bridge has been a "success" because it is still present after 6 years, though it has debonded twice.
What is the most accurate explanation?

A	It is a complete success
B	It is a failure
C	Debonding does not count
D	Survival and success are identical
E	It has survived, but success means in place and free of complications

See answer and justification for Q992 →

Q997 FOUNDATION · Block N

A patient on dual antiplatelet therapy needs a crown preparation with retraction cord.
What is correct?

A	Stop clopidogrel for 7 days
B	Continue both antiplatelets
C	Stop aspirin only
D	Refer to hospital
E	Give vitamin K

See answer and justification for Q997 →

Q999 FOUNDATION · Block N

An audit showed incomplete RPD prescriptions. The practice introduced a template.
What is the next step?

A	Re-audit to close the loop
B	Publish the results
C	Stop auditing
D	Change laboratory
E	Discipline the staff

See answer and justification for Q999 →

Intermediate questions · 730 items

BLOCK A · Applied anatomy (35)

Q1 INTERMEDIATE · Block A

A 58-year-old bruxist has firm, bilateral swellings at the angles of the mandible that bulge on clenching. You suspect masseteric hypertrophy.
Which nerve supplies the affected muscle?

A	Buccal nerve
B	Masseteric nerve, from the anterior division of V3, passing through the mandibular notch
C	Facial nerve
D	Auriculotemporal nerve
E	Nerve to medial pterygoid

See answer and justification for Q1 →

Q4 INTERMEDIATE · Block A

A patient has lost function of the right nerve to lateral pterygoid after surgery.
On wide opening, how will the mandible move?

A	It deviates to the left
B	It opens straight
C	It deviates to the right
D	It cannot open at all
E	It protrudes without opening

See answer and justification for Q4 →

Q6 INTERMEDIATE · Block A

A 41-year-old has 22 mm of pain-free opening, then a click and further opening to 45 mm.
Which movement occurs in the lower (condyle–disc) compartment during the first 20–25 mm?

A	Translation
B	Rotation (hinge movement)
C	Lateral shift only
D	Protrusion
E	No movement

See answer and justification for Q6 →

Q7 INTERMEDIATE · Block A

A patient with TMD describes earache with no ear disease on examination.
Which nerve provides the main sensory supply to the TMJ and explains the referral?

A	Facial nerve
B	Great auricular nerve
C	Chorda tympani
D	Masseteric nerve alone
E	Auriculotemporal nerve

See answer and justification for Q7 →

Q8 INTERMEDIATE · Block A

A 46-year-old with a unilateral ache just anterior to the ear reports pain when chewing and tenderness on palpation over the joint, but no dental pain. The otoscopic examination is normal. Palpation of the temporalis is also mildly tender. Which finding would most strongly support a primary TMJ rather than solely myofascial source?

A	Pain aggravated by ipsilateral joint loading with reproducible preauricular tenderness
B	A history of intermittent tension headache without pain on jaw movement
C	A tender temporalis belly but painless joint compression
D	Pain triggered only by cold drinks to a restored molar
E	A brief electric shock from light cheek contact

See answer and justification for Q8 →

Q9 INTERMEDIATE · Block A

You are reviewing TMJ anatomy before examining a patient with a dislocation history.
Which ligament is the true stabiliser of the joint, limiting posterior and inferior movement of the condyle?

A	Sphenomandibular ligament
B	Stylomandibular ligament
C	Pterygomandibular raphe
D	Lateral (temporomandibular) ligament
E	Stylohyoid ligament

See answer and justification for Q9 →

Q10 INTERMEDIATE · Block A

A dental student asks why the TMJ heals and remodels differently from other synovial joints.
What covers the articular surfaces of the TMJ?

A	Hyaline cartilage
B	Elastic cartilage
C	Fibrocartilage
D	Synovial membrane
E	Periosteum only

See answer and justification for Q10 →

Q12 INTERMEDIATE · Block A

A new lower complete denture lifts every time the patient protrudes the tongue. The anterior lingual flange is over-extended.
Which muscle is displacing it, and what is its nerve supply?

A	Mylohyoid; nerve to mylohyoid
B	Genioglossus; hypoglossal nerve
C	Geniohyoid; C1 via the hypoglossal nerve
D	Palatoglossus; vagus
E	Buccinator; facial nerve

See answer and justification for Q12 →

Q13 INTERMEDIATE · Block A

A lower denture lifts when the patient swallows and the lingual flange is excessively deep in the posterior molar region. Which structure is most implicated?

A	Loss of posterior palatal seal at the vibrating line
B	Coronoid interference with a maxillary distobuccal flange
C	Masseteric action against the mandibular distobuccal flange
D	Pressure over a superficial mental foramen on a resorbed ridge
E	Mylohyoid elevation affecting lingual flange movement

See answer and justification for Q13 →

Q17 INTERMEDIATE · Block A

On the master cast of an edentulous maxilla, you identify the hamular (pterygomaxillary) notches. What do they mark?

A	The posterior limit of the tuberosity and the lateral extent of the post-dam
B	The anterior limit of the soft palate
C	The parotid duct opening
D	The position of the greater palatine foramen
E	The buccal frenum

See answer and justification for Q17 →

Q18 INTERMEDIATE · Block A

A 76-year-old with a severely resorbed mandible has tingling in the lower lip when chewing with a new denture.
What is the most likely cause?

A	Pressure on the lingual nerve
B	Pressure on the mental nerve where the foramen now lies near the ridge crest
C	Pressure on the buccal nerve
D	Pressure on the nerve to mylohyoid
E	An allergy to the denture base

See answer and justification for Q18 →

Q19 INTERMEDIATE · Block A

A patient has a burning pain behind the upper incisors under a new upper denture. The mucosa there is slightly red.
Which area needs relief?

A	Rugae
B	Fovea palatini
C	Midline raphe posteriorly
D	Zygomatic buttress
E	Incisive papilla (nasopalatine nerve)

See answer and justification for Q19 →

Q21 INTERMEDIATE · Block A

A 45-year-old has an abscess on a lower second molar, with swelling below the lower border of the mandible. Which space is involved?

A	Sublingual space
B	Buccal space
C	Canine space
D	Submandibular space
E	Submental space

See answer and justification for Q21 →

Q22 INTERMEDIATE · Block A

A lower first premolar abscess has produced a swelling in the floor of the mouth, above the mylohyoid. Which space is involved?

A	Sublingual space
B	Submandibular space
C	Submasseteric space
D	Pterygomandibular space
E	Parapharyngeal space

See answer and justification for Q22 →

Q23 INTERMEDIATE · Block A

A 50-year-old presents with bilateral swelling below the mandible and under the chin, a raised floor of the mouth, drooling and a muffled voice.
What is the most appropriate action?

A	Antibiotics and review in 48 hours
B	Incise and drain intraorally in the practice
C	Extract the cause and send the patient home
D	Warm saline mouthwashes
E	Emergency referral for airway management

See answer and justification for Q23 →

Q25 INTERMEDIATE · Block A

You explain to a colleague why infection in the infratemporal region is dangerous.
How does the pterygoid venous plexus communicate with the cavernous sinus?

A	Through the facial artery
B	Through the parotid gland
C	Through emissary veins
D	Through the external jugular vein
E	There is no communication

See answer and justification for Q25 →

Q27 INTERMEDIATE · Block A

After an infected lower third molar, a patient has marked trismus and swelling over the angle of the mandible.
Which space is most likely involved?

A	Submasseteric space
B	Canine space
C	Sublingual space
D	Submental space
E	Space of the body of the mandible

See answer and justification for Q27 →

Q33 INTERMEDIATE · Block A

During palatal surgery, brisk bleeding occurs from the greater palatine foramen.
From which vessel does the greater palatine artery arise?

A	Facial artery
B	Lingual artery
C	First part of the maxillary artery
D	Descending palatine artery (third part of the maxillary artery)
E	Ascending pharyngeal artery

See answer and justification for Q33 →

Q36 INTERMEDIATE · Block A

A patient with a fractured mandibular angle has displacement resisted on both surfaces of the ramus by two muscles.
Which pair forms the sling around the angle of the mandible?

A	Masseter and medial pterygoid
B	Temporalis and lateral pterygoid
C	Masseter and temporalis
D	Medial and lateral pterygoids
E	Mylohyoid and digastric

See answer and justification for Q36 →

Q42 INTERMEDIATE · Block A

You are explaining soft-palate anatomy before a post-dam is cut.
Which soft palate muscle is not supplied by the pharyngeal plexus?

A	Levator veli palatini
B	Musculus uvulae
C	Tensor veli palatini (nerve to medial pterygoid, V3)
D	Palatopharyngeus
E	Palatoglossus

See answer and justification for Q42 →

Q43 INTERMEDIATE · Block A

After a Bell's palsy, a lower denture wearer finds food collecting in the buccal sulcus and the denture less stable.
Which weakened muscle explains this?

A	Buccinator
B	Masseter
C	Mylohyoid
D	Medial pterygoid
E	Temporalis

See answer and justification for Q43 →

Q44 INTERMEDIATE · Block A

A patient with a severely atrophic mandible has focal lower-lip paraesthesia while wearing a new denture; palpation reveals a prominent foramen near the crest. Which structure needs immediate attention?

A	Mylohyoid elevation affecting lingual flange movement
B	Pressure over a superficial mental foramen on a resorbed ridge
C	Loss of posterior palatal seal at the vibrating line
D	Coronoid interference with a maxillary distobuccal flange
E	Masseteric action against the mandibular distobuccal flange

See answer and justification for Q44 →

Q46 INTERMEDIATE · Block A

A patient with facial nerve palsy is examined for swallowing problems.
Which hyoid muscle is supplied by the facial nerve?

A	Mylohyoid
B	Anterior belly of digastric
C	Geniohyoid
D	Thyrohyoid
E	Posterior belly of digastric (and stylohyoid)

See answer and justification for Q46 →

Q48 INTERMEDIATE · Block A

A 72-year-old has a moderately resorbed mandible.
Which area provides the primary support for the lower complete denture?

A	The buccal shelf
B	The genial tubercles
C	The lingual sulcus
D	The mylohyoid ridge
E	The labial vestibule

See answer and justification for Q48 →

Q50 INTERMEDIATE · Block A

A new lower denture lifts at the back each time the patient opens wide. The base extends over the whole retromolar pad.
Which structure is tightening against the over-extension?

A	Mylohyoid
B	Pterygomandibular raphe
C	Masseter
D	Buccinator
E	Genioglossus

See answer and justification for Q50 →

Q56 INTERMEDIATE · Block A

A patient has reduced submandibular flow after a middle-ear operation.
Which parasympathetic pathway supplies the submandibular gland?

A	Glossopharyngeal nerve, otic ganglion
B	Vagus nerve
C	Facial nerve, chorda tympani, lingual nerve, submandibular ganglion
D	Lesser petrosal nerve, auriculotemporal nerve
E	Hypoglossal nerve

See answer and justification for Q56 →

Q57 INTERMEDIATE · Block A

A patient has a ranula-like swelling in the floor of the mouth.
Which statement about the sublingual gland is correct?

A	It is purely serous
B	It drains through a single duct opposite the upper second molar
C	It is supplied by the otic ganglion
D	It produces most of the stimulated saliva
E	It is mainly mucous and drains through multiple ducts of Rivinus on the sublingual fold

See answer and justification for Q57 →

Q61 INTERMEDIATE · Block A

A new lower denture is retentive at rest but is displaced whenever the patient speaks or smiles. What should you do first?

A	Recommend a denture adhesive
B	Assess the border extension and polished-surface contours against muscle action
C	Adjust the occlusion
D	Remake the denture at once
E	Increase the OVD

See answer and justification for Q61 →

Q62 INTERMEDIATE · Block A

An upper denture is retentive until the patient pronounces “ah”; the post-dam does not correspond to the functional soft-palate transition. What area requires reassessment?

A	Mylohyoid elevation affecting lingual flange movement
B	Loss of posterior palatal seal at the vibrating line
C	Masseteric action against the mandibular distobuccal flange
D	Coronoid interference with a maxillary distobuccal flange
E	Pressure over a superficial mental foramen on a resorbed ridge

See answer and justification for Q62 →

Q63 INTERMEDIATE · Block A

A 69-year-old wearing a lower denture has new unilateral lower-lip paraesthesia for five weeks. The denture has been relieved over the mental foramen without improvement, and there is no clear traumatic lesion. Which next step is most appropriate?

A	Repeat relief and reassess in six months
B	Prescribe antifungal treatment for presumed denture stomatitis
C	Arrange prompt assessment for unexplained neurosensory change and potential serious pathology
D	Replace the lower denture without further diagnostic investigation
E	Label the finding as a normal consequence of age-related ridge resorption

See answer and justification for Q63 →

Q64 INTERMEDIATE · Block A

A patient with a lower molar infection has a temperature of 38.6°C, trismus of 15 mm and difficulty swallowing.
What is the most appropriate action?

A	Prescribe amoxicillin and review in a week
B	Extract the tooth under LA in the practice
C	Advise hot mouthwashes
D	Refer urgently to the maxillofacial team
E	Prescribe metronidazole only

See answer and justification for Q64 →

Q65 INTERMEDIATE · Block A

An upper complete denture dislodges only when the patient opens widely; the distobuccal flange is thick and close to the ascending ramus. Which structure is most likely?

A	Mylohyoid elevation affecting lingual flange movement
B	Coronoid interference with a maxillary distobuccal flange
C	Loss of posterior palatal seal at the vibrating line
D	Pressure over a superficial mental foramen on a resorbed ridge
E	Masseteric action against the mandibular distobuccal flange

See answer and justification for Q65 →

Q66 INTERMEDIATE · Block A

You are deciding the posterior extent of a new upper complete denture.
What should be assessed before planning the posterior palatal seal?

A	Rugae pattern
B	Tongue size
C	Soft-palate form and movement at the vibrating line
D	Lip length
E	Shade of the mucosa

See answer and justification for Q66 →

Q69 INTERMEDIATE · Block A

A patient with an inferior alveolar block infection develops severe trismus and dysphagia. Infection in the pterygomandibular space can spread directly to which space?

A	Canine space
B	Submental space
C	Sublingual space
D	Buccal space
E	Lateral pharyngeal space

See answer and justification for Q69 →

Q70 INTERMEDIATE · Block A

A mandibular denture is displaced when the patient clenches and has soreness at its distobuccal corner. Which muscle is acting on the border?

A	Pressure over a superficial mental foramen on a resorbed ridge
B	Loss of posterior palatal seal at the vibrating line
C	Masseteric action against the mandibular distobuccal flange
D	Mylohyoid elevation affecting lingual flange movement
E	Coronoid interference with a maxillary distobuccal flange

See answer and justification for Q70 →

BLOCK B · Physiology and pathology (35)

Q71 INTERMEDIATE · Block B

A 62-year-old reports bilateral burning of the tongue for nine months, often worsening through the day. The oral mucosa appears normal, but the patient has recently started an anticholinergic antidepressant and has reduced salivary flow. Which approach is most defensible before diagnosing primary burning mouth disorder?

A	Begin repeated empirical antifungal courses without examining saliva or medications
B	Exclude secondary causes including xerostomia, medications, haematinic deficiency and metabolic disease before diagnosing a primary neuropathic disorder
C	Perform irreversible occlusal adjustment because the symptom affects the tongue
D	Make an immediate complete denture irrespective of current dentition
E	Diagnose pemphigus vulgaris despite no oral erosions or blisters

See answer and justification for Q71 →

Q73 INTERMEDIATE · Block B

A 58-year-old has painful widespread gingival and palatal erosions, fragile blisters, and new ocular irritation. A dentist is considering erosive lichen planus, mucous membrane pemphigoid and pemphigus vulgaris. What is the most appropriate next action?

A	Prescribe a new denture and review after six months
B	Assume that all three conditions have identical risk and review requirements
C	Arrange timely specialist oral-medicine assessment with consideration of biopsy/immunopathology and urgent evaluation of ocular involvement
D	Provide long-term systemic corticosteroids before obtaining a diagnosis
E	Observe without investigations because oral blistering diseases do not scar

See answer and justification for Q73 →

Q74 INTERMEDIATE · Block B

A 28-year-old with asthma reports a white removable palatal coating and mild soreness. Medication reconciliation finds daily inhaled beclometasone and as-needed salbutamol. The patient does not rinse after the preventer inhaler. What is the best initial integrated management?

A	Replace the daily steroid preventer with more frequent salbutamol without involving the asthma clinician
B	Prescribe systemic antifungal therapy without addressing steroid deposition or examining the lesion
C	Recommend rinsing after each salbutamol dose only, since the preventer is non-irritating
D	Correct steroid inhaler technique and rinsing, and evaluate/manage the confirmed candidiasis at the same time
E	Provide antifungal therapy alone while maintaining the same inhaler technique and risk exposure

See answer and justification for Q74 →

Q75 INTERMEDIATE · Block B

A 75-year-old denture wearer has a uniformly erythematous hard palate beneath an upper denture worn overnight. There are no removable white plaques, and the patient is otherwise well. Which diagnosis and first plan fit best?

A	Pseudomembranous candidiasis requiring immediate systemic antifungal therapy
B	Primary herpetic gingivostomatitis requiring antivirals
C	Erythematous denture-associated stomatitis: correct denture hygiene, night removal and fit, and add appropriate antifungal treatment if indicated
D	Erosive lichen planus diagnosed purely from its distribution under the denture base
E	Leukoplakia requiring excision without considering denture trauma

See answer and justification for Q75 →

Q78 INTERMEDIATE · Block B

A 70-year-old with new complete dentures says they cannot chew steak as they did with natural teeth. About what proportion of dentate bite force and chewing efficiency do complete denture wearers achieve?

A	The same
B	About half
C	About one-fifth to one-sixth
D	About one-third
E	About one-twentieth

See answer and justification for Q78 →

Q79 INTERMEDIATE · Block B

You are explaining why complete denture wearers find fine biting control harder than dentate patients. On which receptors does a complete denture wearer depend for feedback?

A	Periodontal ligament mechanoreceptors
B	Mucosal and TMJ receptors
C	Pulpal receptors
D	Osseoperception
E	Muscle spindles alone

See answer and justification for Q79 →

Q80 INTERMEDIATE · Block B

A patient with an implant crown asks whether the implant will "feel" like a tooth. Compared with a natural tooth, feedback from an implant (osseoperception) is:

A	Identical
B	More sensitive
C	Absent
D	Less sensitive
E	Thermal only

See answer and justification for Q80 →

Q81 INTERMEDIATE · Block B

A healthy 72-year-old has lost all four first and second molars but has sound premolars and anterior teeth in both arches. She is comfortable and chews well.
What is the most appropriate advice?

A	A removable partial denture to replace the molars
B	Implants to replace the molars
C	Cantilever bridges from the premolars
D	Extraction of the remaining teeth for complete dentures
E	No replacement

See answer and justification for Q81 →

Q83 INTERMEDIATE · Block B

An 84-year-old with Parkinson's disease has worn the same well-liked complete dentures for 12 years. They are worn and loose.
Which approach best suits her ability to adapt?

A	New dentures of a completely new design
B	Implant-retained overdentures
C	Immediate dentures
D	Copy (replica) dentures
E	No dentures

See answer and justification for Q83 →

Q84 INTERMEDIATE · Block B

A patient edentulous for 30 years appears Class III, with the upper denture teeth set inside the lower ridge.
What explains this appearance?

A	The maxilla resorbs upwards and inwards, and the mandible downwards and outwards
B	The maxilla keeps growing
C	The mandible resorbs inwards (centripetally)
D	TMJ remodelling only
E	Wear of the denture teeth

See answer and justification for Q84 →

Q88 INTERMEDIATE · Block B

A patient has worn the same ill-fitting dentures day and night for 15 years.
Which habit is most likely to have accelerated ridge resorption?

A	Continuous 24-hour wear of ill-fitting dentures
B	Removing the dentures at night
C	Regular relines
D	Well-fitting dentures
E	Implants

See answer and justification for Q88 →

Q90 INTERMEDIATE · Block B

An 80-year-old drinks sweet tea through the day and has several root caries lesions.
At about what pH does root dentine begin to demineralise?

A	4.5
B	5.5
C	6.2–6.7
D	7.4
E	3.0

See answer and justification for Q90 →

Q94 INTERMEDIATE · Block B

A deep carious lesion on 36 is close to the pulp. The tooth is vital with no symptoms of irreversible pulpitis.
What is the most appropriate caries removal approach?

A	Remove all softened dentine to hard dentine over the pulp
B	Extract the tooth
C	Root canal treatment
D	Leave infected dentine at the periphery
E	Remove infected dentine, leave affected dentine near the pulp, and seal

See answer and justification for Q94 →

Q95 INTERMEDIATE · Block B

A patient begins antimuscarinic therapy and develops rapid root caries plus poor denture retention. What oral physiologic effect most directly links the problems?

A	Xerostomia reduces buffering and clearance
B	Occlusal wear may be compensated by dentoalveolar change
C	RPD components can increase local plaque retention
D	Repeated sugar exposure increases time below the critical pH
E	Exposed root surfaces demineralise at a higher pH than enamel

See answer and justification for Q95 →

Q98 INTERMEDIATE · Block B

A 21-year-old has palatal erosion of the upper anterior teeth, bilateral parotid enlargement and calluses on her knuckles.
What is the most likely cause?

A	Dietary erosion from fruit juice
B	Bulimia nervosa
C	Toothbrush abrasion
D	Bruxism
E	Abfraction

See answer and justification for Q98 →

Q102 INTERMEDIATE · Block B

A bruxist has wedge-shaped cervical lesions with sharp internal angles, some extending below the gingival margin, on teeth with heavy lateral contacts.
Which process is proposed to explain these lesions?

A	Erosion
B	Abrasion
C	Root caries
D	Abfraction (flexural stress at the cervical margin)
E	Attrition

See answer and justification for Q102 →

Q103 INTERMEDIATE · Block B

A 55-year-old has slow, localised wear of the upper anterior teeth over many years, yet the posterior teeth still meet and there is no space to restore the worn teeth.
Why is there no interocclusal space?

A	Compensatory alveolar growth has kept the worn teeth in contact
B	The OVD has been lost
C	TMD has developed
D	The bruxism has stopped
E	The posterior teeth have intruded

See answer and justification for Q103 →

Q106 INTERMEDIATE · Block B

You suspect an eating disorder in a 19-year-old with palatal erosion, who denies vomiting.
What is the most appropriate approach?

A	Confront her with the diagnosis
B	Approach with empathy, liaise with her GMP with consent, and focus on prevention and monitoring before restoration
C	Place crowns immediately
D	Contact the police
E	Take no action

See answer and justification for Q106 →

Q107 INTERMEDIATE · Block B

An upper denture wearer has nodular, papillary tissue in the centre of the palate under the denture.
Which Newton type of denture stomatitis is this?

A	Type III (papillary hyperplasia)
B	Type I (pinpoint hyperaemia)
C	Type II (diffuse erythema)
D	Type IV
E	It is not denture stomatitis

See answer and justification for Q107 →

Q108 INTERMEDIATE · Block B

A 75-year-old on warfarin has diffuse asymptomatic erythema under a maxillary acrylic complete denture. The denture is worn continuously, is visibly plaque-contaminated and has a poor fit; no extraoral features are present. Which intervention is most appropriate first?

A	Prescribe miconazole oral gel as sole treatment without reviewing interacting medicines
B	Arrange urgent systemic fluconazole treatment before checking hygiene
C	Treat the erythema with topical corticosteroids alone
D	Replace the denture without addressing continuous wear or cleaning
E	Institute denture hygiene, overnight removal and fit correction; use a suitable antifungal if indicated after interaction review

See answer and justification for Q108 →

Q109 INTERMEDIATE · Block B

A patient reports sore white patches on the palate after a change in asthma medication. The list includes salbutamol for relief and newly added beclometasone for maintenance; the patches wipe off to expose erythema. Which prevention advice is most directly relevant to the likely medication-related mechanism?

A	Rinse and gargle with water, then spit, after the corticosteroid inhaler and assess technique or spacer use
B	Stop salbutamol, as it directly causes steroid-associated thrush
C	Use antibiotics following every inhaled dose
D	Replace all topical treatment with an immediate systemic antifungal
E	Avoid rinsing because the inhaled steroid must remain on the oral mucosa

See answer and justification for Q109 →

Q111 INTERMEDIATE · Block B

A patient with angular cheilitis grows both Candida and Staphylococcus aureus from swabs. He is not on any interacting drugs.
Which topical treatment covers both organisms?

A	Miconazole cream
B	Nystatin cream alone
C	Hydrocortisone cream alone
D	Aciclovir cream
E	Chlorhexidine gel

See answer and justification for Q111 →

Q112 INTERMEDIATE · Block B

A leaf of firm tissue lies in the lower labial sulcus under an over-extended flange of a 10-year-old denture. What is the most appropriate management?

A	Excise it immediately without adjusting the denture
B	Ignore it
C	Prescribe an antifungal
D	Trim the flange, use a tissue conditioner, and excise with histology if it does not resolve
E	Prescribe a topical steroid

See answer and justification for Q112 →

Q113 INTERMEDIATE · Block B

A patient with recession develops root caries at a near-neutral pH that would not usually dissolve sound enamel. Why?

A	Occlusal wear may be compensated by dentoalveolar change
B	Exposed root surfaces demineralise at a higher pH than enamel
C	RPD components can increase local plaque retention
D	Repeated sugar exposure increases time below the critical pH
E	Xerostomia reduces buffering and clearance

See answer and justification for Q113 →

Q114 INTERMEDIATE · Block B

An edentulous upper anterior ridge is mobile and displaceable. The patient has natural lower anterior teeth. Which statement is correct?

A	It should always be excised
B	It is bone hypertrophy
C	It is caused by Candida
D	It should be compressed in a putty impression
E	It is fibrous replacement of bone, best recorded with a mucostatic or selective-pressure impression

See answer and justification for Q114 →

Q115 INTERMEDIATE · Block B

A 62-year-old smoker has a well-demarcated red patch on the floor of the mouth that does not wipe off. Which lesion carries the highest risk of malignant transformation?

A	Homogeneous leukoplakia
B	Frictional keratosis
C	Erythroplakia
D	Linea alba
E	Geographic tongue

See answer and justification for Q115 →

Q116 INTERMEDIATE · Block B

You are performing a soft-tissue examination on a high-risk patient.
Which sites carry the highest risk of oral squamous cell carcinoma?

A	Hard palate and gingiva
B	Dorsum of the tongue and lips
C	Buccal mucosa and palate
D	Lateral border of the tongue and floor of the mouth
E	Retromolar pad only

See answer and justification for Q116 →

Q119 INTERMEDIATE · Block B

A 79-year-old's teeth respond weakly to cold testing, and root canal access on a molar proves difficult. What explains both findings?

A	Pulp necrosis
B	Secondary dentine deposition
C	Allergy
D	Root resorption
E	Enamel hypoplasia

See answer and justification for Q119 →

Q120 INTERMEDIATE · Block B

A patient has substantial tooth wear but retains familiar OVD and no interocclusal restorative space. What explains the apparently preserved bite height?

A	Exposed root surfaces demineralise at a higher pH than enamel
B	Repeated sugar exposure increases time below the critical pH
C	RPD components can increase local plaque retention
D	Xerostomia reduces buffering and clearance
E	Occlusal wear may be compensated by dentoalveolar change

See answer and justification for Q120 →

Q122 INTERMEDIATE · Block B

A recently fitted RPD wearer develops a carious lesion adjacent to a clasped abutment while other teeth remain stable. What local factor should be assessed?

A	Occlusal wear may be compensated by dentoalveolar change
B	RPD components can increase local plaque retention
C	Repeated sugar exposure increases time below the critical pH
D	Exposed root surfaces demineralise at a higher pH than enamel
E	Xerostomia reduces buffering and clearance

See answer and justification for Q122 →

Q124 INTERMEDIATE · Block B

An 86-year-old care-home resident with poor dexterity needs replacement of missing teeth. Attendance and cleaning are unreliable.
Which approach is most appropriate?

A	A long-span fixed bridge
B	Multiple implants
C	Precision-attachment RPD
D	A simple, hygienic, maintainable design
E	Full-mouth crowns

See answer and justification for Q124 →

Q126 INTERMEDIATE · Block B

A 60-year-old has lost 47 only. There is no drifting or over-eruption and he has no complaints.
What is the most appropriate advice?

A	No replacement is a valid option
B	An implant is essential
C	A bridge is essential
D	An RPD is essential
E	Extract 17

See answer and justification for Q126 →

Q128 INTERMEDIATE · Block B

A 77-year-old taking oxybutynin and amitriptyline has dry mouth, new active root caries and an otherwise serviceable cobalt–chromium RPD. She asks for replacement of the denture because it feels less retentive.
What is the most defensible initial plan?

A	Replace the Co–Cr RPD immediately with an acrylic appliance
B	Start an irreversible full-mouth crown programme to protect exposed roots
C	Use an acidic saliva substitute to increase flow before any risk assessment
D	Assess and address medication-related xerostomia, caries risk, hygiene and salivary support before revising the prosthesis
E	Stop both medicines without contacting the prescribing clinician

See answer and justification for Q128 →

Q129 INTERMEDIATE · Block B

Two patients consume similar total sugar, but one sips sweet tea every 15 minutes. Why is that patient at higher caries risk?

A	RPD components can increase local plaque retention
B	Repeated sugar exposure increases time below the critical pH
C	Exposed root surfaces demineralise at a higher pH than enamel
D	Xerostomia reduces buffering and clearance
E	Occlusal wear may be compensated by dentoalveolar change

See answer and justification for Q129 →

Q130 INTERMEDIATE · Block B

A patient with frequent sugar intake and low saliva develops rapidly progressive caries.
Which concept explains the shift towards acid-producing species?

A	The specific plaque hypothesis alone
B	The non-specific plaque hypothesis
C	The ecological plaque hypothesis
D	The Stephan curve
E	The focal infection theory

See answer and justification for Q130 →

BLOCK C · Dental materials science (70)

Q131 INTERMEDIATE · Block C

A surveyed mandibular premolar has a favourable mesial rest and guide plane. The planned retentive arm is a conventional short cast cobalt–chromium circumferential clasp, with a measured buccal undercut of 0.50 mm. There is adequate space to revise the survey line by conservative enamel recontouring. Which design decision is most defensible?

A	Retain the 0.50 mm undercut but make the same cast Co–Cr arm slightly shorter to control flexibility
B	Engage the 0.50 mm undercut with the existing cast arm, provided the reciprocal component is moved closer to the gingiva
C	Re-survey and make a conservative contour change to achieve a shallower usable undercut for the specified cast arm
D	Use the 0.50 mm undercut with the same Co–Cr arm but compensate with an additional indirect retainer
E	Increase the terminal taper of the same arm without evaluating the resultant peak strain or altered clasp path

See answer and justification for Q131 →

Q132 INTERMEDIATE · Block C

A complex wax pattern is left overnight before investing and the casting subsequently rocks on the die. Which mechanism is most likely?

A	Excess spatulation alters setting kinetics and expansion
B	Insufficient investment expansion produces an undersized casting
C	Delayed investing permits wax pattern stress relaxation
D	Excess water–powder ratio reduces cast strength
E	High-temperature breakdown of gypsum-bonded investment

See answer and justification for Q132 →

Q134 INTERMEDIATE · Block C

An acrylic batch was left uncovered on the bench beyond its appropriate dough stage before packing. The finished base has a grainy, chalky surface containing numerous small defects. Which mechanism best fits?

A	Gaseous porosity from excessive temperature rise during cure
B	Granular porosity from loss of monomer during the dough stage
C	Residual monomer from incomplete polymerisation
D	Contraction porosity from inadequate packing or pressure
E	Fatigue crack growth under repeated functional flexure

See answer and justification for Q134 →

Q135 INTERMEDIATE · Block C

A cobalt–chromium occlusally approaching clasp originally engaged a 0.25 mm undercut. After repeated chairside bending to increase retention, the patient describes a loose RPD and the clasp has a whitish line at the origin of the retentive arm. Which explanation best integrates the loss of function and risk?

A	Repeated adjustment improves elastic recovery but decreases surface hardness
B	The line is a sign of galvanic corrosion that is reversed by polishing alone
C	A longer guiding plane increases clasp fatigue without changing its deflection
D	Plastic deformation and fatigue damage at a stress concentration have reduced reliable springback
E	The acrylic saddle is necessarily responsible because cobalt–chromium cannot fatigue

See answer and justification for Q135 →

Q137 INTERMEDIATE · Block C

A deep dentine surface is acid-etched then blown dry until visibly chalky; the bonded restoration debonds early. Which mechanism best explains the interfacial problem in a conventional etch-and-rinse system?

A	Lack of silanisation after conditioning a silica-based ceramic
B	Inadequate micromechanical/MDP protocol for zirconia
C	Poor resistance form of an over-tapered, short preparation
D	Isolation failure with salivary contamination at bonding
E	Moist dentine collapse after over-drying etched collagen

See answer and justification for Q137 →

Q138 INTERMEDIATE · Block C

A cold-cure repair separates after a few weeks; the technician did not prepare or wet the old fracture surface according to the repair system instructions. What change is most directly indicated?

A	Review the wax design and invest without unnecessary delay
B	Repeat the cure with the specified validated time-temperature profile
C	Reinvest/reprocess after correcting inadequate packing conditions
D	Adjust repair surface preparation and appropriate acrylic technique
E	Make a new impression with a rigid tray and suitable adhesive

See answer and justification for Q138 →

Q139 INTERMEDIATE · Block C

A wax pattern left for days before investing has changed contour. What procedural modification best prevents recurrence?

A	Review the wax design and invest without unnecessary delay
B	Repeat the cure with the specified validated time-temperature profile
C	Adjust repair surface preparation and appropriate acrylic technique
D	Make a new impression with a rigid tray and suitable adhesive
E	Reinvest/reprocess after correcting inadequate packing conditions

See answer and justification for Q139 →

Q140 INTERMEDIATE · Block C

A heat-cured maxillary denture is returned from the laboratory after a markedly shortened low-temperature polymerisation cycle. The fit is satisfactory, but the patient develops diffuse mucosal burning and the denture has a strong monomer odour. No occlusal pressure spots are evident. What is the most plausible primary manufacturing problem?

A	Residual methyl methacrylate caused by incomplete conversion during the shortened polymerisation cycle
B	Gaseous porosity caused by local monomer vaporisation during prolonged overheating of the thicker palate
C	Excess initiator decomposition causing a finished polymer with lower than expected residual monomer
D	Water sorption occurring after polymerisation causing a temporary change in fitting-surface contact
E	Increased surface roughness caused by the final polishing regime, without a conversion problem

See answer and justification for Q140 →

Q141 INTERMEDIATE · Block C

Two dentures are processed from the same dough mix. The first receives the specified controlled heating cycle; the second is heated rapidly to boiling temperature. Only the second has numerous rounded voids concentrated in its thick palatal section after deflasking. Which laboratory change most directly addresses the defect?

A	Increase the pressure during trial packing while preserving the rapid heating cycle
B	Allow the flask to bench-cool more slowly after the high-temperature hold
C	Apply a secondary polymerisation cycle after the first overheating event to close the voids
D	Reduce the palate thickness at the same time as using the original uncontrolled heating schedule
E	Use a validated heating cycle that limits local monomer vaporisation in thick acrylic

See answer and justification for Q141 →

Q142 INTERMEDIATE · Block C

A denture laboratory record shows that a heat-cure flask was closed after insufficient dough was packed. After curing with an otherwise correct cycle, irregular voids are found near the upper denture’s posterior border and thin flange regions. Which process best explains the defect?

A	Vapour formation from the monomer as the internal dough temperature exceeds its boiling range
B	Insufficient dough bulk and packing compensation during polymerisation contraction
C	Intermittent hydration of the finished resin before processing was complete
D	Local stress release from differential cooling after an otherwise adequately packed cycle
E	Delayed investing of a wax trial that deformed the intended border configuration

See answer and justification for Q142 →

Q143 INTERMEDIATE · Block C

A mandibular acrylic denture fractures through the midline twice within six months. It has been repaired with a generous bulk of cold-cure resin on each occasion. The fracture repeatedly begins at the original repair margin, while an uneven posterior occlusal contact remains unchanged. Which plan most directly reduces the likelihood of another failure?

A	Replace the previous repair with a wider cold-cure resin strip while preserving the same occlusion
B	Add surface polish and a flexible soft liner to the repair site without modifying the force pattern
C	Reassess support and loading, remove the recurring flexural cause, and plan a suitable structural repair or remake
D	Replace the fractured teeth first and preserve the original baseplate design until further review
E	Move the repaired denture into slightly heavier anterior guidance to shift load away from the posterior contacts

See answer and justification for Q143 →

Q144 INTERMEDIATE · Block C

A base-metal framework is cast using gypsum-bonded investment at temperatures intended for high-fusing alloys; surface contamination and breakdown occur. Which mechanism is most relevant?

A	Excess water–powder ratio reduces cast strength
B	High-temperature breakdown of gypsum-bonded investment
C	Insufficient investment expansion produces an undersized casting
D	Excess spatulation alters setting kinetics and expansion
E	Delayed investing permits wax pattern stress relaxation

See answer and justification for Q144 →

Q145 INTERMEDIATE · Block C

A complete maxillary denture opposing natural mandibular anterior teeth splits gradually along its palatal midline after three years, without any known fall. Examination shows anterior loading and a prominent labial frenal notch. What is the dominant failure mechanism?

A	A single impact exceeding the material’s ultimate strength
B	Cyclic flexural fatigue with crack initiation and propagation at a stress concentrator
C	Direct chemical dissolution by saliva
D	Permanent dimensional change caused only by gypsum setting expansion
E	Creep of a polyether impression left under water

See answer and justification for Q145 →

Q146 INTERMEDIATE · Block C

A heat-cure batch is under-processed by stopping halfway through the prescribed polymerisation cycle. What process correction should the lab adopt for subsequent cases?

A	Reinvest/reprocess after correcting inadequate packing conditions
B	Make a new impression with a rigid tray and suitable adhesive
C	Repeat the cure with the specified validated time-temperature profile
D	Review the wax design and invest without unnecessary delay
E	Adjust repair surface preparation and appropriate acrylic technique

See answer and justification for Q146 →

Q147 INTERMEDIATE · Block C

A patient has a firm posterior maxillary ridge but a displaceable anterior flabby segment. The first definitive impression used uniform heavy-bodied material and the completed denture rocks on seating. Which impression change most directly targets the causal error?

A	Apply greater pressure to the flabby segment to record the ridge under maximum functional load
B	Record posterior support selectively while recording the flabby anterior tissue without displacement, using a suitable relieved/window technique
C	Use a rigid compound wash over the entire arch to maximise compressive recording
D	Shorten the anterior border of the denture without revisiting the impression
E	Increase the vertical dimension to compensate for tissue rebound

See answer and justification for Q147 →

Q149 INTERMEDIATE · Block C

An alginate impression is left overnight unpoured and an otherwise competent technician reports a distorted study cast. Which mechanism is most likely?

A	Syneresis or imbibition during delayed alginate pouring
B	Latex/sulphur inhibition of addition-silicone polymerisation
C	Excessive water uptake by polyether during prolonged immersion
D	Ethanol loss during delayed condensation-silicone pouring
E	Tray flexure or inadequate tray support

See answer and justification for Q149 →

Q150 INTERMEDIATE · Block C

The technician adds extra water to make a Type IV stone easier to pour. The resulting dies abrade and chip during crown waxing. Which process variable best explains the weakness?

A	Delayed investing permits wax pattern stress relaxation
B	High-temperature breakdown of gypsum-bonded investment
C	Excess water–powder ratio reduces cast strength
D	Excess spatulation alters setting kinetics and expansion
E	Insufficient investment expansion produces an undersized casting

See answer and justification for Q150 →

Q151 INTERMEDIATE · Block C

A cobalt–chromium RPD framework repeatedly fails to seat on the master cast. The technician used a conventional gypsum-bonded investment during high-temperature casting, with evidence of investment degradation. Which correction best targets the metallurgical source of error?

A	Retain gypsum investment but increase the water-to-powder ratio to improve refractory strength
B	Switch to a suitable phosphate-bonded refractory investment and validate burnout/casting expansion for the alloy
C	Reduce all occlusal rests to avoid detecting casting distortion
D	Extend the wax pattern beyond finish lines to compensate for framework seating errors
E	Sandblast the completed framework until it seats without assessing whether rest seats still fit

See answer and justification for Q151 →

Q152 INTERMEDIATE · Block C

Casts poured immediately from an addition silicone impression have bubbly, porous die surfaces. What explains this?

A	Syneresis
B	Ethanol release
C	Hydrogen release from the set material
D	Imbibition
E	Latex contamination

See answer and justification for Q152 →

Q153 INTERMEDIATE · Block C

A polyether impression is taken for a lower implant case with large embrasure spaces. The stone dies break on removal.
Which property of polyether caused the problem?

A	Hydrophobicity
B	High stiffness
C	Low tear strength
D	Hydrogen release
E	Low dimensional stability

See answer and justification for Q153 →

Q154 INTERMEDIATE · Block C

A clinician wants an elastomer that wets a moist preparation well without added surfactant.
Which material is inherently hydrophilic?

A	Polyether
B	Addition silicone
C	Condensation silicone
D	Impression compound
E	Polysulphide

See answer and justification for Q154 →

Q155 INTERMEDIATE · Block C

A denture survives many normal chewing cycles then breaks from an enlarging fine crack at a palatal midline notch. No single traumatic bite occurred. What is the best failure model?

A	Cyclic stress propagates a crack below the single-load fracture threshold
B	A pre-existing flaw creates a stress-intensity concentration
C	An abrupt overload causes an impact-type catastrophic fracture
D	Water uptake and thermal cycling change properties over time
E	A weak repair junction concentrates load at its boundary

See answer and justification for Q155 →

Q156 INTERMEDIATE · Block C

A repaired acrylic fracture opens along the exact old repair line after several weeks even though the remaining base has adequate thickness. Examination shows a glossy unprepared interface at the repair. What is the most likely local mechanism?

A	Early processing dimensional change due to premature deflasking
B	Stress concentration at the frenum notch with cyclic flexure
C	Impact loading from a one-off fall
D	Inadequate bonding at a self-cure repair interface
E	Thermal distortion of PMMA after hot-water cleaning

See answer and justification for Q156 →

Q157 INTERMEDIATE · Block C

Several casts from one practice are distorted. The practice uses thin plastic stock trays without adhesive for elastomers.
What is the main cause?

A	The material brand
B	The disinfectant
C	The stone
D	Tray flexure and lack of adhesive
E	Room temperature

See answer and justification for Q157 →

Q158 INTERMEDIATE · Block C

A recently delivered heat-cured upper denture has a persistent acrid odour and diffuse mucosal burning. Records show that the prescribed curing cycle was shortened substantially; the base is not visibly porous. Which primary processing mechanism best explains the complaint?

A	Gaseous porosity from excessive temperature rise during cure
B	Contraction porosity from inadequate packing or pressure
C	Fatigue crack growth under repeated functional flexure
D	Granular porosity from loss of monomer during the dough stage
E	Residual monomer from incomplete polymerisation

See answer and justification for Q158 →

Q163 INTERMEDIATE · Block C

A laboratory reports repeated irritation with newly fabricated heat-cured bases after deliberately reducing the manufacturer’s cure time to increase throughput. Which corrective measure directly addresses the process defect?

A	Autopolymerising repair with counselling about residual monomer and lower bulk strength
B	Correct the base thickness and occlusal loading before repair
C	Repeat processing with a validated controlled curing cycle
D	Investigate possible contact irritation and material exposure before elective replacement
E	Laboratory heat-polymerised repair with appropriate reinforcement/design review

See answer and justification for Q163 →

Q164 INTERMEDIATE · Block C

All margins in a full-arch elastomer impression are distorted in a consistent direction, and a visibly flexible stock tray was used. What is the most likely source?

A	Latex/sulphur inhibition of addition-silicone polymerisation
B	Tray flexure or inadequate tray support
C	Excessive water uptake by polyether during prolonged immersion
D	Ethanol loss during delayed condensation-silicone pouring
E	Syneresis or imbibition during delayed alginate pouring

See answer and justification for Q164 →

Q165 INTERMEDIATE · Block C

A setting stone cast is left under water in a soaking bath and comes out oversized.
What explains this?

A	Thermal expansion
B	Syneresis
C	Imbibition of the impression
D	Hygroscopic expansion
E	Polymerisation shrinkage

See answer and justification for Q165 →

Q166 INTERMEDIATE · Block C

A repaired denture repeatedly separates at the junction between old heat-cured PMMA and a thick self-curing patch, while adjacent base remains intact. What mechanism is most likely?

A	Water uptake and thermal cycling change properties over time
B	Cyclic stress propagates a crack below the single-load fracture threshold
C	A pre-existing flaw creates a stress-intensity concentration
D	A weak repair junction concentrates load at its boundary
E	An abrupt overload causes an impact-type catastrophic fracture

See answer and justification for Q166 →

Q167 INTERMEDIATE · Block C

An intact denture is only a week old and the patient reports burning that began immediately after insertion. It was made with a high proportion of room-temperature cured repair acrylic at the fitting surface. What is the most appropriate assessment?

A	Repeat processing with a validated controlled curing cycle
B	Laboratory heat-polymerised repair with appropriate reinforcement/design review
C	Autopolymerising repair with counselling about residual monomer and lower bulk strength
D	Correct the base thickness and occlusal loading before repair
E	Investigate possible contact irritation and material exposure before elective replacement

See answer and justification for Q167 →

Q168 INTERMEDIATE · Block C

A patient immersed a PMMA denture in very hot water; it now rocks despite intact borders and no fracture. Which mechanism is most likely?

A	Stress concentration at the frenum notch with cyclic flexure
B	Early processing dimensional change due to premature deflasking
C	Inadequate bonding at a self-cure repair interface
D	Impact loading from a one-off fall
E	Thermal distortion of PMMA after hot-water cleaning

See answer and justification for Q168 →

Q169 INTERMEDIATE · Block C

A newly cast Co–Cr RPD framework has a porous region at the junction of a clasp shoulder and minor connector. The framework has not yet been finished; a technician suggests polishing away the defect. What is the most appropriate action?

A	Increase retention by adding acrylic over the porous region
B	Finish and polish aggressively because porosity is superficial by definition
C	Put the framework into function and review in 12 months
D	Heat the clasp manually to close casting voids
E	Reject and recast the defective framework after reviewing sprueing, investment, burnout and casting practice

See answer and justification for Q169 →

Q170 INTERMEDIATE · Block C

A localised region of a PVS wash never sets adjacent to a cord handled with latex gloves. Which mechanism is most likely?

A	Excessive water uptake by polyether during prolonged immersion
B	Syneresis or imbibition during delayed alginate pouring
C	Latex/sulphur inhibition of addition-silicone polymerisation
D	Ethanol loss during delayed condensation-silicone pouring
E	Tray flexure or inadequate tray support

See answer and justification for Q170 →

Q171 INTERMEDIATE · Block C

A previously intact denture separates abruptly into three pieces immediately after it strikes a hard tiled floor. The patient reports no prior visible crack. Which loading mechanism is most likely?

A	Thermal distortion of PMMA after hot-water cleaning
B	Impact loading from a one-off fall
C	Stress concentration at the frenum notch with cyclic flexure
D	Inadequate bonding at a self-cure repair interface
E	Early processing dimensional change due to premature deflasking

See answer and justification for Q171 →

Q174 INTERMEDIATE · Block C

A resin-bonded bridge fails soon after cementation; the field was visibly wet with saliva and isolation was abandoned. Which factor is most directly implicated?

A	Inadequate micromechanical/MDP protocol for zirconia
B	Moist dentine collapse after over-drying etched collagen
C	Lack of silanisation after conditioning a silica-based ceramic
D	Isolation failure with salivary contamination at bonding
E	Poor resistance form of an over-tapered, short preparation

See answer and justification for Q174 →

Q175 INTERMEDIATE · Block C

Acrylic denture teeth from one manufacturer resist crazing around the necks better than another brand. Which component improves craze resistance?

A	Hydroquinone
B	A cross-linking agent (ethylene glycol dimethacrylate)
C	Pigments
D	Plasticiser
E	Water

See answer and justification for Q175 →

Q176 INTERMEDIATE · Block C

A heat-cured acrylic maxillary denture is returned from the laboratory with bubbles concentrated in the thick palatal region. The packing records show adequate dough and pressure, but the flask was taken rapidly to a high-temperature boil. Which process change best addresses the most likely cause?

A	Increase monomer concentration while retaining rapid heating
B	Use the validated controlled polymerisation cycle to reduce internal monomer vaporisation
C	Decrease packing pressure so volatile gases can escape
D	Increase the polishing time until internal pores are removed
E	Replace heat-cure PMMA with unverified self-cure resin for the definitive base

See answer and justification for Q176 →

Q177 INTERMEDIATE · Block C

A denture shows irregular contraction voids where too little acrylic dough was packed and flask pressure was inadequate. What production stage should be corrected?

A	Repeat the cure with the specified validated time-temperature profile
B	Reinvest/reprocess after correcting inadequate packing conditions
C	Review the wax design and invest without unnecessary delay
D	Adjust repair surface preparation and appropriate acrylic technique
E	Make a new impression with a rigid tray and suitable adhesive

See answer and justification for Q177 →

Q178 INTERMEDIATE · Block C

A lithium-disilicate veneer is etched appropriately but fitted without a silane coupling step; most of the failure occurs at the ceramic–resin interface. What is the most direct deficiency?

A	Lack of silanisation after conditioning a silica-based ceramic
B	Inadequate micromechanical/MDP protocol for zirconia
C	Poor resistance form of an over-tapered, short preparation
D	Moist dentine collapse after over-drying etched collagen
E	Isolation failure with salivary contamination at bonding

See answer and justification for Q178 →

Q179 INTERMEDIATE · Block C

A mandibular acrylic RPD has fractured repeatedly across a previous self-cure repair. The newest fracture follows the original repair interface rather than the middle of an intact base. Occlusion shows a distal-extension saddle rocking on function. Which approach best addresses the problem?

A	Repeat the same cold-cure repair without assessing the saddle
B	Add a thicker temporary self-cure strip to the fitting surface and ignore fit
C	Investigate fit and functional loading, then redesign or rebase as appropriate with a durable laboratory repair or replacement
D	Polish the crack until it disappears without joining the segments
E	Use an acrylic adhesive at home and review when the fracture recurs

See answer and justification for Q179 →

Q180 INTERMEDIATE · Block C

A crown made from a condensation-silicone impression poured after two days fails to seat, despite adequate preparation clearance. The material was otherwise handled correctly. Which error is most likely?

A	Excessive water uptake by polyether during prolonged immersion
B	Tray flexure or inadequate tray support
C	Ethanol loss during delayed condensation-silicone pouring
D	Latex/sulphur inhibition of addition-silicone polymerisation
E	Syneresis or imbibition during delayed alginate pouring

See answer and justification for Q180 →

Q181 INTERMEDIATE · Block C

Two identical gypsum batches use the same water–powder ratio, but one is spatulated much more vigorously. That cast sets earlier and exhibits different expansion. What is the most likely cause?

A	Excess water–powder ratio reduces cast strength
B	Insufficient investment expansion produces an undersized casting
C	Delayed investing permits wax pattern stress relaxation
D	Excess spatulation alters setting kinetics and expansion
E	High-temperature breakdown of gypsum-bonded investment

See answer and justification for Q181 →

Q183 INTERMEDIATE · Block C

A patient may have swallowed a fragment of a fractured acrylic partial denture, but it is not visible on the chest radiograph.
Why?

A	It has dissolved
B	The exposure was too high
C	Acrylic absorbs X-rays strongly
D	The fragment was too small to matter
E	PMMA is radiolucent

See answer and justification for Q183 →

Q186 INTERMEDIATE · Block C

An upper denture has fractured twice across the midline and examination shows an extremely thin palatal section with heavy opposing contacts. Which step is most important before simply repeating the repair?

A	Repeat processing with a validated controlled curing cycle
B	Autopolymerising repair with counselling about residual monomer and lower bulk strength
C	Investigate possible contact irritation and material exposure before elective replacement
D	Laboratory heat-polymerised repair with appropriate reinforcement/design review
E	Correct the base thickness and occlusal loading before repair

See answer and justification for Q186 →

Q187 INTERMEDIATE · Block C

A silicone long-term soft liner has partly separated from the base and is colonised by Candida.
Which statement about silicone soft liners is correct?

A	They bond less well to the base and support Candida growth
B	They harden within days
C	They are best trimmed with hot burs
D	They release ethanol
E	They never need replacing

See answer and justification for Q187 →

Q188 INTERMEDIATE · Block C

Two otherwise identical denture bases experience comparable loading, but only the base with an internal air void begins to crack at that void. Which explanation best relates flaw size to failure?

A	A weak repair junction concentrates load at its boundary
B	Water uptake and thermal cycling change properties over time
C	An abrupt overload causes an impact-type catastrophic fracture
D	A pre-existing flaw creates a stress-intensity concentration
E	Cyclic stress propagates a crack below the single-load fracture threshold

See answer and justification for Q188 →

Q189 INTERMEDIATE · Block C

A monolithic zirconia crown is bonded after conventional hydrofluoric acid etching alone, and debonds despite good field isolation. Which problem is most plausible?

A	Lack of silanisation after conditioning a silica-based ceramic
B	Inadequate micromechanical/MDP protocol for zirconia
C	Poor resistance form of an over-tapered, short preparation
D	Moist dentine collapse after over-drying etched collagen
E	Isolation failure with salivary contamination at bonding

See answer and justification for Q189 →

Q191 INTERMEDIATE · Block C

A diagnostic cast made from an elastomer impression has all distal surfaces distorted because the impression tray flexed on removal. What should be altered for the next impression?

A	Adjust repair surface preparation and appropriate acrylic technique
B	Repeat the cure with the specified validated time-temperature profile
C	Reinvest/reprocess after correcting inadequate packing conditions
D	Make a new impression with a rigid tray and suitable adhesive
E	Review the wax design and invest without unnecessary delay

See answer and justification for Q191 →

Q193 INTERMEDIATE · Block C

An acrylic lower denture shatters immediately when accidentally run over by a heavy trolley, with no earlier cracks. What load mechanism dominates?

A	Water uptake and thermal cycling change properties over time
B	A pre-existing flaw creates a stress-intensity concentration
C	An abrupt overload causes an impact-type catastrophic fracture
D	Cyclic stress propagates a crack below the single-load fracture threshold
E	A weak repair junction concentrates load at its boundary

See answer and justification for Q193 →

Q194 INTERMEDIATE · Block C

A patient notices a slowly developing midline white line starting at the sharp labial frenal notch of an upper denture. There is no history of falling. Which mechanism best accounts for the crack origin?

A	Thermal distortion of PMMA after hot-water cleaning
B	Early processing dimensional change due to premature deflasking
C	Stress concentration at the frenum notch with cyclic flexure
D	Impact loading from a one-off fall
E	Inadequate bonding at a self-cure repair interface

See answer and justification for Q194 →

Q195 INTERMEDIATE · Block C

A patient feels a sharp twinge when foil touches a new gold crown that contacts an amalgam in the opposing tooth.
What explains this?

A	Crevice corrosion
B	Galvanic current between dissimilar metals in saliva
C	Stress corrosion
D	Tarnish
E	Allergy to gold

See answer and justification for Q195 →

Q196 INTERMEDIATE · Block C

A technician explains how porcelain bonds to a metal coping.
Which bond mechanism is most important?

A	Mechanical retention from air abrasion
B	Van der Waals forces
C	Chemical bonding through the metal oxide layer
D	Compressive fit
E	Cement

See answer and justification for Q196 →

Q197 INTERMEDIATE · Block C

Porcelain repeatedly delaminates from a base-metal coping fired with a thick, dark oxide layer.
What explains the failure?

A	Too little oxide
B	Porcelain CTE too low
C	Gold contamination
D	Insufficient firing
E	An oxide layer that is too thick weakens the bond

See answer and justification for Q197 →

Q198 INTERMEDIATE · Block C

A cast metal restoration is consistently too tight because the investment expansion did not compensate for alloy solidification shrinkage. Which cause is most direct?

A	Insufficient investment expansion produces an undersized casting
B	Excess water–powder ratio reduces cast strength
C	Delayed investing permits wax pattern stress relaxation
D	High-temperature breakdown of gypsum-bonded investment
E	Excess spatulation alters setting kinetics and expansion

See answer and justification for Q198 →

Q199 INTERMEDIATE · Block C

A base retained in a hot humid environment gradually changes dimensions, while there is no fracture line or repaired joint. Which material consideration is most relevant?

A	An abrupt overload causes an impact-type catastrophic fracture
B	Cyclic stress propagates a crack below the single-load fracture threshold
C	Water uptake and thermal cycling change properties over time
D	A weak repair junction concentrates load at its boundary
E	A pre-existing flaw creates a stress-intensity concentration

See answer and justification for Q199 →

Q200 INTERMEDIATE · Block C

A full-coverage molar crown debonds repeatedly; the preparation is 2.5 mm high with marked taper, and bonding was done correctly. What is the most fundamental explanation?

A	Poor resistance form of an over-tapered, short preparation
B	Isolation failure with salivary contamination at bonding
C	Lack of silanisation after conditioning a silica-based ceramic
D	Inadequate micromechanical/MDP protocol for zirconia
E	Moist dentine collapse after over-drying etched collagen

See answer and justification for Q200 →

Q201 INTERMEDIATE · Block C

A monolithic zirconia crown with limited retention is to be cemented.
What surface treatment is recommended?

A	Hydrofluoric acid and silane
B	Phosphoric acid and silane
C	No treatment and zinc oxide eugenol
D	Low-pressure air abrasion (about 50 µm alumina) with an MDP-containing primer or cement
E	Coarse diamond roughening and silane

See answer and justification for Q201 →

Q202 INTERMEDIATE · Block C

A clinician applies silane alone to the fitting surface of a zirconia crown before resin cementation. It debonds. Why did silane fail?

A	Silane was out of date
B	Zirconia is too rough
C	Silane needs a silica-based surface
D	Resin cement cannot bond to tooth
E	The crown was too thick

See answer and justification for Q202 →

Q205 INTERMEDIATE · Block C

A patient needs an urgent temporary repair of a fractured denture and the laboratory cannot perform a controlled heat cure today. Which option best describes the acceptable interim material decision?

A	Laboratory heat-polymerised repair with appropriate reinforcement/design review
B	Investigate possible contact irritation and material exposure before elective replacement
C	Correct the base thickness and occlusal loading before repair
D	Repeat processing with a validated controlled curing cycle
E	Autopolymerising repair with counselling about residual monomer and lower bulk strength

See answer and justification for Q205 →

Q206 INTERMEDIATE · Block C

A complete denture develops an initially fine midline crack over several years, which extends despite no reported drop. The denture repeatedly flexes over a high palatal vault. Which mechanism best explains the propagation?

A	Residual monomer from incomplete polymerisation
B	Granular porosity from loss of monomer during the dough stage
C	Fatigue crack growth under repeated functional flexure
D	Gaseous porosity from excessive temperature rise during cure
E	Contraction porosity from inadequate packing or pressure

See answer and justification for Q206 →

Q207 INTERMEDIATE · Block C

A patient's lower incisors are wearing rapidly against new upper ceramic crowns that were adjusted chairside and not repolished.
Which surface is most abrasive to opposing enamel?

A	Polished monolithic zirconia
B	Unglazed or roughened feldspathic porcelain
C	Gold
D	Glazed lithium disilicate
E	Composite

See answer and justification for Q207 →

Q208 INTERMEDIATE · Block C

A patient feels sensitivity after a gold crown is cemented with zinc phosphate.
Which property of zinc phosphate explains this, and how is it mixed?

A	High pH; mixed quickly in bulk
B	Fluoride release; mixed on a warm slab
C	Eugenol content; mixed on paper
D	Moisture sensitivity; mixed on a pad
E	Low initial pH

See answer and justification for Q208 →

Q209 INTERMEDIATE · Block C

A leucite-reinforced glass-ceramic crown cracked a month after it was cemented with resin-modified glass ionomer.
What is the likely cause?

A	Too thin a cement film
B	Eugenol contamination
C	Hygroscopic expansion of the RMGIC fracturing a weak glass ceramic
D	Galvanic corrosion
E	Silane failure

See answer and justification for Q209 →

Q211 INTERMEDIATE · Block C

A high-impact repair is requested after a repeated base fracture, and clinical loading has been assessed and corrected. Which definitive processing approach is most defensible for a durable repair?

A	Repeat processing with a validated controlled curing cycle
B	Laboratory heat-polymerised repair with appropriate reinforcement/design review
C	Autopolymerising repair with counselling about residual monomer and lower bulk strength
D	Correct the base thickness and occlusal loading before repair
E	Investigate possible contact irritation and material exposure before elective replacement

See answer and justification for Q211 →

Q212 INTERMEDIATE · Block C

A thick posterior palatal section of a newly processed PMMA denture contains smooth, roughly spherical internal voids concentrated in its deepest part. The flask was brought rapidly to boiling. What is the most likely mechanism?

A	Residual monomer from incomplete polymerisation
B	Gaseous porosity from excessive temperature rise during cure
C	Granular porosity from loss of monomer during the dough stage
D	Contraction porosity from inadequate packing or pressure
E	Fatigue crack growth under repeated functional flexure

See answer and justification for Q212 →

Q215 INTERMEDIATE · Block C

A denture base exhibits irregular distributed voids around the last-filled area; flask closure notes show insufficient dough and low packing pressure, but the temperature cycle was otherwise appropriate. Which defect is most likely?

A	Fatigue crack growth under repeated functional flexure
B	Contraction porosity from inadequate packing or pressure
C	Residual monomer from incomplete polymerisation
D	Granular porosity from loss of monomer during the dough stage
E	Gaseous porosity from excessive temperature rise during cure

See answer and justification for Q215 →

Q216 INTERMEDIATE · Block C

A polyether impression is left immersed in disinfectant well beyond the specified contact time; dies are subsequently enlarged. Which mechanism fits best?

A	Syneresis or imbibition during delayed alginate pouring
B	Excessive water uptake by polyether during prolonged immersion
C	Tray flexure or inadequate tray support
D	Ethanol loss during delayed condensation-silicone pouring
E	Latex/sulphur inhibition of addition-silicone polymerisation

See answer and justification for Q216 →

Q218 INTERMEDIATE · Block C

A laboratory deflasks an acrylic base before it has cooled as recommended. The new dentures show altered fit from the outset, without porosity. What process-related mechanism should be suspected?

A	Inadequate bonding at a self-cure repair interface
B	Stress concentration at the frenum notch with cyclic flexure
C	Thermal distortion of PMMA after hot-water cleaning
D	Early processing dimensional change due to premature deflasking
E	Impact loading from a one-off fall

See answer and justification for Q218 →

Q219 INTERMEDIATE · Block C

A root-filled premolar is to be crowned with a post-retained core.
Which factor is most important for the long-term survival of the tooth?

A	A ferrule of 1.5–2 mm of sound dentine all round
B	A wide, long post
C	A metal post
D	Leaving no root filling apically
E	A threaded post

See answer and justification for Q219 →

Q220 INTERMEDIATE · Block C

A clinician plans a full-arch zirconia bridge from an intraoral scan of an edentulous maxilla.
Which statement about intraoral scanner accuracy is correct?

A	It is greatest over full arches
B	It is greatest over short spans and lower over full arches and mobile, featureless edentulous mucosa
C	It is unaffected by saliva
D	It equals conventional impressions in every situation
E	It is best for edentulous ridges

See answer and justification for Q220 →

BLOCK D · Pharmacology, pain control, emergencies and the medically compromised patient (48)

Q221 INTERMEDIATE · Block D

A 73-year-old with a mechanical heart valve is taking warfarin. Persistent erythematous denture stomatitis remains after attention to hygiene and night removal. Which prescribing consideration best avoids an important interaction while still addressing the condition?

A	Choose miconazole oral gel without checking warfarin because topical products are never absorbed
B	Consider a suitable alternative such as nystatin and coordinate prescribing according to current guidance rather than exposing the patient to a known azole–warfarin interaction
C	Stop warfarin for two weeks to permit routine antifungal treatment
D	Prescribe oral fluconazole as an automatic first line in all warfarin patients
E	Use chlorhexidine as the sole treatment without assessing the denture or mucosal response

See answer and justification for Q221 →

Q222 INTERMEDIATE · Block D

A 61-year-old who smokes 15 cigarettes daily and has an HbA1c of 9% is being assessed for a complex RPD and implant options. There is unresolved periodontal inflammation and the patient has missed supportive-care appointments. What is the most defensible sequence?

A	Place implants immediately because mechanical stability makes systemic risks irrelevant
B	Proceed with definitive extensive bridgework while periodontal disease is active
C	Treat smoking and glycaemic status as absolute contraindications to every possible restorative intervention
D	Focus only on adding prosthetic retention, because smoking affects gingival colour rather than treatment response
E	Control active periodontal disease and modifiable risk factors, assess maintenance capacity and reassess before definitive treatment or referral

See answer and justification for Q222 →

Q223 INTERMEDIATE · Block D

A patient who is 36 weeks pregnant needs an urgent extraction for a painful tooth.
Which local anaesthetic is best avoided?

A	Prilocaine with felypressin
B	Lidocaine with adrenaline
C	Articaine with adrenaline
D	Mepivacaine plain
E	None; all are equally suitable

See answer and justification for Q223 →

Q225 INTERMEDIATE · Block D

A patient taking propranolol receives several cartridges of lidocaine with adrenaline.
What interaction may follow?

A	Hypotension with tachycardia
B	Methaemoglobinaemia
C	Hypertension with reflex bradycardia
D	Bronchospasm only
E	No interaction

See answer and justification for Q225 →

Q227 INTERMEDIATE · Block D

A patient with trismus of 15 mm needs anaesthesia for a lower molar.
Which technique is designed for limited opening?

A	Standard inferior alveolar block
B	Mental nerve block
C	Akinosi (closed-mouth) block
D	Palatal infiltration
E	Long buccal block only

See answer and justification for Q227 →

Q229 INTERMEDIATE · Block D

An anxious patient asks for inhalation sedation for impressions.
Which of these is a relative contraindication?

A	Mild hypertension
B	First-trimester pregnancy
C	Controlled asthma
D	Age over 60
E	Dental anxiety

See answer and justification for Q229 →

Q230 INTERMEDIATE · Block D

During IV sedation with midazolam, a patient's respiratory rate falls and they become unresponsive.
Which drug reverses midazolam?

A	Naloxone
B	Glucagon
C	Atropine
D	Flumazenil
E	Adrenaline

See answer and justification for Q230 →

Q231 INTERMEDIATE · Block D

A patient is to have IV midazolam sedation for a lengthy procedure.
What post-operative instruction is correct?

A	They may drive home after an hour
B	No escort is needed
C	They may return to work that afternoon
D	They may drink alcohol that evening
E	A responsible adult escort, and no driving or operating machinery until the next day

See answer and justification for Q231 →

Q232 INTERMEDIATE · Block D

A patient with poorly controlled heart failure (ASA III) asks for IV sedation in your practice.
Which patients are generally suitable for sedation in primary care?

A	Any ASA grade
B	ASA I–II
C	ASA III–IV
D	ASA IV only
E	Only patients over 70

See answer and justification for Q232 →

Q233 INTERMEDIATE · Block D

A patient has a dental abscess with facial swelling, a temperature of 38.5°C and malaise. You have drained it.
Which antibiotic regimen does SDCEP recommend?

A	Amoxicillin 3 g single dose
B	Clindamycin 150 mg four times daily for 14 days
C	Metronidazole 200 mg once daily
D	Phenoxymethylpenicillin 500 mg four times daily for 5 days (or amoxicillin 500 mg three times daily)
E	Doxycycline 100 mg daily

See answer and justification for Q233 →

Q235 INTERMEDIATE · Block D

A smoker has painful, bleeding, punched-out interdental papillae and halitosis.
What antimicrobial, if indicated, is recommended?

A	Amoxicillin 500 mg for 14 days
B	Doxycycline 20 mg for 3 months
C	Metronidazole 400 mg three times daily for 3 days
D	Fluconazole 50 mg for 7 days
E	Aciclovir 200 mg five times daily

See answer and justification for Q235 →

Q236 INTERMEDIATE · Block D

A 52-year-old has severe spontaneous nocturnal pain from a heavily restored lower molar, lingering for over a minute after cold testing. The tooth has no diffuse swelling or systemic illness, and a periapical radiograph shows no apical change. Which plan best addresses the likely source of pain?

A	Prescribe antibiotics alone and review after one week
B	Confirm the diagnosis and provide definitive pulpal intervention or arrange appropriate timely referral with analgesia advice
C	Extract the tooth without assessing restorability or patient preference
D	Use a long course of antifungal treatment before considering pulp therapy
E	Diagnose a primary TMD solely because no radiolucency is present

See answer and justification for Q236 →

Q237 INTERMEDIATE · Block D

A 73-year-old taking warfarin for a mechanical heart valve is diagnosed with mild localised denture-associated candidosis. Denture hygiene measures have been started. The patient requests miconazole oral gel because a friend used it. What is the safest next action?

A	Check current interaction guidance and choose an appropriate treatment without exposing the patient to a major warfarin interaction
B	Stop warfarin for one week so miconazole can be prescribed
C	Prescribe miconazole because it is applied locally and has no systemic absorption
D	Prescribe systemic fluconazole without medication review
E	Delay the treatment of every oral fungal infection until the next routine INR visit

See answer and justification for Q237 →

Q239 INTERMEDIATE · Block D

A frail patient has facial swelling, fever, trismus and dysphagia from a mandibular dental infection. What is the most urgent action?

A	Identify interaction between oral miconazole and warfarin
B	Refer high-risk sedation cases to an appropriate service
C	Urgent medical evaluation of significant unexplained or progressive infection
D	Review dental bleeding risk and local haemostatic strategy without routine drug cessation
E	Adapt treatment and preventive planning to xerostomia-inducing polypharmacy

See answer and justification for Q239 →

Q240 INTERMEDIATE · Block D

You plan analgesia after a difficult extraction for several patients.
Which patient should not be given ibuprofen?

A	A 30-year-old with no medical history
B	A patient with NSAID-sensitive asthma
C	A patient taking a statin
D	A patient with controlled hypothyroidism
E	A patient with a penicillin allergy

See answer and justification for Q240 →

Q242 INTERMEDIATE · Block D

A denture wearer on warfarin requires antifungal treatment for confirmed candidosis. The clinician selects miconazole oral gel. What risk is most important?

A	Adapt treatment and preventive planning to xerostomia-inducing polypharmacy
B	Review dental bleeding risk and local haemostatic strategy without routine drug cessation
C	Urgent medical evaluation of significant unexplained or progressive infection
D	Refer high-risk sedation cases to an appropriate service
E	Identify interaction between oral miconazole and warfarin

See answer and justification for Q242 →

Q244 INTERMEDIATE · Block D

A patient on warfarin has an INR of 4.6 on the day of a planned extraction.
What should you do?

A	Proceed with extra sutures
B	Proceed with tranexamic acid only
C	Proceed under IV sedation
D	Halve the next warfarin dose and proceed
E	Defer treatment and seek advice from the anticoagulation service

See answer and justification for Q244 →

Q249 INTERMEDIATE · Block D

A patient on apixaban needs crown preparation with retraction cord and impressions.
How should the apixaban be managed?

A	Miss the morning dose
B	Stop for 48 hours
C	No interruption
D	Switch to aspirin
E	Refer to hospital

See answer and justification for Q249 →

Q250 INTERMEDIATE · Block D

A patient has exposed bone in the lower lingual region after a sore spot from a denture. She has taken alendronic acid for 7 years and has never had radiotherapy.
When does this meet the definition of MRONJ?

A	Immediately
B	After 2 weeks
C	After 4 weeks
D	When the bone has been exposed for more than 8 weeks
E	Only if it is painful

See answer and justification for Q250 →

Q251 INTERMEDIATE · Block D

A patient stopped denosumab 18 months ago; another stopped IV zoledronic acid 18 months ago.
Which statement is correct?

A	Denosumab's effect wears off within months
B	Both effects stop at once
C	Bisphosphonate effects wear off in weeks
D	Denosumab persists for decades
E	Neither drug is associated with MRONJ

See answer and justification for Q251 →

Q253 INTERMEDIATE · Block D

A cancer patient on IV zoledronic acid has a lower canine with a heavily broken-down crown but a sound root. A denture is planned.
What is the preferred management of the canine?

A	Extract it before making the denture
B	Extract it with antibiotic cover
C	Leave it untreated
D	Crown it
E	Retain the root (root treatment and decoronation as an overdenture abutment)

See answer and justification for Q253 →

Q255 INTERMEDIATE · Block D

An oncologist plans IV zoledronic acid for a patient with bone metastases. The patient has an ill-fitting partial denture and a broken-down tooth.
What should happen first?

A	Start the drug and see the dentist later
B	Extract every tooth
C	Give antibiotics only
D	A dental assessment, getting the mouth healthy before treatment starts if possible
E	Implants to support the denture

See answer and justification for Q255 →

Q256 INTERMEDIATE · Block D

A 58-year-old will start radiotherapy for an oropharyngeal cancer, with both parotids in the field. A lower molar has a poor prognosis.
When should it ideally be extracted?

A	During radiotherapy
B	Six months after radiotherapy
C	Only if it becomes painful
D	The day before radiotherapy starts
E	At least 10–14 days before radiotherapy, to allow healing

See answer and justification for Q256 →

Q257 INTERMEDIATE · Block D

A dentate patient finished head and neck radiotherapy 6 months ago and now has rampant cervical caries, a dry mouth and limited opening.
Which measures are appropriate?

A	Routine 1450 ppm toothpaste only
B	Extraction of all carious teeth
C	Antibiotics for the caries
D	5000 ppm toothpaste, fluoride gel in custom trays, saliva substitutes and trismus exercises
E	Complete dentures at once

See answer and justification for Q257 →

Q258 INTERMEDIATE · Block D

A patient who had mandibular radiotherapy 8 years ago asks whether the risk of osteoradionecrosis has gone.
Which statement is correct?

A	The risk ends after 5 years
B	The risk is lifelong, dose-dependent and much higher in the mandible than the maxilla
C	The maxilla is at higher risk
D	Only implants carry a risk
E	The risk applies only during radiotherapy

See answer and justification for Q258 →

Q259 INTERMEDIATE · Block D

A patient who had a maxillectomy and radiotherapy needs an obturator and asks about implants in the irradiated bone.
At which NHS complexity level is this care?

A	Level 3
B	Level 1
C	Level 2
D	Not provided by the NHS
E	Hygienist level

See answer and justification for Q259 →

Q262 INTERMEDIATE · Block D

A patient with severe chronic respiratory compromise and multiple systemic conditions requests deep sedation for elective restorative treatment. What is safest?

A	Adapt treatment and preventive planning to xerostomia-inducing polypharmacy
B	Urgent medical evaluation of significant unexplained or progressive infection
C	Review dental bleeding risk and local haemostatic strategy without routine drug cessation
D	Identify interaction between oral miconazole and warfarin
E	Refer high-risk sedation cases to an appropriate service

See answer and justification for Q262 →

Q266 INTERMEDIATE · Block D

The chest pain in the previous patient is unrelieved after GTN and lasts over 20 minutes. He is pale and sweaty.
What should you do?

A	Give more GTN and send him home
B	Give glucose
C	Call 999, give aspirin 300 mg chewed, and give oxygen if he is hypoxic
D	Give adrenaline
E	Lie him flat with legs raised and wait

See answer and justification for Q266 →

Q267 INTERMEDIATE · Block D

A patient with asthma becomes wheezy and breathless during an appointment but can still speak.
What is the first treatment?

A	Adrenaline IM
B	Salbutamol 100 micrograms per puff, 4–6 puffs via a spacer, repeated up to 10 puffs
C	Glucagon
D	GTN
E	Hydrocortisone orally

See answer and justification for Q267 →

Q268 INTERMEDIATE · Block D

A diabetic patient becomes confused, tremulous and sweaty midway through a lengthy denture appointment and can still swallow. The patient took insulin but missed lunch. A capillary blood glucose reading is 3.0 mmol/L. What is the best immediate action?

A	Continue the impression because completing treatment avoids further fasting
B	Administer insulin to correct the low measurement
C	Stop dental treatment and provide fast-acting oral glucose in accordance with emergency guidance, then recheck and monitor
D	Give intramuscular adrenaline routinely
E	Arrange a new RPD fitting before checking the patient’s medical status

See answer and justification for Q268 →

Q269 INTERMEDIATE · Block D

The same patient becomes unconscious and cannot swallow.
What should you give?

A	Oral glucose gel between the cheeks
B	Insulin
C	Adrenaline
D	Midazolam
E	Glucagon 1 mg IM

See answer and justification for Q269 →

Q270 INTERMEDIATE · Block D

A patient with epilepsy has a tonic–clonic seizure in the chair that is still continuing after 5 minutes.
What should you give?

A	Buccal midazolam 10 mg
B	Glucagon 1 mg
C	Adrenaline 500 micrograms
D	Diazepam tablets
E	Nothing; wait for it to stop

See answer and justification for Q270 →

Q271 INTERMEDIATE · Block D

A young patient becomes pale, sweaty and dizzy, then briefly loses consciousness during an injection. The pulse is slow but present.
What is the immediate management?

A	Adrenaline IM
B	CPR
C	Glucagon
D	Sit the patient upright
E	Lie the patient flat, raise the legs and give oxygen

See answer and justification for Q271 →

Q274 INTERMEDIATE · Block D

A patient inhales part of a crown preparation's debris, coughs ineffectively and cannot speak. She is conscious.
What is the first action?

A	Start CPR
B	Give adrenaline
C	Up to 5 back blows, then up to 5 abdominal thrusts
D	Encourage her to lie down
E	Give water

See answer and justification for Q274 →

Q276 INTERMEDIATE · Block D

A 30-year-old with poorly controlled epilepsy has lost 11 and 21 and takes phenytoin.
Which prosthodontic approach is most appropriate?

A	Prefer a fixed option, avoiding small removable prostheses that could be inhaled or fractured
B	A small acrylic partial denture worn at night
C	Porcelain teeth on an acrylic base
D	No replacement because of the epilepsy
E	A spoon denture

See answer and justification for Q276 →

Q277 INTERMEDIATE · Block D

A patient is 10 weeks pregnant and asks when to have elective bridge work.
When is the safest time for necessary dental treatment?

A	First trimester
B	Immediately, regardless of stage
C	Third trimester
D	During labour
E	Second trimester

See answer and justification for Q277 →

Q278 INTERMEDIATE · Block D

A patient on apixaban needs a straightforward low-risk dental procedure; the practitioner proposes automatic multi-day discontinuation without assessment. What principle is preferable?

A	Review dental bleeding risk and local haemostatic strategy without routine drug cessation
B	Urgent medical evaluation of significant unexplained or progressive infection
C	Identify interaction between oral miconazole and warfarin
D	Refer high-risk sedation cases to an appropriate service
E	Adapt treatment and preventive planning to xerostomia-inducing polypharmacy

See answer and justification for Q278 →

Q279 INTERMEDIATE · Block D

A patient taking multiple anticholinergic medications has active root caries and a poorly retentive denture. What systems-based approach is most appropriate?

A	Identify interaction between oral miconazole and warfarin
B	Adapt treatment and preventive planning to xerostomia-inducing polypharmacy
C	Urgent medical evaluation of significant unexplained or progressive infection
D	Refer high-risk sedation cases to an appropriate service
E	Review dental bleeding risk and local haemostatic strategy without routine drug cessation

See answer and justification for Q279 →

Q280 INTERMEDIATE · Block D

A denture wearer has a confirmed methyl methacrylate allergy on patch testing.
Which base material may be considered?

A	Self-cured acrylic
B	A tissue conditioner
C	Fully cured heat-cured PMMA, polyamide or polycarbonate
D	Chairside reline resin
E	Light-cured composite base

See answer and justification for Q280 →

Q281 INTERMEDIATE · Block D

A patient with an implanted cardiac defibrillator needs gingival management before an impression.
Which instrument needs caution?

A	Hand curettes
B	Electrosurgery (and some ultrasonic devices)
C	Retraction cord
D	A slow-speed handpiece
E	Air–water syringe

See answer and justification for Q281 →

Q283 INTERMEDIATE · Block D

A 52-year-old wants a 3-unit bridge. She has active, untreated periodontitis with bleeding and 6 mm pockets around the proposed abutments.
What is the most appropriate plan?

A	Prepare the bridge now and treat the periodontitis later
B	Use resin-bonded bridges to avoid preparation
C	Extract the abutments
D	Stabilise the periodontal disease first, then reassess before any definitive bridge
E	Prescribe antibiotics and proceed

See answer and justification for Q283 →

Q284 INTERMEDIATE · Block D

Before providing an implant-retained overdenture in a patient with a history of antiresorptive medication, the patient asks only about the expected functional benefits. Which consent discussion is most appropriate?

A	Discuss likely benefit but omit long-term cleaning obligations
B	Explain benefit, maintenance, reasonable alternatives and relevant medical risks including MRONJ where applicable
C	Assure the patient that implants eliminate all future biological complications
D	Present the signed treatment estimate as the complete consent record
E	Discuss only immediate surgical complications and no non-surgical options

See answer and justification for Q284 →

Q285 INTERMEDIATE · Block D

An 85-year-old care-home resident with Parkinson's disease needs replacement complete dentures.
Which approach is most appropriate?

A	An entirely new design to improve appearance
B	Implant-retained fixed bridges
C	Copy dentures with minimal changes, and labelling for identification
D	Immediate dentures
E	No dentures

See answer and justification for Q285 →

Q286 INTERMEDIATE · Block D

A patient on long-term oral prednisolone for arthritis needs a new partial denture.
Which consequence is most relevant to the prosthodontic care?

A	Steroid cover is needed for impressions
B	Steroids prevent denture stomatitis
C	Increased risk of candidosis and delayed healing
D	Steroids cause gingival overgrowth
E	No relevance at all

See answer and justification for Q286 →

Q287 INTERMEDIATE · Block D

An edentulous patient with Sjögren's syndrome has poor retention despite well-made complete dentures.
Which option should be considered?

A	A thicker denture base
B	An implant-retained overdenture, after assessment
C	Porcelain teeth
D	A higher OVD
E	Removing the post-dam

See answer and justification for Q287 →

Q288 INTERMEDIATE · Block D

A 35-year-old has erosion from gastro-oesophageal reflux that is not yet controlled. She wants her worn teeth crowned.
What is the most appropriate approach?

A	Crown all affected teeth now
B	Extract the worst teeth
C	Refuse all treatment
D	Prescribe antibiotics
E	Manage the cause with her GMP first

See answer and justification for Q288 →

Q289 INTERMEDIATE · Block D

A patient with recently diagnosed, unstable angina asks for a complex bridge.
What is the most appropriate approach?

A	Stabilise oral disease, liaise with her physician, and defer elective definitive prosthodontics until her condition is stable
B	Proceed with all treatment in one long session
C	Refuse all care permanently
D	Use IV sedation and proceed
E	Prescribe GTN and proceed

See answer and justification for Q289 →

Q290 INTERMEDIATE · Block D

An ASA III patient with severe COPD requests IV sedation for implant-supported overdenture work.
What is most appropriate?

A	Sedate in the practice
B	Use inhalation sedation at 80% nitrous oxide
C	Treat without sedation regardless of anxiety
D	Refer to a secondary care setting
E	Give oral diazepam and proceed

See answer and justification for Q290 →

BLOCK E · Assessment, diagnosis, imaging, classifications and planning (60)

Q291 INTERMEDIATE · Block E

A 45-year-old has generalised interdental attachment loss consistent with Stage III periodontitis and radiographic bone loss affecting approximately 40% of root length. No reliable records show progression. The patient smokes 14 cigarettes/day. Which grading decision is best justified?

A	Apply Grade A to reflect little current plaque despite historical attachment loss
B	Retain Grade B from the indirect estimate and handle smoking exclusively as a separate prognostic note
C	Apply Grade C because smoking at least 10 cigarettes daily is a major grade modifier
D	Defer grading until five years of direct bone-loss measurements are available
E	Reclassify the stage, rather than grade, based on the tobacco exposure

See answer and justification for Q291 →

Q292 INTERMEDIATE · Block E

A 51-year-old with Stage II periodontitis has no reliable longitudinal bone-loss films. The indirect progression estimate suggests Grade B, but the patient has type 2 diabetes and an HbA1c of 8.2%. How should this information affect grading and restorative planning?

A	Preserve Grade B and document diabetes only in the medical-history section
B	Use Grade C as a risk-modified classification and prioritise glycaemic/periodontal control before complex reconstruction
C	Increase stage from II to IV because glycaemic status alone determines complexity
D	Proceed to definitive implants once the crown design is acceptable despite persistent active inflammation
E	Replace non-surgical periodontal therapy with systemic antibiotic treatment

See answer and justification for Q292 →

Q293 INTERMEDIATE · Block E

A 56-year-old has a mandibular molar with a Class II buccal furcation, residual 6-mm probing after subgingival instrumentation and good home plaque control. The molar is restorable, and the patient strongly wishes to retain it. Which plan best reflects evidence and scope?

A	Recommend extraction because Class II furcation indicates inherently unrecoverable tooth prognosis
B	Continue routine supportive recall only despite a persistent 6-mm active site and good hygiene
C	Complete definitive prosthodontics before addressing residual furcation inflammation
D	Offer further periodontal treatment discussion and referral for surgical/regenerative assessment when appropriate
E	Repeat nonsurgical instrumentation indefinitely without discussing additional therapy or prognosis

See answer and justification for Q293 →

Q295 INTERMEDIATE · Block E

A patient presents with worn complete dentures and requests a duplicate pair. Examination shows the old bases are retentive but the OVD and occlusal contacts have changed significantly. What is the most appropriate use of the old prostheses?

A	Replicate every aspect without review because the patient likes them
B	Use only their acrylic colour for reference
C	Discard them before history-taking
D	Use them as a reference for accepted contours while re-assessing fit, jaw relation, aesthetics and wear
E	Select new teeth solely from the existing tooth mould

See answer and justification for Q295 →

Q296 INTERMEDIATE · Block E

A denture wearer has episodic burning/electric pain without local mucosal changes, and symptoms persist when the appliance is removed. What should be considered?

A	Periodontal disease of a poorly cleansable abutment
B	Denture-associated candidosis requiring hygiene/fit review
C	Potential oral malignancy requiring urgent referral
D	Traumatic ulcer from denture flange movement
E	Neuropathic pain requiring further assessment

See answer and justification for Q296 →

Q297 INTERMEDIATE · Block E

A patient asks for “Hollywood teeth” but gives conflicting descriptions of what success would mean and has little understanding of maintenance. What should be addressed first?

A	Reconcile medical risks and drug interactions before invasive care
B	Establish periodontal, endodontic and restorative prognosis of proposed abutments
C	Investigate persistent unexplained soft-tissue lesions urgently
D	Use diagnostic wax-up/mock-up to test functional and aesthetic changes
E	Clarify the patient's main concerns and expectations

See answer and justification for Q297 →

Q300 INTERMEDIATE · Block E

A fixed bridge is proposed across a periodontally compromised arch with previous endodontic treatment of uncertain status. What is the most important diagnostic step?

A	Use diagnostic wax-up/mock-up to test functional and aesthetic changes
B	Reconcile medical risks and drug interactions before invasive care
C	Investigate persistent unexplained soft-tissue lesions urgently
D	Clarify the patient's main concerns and expectations
E	Establish periodontal, endodontic and restorative prognosis of proposed abutments

See answer and justification for Q300 →

Q301 INTERMEDIATE · Block E

A 10-year-old has severe early periodontal destruction with premature loss of primary teeth and hyperkeratotic lesions on the palms and soles. Several family members are unaffected. Which condition is the most appropriate diagnostic consideration?

A	Papillon–Lefèvre syndrome
B	Simple plaque-induced gingivitis without systemic association
C	Isolated enamel fluorosis
D	Ehlers–Danlos syndrome of unspecified subtype
E	Acute necrotising gingivitis as the sole explanation

See answer and justification for Q301 →

Q302 INTERMEDIATE · Block E

A child presents with marked gingival inflammation and early periodontal breakdown, recurrent severe infections, partial oculocutaneous hypopigmentation and easy bruising. Which associated disorder most warrants medical evaluation?

A	Isolated TMD with secondary plaque accumulation
B	Hereditary enamel hypoplasia alone
C	Uncomplicated fluorosis
D	Papillon–Lefèvre syndrome without skin lesions
E	Chediak–Higashi syndrome involving defective lysosomal trafficking and immune dysfunction

See answer and justification for Q302 →

Q303 INTERMEDIATE · Block E

A patient on anticoagulants and antiresorptive therapy requires extractions to create an overdenture. What assessment should occur before scheduling surgery?

A	Reconcile medical risks and drug interactions before invasive care
B	Investigate persistent unexplained soft-tissue lesions urgently
C	Establish periodontal, endodontic and restorative prognosis of proposed abutments
D	Use diagnostic wax-up/mock-up to test functional and aesthetic changes
E	Clarify the patient's main concerns and expectations

See answer and justification for Q303 →

Q304 INTERMEDIATE · Block E

A non-healing red-and-white ulcerated lesion is found under an otherwise serviceable partial denture. What should take precedence?

A	Investigate persistent unexplained soft-tissue lesions urgently
B	Use diagnostic wax-up/mock-up to test functional and aesthetic changes
C	Establish periodontal, endodontic and restorative prognosis of proposed abutments
D	Reconcile medical risks and drug interactions before invasive care
E	Clarify the patient's main concerns and expectations

See answer and justification for Q304 →

Q308 INTERMEDIATE · Block E

A patient with Stage III periodontitis attends three months after non-surgical treatment. Plaque control has improved, but a mandibular molar retains a bleeding 7-mm site beside a Class II furcation. The patient wishes to retain the tooth as a future RPD abutment. Which decision is best?

A	Proceed with final RPD impressions because initial treatment was already completed
B	Provide indefinite supragingival polishing only without discussing the remaining deep pocket
C	Extract immediately because a Class II furcation cannot be treated
D	Reassess overall risk and prognosis, pursue further site-specific periodontal therapy and discuss referral for advanced furcation treatment before committing to the RPD design
E	Use a clasp to splint the molar and treat the periodontal defect indirectly

See answer and justification for Q308 →

Q309 INTERMEDIATE · Block E

A full upper acrylic partial denture is worn overnight and the underlying palatal mucosa is diffusely erythematous with plaque on the fitting surface. Which diagnosis is most likely?

A	Periodontal disease of a poorly cleansable abutment
B	Neuropathic pain requiring further assessment
C	Denture-associated candidosis requiring hygiene/fit review
D	Traumatic ulcer from denture flange movement
E	Potential oral malignancy requiring urgent referral

See answer and justification for Q309 →

Q313 INTERMEDIATE · Block E

A patient's periodontitis affects 6 of 26 teeth.
Is the extent localised or generalised?

A	Generalised, because more than 5 teeth are affected
B	Localised, because fewer than 30% of teeth are affected
C	Generalised, because more than 25% are affected
D	Molar–incisor pattern
E	Extent is not recorded

See answer and justification for Q313 →

Q314 INTERMEDIATE · Block E

After treatment, a patient has bleeding on probing at 6% of sites, all pockets 4 mm or less, and no bleeding at 4 mm sites.
What is the periodontal status?

A	Currently stable
B	In remission
C	Unstable
D	Stage IV
E	Grade C

See answer and justification for Q314 →

Q315 INTERMEDIATE · Block E

A treated patient has bleeding on probing at 15% of sites, all pockets 4 mm or less, and no bleeding at 4 mm sites.
What is the periodontal status?

A	Currently stable
B	Unstable
C	Currently in remission
D	Untreated
E	Stage IV

See answer and justification for Q315 →

Q317 INTERMEDIATE · Block E

A 58-year-old has lost 6 teeth to periodontitis, has fewer than 20 teeth, and has drifting with masticatory dysfunction.
Which stage is this likely to be?

A	Stage IV
B	Stage I
C	Stage II
D	Stage III Grade A
E	It cannot be staged

See answer and justification for Q317 →

Q318 INTERMEDIATE · Block E

After steps 1 and 2 of periodontal therapy in primary care, a patient has Stage III Grade A periodontitis.
At what NHS complexity level does further care sit?

A	Level 1
B	Level 3
C	Not provided
D	Hygienist only
E	Level 2

See answer and justification for Q318 →

Q319 INTERMEDIATE · Block E

A 35-year-old has Stage III Grade C periodontitis.
At what NHS complexity level does care sit?

A	Level 1
B	Level 2
C	Hygienist only
D	Level 3
E	Not provided

See answer and justification for Q319 →

Q320 INTERMEDIATE · Block E

During a BPE, the black band of the probe is partly visible in a sextant.
What code is this?

A	Code 2
B	Code 3 (pocket 3.5–5.5 mm)
C	Code 4
D	Code *
E	Code 1

See answer and justification for Q320 →

Q322 INTERMEDIATE · Block E

A lower first molar has a furcation lesion that a probe passes right through from buccal to lingual.
What is the Hamp furcation grade?

A	Grade I
B	Grade II
C	Grade 0
D	Grade III
E	Grade IV

See answer and justification for Q322 →

Q323 INTERMEDIATE · Block E

An irregular indurated ulcer on the lateral tongue persists three weeks after all possible denture trauma has been removed. What must be excluded urgently?

A	Potential oral malignancy requiring urgent referral
B	Traumatic ulcer from denture flange movement
C	Periodontal disease of a poorly cleansable abutment
D	Neuropathic pain requiring further assessment
E	Denture-associated candidosis requiring hygiene/fit review

See answer and justification for Q323 →

Q325 INTERMEDIATE · Block E

A patient's upper central incisors are retroclined with a deep overbite, and the lower incisal edges lie behind the cingulum plateau.
What is the incisor relationship?

A	Class I
B	Class II division 1
C	Class III
D	Class I with crowding
E	Class II division 2

See answer and justification for Q325 →

Q326 INTERMEDIATE · Block E

A patient considering an increase in OVD has uncertain speech and aesthetics with several competing fixed options. What diagnostic step can reduce irreversible risk?

A	Investigate persistent unexplained soft-tissue lesions urgently
B	Clarify the patient's main concerns and expectations
C	Reconcile medical risks and drug interactions before invasive care
D	Establish periodontal, endodontic and restorative prognosis of proposed abutments
E	Use diagnostic wax-up/mock-up to test functional and aesthetic changes

See answer and justification for Q326 →

Q329 INTERMEDIATE · Block E

You need to assess bone level, root form and periapical status of a proposed bridge abutment.
Which radiograph is most appropriate?

A	Panoramic radiograph
B	Periapical radiograph using the paralleling technique
C	Occlusal radiograph
D	Lateral cephalogram
E	CBCT

See answer and justification for Q329 →

Q330 INTERMEDIATE · Block E

A patient with moderate periodontitis needs bone levels recorded in the posterior segments.
Which radiograph is best?

A	Horizontal bitewings
B	Lateral cephalogram
C	Vertical bitewings
D	Occlusal radiographs
E	CBCT

See answer and justification for Q330 →

Q331 INTERMEDIATE · Block E

A panoramic radiograph shows a hazy radiopacity over the right ramus. The patient has an earring in the left ear.
Which statement explains the artefact?

A	It is a calcified lymph node
B	It appears on the same side, smaller and lower
C	It is a sialolith
D	It indicates a retained root
E	It is a ghost image, projected on the opposite side, higher and larger

See answer and justification for Q331 →

Q333 INTERMEDIATE · Block E

Under IR(ME)R 2017, a dental nurse takes a radiograph requested by the dentist.
Who justifies the exposure?

A	The practitioner (here, the dentist)
B	The operator
C	The employer
D	The radiation protection adviser
E	The patient

See answer and justification for Q333 →

Q335 INTERMEDIATE · Block E

A patient worried about radiation asks which dental radiograph gives the highest dose.
What is the order from lowest to highest effective dose?

A	CBCT, panoramic, intraoral
B	Intraoral, panoramic, CBCT
C	Panoramic, intraoral, CBCT
D	All are the same
E	Intraoral, CBCT, panoramic

See answer and justification for Q335 →

Q336 INTERMEDIATE · Block E

You need a reliable sensibility test on a lower premolar abutment.
Which thermal test is more reliable?

A	Hot gutta-percha
B	Warm water bath
C	Ice from the freezer
D	Cold, using tetrafluoroethane or ethyl chloride on a pledget
E	Hot air

See answer and justification for Q336 →

Q337 INTERMEDIATE · Block E

An upper central incisor showed no response to the electric pulp test 2 weeks after a sporting injury. There is no discolouration or tenderness.
What is the correct interpretation?

A	The pulp is necrotic; start root canal treatment now
B	The EPT measures nerve response, not blood flow
C	The tooth must be extracted
D	The EPT is always accurate after trauma
E	The tooth is ankylosed

See answer and justification for Q337 →

Q338 INTERMEDIATE · Block E

A researcher wants to measure true pulpal vitality (blood flow) in traumatised teeth.
Which test assesses blood flow?

A	Electric pulp testing
B	Cold testing
C	Heat testing
D	Percussion
E	Laser Doppler flowmetry or pulse oximetry

See answer and justification for Q338 →

Q339 INTERMEDIATE · Block E

A patient has sharp pain on biting with a lower molar that has a large amalgam; the pain stops on release.
Which test helps confirm a cracked cusp?

A	Electric pulp test
B	Periapical radiograph alone
C	A bite test on individual cusps (for example a Tooth Slooth)
D	Percussion of the opposite tooth
E	Mobility testing

See answer and justification for Q339 →

Q340 INTERMEDIATE · Block E

You are planning a lower Co-Cr RPD and a crown on an abutment.
What is the main value of mounted study casts on a semi-adjustable articulator with a facebow and RCP record?

A	To assess interocclusal space, plan a wax-up and survey the RPD design
B	To measure caries risk
C	To assess pulp vitality
D	To replace radiographs
E	To check periodontal pockets

See answer and justification for Q340 →

Q341 INTERMEDIATE · Block E

A patient wants composite build-ups for worn incisors but is unsure about the look.
What is the best way to test aesthetics and the proposed OVD before treatment?

A	A diagnostic wax-up and intraoral mock-up
B	A panoramic radiograph
C	A photograph only
D	A verbal description
E	Proceeding directly with composite

See answer and justification for Q341 →

Q342 INTERMEDIATE · Block E

A denture wearer with angular cheilitis and glossitis has a full blood count showing a low MCV.
What deficiency is most likely?

A	Vitamin B12
B	Iron
C	Folate
D	Vitamin D
E	Zinc

See answer and justification for Q342 →

Q345 INTERMEDIATE · Block E

A patient asks for root canal treatment of a lower molar under the NHS before a crown.
What does the NHS standard require for the tooth to enter endodontic care?

A	Any tooth may be treated
B	The tooth must already be crowned
C	The patient must be over 50
D	Sound coronal tissue above the crest (about 2 mm high and 1 mm wide, or over 30% of the original crown)
E	A periapical lesion must be present

See answer and justification for Q345 →

Q346 INTERMEDIATE · Block E

A maxillary premolar has an isolated narrow 10-mm pocket, pain on biting, and an inconsistent response to cold. General periodontal charting shows only mild disease elsewhere. A recent crown was fitted after root treatment. Which investigation is most important before labelling the lesion isolated periodontal disease?

A	Refer for new complete dentures without further testing
B	Treat only with repeated subgingival instrumentation
C	Prescribe antibiotics and assume periodontal remission
D	Increase the occlusal vertical dimension to reduce axial load
E	Assess for vertical root fracture and endodontic pathology using careful history, probing pattern, tests and appropriate imaging

See answer and justification for Q346 →

Q347 INTERMEDIATE · Block E

A colleague orders a full radiographic survey for a patient attending only for denture adjustment, with no clinical indication.
Which principle does this break?

A	Imaging should always be routine
B	Radiographs must be taken every visit
C	Investigations should be chosen only when their results may change the diagnosis, prognosis or plan
D	Imaging is optional if the patient pays
E	None

See answer and justification for Q347 →

Q348 INTERMEDIATE · Block E

A denture wearer has a persistent, indurated ulcer under the lower flange. A panoramic radiograph looks normal.
What should you do?

A	Reassure, because the radiograph is normal
B	Refer urgently
C	Adjust the flange and review in 3 months
D	Take a CBCT
E	Prescribe antibiotics

See answer and justification for Q348 →

Q349 INTERMEDIATE · Block E

A 45-year-old presents in pain from a periapical abscess. She also has active caries, untreated periodontitis and missing teeth she wants replaced.
What is the correct sequence of care?

A	Definitive bridges, then caries control
B	Implants first
C	Maintenance, then urgent care
D	Urgent care, stabilisation, re-evaluation, definitive treatment, supportive maintenance
E	Re-evaluation, then urgent care

See answer and justification for Q349 →

Q350 INTERMEDIATE · Block E

A 40-year-old has lost 25 and asks what can be done.
Which options should be discussed?

A	Leaving the space, a resin-bonded bridge, a conventional bridge, an RPD or an implant, with their pros and cons
B	An implant only
C	A conventional bridge only
D	An RPD only
E	Only the cheapest option

See answer and justification for Q350 →

Q351 INTERMEDIATE · Block E

A patient needs a single crown on 36. She has a stable, comfortable intercuspal position with no symptoms.
Which occlusal approach is appropriate?

A	Reorganise to RCP
B	Increase the OVD
C	Full-mouth equilibration first
D	A conformative approach, matching the existing ICP
E	Splint therapy first

See answer and justification for Q351 →

Q352 INTERMEDIATE · Block E

A clasped canine bleeds on probing and has deep pockets with significant plaque retention; a new partial denture is requested. What diagnosis/risk dominates?

A	Neuropathic pain requiring further assessment
B	Periodontal disease of a poorly cleansable abutment
C	Potential oral malignancy requiring urgent referral
D	Denture-associated candidosis requiring hygiene/fit review
E	Traumatic ulcer from denture flange movement

See answer and justification for Q352 →

Q353 INTERMEDIATE · Block E

You are consenting a patient for a conventional bridge.
Which principle does Montgomery establish?

A	The dentist decides what risks to mention
B	Material risks and reasonable alternatives, including no treatment, must be discussed
C	Only risks over 10% need mentioning
D	Written consent is enough
E	The Bolam test alone applies

See answer and justification for Q353 →

Q354 INTERMEDIATE · Block E

You refer a patient with a complex failing dentition to a restorative consultant.
What should the referral contain?

A	The patient's name only
B	A request to "see and treat"
C	The diagnosis, key-tooth prognosis, proposed phases, responsibilities and relevant records
D	Only radiographs
E	The patient's payment details

See answer and justification for Q354 →

Q355 INTERMEDIATE · Block E

A patient with active periodontitis requests four new crowns on the upper anteriors.
What is the main flaw in starting with the crowns?

A	Crowns are not available on the NHS
B	The patient is too young
C	Crowns need no periodontal support
D	Crowns cure periodontitis
E	Definitive restorations before disease control are unsound

See answer and justification for Q355 →

Q356 INTERMEDIATE · Block E

A complete denture wearer has angular cheilitis, deep nasolabial folds and a prominent chin when the dentures are in occlusion.
What does this suggest, and how would you confirm it?

A	Excessive OVD; measure the overjet
B	Allergy; patch test
C	Reduced OVD
D	Candida only; swab
E	TMD; MRI

See answer and justification for Q356 →

Q358 INTERMEDIATE · Block E

Before preparing several crowns, you find a 2 mm slide from RCP to ICP.
Why should this be recorded?

A	It may affect whether a conformative or reorganised approach is needed
B	It indicates caries
C	It shows periodontitis
D	It has no relevance
E	It means the patient has TMD

See answer and justification for Q358 →

Q360 INTERMEDIATE · Block E

A complete denture wearer complains of poor retention and a dry mouth.
How is unstimulated whole saliva flow measured?

A	A blood test
B	Timed collection by spitting or drooling for 5–15 minutes
C	A panoramic radiograph
D	Chewing paraffin wax only
E	A swab

See answer and justification for Q360 →

Q361 INTERMEDIATE · Block E

A new partial denture rubs a sulcus at a reproducible flange contact; the ulcer is shallow and appeared after insertion. Which initial hypothesis best fits?

A	Denture-associated candidosis requiring hygiene/fit review
B	Neuropathic pain requiring further assessment
C	Traumatic ulcer from denture flange movement
D	Periodontal disease of a poorly cleansable abutment
E	Potential oral malignancy requiring urgent referral

See answer and justification for Q361 →

Q362 INTERMEDIATE · Block E

A 19-year-old has bone loss confined to the incisors and first molars.
How is the extent described in the 2017 classification?

A	Generalised
B	Localised
C	Stage IV
D	Necrotising
E	Molar–incisor pattern

See answer and justification for Q362 →

Q363 INTERMEDIATE · Block E

A 40-year-old has localised Stage III Grade B periodontitis after initial therapy.
At what NHS complexity level is further care?

A	Level 1
B	Level 2
C	Level 3
D	Not provided
E	Hygienist only

See answer and justification for Q363 →

Q364 INTERMEDIATE · Block E

You need to check for proximal caries on the abutments of an RPD.
Which radiograph is best?

A	Horizontal bitewings
B	Panoramic radiograph
C	Occlusal radiograph
D	Lateral cephalogram
E	CBCT

See answer and justification for Q364 →

Q365 INTERMEDIATE · Block E

A colleague uses panoramic radiographs alone to screen for early caries.
Why is this inadequate?

A	Panoramic images have no dose
B	They are three-dimensional
C	They show enamel detail perfectly
D	Focal trough distortion and overlap limit fine detail
E	They cannot show teeth

See answer and justification for Q365 →

Q366 INTERMEDIATE · Block E

A patient asks how much radiation a CBCT gives compared with a periapical.
Which approximate figures are correct?

A	Intraoral about 0.001–0.008 mSv
B	Both about 1 Sv
C	Intraoral higher than CBCT
D	CBCT about 10 mSv always
E	Both negligible and equal

See answer and justification for Q366 →

Q367 INTERMEDIATE · Block E

An anxious patient reports a response to the electric pulp test on a tooth with a periapical radiolucency.
What could explain a false-positive result?

A	Recent trauma
B	An immature apex
C	Anxiety, or liquefactive necrosis conducting the current
D	A calcified pulp
E	Old age

See answer and justification for Q367 →

Q368 INTERMEDIATE · Block E

An 82-year-old's lower canine, with a sound restoration and no radiographic lesion, does not respond to the EPT.
What is a likely explanation?

A	Liquefactive necrosis
B	Anxiety
C	Too much current
D	Calcification and reduced pulp size with age
E	Pulpitis

See answer and justification for Q368 →

Q369 INTERMEDIATE · Block E

An arch has one large free-end saddle and two small bounded spaces.
How are the bounded spaces handled in Kennedy classification?

A	They change the class
B	They are counted as modification spaces by number, whatever their size
C	Only the largest counts
D	They are ignored
E	Each is a separate class

See answer and justification for Q369 →

Q370 INTERMEDIATE · Block E

An RPD is being redesigned around an abutment with declining periodontal support and a repeatedly failing core restoration. Which factor most directly determines whether that tooth remains strategically suitable?

A	Patient satisfaction with old clasp colour
B	Occlusal plane photographs alone
C	Clasp survey line alone
D	Integrated periodontal, endodontic and restorability prognosis
E	Presence of a posterior implant at another site

See answer and justification for Q370 →

BLOCK F · Removable partial dentures (75)

Q372 INTERMEDIATE · Block F

An RPD is tight but the supporting tooth tips laterally each time the clasp passes the height of contour. What design defect is most likely?

A	An ill-fitting distal saddle increases torque on the terminal abutment
B	Loss of an occlusal rest shifts load to soft tissue
C	Excessive undercut increases strain at insertion/removal
D	A short thick arm has excessive bending stiffness
E	Failure of reciprocation transmits lateral force to the abutment

See answer and justification for Q372 →

Q373 INTERMEDIATE · Block F

At framework try-in, a minor connector binds against a restoration and prevents complete seating. The patient asks that the clasps be tightened. What is the best next step?

A	Review occlusal rest-seat form to obtain tooth-supported load
B	Correct an interfering framework component rather than increasing retention
C	Change the insertion path and survey design
D	Select a longer or more flexible retentive arm in a suitable undercut
E	Modify a guide plane/reciprocal element to improve stability

See answer and justification for Q373 →

Q374 INTERMEDIATE · Block F

A mandibular Kennedy Class I RPD has rigid circumferential clasps on both terminal premolars and distal rests. The mucosa-supported saddles sink during chewing and the abutments exhibit increasing distal tipping. Which redesign has the strongest mechanical rationale?

A	Retain distal rests but use longer, more flexible circumferential arms while keeping the same base support
B	Keep the rigid retentive arm but add indirect retention alone to control displacement of the saddle
C	Rebase both saddles without changing the original terminal-rest and retainer geometry
D	Reconsider mesial rest support, guide planes, appropriate stress release and tissue support as an integrated redesign
E	Add more rigid splinting to the terminal premolars so the original distal-rest lever system can be maintained

See answer and justification for Q374 →

Q375 INTERMEDIATE · Block F

A Co–Cr clasp engages a 0.75-mm undercut rather than the planned 0.25 mm, and becomes permanently distorted after repeated use. Which mechanism is most plausible?

A	Excessive undercut increases strain at insertion/removal
B	Failure of reciprocation transmits lateral force to the abutment
C	A short thick arm has excessive bending stiffness
D	Loss of an occlusal rest shifts load to soft tissue
E	An ill-fitting distal saddle increases torque on the terminal abutment

See answer and justification for Q375 →

Q378 INTERMEDIATE · Block F

A mandibular bilateral free-end saddle RPD has firm terminal premolar abutments but compressible posterior residual ridges. On the refractory framework try-in the rests seat fully, yet the saddles sink under functional loading. What is the best rationale for considering an altered-cast technique?

A	Record the displaceable distal saddle tissues with the metal framework fully seated on stable tooth rests
B	Re-record tooth guide planes and rest-seat location with a ridge impression after the framework try-in
C	Increase the compression of the tooth-supported rests while avoiding any change in saddle support
D	Capture the distal-extension tissue at rest while leaving the metal framework off the teeth
E	Reposition the fulcrum line of an incorrectly designed framework through a new master cast

See answer and justification for Q378 →

Q379 INTERMEDIATE · Block F

A patient has a severe undercut at a lingually tilted mandibular molar and a large lingual bony prominence. The missing anterior teeth need replacement, but a conventional rigid RPD cannot be inserted along a single acceptable path. The patient requests a flexible polyamide denture. What is the best next planning step?

A	Choose polyamide if retention is prioritised over rest seats, because greater base flexibility can address any insertion conflict
B	Use a metal framework with a shortened lingual connector so that it flexes around the bony prominence
C	Select the conventional framework and prepare the molar for a deeper clasp undercut without re-surveying the cast
D	Use a hinged sectional prosthesis immediately without testing joint clearance, hygiene access or maintenance burden
E	Re-survey available paths and evaluate tooth modification, sectional insertion and alternative prostheses against support and maintainability

See answer and justification for Q379 →

Q380 INTERMEDIATE · Block F

A partially dentate 68-year-old needs replacement of lower incisors. Both lower second molars are tipped severely lingually and bilateral mandibular tori reduce the lingual sulcus. The preliminary design proposes a conventional lingual bar that would interfere with the tori. Which action best precedes choosing a connector?

A	Ignore the posterior tilt because only anterior teeth are to be replaced
B	Use a lower lingual bar at the same height and relieve the metal until it is flexible
C	Decide a surveyed path of insertion and evaluate available lingual clearance, tooth modifications and alternative major connectors
D	Remove rests from the anterior replacement to increase connector freedom
E	Select a clasp undercut solely by whether the patient prefers gold or cobalt–chromium

See answer and justification for Q380 →

Q381 INTERMEDIATE · Block F

A new mandibular RPD framework looks satisfactory on the refractory cast but rocks on the patient: one rest lifts when the opposite rest is seated. The technician proposes extending the acrylic saddles to stabilise it before the metal fit is corrected. Which decision is most appropriate?

A	Record an altered cast impression immediately while accepting the unseated metal rests
B	Increase retentive clasp engagement until the framework is forced into position
C	Add tissue conditioner beneath both saddles before checking whether the framework passively fits
D	Identify framework distortion, casting nodules, guide-plane interferences and rest seating; correct or remake the framework before jaw registration and acrylic processing
E	Process the acrylic bases first and adjust the occlusion until all rests appear level

See answer and justification for Q381 →

Q382 INTERMEDIATE · Block F

15 has no retentive undercut on the buccal surface for the chosen path.
How can an undercut be created conservatively?

A	Add composite to create a contour with a suitable undercut
B	Crown the tooth
C	Grind the occlusal surface
D	Extract it
E	Tilt the cast until any undercut appears

See answer and justification for Q382 →

Q383 INTERMEDIATE · Block F

A Co–Cr RPD framework cannot seat completely because one proximal plate binds on a nonparallel tooth surface; the survey design permits an alternative path. What should be reassessed first?

A	Modify a guide plane/reciprocal element to improve stability
B	Change the insertion path and survey design
C	Review occlusal rest-seat form to obtain tooth-supported load
D	Correct an interfering framework component rather than increasing retention
E	Select a longer or more flexible retentive arm in a suitable undercut

See answer and justification for Q383 →

Q384 INTERMEDIATE · Block F

A gingivally approaching clasp arm would cross a deep buccal soft-tissue undercut and impinge during insertion. What design choice is most important?

A	Recontour the survey crown for a suitable guide plane
B	Alter clasp approach to avoid a soft-tissue undercut
C	Reassess indirect retention about the functional fulcrum line
D	Reassess periodontal prognosis before using an abutment
E	Re-survey with a corrected path of insertion

See answer and justification for Q384 →

Q385 INTERMEDIATE · Block F

A short cast clasp has fractured repeatedly near its origin while engaging a deep undercut on a premolar. Which design modification most directly addresses the fatigue issue?

A	Review occlusal rest-seat form to obtain tooth-supported load
B	Modify a guide plane/reciprocal element to improve stability
C	Select a longer or more flexible retentive arm in a suitable undercut
D	Change the insertion path and survey design
E	Correct an interfering framework component rather than increasing retention

See answer and justification for Q385 →

Q386 INTERMEDIATE · Block F

A rest is needed on a lower canine for an RPD.
Which rest is preferred?

A	An incisal rest
B	A cingulum rest (on a natural or composite-built cingulum)
C	A rest on the labial surface
D	A rest on the cusp tip
E	No rest

See answer and justification for Q386 →

Q387 INTERMEDIATE · Block F

A new RPD is retentive but shifts horizontally with chewing; there is no effective reciprocal contact opposite the retentive clasp. What should be corrected?

A	Change the insertion path and survey design
B	Select a longer or more flexible retentive arm in a suitable undercut
C	Modify a guide plane/reciprocal element to improve stability
D	Review occlusal rest-seat form to obtain tooth-supported load
E	Correct an interfering framework component rather than increasing retention

See answer and justification for Q387 →

Q388 INTERMEDIATE · Block F

A mesial rest is considered on a premolar adjacent to a free-end saddle, with an appropriate RPI/RPA concept under review. What is the principal reason to consider this modification?

A	Provide appropriately placed indirect retention
B	Reassess the distal-extension design and abutment prognosis before altering clasp tightness
C	Review clasp flexibility and location of the retentive tip
D	Alter the rest position/design to accommodate tissueward rotation
E	Improve broad, functional saddle support while maintaining hygiene

See answer and justification for Q388 →

Q390 INTERMEDIATE · Block F

A new RPD will not insert because its rigid proximal plates bind when the dentist follows an arbitrary path. What design decision should be revisited?

A	Reassess periodontal prognosis before using an abutment
B	Recontour the survey crown for a suitable guide plane
C	Re-survey with a corrected path of insertion
D	Alter clasp approach to avoid a soft-tissue undercut
E	Reassess indirect retention about the functional fulcrum line

See answer and justification for Q390 →

Q391 INTERMEDIATE · Block F

A Kennedy Class II lower arch has no modification space on the dentate side.
How is retention usually gained on that side?

A	An embrasure clasp, with the embrasure prepared for space
B	An I-bar on the canine
C	A ring clasp on the canine
D	No retention
E	A labial bar

See answer and justification for Q391 →

Q392 INTERMEDIATE · Block F

A patient with progressive recession around a distal-extension abutment is referred solely for retightening the clasp. The base is also unstable and recent radiographs show reduced support. Which response is best?

A	Provide appropriately placed indirect retention
B	Improve broad, functional saddle support while maintaining hygiene
C	Reassess the distal-extension design and abutment prognosis before altering clasp tightness
D	Review clasp flexibility and location of the retentive tip
E	Alter the rest position/design to accommodate tissueward rotation

See answer and justification for Q392 →

Q393 INTERMEDIATE · Block F

A free-end abutment needs a flexible clasp engaging a 0.5–0.75 mm undercut.
Which clasp is suitable?

A	Cast Co-Cr
B	Wrought wire
C	Cast gold
D	Acrylic
E	Cast titanium

See answer and justification for Q393 →

Q394 INTERMEDIATE · Block F

A bounded-saddle RPD settles into the gingiva and the intended rests have no positive seating on enamel. What structural feature must be revised?

A	Change the insertion path and survey design
B	Modify a guide plane/reciprocal element to improve stability
C	Select a longer or more flexible retentive arm in a suitable undercut
D	Review occlusal rest-seat form to obtain tooth-supported load
E	Correct an interfering framework component rather than increasing retention

See answer and justification for Q394 →

Q395 INTERMEDIATE · Block F

A Kennedy I lower RPD lifts anteriorly when sticky food is removed from distal saddles; supports otherwise appear satisfactory. Which design aspect is most directly relevant?

A	Reassess the distal-extension design and abutment prognosis before altering clasp tightness
B	Review clasp flexibility and location of the retentive tip
C	Improve broad, functional saddle support while maintaining hygiene
D	Alter the rest position/design to accommodate tissueward rotation
E	Provide appropriately placed indirect retention

See answer and justification for Q395 →

Q397 INTERMEDIATE · Block F

A patient asks why her lower free-end RPD moves while her other partial denture does not.
What explains the problem of free-end saddles?

A	Teeth move about 0.02–0.1 mm under load, while mucosa compresses about 0.4–2 mm
B	Mucosa is less compressible than teeth
C	Teeth move more than mucosa
D	Clasps are always loose
E	Saliva is thinner posteriorly

See answer and justification for Q397 →

Q398 INTERMEDIATE · Block F

You are planning the extension of a lower free-end saddle.
What should it cover?

A	Only the ridge crest
B	Half the length of the ridge
C	The retromolar pad and buccal shelf
D	The floor of the mouth
E	The external oblique ridge only

See answer and justification for Q398 →

Q399 INTERMEDIATE · Block F

A lower Kennedy Class I RPD is planned for a patient with a resorbed, displaceable ridge.
Which impression technique records the saddle area functionally?

A	Alginate in a stock tray
B	Putty only
C	A mucostatic plaster impression
D	A one-stage impression of the teeth and ridge in compound
E	The altered cast technique (Applegate)

See answer and justification for Q399 →

Q401 INTERMEDIATE · Block F

A patient bites on the free-end saddle of a lower RPD with an RPI assembly on 34.
What happens to the I-bar and proximal plate?

A	They tighten against the tooth
B	They move gingivally and mesially, disengaging from the tooth and reducing torque
C	They move occlusally
D	They lock the denture in place
E	Nothing

See answer and justification for Q401 →

Q402 INTERMEDIATE · Block F

44 is the terminal abutment of a free-end saddle, but the buccal sulcus is 3 mm deep with a soft-tissue undercut.
Which clasp assembly is appropriate?

A	RPI
B	Ring clasp
C	Gingivally approaching T-bar
D	No clasp
E	RPA, replacing the I-bar with an Akers-type arm

See answer and justification for Q402 →

Q403 INTERMEDIATE · Block F

You are designing indirect retention for a lower Kennedy Class I RPD.
Where does the fulcrum line run?

A	Along the lingual bar
B	Across the incisors
C	Through the clasp tips
D	Through the rests nearest to the free-end saddles
E	Along the ridge crest

See answer and justification for Q403 →

Q404 INTERMEDIATE · Block F

A patient's lower bilateral free-end RPD lifts at the back when she eats toffee.
Where should an indirect retainer be placed?

A	On the fulcrum line
B	Behind the fulcrum line
C	As far anterior to the fulcrum line as possible
D	On the saddle
E	On the retromolar pad

See answer and justification for Q404 →

Q406 INTERMEDIATE · Block F

A long lower free-end saddle on a resorbed ridge is being set up.
How should the saddle teeth be arranged?

A	Wide, cusped teeth
B	Extra teeth to the retromolar pad
C	Porcelain teeth
D	High occlusal contacts
E	Narrow, reduced occlusal tables in harmonious occlusion

See answer and justification for Q406 →

Q408 INTERMEDIATE · Block F

A 70-year-old has only 13–23 remaining. The palate is flat and resorbed.
Which maxillary connector is appropriate?

A	Full palatal plate
B	Narrow palatal bar
C	Horseshoe
D	Labial bar
E	Mid-palatal strap

See answer and justification for Q408 →

Q409 INTERMEDIATE · Block F

An upper arch has short bounded saddles on both sides (16 and 26 missing).
Which maxillary connector is appropriate?

A	Full palatal plate
B	Mid-palatal strap, at least 8 mm wide
C	Horseshoe
D	Anterior palatal bar
E	Labial bar

See answer and justification for Q409 →

Q410 INTERMEDIATE · Block F

An upper arch has anterior and posterior saddles and a large torus that the patient will not have removed.
Which connector is appropriate?

A	Full palatal plate over the torus
B	Narrow mid-palatal strap over the torus
C	Horseshoe
D	Anteroposterior strap avoiding the torus
E	Lingual bar

See answer and justification for Q410 →

Q411 INTERMEDIATE · Block F

A very short cast clasp on a premolar fractures during insertion even with modest undercut engagement. Which geometry variable best explains the higher stress?

A	A short thick arm has excessive bending stiffness
B	Loss of an occlusal rest shifts load to soft tissue
C	Excessive undercut increases strain at insertion/removal
D	Failure of reciprocation transmits lateral force to the abutment
E	An ill-fitting distal saddle increases torque on the terminal abutment

See answer and justification for Q411 →

Q412 INTERMEDIATE · Block F

A lower RPD is planned. From the gingival margin to the functional floor of the mouth there are 9 mm. Which connector and dimensions are correct?

A	Lingual plate, because the sulcus is deep
B	Labial bar
C	Sublingual bar
D	Lingual bar: half-pear, about 4 mm high, at least 3 mm clear of the gingival margin
E	No connector

See answer and justification for Q412 →

Q413 INTERMEDIATE · Block F

A 66-year-old has mobile lower incisors, a 6 mm lingual sulcus, and may lose more teeth. Which lower connector is appropriate?

A	Lingual plate
B	Lingual bar
C	Labial bar
D	Sublingual bar
E	Dental bar

See answer and justification for Q413 →

Q414 INTERMEDIATE · Block F

A shallow lingual sulcus would place a lingual bar on the gingivae, but the incisors are healthy. Which connector lies deeper in the sulcus and needs a functional impression?

A	Lingual plate
B	Sublingual bar
C	Labial bar
D	Horseshoe
E	Continuous clasp

See answer and justification for Q414 →

Q415 INTERMEDIATE · Block F

A patient has markedly lingually inclined lower premolars and large tori.
Which connector is indicated?

A	Lingual bar
B	Lingual plate
C	Sublingual bar
D	Dental bar
E	Labial bar

See answer and justification for Q415 →

Q417 INTERMEDIATE · Block F

A restoration under an RPD rest seat fractures and the rest no longer contacts the abutment. The saddle becomes painful during chewing. Why?

A	An ill-fitting distal saddle increases torque on the terminal abutment
B	Failure of reciprocation transmits lateral force to the abutment
C	A short thick arm has excessive bending stiffness
D	Excessive undercut increases strain at insertion/removal
E	Loss of an occlusal rest shifts load to soft tissue

See answer and justification for Q417 →

Q418 INTERMEDIATE · Block F

A free-end saddle RPD lifts around the line through its terminal rests when the patient pulls sticky food. What component should be reviewed?

A	Recontour the survey crown for a suitable guide plane
B	Reassess periodontal prognosis before using an abutment
C	Re-survey with a corrected path of insertion
D	Alter clasp approach to avoid a soft-tissue undercut
E	Reassess indirect retention about the functional fulcrum line

See answer and justification for Q418 →

Q419 INTERMEDIATE · Block F

A technician proposes a thin, flexible lingual bar to improve comfort.
Why is this wrong?

A	Thin bars are more expensive to cast
B	Rigidity does not matter
C	Major connectors must be rigid so that forces spread across the arch
D	Flexible bars improve retention
E	Lingual bars must be acrylic

See answer and justification for Q419 →

Q421 INTERMEDIATE · Block F

Several teeth of uncertain prognosis remain, and further extractions are likely within a year.
Which RPD is most appropriate?

A	A Co-Cr framework with precision attachments
B	A swing-lock
C	A flexible nylon denture
D	Implants
E	An acrylic interim (transitional) denture

See answer and justification for Q421 →

Q422 INTERMEDIATE · Block F

An upper acrylic RPD is designed with point contacts on the teeth, open embrasures, free gingival margins and distal stabilising wires.
What design is this?

A	A spoon denture
B	An Every denture
C	A swing-lock
D	A horseshoe Co-Cr
E	An immediate denture

See answer and justification for Q422 →

Q423 INTERMEDIATE · Block F

The proposed terminal abutment has active grade C periodontal disease and progressive mobility. What takes precedence over choosing a clasp?

A	Reassess periodontal prognosis before using an abutment
B	Alter clasp approach to avoid a soft-tissue undercut
C	Re-survey with a corrected path of insertion
D	Recontour the survey crown for a suitable guide plane
E	Reassess indirect retention about the functional fulcrum line

See answer and justification for Q423 →

Q424 INTERMEDIATE · Block F

A patient has missing anterior teeth and periodontally compromised remaining teeth with labial undercuts, where conventional clasps work poorly.
Which design is indicated?

A	An Every denture
B	A spoon denture
C	A flexible nylon denture
D	A swing-lock denture
E	A horseshoe

See answer and justification for Q424 →

Q428 INTERMEDIATE · Block F

A cobalt–chromium framework is inserted for trial. Occlusal rests appear high, the framework rocks and the clasp arms seem tight. What is the best first technical step?

A	Bend every clasp inward immediately
B	Add wax beneath all rests
C	Use a disclosing medium to locate interference and verify full rest seating before evaluating retention
D	Force the framework into place under bite pressure
E	Reduce all occlusal rests without identifying the binding site

See answer and justification for Q428 →

Q429 INTERMEDIATE · Block F

A surveyed crown has a bulbous proximal surface inconsistent with the agreed guide plane; the framework is stable on the other abutments. What is most appropriate?

A	Reassess periodontal prognosis before using an abutment
B	Reassess indirect retention about the functional fulcrum line
C	Recontour the survey crown for a suitable guide plane
D	Re-survey with a corrected path of insertion
E	Alter clasp approach to avoid a soft-tissue undercut

See answer and justification for Q429 →

Q433 INTERMEDIATE · Block F

A laboratory returns a Co-Cr framework with clasps on different teeth from those planned.
Who is responsible for the design?

A	The technician
B	The patient
C	The practice manager
D	The dental nurse
E	The dentist, whose prescription must specify every component

See answer and justification for Q433 →

Q434 INTERMEDIATE · Block F

A student designs a retentive clasp on 45 with no reciprocal element.
Why is the design incomplete?

A	Retention is too high
B	Without reciprocation, the retentive arm pushes the tooth as it flexes
C	It is too rigid
D	It looks bad
E	Reciprocation is optional

See answer and justification for Q434 →

Q435 INTERMEDIATE · Block F

A patient has pain on biting on 36 a week after fitting a Co-Cr RPD. A rest on 36 marks heavily with articulating paper.
What is the likely cause and remedy?

A	A high rest
B	A loose clasp; tighten it
C	Candida; antifungal
D	Allergy; remake in acrylic
E	Caries; restore

See answer and justification for Q435 →

Q436 INTERMEDIATE · Block F

An upper arch is missing 24, 25 and 26 only, with sound 23 and 27.
What support and indirect retention does this design need?

A	Mucosal support with indirect retainers
B	Tooth support from rests at both ends
C	Free-end design with an RPI
D	A full palatal plate
E	No rests

See answer and justification for Q436 →

Q438 INTERMEDIATE · Block F

A lower Kennedy Class I RPD is planned with 33 to 44 remaining and a deep lingual sulcus.
Which design is most appropriate?

A	Mesial rests and RPI clasps on 34 and 44, cingulum rests on the canines, a lingual bar and maximum saddle extension
B	Distal rests and Akers clasps on the premolars
C	An acrylic mucosa-borne plate
D	A labial bar with no rests
E	Ring clasps on the canines

See answer and justification for Q438 →

Q439 INTERMEDIATE · Block F

An upper arch has 13, 12, 22 and 23 remaining, with bilateral free-end saddles and a high, well-formed vault.
Which connector gives the best support and retention?

A	Horseshoe
B	Narrow palatal bar
C	Full palatal plate
D	Mid-palatal strap
E	Labial bar

See answer and justification for Q439 →

Q442 INTERMEDIATE · Block F

A bilateral distal extension RPD causes early soreness and sinking under occlusal load despite well-adjusted clasps. The record base covers only a narrow strip of each ridge. What should be prioritised?

A	Improve broad, functional saddle support while maintaining hygiene
B	Provide appropriately placed indirect retention
C	Review clasp flexibility and location of the retentive tip
D	Reassess the distal-extension design and abutment prognosis before altering clasp tightness
E	Alter the rest position/design to accommodate tissueward rotation

See answer and justification for Q442 →

Q443 INTERMEDIATE · Block F

You tilt a lower cast on the surveyor so that its heels are lowered.
What happens to the undercuts?

A	Mesial undercuts increase
B	Distal undercuts increase
C	Lingual undercuts vanish
D	Nothing changes
E	All undercuts double

See answer and justification for Q443 →

Q444 INTERMEDIATE · Block F

A distal-extension saddle has resorbed away from the tissue and the abutment’s mobility is increasing, while clasp geometry is unchanged. Which mechanism is most likely?

A	Loss of an occlusal rest shifts load to soft tissue
B	An ill-fitting distal saddle increases torque on the terminal abutment
C	A short thick arm has excessive bending stiffness
D	Excessive undercut increases strain at insertion/removal
E	Failure of reciprocation transmits lateral force to the abutment

See answer and justification for Q444 →

Q445 INTERMEDIATE · Block F

A patient dislikes the buccal circumferential clasp showing on 13.
Which is a more aesthetic option if the sulcus allows?

A	A longer circumferential arm
B	A ring clasp
C	An embrasure clasp
D	A gingivally approaching I-bar into a mesiobuccal undercut, or moving retention to a premolar
E	No retention at all

See answer and justification for Q445 →

Q447 INTERMEDIATE · Block F

A patient with xerostomia and high caries risk needs an RPD.
Which design principle is most important?

A	Maximum tooth coverage
B	Acrylic, mucosa-borne design
C	Plates covering all gingival margins
D	A minimal, open design with little tooth and gingival coverage
E	Precision attachments

See answer and justification for Q447 →

Q450 INTERMEDIATE · Block F

A lower Kennedy Class II RPD fitted 6 years ago now rocks, and the indirect retainer lifts when the saddle is pressed.
What is the correct management?

A	Tighten the clasps
B	Add another clasp
C	Reline the free-end saddle
D	Remove the indirect retainer
E	Extract the abutment

See answer and justification for Q450 →

Q451 INTERMEDIATE · Block F

An abutment tooth has over-erupted slightly since the patient stopped wearing her RPD.
Which function of rests helps prevent this?

A	Retention
B	Preventing over-eruption and maintaining tooth position
C	Bracing only
D	Aesthetics
E	Reciprocation

See answer and justification for Q451 →

Q452 INTERMEDIATE · Block F

A patient requires replacement of the mandibular incisors. Both molars have significant lingual inclination and there are bilateral lingual tori. An impression shows that a planned rigid lingual plate cannot be inserted without interfering with the tori and undercuts. What is the most defensible next step?

A	Use flexible nylon because it automatically provides rigid tooth-borne support
B	Re-survey the cast and evaluate an alternative path, relief, connector design or multidisciplinary management before selecting a definitive prosthesis
C	Relieve all lingual surfaces sufficiently to eliminate every guide plane
D	Use an unmodified lingual bar directly over the tori
E	Extend a rigid connector into the floor of mouth to bypass surveying

See answer and justification for Q452 →

Q453 INTERMEDIATE · Block F

Food collects between a saddle and the abutment of a Co-Cr RPD.
What design feature is likely missing?

A	A rest
B	A reciprocal arm
C	A lingual bar
D	A proximal plate on a guide plane
E	A labial flange

See answer and justification for Q453 →

Q454 INTERMEDIATE · Block F

A Co-Cr RPD is being set up for a patient with a stable natural occlusion.
How should the denture occlusion be arranged?

A	Balanced occlusion like a complete denture
B	At an increased OVD
C	With heavy contacts on the saddle teeth
D	Without contact on the saddle teeth
E	In harmony with the existing ICP

See answer and justification for Q454 →

Q455 INTERMEDIATE · Block F

A tilted lower molar and over-erupted upper molar complicate an RPD design.
How is correcting them before design classed?

A	Level 1, always simple
B	Level 3 only
C	Pre-prosthetic optimisation, within Level 2
D	Unnecessary
E	Specialist orthodontics only

See answer and justification for Q455 →

Q456 INTERMEDIATE · Block F

A 70-year-old has grade 2 mobile lower incisors with stable periodontal disease and needs a free-end RPD.
Which lower connector helps?

A	Lingual bar
B	Labial bar
C	Sublingual bar
D	Lingual plate, splinting the incisors
E	No connector

See answer and justification for Q456 →

Q457 INTERMEDIATE · Block F

A lower lingual bar has sunk onto the lingual gingiva of the incisors.
What is the most likely cause?

A	The bar is too thick
B	The bar is too rigid
C	Inadequate rest support, so the denture settles
D	Allergy
E	Saliva

See answer and justification for Q457 →

Q458 INTERMEDIATE · Block F

A mandibular Kennedy Class I RPD rotates away from its distal extension bases when sticky food pulls on them. The terminal abutment rests are mesial; the framework and clasp arms are correctly seated. What design feature principally counteracts lifting around the fulcrum line?

A	A thicker acrylic saddle without rests
B	Maximum retentive clasp engagement on the terminal abutments
C	A properly located indirect retainer anterior to the fulcrum line, supported by suitable prepared rests
D	A shorter distal saddle in a more anterior position
E	Elimination of all bracing elements

See answer and justification for Q458 →

Q460 INTERMEDIATE · Block F

A patient's RPD lifts at the back when she eats sticky food. The only rests are on the premolars next to the saddles.
Where should an extra rest be added?

A	On the premolar distal surface
B	As far forward of the fulcrum line as possible, such as a cingulum rest on a canine
C	On the saddle
D	On the retromolar pad
E	Nowhere

See answer and justification for Q460 →

Q463 INTERMEDIATE · Block F

A patient has lost several anterior teeth and the remaining teeth have reduced support with labial undercuts. Which design gives retention without conventional clasps?

A	An Every denture
B	A swing-lock denture
C	A spoon denture
D	A lingual plate
E	A flexible denture

See answer and justification for Q463 →

Q464 INTERMEDIATE · Block F

An RPD is needed while a patient's periodontal treatment outcome is uncertain.
Which type is most appropriate?

A	An acrylic interim RPD
B	A Co-Cr RPD with precision attachments
C	Implants
D	A swing-lock
E	A conventional bridge

See answer and justification for Q464 →

Q465 INTERMEDIATE · Block F

A patient requests an RPD with precision attachments on crowned canines.
At what level is this care usually provided?

A	Level 1
B	Hygienist level
C	Not provided
D	Level 2
E	Beyond Level 2 (specialist)

See answer and justification for Q465 →

Q466 INTERMEDIATE · Block F

A Co-Cr framework will not seat. Disclosing medium shows contact on the inner surface of a minor connector.
What should you do?

A	Adjust only the marked area and re-check
B	Remake immediately
C	Bend the clasps
D	Force it into place
E	Reline it

See answer and justification for Q466 →

Q467 INTERMEDIATE · Block F

A week after fitting, a patient reports a sore abutment. The rest on it marks heavily in occlusion.
What should you do?

A	Tighten the clasp
B	Adjust the high rest
C	Extract the tooth
D	Remake the framework
E	Prescribe analgesics only

See answer and justification for Q467 →

Q468 INTERMEDIATE · Block F

A distal-extension denture has a rigid short cast clasp engaging a deep undercut and the terminal abutment is becoming mobile. What component is most directly suspect?

A	Alter the rest position/design to accommodate tissueward rotation
B	Reassess the distal-extension design and abutment prognosis before altering clasp tightness
C	Improve broad, functional saddle support while maintaining hygiene
D	Review clasp flexibility and location of the retentive tip
E	Provide appropriately placed indirect retention

See answer and justification for Q468 →

Q469 INTERMEDIATE · Block F

A patient lisps on "s" with a new upper Co-Cr RPD whose anterior strap is thick over the rugae.
What should be adjusted?

A	The clasps
B	The rests
C	The saddle teeth
D	The thickness and position of the anterior strap
E	The OVD

See answer and justification for Q469 →

Q470 INTERMEDIATE · Block F

An RPD wearer develops recurrent gingival inflammation beneath a broad connector. The rests and clasps are clinically serviceable and the patient struggles with cleaning. Which redesign principle most directly reduces preventable biological risk?

A	Maximise gingival coverage to increase retention
B	Use more adhesive rather than altered design
C	Increase clasp undercut depth
D	Maintain necessary tooth support but minimise unnecessary soft-tissue coverage and plan hygiene review
E	Remove all rests to reduce metal visibility

See answer and justification for Q470 →

BLOCK G · Complete dentures (75)

Q473 INTERMEDIATE · Block G

A patient with a severely resorbed mandibular ridge has a new lower denture that is stable at rest but dislodges when talking and swallowing. Border extensions appear clinically acceptable. The posterior teeth lie markedly buccal to the position adopted by the tongue during speech. Which intervention most directly addresses the principal instability?

A	Increase OVD enough to improve muscular contact and then correct tooth positions only if it remains unstable
B	Extend the distolingual flanges further to compensate for the buccal tooth position even when existing borders are adequate
C	Rebase the denture against the flat ridge without altering polished surfaces or tooth positions
D	Use a functional neutral-zone impression to reorganise the arch position and polished contours within the muscle envelope
E	Change the posterior occlusal scheme alone while retaining the same buccal arrangement of teeth

See answer and justification for Q473 →

Q474 INTERMEDIATE · Block G

A 79-year-old has long-term complete dentures with severe posterior wear, angular cheilitis and an uncertain habitual mandibular position. The interocclusal rest space varies between appointments, and the patient has a neuromuscular disorder. What is the most defensible approach to prescribing the new occlusal vertical dimension?

A	Prescribe OVD using a single rest-position measurement repeated twice at the same visit
B	Triangulate reproducible jaw records, facial appearance, phonetics, interocclusal space and functional tolerance at trial
C	Copy the old OVD because it is most likely to be neurologically adapted despite marked wear
D	Set the anterior pin to remove all interocclusal space because it stabilises the mandibular position
E	Use the anatomical cranial orientation reference to determine the OVD without other measurements

See answer and justification for Q474 →

Q475 INTERMEDIATE · Block G

At an upper complete-denture try-in, a patient produces a whistling /s/ sound and the closest-speaking space appears almost closed, despite an acceptable resting freeway space and lip support. Which adjustment should be evaluated first?

A	Assess anterior tooth position and palatal contour during sibilants before making irreversible OVD changes
B	Increase posterior tooth height automatically because all /s/ problems mean excessive freeway space
C	Shorten the posterior palatal seal as the first correction regardless of retention
D	Replace all upper teeth with porcelain to improve the speech sound
E	Reduce every labial flange equally so that the lip can move freely

See answer and justification for Q475 →

Q477 INTERMEDIATE · Block G

A dentist uses a facebow to mount a maxillary complete-denture record on a semi-adjustable articulator. The laboratory asks whether the Frankfort horizontal plane should itself determine the anterior tooth display and occlusal plane. What is the best response?

A	The plane is identical to the patient’s occlusal plane in every edentulous patient
B	It sets the freeway space directly
C	It replaces clinical evaluation of the interpupillary line and smile
D	It predicts posterior palatal seal width from the ear reference points
E	It supplies an anatomical cranial reference for orientation; occlusal-plane position still requires intraoral, facial and functional assessment

See answer and justification for Q477 →

Q478 INTERMEDIATE · Block G

A maxillary complete denture has adequate peripheral extensions and a stable occlusion but loses retention during light speech. The posterior border stops anterior to the hamular notches and no functional assessment of the vibrating line was made. What is the most appropriate next evaluation?

A	Increase the incisal guidance angle to improve retention
B	Assess the hamular notches, soft-palate mobility and posterior palatal seal before modifying the border
C	Move the anterior teeth palatally to compensate for loss of posterior seal
D	Remove the posterior border to reduce functional tissue displacement
E	Convert the denture to a fully mucocompressive impression regardless of ridge tissue type

See answer and justification for Q478 →

Q479 INTERMEDIATE · Block G

A patient with a flat mandibular residual ridge complains of pain beneath the sharp midcrest and instability despite adequate anterior extension. The clinician plans to increase load on the crest by shortening the buccal flanges. Which approach is more biomechanically appropriate?

A	Use the knife-edge crest as the sole support because it is closest to the occlusal load
B	Move the posterior teeth lingually without checking the functional neutral zone
C	Optimise broad stress distribution over suitable support areas such as the buccal shelf, with selective relief of painful prominences and functional flange shaping
D	Discard the need to check occlusion because ridge anatomy is the only determinant
E	Raise OVD until the denture presses more heavily against the crest

See answer and justification for Q479 →

Q481 INTERMEDIATE · Block G

An 85-year-old with Parkinson disease likes the polished contours and appearance of her existing complete dentures but has lost posterior occlusal support and reports increasing instability. She struggled for months to adapt to the original dentures. What is the most justified replacement strategy?

A	Replicate useful established tooth position and polished-surface features while selectively correcting occlusal wear, fitting surface and border defects
B	Copy the fitting surface and OVD exactly without checking for present tissue change
C	Make an entirely different arrangement to maximise aesthetic improvement even though the patient has requested familiarity
D	Increase OVD using a single estimated freeway-space value before any trial
E	Replace only the posterior teeth and assume the original base will remain stable without assessment

See answer and justification for Q481 →

Q482 INTERMEDIATE · Block G

Trial bases shift and flex on the ridges during registration, so repeated bite records vary greatly. What should be improved?

A	Correct an early contact after remount verification
B	Re-record the jaw relationship before extensive grinding
C	Trial a reversible OVD change where adaptation is uncertain
D	Correct a peripheral extension interference first
E	Evaluate unstable record bases prior to jaw registration

See answer and justification for Q482 →

Q483 INTERMEDIATE · Block G

A complete denture with widespread supporting-mucosa soreness has a narrow bearing area and heavy occlusal contacts. What should be assessed before adding another liner?

A	Functional border moulding to capture muscular limits
B	Review occlusal load and tissue support before adding soft liner
C	Selective-pressure/window impression for a displaceable anterior ridge
D	Adjust an overextended border using pressure-indicating and functional movements
E	Use a copy technique to preserve accepted contours for a poor adaptor

See answer and justification for Q483 →

Q484 INTERMEDIATE · Block G

At the wax try-in of maxillary and mandibular complete dentures, the patient whistles during /s/ sounds. Vertical rest space is within the patient’s established range, and the rims were previously accepted for lip support. What should be checked first?

A	Anterior tooth arrangement, closest-speaking space and palatal contour during repeated phonetic testing
B	Increase OVD until the patient’s teeth almost contact at rest
C	Thicken the palatal acrylic in all regions without testing speech
D	Reduce posterior palatal border length regardless of retention
E	Ignore the problem because phonetics cannot be assessed before processing

See answer and justification for Q484 →

Q486 INTERMEDIATE · Block G

You check the upper occlusal rim with a Fox's plane guide.
How should the occlusal plane be oriented?

A	Parallel to the Frankfort plane
B	Tilted down anteriorly by 10°
C	Parallel to the lower ridge only
D	Parallel to the interpupillary line anteriorly and to the ala–tragal line (Camper's plane) posteriorly
E	Perpendicular to the facial midline

See answer and justification for Q486 →

Q488 INTERMEDIATE · Block G

An 85-year-old with reduced neuromuscular adaptability finds a new denture contour intolerable although the older denture is familiar and still clinically useful as a reference. What approach is most suitable?

A	Use a copy technique to preserve accepted contours for a poor adaptor
B	Functional border moulding to capture muscular limits
C	Adjust an overextended border using pressure-indicating and functional movements
D	Review occlusal load and tissue support before adding soft liner
E	Selective-pressure/window impression for a displaceable anterior ridge

See answer and justification for Q488 →

Q489 INTERMEDIATE · Block G

A patient after head-and-neck radiotherapy has scant ropy saliva and a loose upper complete denture despite good tissue fit and border extension. What factor is most likely?

A	Deficient border seal despite adequate adaptation
B	Poor tissue adaptation requiring reline or remake assessment
C	Excessive flange length displaced by a muscle
D	Occlusal interference that destabilises the base during contact
E	Salivary deficit that reduces the interfacial film

See answer and justification for Q489 →

Q492 INTERMEDIATE · Block G

Rest vertical dimension readings vary between visits for the same patient.
Why, and how should you respond?

A	The gauge is faulty
B	Rest position is fixed for life
C	Measurement is pointless
D	Rest position varies with posture, fatigue and emotion
E	Use the old denture's OVD only

See answer and justification for Q492 →

Q493 INTERMEDIATE · Block G

A week after fitting, a patient complains of generalised soreness of the lower ridge. Her teeth click when she speaks and her lips look strained.
What is the most likely cause?

A	OVD too low
B	Candida
C	OVD too high (freeway space too small)
D	An over-extended upper flange
E	Porcelain teeth only

See answer and justification for Q493 →

Q494 INTERMEDIATE · Block G

A patient's old dentures give him a collapsed appearance, angular cheilitis and a prominent chin.
What do these signs indicate?

A	OVD too high
B	Over-extension
C	Candida only
D	Allergy
E	OVD too low (freeway space too large)

See answer and justification for Q494 →

Q495 INTERMEDIATE · Block G

A colleague builds the OVD of new dentures well above the rest position.
Which long-term consequences may follow?

A	Improved chewing
B	Muscle fatigue, soreness and faster ridge resorption
C	Better retention
D	Angular cheilitis
E	Reduced clicking

See answer and justification for Q495 →

Q497 INTERMEDIATE · Block G

A new denture produces a reproducible linear ulcer beside a labial frenum after insertion. The border is visibly overextended. What is the first corrective action?

A	Functional border moulding to capture muscular limits
B	Use a copy technique to preserve accepted contours for a poor adaptor
C	Selective-pressure/window impression for a displaceable anterior ridge
D	Review occlusal load and tissue support before adding soft liner
E	Adjust an overextended border using pressure-indicating and functional movements

See answer and justification for Q497 →

Q498 INTERMEDIATE · Block G

You use a Gothic arch (needle-point) tracer because registration is unreliable.
Where is RCP recorded?

A	At the end of the right lateral path
B	In the centre of the tracing
C	At the most protruded point
D	At the apex of the arrowhead
E	Anywhere on the left path

See answer and justification for Q498 →

Q501 INTERMEDIATE · Block G

An upper denture is closely adapted to the palate but has a posterior border ending short of the vibrating line. It loses suction with speech. Which factor dominates?

A	Excessive flange length displaced by a muscle
B	Deficient border seal despite adequate adaptation
C	Poor tissue adaptation requiring reline or remake assessment
D	Salivary deficit that reduces the interfacial film
E	Occlusal interference that destabilises the base during contact

See answer and justification for Q501 →

Q502 INTERMEDIATE · Block G

A technician needs the Bennett angle but has no lateral records.
Which formula estimates it from condylar guidance (H)?

A	L = H × 2
B	L = 30 − H
C	L = H/8 + 12 (Hanau)
D	L = H/2
E	L = H + 15

See answer and justification for Q502 →

Q503 INTERMEDIATE · Block G

The denture tips reproducibly at the first chewing contact despite correct borders and an acceptable OVD. Which finding should be pursued?

A	An erroneous horizontal jaw relation despite acceptable vertical dimension
B	An excessively increased OVD producing muscular symptoms
C	A reduced OVD with loss of facial support and commissural folding
D	The need for a cautious reversible trial due to limited adaptive capacity
E	Local traumatic denture instability from a deflective occlusal contact

See answer and justification for Q503 →

Q505 INTERMEDIATE · Block G

An upper complete denture loses retention mainly when the patient says “ah”; the posterior border is short of the functional seal and the ridge is sound. What should be reassessed?

A	Review polished-surface contours and the neutral zone
B	Adjust the masseteric notch/distobuccal flange
C	Reassess posterior palatal seal and soft-palate movement
D	Reassess occlusal contacts and jaw relation after verifying border fit
E	Correct overextension along the mylohyoid region

See answer and justification for Q505 →

Q508 INTERMEDIATE · Block G

An elderly patient shows commissural folds, reduced lower facial height and overclosed appearance; the denture teeth are markedly worn. Which explanation is best?

A	A reduced OVD with loss of facial support and commissural folding
B	Local traumatic denture instability from a deflective occlusal contact
C	An erroneous horizontal jaw relation despite acceptable vertical dimension
D	An excessively increased OVD producing muscular symptoms
E	The need for a cautious reversible trial due to limited adaptive capacity

See answer and justification for Q508 →

Q509 INTERMEDIATE · Block G

A patient has a severely resorbed mandible, a Class III jaw relationship and poor neuromuscular control. Which posterior teeth are most suitable?

A	Steep-cusped teeth
B	Porcelain anatomic teeth
C	33° cusped teeth
D	Monoplane (zero-degree) teeth
E	No posterior teeth

See answer and justification for Q509 →

Q510 INTERMEDIATE · Block G

You estimate the width of an upper central incisor for a patient without pre-extraction records. Which guide (Berry) relates it to facial width?

A	One-quarter of the inter-alar width
B	Equal to the canine lines
C	One-tenth of the face height
D	One-half of the interpupillary distance
E	About one-sixteenth of the bizygomatic width

See answer and justification for Q510 →

Q511 INTERMEDIATE · Block G

A severely resorbed lower denture displaces whenever the patient speaks, despite acceptable intaglio fit and stable static occlusion. A bulky lingual polished surface crowds the tongue. What is the best design focus?

A	Review polished-surface contours and the neutral zone
B	Adjust the masseteric notch/distobuccal flange
C	Correct overextension along the mylohyoid region
D	Reassess posterior palatal seal and soft-palate movement
E	Reassess occlusal contacts and jaw relation after verifying border fit

See answer and justification for Q511 →

Q512 INTERMEDIATE · Block G

An 80-year-old wants new dentures and asks for "bright white, perfect teeth".
Which approach to shade is best?

A	Always the lightest shade
B	Involve her in the choice, usually favouring a darker, characterised shade that suits her age
C	Choose without discussion
D	Grey teeth
E	Match her old dentures regardless

See answer and justification for Q512 →

Q514 INTERMEDIATE · Block G

A patient has a severely resorbed maxilla that is much narrower than the mandible.
How are the posterior teeth usually set?

A	Lingual to the upper ridge
B	In normal overjet regardless
C	Omitted
D	On the palate
E	In crossbite, with the upper teeth possibly buccal to the resorbed ridge

See answer and justification for Q514 →

Q515 INTERMEDIATE · Block G

A 79-year-old with severely resorbed mandibular ridges has repeated lower denture instability despite satisfactory fit at rest. The lower anterior teeth are placed labially to obtain lip fullness, and the denture dislodges during speech. Which approach best addresses the functional cause?

A	Increase the posterior palatal seal of the maxillary denture
B	Use stronger occlusal contacts to seat the lower denture
C	Set all lower teeth exactly over the ridge crest without a functional assessment
D	Record the neutral zone and reshape tooth position and polished surfaces to the patient’s muscular balance
E	Thicken all buccal and lingual polished surfaces to improve rigidity

See answer and justification for Q515 →

Q516 INTERMEDIATE · Block G

An older complete denture shows a wide internal gap over a resorbed ridge but its functional borders remain acceptable. It loses retention at rest. What mechanism fits?

A	Excessive flange length displaced by a muscle
B	Salivary deficit that reduces the interfacial film
C	Poor tissue adaptation requiring reline or remake assessment
D	Occlusal interference that destabilises the base during contact
E	Deficient border seal despite adequate adaptation

See answer and justification for Q516 →

Q517 INTERMEDIATE · Block G

The patient cannot close reproducibly in the intended relation and the new denture shows multiple inconsistent contacts; the recorded jaw relation appears unreliable. What should come first?

A	Trial a reversible OVD change where adaptation is uncertain
B	Evaluate unstable record bases prior to jaw registration
C	Correct an early contact after remount verification
D	Re-record the jaw relationship before extensive grinding
E	Correct a peripheral extension interference first

See answer and justification for Q517 →

Q518 INTERMEDIATE · Block G

An upper complete denture patient has a highly displaceable anterior ridge that becomes distorted when conventional heavy-body material is used. What technique is most appropriate?

A	Functional border moulding to capture muscular limits
B	Selective-pressure/window impression for a displaceable anterior ridge
C	Use a copy technique to preserve accepted contours for a poor adaptor
D	Adjust an overextended border using pressure-indicating and functional movements
E	Review occlusal load and tissue support before adding soft liner

See answer and justification for Q518 →

Q519 INTERMEDIATE · Block G

A patient with Parkinsonism has worn the same familiar dentures for many years. A major increase in OVD is proposed but adaptation is uncertain. What is the most appropriate principle?

A	A reduced OVD with loss of facial support and commissural folding
B	The need for a cautious reversible trial due to limited adaptive capacity
C	An excessively increased OVD producing muscular symptoms
D	Local traumatic denture instability from a deflective occlusal contact
E	An erroneous horizontal jaw relation despite acceptable vertical dimension

See answer and justification for Q519 →

Q520 INTERMEDIATE · Block G

You mark the vibrating line for an upper complete denture.
How are the posterior and anterior vibrating lines identified?

A	By palpation only
B	With a radiograph
C	At the incisive papilla
D	Posterior: by asking the patient to say "ah"
E	By the fovea alone

See answer and justification for Q520 →

Q522 INTERMEDIATE · Block G

A new denture has apparently acceptable OVD and aesthetics but the patient is guided into an unstable retruded contact with a marked slide. What is the primary jaw-registration issue?

A	Local traumatic denture instability from a deflective occlusal contact
B	The need for a cautious reversible trial due to limited adaptive capacity
C	An excessively increased OVD producing muscular symptoms
D	An erroneous horizontal jaw relation despite acceptable vertical dimension
E	A reduced OVD with loss of facial support and commissural folding

See answer and justification for Q522 →

Q523 INTERMEDIATE · Block G

A mandibular denture remains seated at rest but is repeatedly dislodged when the patient swallows. Its lingual flange extends below the raised mylohyoid. Which factor dominates?

A	Salivary deficit that reduces the interfacial film
B	Deficient border seal despite adequate adaptation
C	Poor tissue adaptation requiring reline or remake assessment
D	Occlusal interference that destabilises the base during contact
E	Excessive flange length displaced by a muscle

See answer and justification for Q523 →

Q524 INTERMEDIATE · Block G

A new upper complete denture drops when the patient talks or opens wide.
What is the most likely cause?

A	OVD too high
B	An over-extended distobuccal flange (coronoid process)
C	Lingualised occlusion
D	Thin saliva
E	An under-extended lower flange

See answer and justification for Q524 →

Q525 INTERMEDIATE · Block G

A new upper denture is loose even at rest, with no displacement on opening.
What is the most likely cause?

A	Excessive OVD
B	Porcelain teeth
C	An inadequate post-dam or peripheral seal
D	Cheek biting
E	An over-extended distobuccal flange

See answer and justification for Q525 →

Q526 INTERMEDIATE · Block G

A denture is retained at rest and during speech but tips precisely as the patient closes onto a heavy right posterior premature contact. What is most likely?

A	Deficient border seal despite adequate adaptation
B	Salivary deficit that reduces the interfacial film
C	Poor tissue adaptation requiring reline or remake assessment
D	Excessive flange length displaced by a muscle
E	Occlusal interference that destabilises the base during contact

See answer and justification for Q526 →

Q527 INTERMEDIATE · Block G

A patient's denture-bearing areas are sore all over after a week. The occlusion and extension look satisfactory, and she has anaemia.
Which causes of generalised soreness should be considered?

A	An over-extended flange only
B	Cheek biting only
C	Excessive OVD, an RCP error, parafunction, inadequate support, or systemic factors such as anaemia or diabetes
D	Monomer allergy only
E	Thin saliva only

See answer and justification for Q527 →

Q528 INTERMEDIATE · Block G

An older patient with neurological impairment is offered a substantial new OVD; they tolerated an old denture for years. What planning strategy is preferable?

A	Evaluate unstable record bases prior to jaw registration
B	Trial a reversible OVD change where adaptation is uncertain
C	Correct a peripheral extension interference first
D	Re-record the jaw relationship before extensive grinding
E	Correct an early contact after remount verification

See answer and justification for Q528 →

Q529 INTERMEDIATE · Block G

A patient has an ulcer on the crest of the lower ridge under a new denture.
What is the most likely cause?

A	An occlusal interference, a pressure spot or a sharp underlying bony crest
B	An over-extended flange
C	A short post-dam
D	Lingualised occlusion
E	A deep labial notch

See answer and justification for Q529 →

Q530 INTERMEDIATE · Block G

A patient with a severely resorbed mandible has pain and numbness of the lower lip with a new denture.
What is the cause?

A	Candida
B	Excessive OVD
C	Allergy
D	The denture pressing on the mental nerve
E	Anaemia

See answer and justification for Q530 →

Q531 INTERMEDIATE · Block G

A patient bites his cheeks with new complete dentures. The posterior teeth meet edge to edge.
What is the cause?

A	Lack of posterior buccal overjet
B	Excessive OVD
C	A short post-dam
D	An over-extended lingual flange
E	Candida

See answer and justification for Q531 →

Q532 INTERMEDIATE · Block G

A new mandibular trial denture is comfortable at rest but moves with tongue and cheek activity; the functional sulcus depth is uncertain. What step adds the most information?

A	Review occlusal load and tissue support before adding soft liner
B	Functional border moulding to capture muscular limits
C	Adjust an overextended border using pressure-indicating and functional movements
D	Selective-pressure/window impression for a displaceable anterior ridge
E	Use a copy technique to preserve accepted contours for a poor adaptor

See answer and justification for Q532 →

Q533 INTERMEDIATE · Block G

A patient gags with a new upper denture. The posterior border is thick and slightly over-extended. What is the management?

A	Add adhesive
B	Increase the OVD
C	Correct the posterior border, and use a training base, desensitisation or palateless (implant) designs if needed
D	Remove the post-dam entirely
E	Prescribe antiemetics only

See answer and justification for Q533 →

Q534 INTERMEDIATE · Block G

A complete denture tips toward one side at closure; a remount reproduces a single premature contact and the jaw record is otherwise consistent. What is most appropriate?

A	Correct an early contact after remount verification
B	Correct a peripheral extension interference first
C	Re-record the jaw relationship before extensive grinding
D	Trial a reversible OVD change where adaptation is uncertain
E	Evaluate unstable record bases prior to jaw registration

See answer and justification for Q534 →

Q535 INTERMEDIATE · Block G

A denture wearer has a burning sensation under the upper denture. Which causes should be considered?

A	Excessive OVD only
B	Denture stomatitis, xerostomia, haematinic deficiency or diabetes, and very rarely monomer allergy
C	Occlusal error only
D	Over-extension only
E	It is always psychological

See answer and justification for Q535 →

Q536 INTERMEDIATE · Block G

A flange impinges on a frenum during smiling, causing loss of stability before any teeth touch. What should be corrected first?

A	Trial a reversible OVD change where adaptation is uncertain
B	Evaluate unstable record bases prior to jaw registration
C	Re-record the jaw relationship before extensive grinding
D	Correct a peripheral extension interference first
E	Correct an early contact after remount verification

See answer and justification for Q536 →

Q537 INTERMEDIATE · Block G

An upper complete denture has a midline fracture. The patient has a large midline torus and natural lower teeth.
What is the failure mechanism?

A	Impact
B	Fatigue from flexure over a midline fulcrum, worsened by opposing natural teeth
C	Creep
D	Corrosion
E	Thermal shock

See answer and justification for Q537 →

Q538 INTERMEDIATE · Block G

A new mandibular denture is displaced when the patient clenches and reports soreness at the distobuccal corner. The rest of the border is comfortable. What correction is most indicated?

A	Correct overextension along the mylohyoid region
B	Adjust the masseteric notch/distobuccal flange
C	Reassess occlusal contacts and jaw relation after verifying border fit
D	Reassess posterior palatal seal and soft-palate movement
E	Review polished-surface contours and the neutral zone

See answer and justification for Q538 →

Q539 INTERMEDIATE · Block G

An 84-year-old who has worn the same dentures for 15 years needs replacements because they are loose and worn.
Which technique best preserves her adaptation?

A	Immediate dentures
B	Implant-retained fixed bridges
C	A conventional new design
D	A neutral-zone denture
E	Copy (replica) dentures, changing only what is necessary

See answer and justification for Q539 →

Q540 INTERMEDIATE · Block G

A patient asks whether an immediate upper denture can be made without a labial flange.
What is the disadvantage of an open-face (socketed) immediate denture?

A	It looks worse
B	It is heavier
C	Poor retention, and it cannot be relined labially
D	It causes more swelling
E	It needs more appointments to fit

See answer and justification for Q540 →

Q542 INTERMEDIATE · Block G

Two lower canines with good support remain under a planned complete lower denture.
How are they prepared as overdenture abutments?

A	Left full height with crowns
B	Extracted
C	Reduced flush with the bone
D	Root-filled and reduced to 2–3 mm above the gingiva with domed surfaces
E	Splinted with a bridge

See answer and justification for Q542 →

Q544 INTERMEDIATE · Block G

A 70-year-old with an atrophic mandible has struggled with lower complete dentures.
What do the McGill (2002) and York (2009) consensus statements recommend?

A	A two-implant mandibular overdenture as the minimum standard of care
B	A conventional denture as the standard
C	Four implants as the minimum
D	A fixed implant bridge
E	Denture adhesive

See answer and justification for Q544 →

Q545 INTERMEDIATE · Block G

An edentulous maxilla opposing mandibular natural anterior teeth and bilateral distal-extension RPD saddles develops an increasingly flabby anterior ridge and posterior ridge changes. Which clinical process best explains the pattern?

A	An isolated change in post-dam depth
B	A local mandibular canal pathology
C	Occlusal load imbalance and characteristic combination-syndrome changes
D	Conventional chronic periapical infection
E	Primary caries beneath the complete denture

See answer and justification for Q545 →

Q546 INTERMEDIATE · Block G

You are planning for the patient in the previous question.
How is combination syndrome best prevented?

A	Higher OVD
B	Porcelain teeth
C	Removing the lower RPD
D	Heavy anterior contact
E	Restoring posterior support (a well-fitting lower RPD) and reducing anterior loading

See answer and justification for Q546 →

Q547 INTERMEDIATE · Block G

A complete-denture patient has a highly mobile anterior maxillary ridge but firm posterior palate and relatively stable tuberosities. A conventional close-fitting tray displaces the flabby tissue and the resulting denture loses support at rest. Which impression concept is most suitable?

A	Compress all tissues uniformly to their maximal displacement
B	Omit the flabby segment from the definitive impression entirely
C	Use a high-pressure putty impression without special relief
D	Use only the preliminary impression as the master impression
E	Record the firm supporting tissues appropriately while using a window or selective relief to record the flabby tissue minimally displaced

See answer and justification for Q547 →

Q549 INTERMEDIATE · Block G

After a closed-mouth laboratory reline, the patient's dentures feel "too tall".
What has likely happened?

A	The teeth have worn
B	The palate is thinner
C	The patient bit too firmly during the reline impression, increasing the OVD
D	The post-dam was lost
E	Candida

See answer and justification for Q549 →

Q550 INTERMEDIATE · Block G

A patient's dentures are unstable because of enlarged tuberosities and a shallow sulcus. Surgery is being considered.
At which level is pre-prosthetic surgery usually provided?

A	Level 1
B	Level 3
C	Level 2
D	By the dental technician
E	It is never done

See answer and justification for Q550 →

Q551 INTERMEDIATE · Block G

A lower denture lifts when the patient raises the floor of the mouth on swallowing; overextension is visible lingual to the molar saddles. What is most appropriate?

A	Correct overextension along the mylohyoid region
B	Review polished-surface contours and the neutral zone
C	Reassess occlusal contacts and jaw relation after verifying border fit
D	Adjust the masseteric notch/distobuccal flange
E	Reassess posterior palatal seal and soft-palate movement

See answer and justification for Q551 →

Q552 INTERMEDIATE · Block G

A new lower denture keeps lifting because its lingual flange is over-extended. A colleague suggests adhesive and a higher OVD.
Why is this wrong?

A	Adhesive is never acceptable
B	A higher OVD always helps
C	Adhesive and OVD changes rarely cure an over-extended flange; correct the border
D	The patient needs implants
E	It is the correct approach

See answer and justification for Q552 →

Q555 INTERMEDIATE · Block G

An upper denture was retentive in wax at the try-in but is loose after processing.
Which step is likely to have been missed?

A	Cutting the post-dam on the master cast along the marked vibrating line
B	Relining
C	Changing the shade
D	Adding porcelain teeth
E	Increasing OVD

See answer and justification for Q555 →

Q557 INTERMEDIATE · Block G

Both complete dentures have comfortable borders and stable static retention, but a premature posterior contact causes tipping on closure. What should be assessed next?

A	Review polished-surface contours and the neutral zone
B	Correct overextension along the mylohyoid region
C	Adjust the masseteric notch/distobuccal flange
D	Reassess posterior palatal seal and soft-palate movement
E	Reassess occlusal contacts and jaw relation after verifying border fit

See answer and justification for Q557 →

Q558 INTERMEDIATE · Block G

An 82-year-old with early dementia has loose but familiar dentures she manages well.
Which approach is most appropriate?

A	Implant-retained dentures
B	A conventional new design
C	Copy dentures with minimal changes
D	No dentures
E	Immediate dentures

See answer and justification for Q558 →

Q559 INTERMEDIATE · Block G

A patient needs an upper complete denture against a full natural lower dentition.
Which design feature reduces the risk of fracture?

A	A thin acrylic palate
B	A metal (Co-Cr) or reinforced palate, with balanced contacts
C	Porcelain teeth
D	No post-dam
E	A soft liner

See answer and justification for Q559 →

Q560 INTERMEDIATE · Block G

A student asks what mechanical retention means for a complete denture.
Which are mechanical retentive factors?

A	Adhesion and cohesion
B	Atmospheric pressure
C	The neutral zone
D	Surface tension
E	Undercuts, implants and adhesives

See answer and justification for Q560 →

Q561 INTERMEDIATE · Block G

With new dentures in, a patient's lips are apart at rest and look strained.
What is the most likely cause?

A	OVD too low
B	OVD too high
C	A short post-dam
D	Candida
E	Over-extension of the lower flange

See answer and justification for Q561 →

Q562 INTERMEDIATE · Block G

A patient has generalised soreness, clicking teeth and no freeway space with new dentures.
What is the definitive management?

A	Adhesive
B	A soft liner
C	Analgesics
D	Leaving them out for a week
E	Re-establish the OVD with a new jaw record and reset or remake the dentures

See answer and justification for Q562 →

Q563 INTERMEDIATE · Block G

A patient who had an immediate denture fitted removed it the same evening and now cannot reinsert it.
Why should an immediate denture be left in for 24 hours?

A	To improve the shade
B	To stop it fracturing
C	To allow relining
D	It acts as a splint and limits swelling
E	To reduce caries

See answer and justification for Q563 →

Q564 INTERMEDIATE · Block G

A 75-year-old, edentulous for 20 years, has a severely resorbed mandible and has failed with several well-made lower dentures.
What is the most appropriate next step?

A	Another conventional denture
B	Denture adhesive only
C	Assessment and referral for a two-implant overdenture
D	A soft liner
E	Accept the situation

See answer and justification for Q564 →

Q566 INTERMEDIATE · Block G

A replacement complete denture causes new muscle fatigue, clicking during speech and almost no freeway space; previous dentures were comfortable. Which error is most likely?

A	An excessively increased OVD producing muscular symptoms
B	A reduced OVD with loss of facial support and commissural folding
C	An erroneous horizontal jaw relation despite acceptable vertical dimension
D	Local traumatic denture instability from a deflective occlusal contact
E	The need for a cautious reversible trial due to limited adaptive capacity

See answer and justification for Q566 →

Q567 INTERMEDIATE · Block G

You are selecting the mould of the upper anterior teeth.
Which anatomical guides help choose the width of the six anterior teeth?

A	The incisive papilla only
B	The canine lines and inter-alar width
C	The lower lip line
D	The palatal vault
E	The post-dam

See answer and justification for Q567 →

Q570 INTERMEDIATE · Block G

A replacement upper complete denture must be made for a patient who values the appearance of their old prosthesis but complains that it loses retention when swallowing. What should be copied and what must be reassessed?

A	Copy tooth display but reassess posterior extension, seal, fit and jaw relation
B	Copy the entire base and post-dam without examination
C	Change tooth shade and do not assess the borders
D	Restore the original post-dam solely by adding adhesive
E	Reproduce all worn occlusal contacts because they are familiar

See answer and justification for Q570 →

BLOCK H · Fixed prosthodontics (76)

Q571 INTERMEDIATE · Block H

A maxillary premolar requires a metal–ceramic crown. The facial surface will carry porcelain for aesthetics; the palatal surface has adequate clearance for metal and is less visible. Periodontal tissues are healthy and margins can be placed supragingivally. Which finish-line prescription best balances material support and tissue preservation?

A	Use a deep chamfer both facially and palatally, even where a thin metal-only palatal margin could conserve more tooth
B	Use a shallow chamfer facially and a broad shoulder on the metal-only palatal surface
C	Use a conservative feather margin facially and a shoulder on the metal-only palatal surface
D	Use a shoulder/deep chamfer where porcelain needs bulk and a more conservative metal-compatible palatal chamfer
E	Place a broad subgingival shoulder circumferentially to optimise aesthetic concealment despite healthy tissues

See answer and justification for Q571 →

Q572 INTERMEDIATE · Block H

An older patient has a short clinical crown, already-thin residual walls and a need for a full-metal crown. A deep shoulder would remove a disproportionate amount of tooth tissue. Which statement best describes the place of a knife-edge preparation?

A	It can be considered selectively for a metal restoration when conservation is critical, but risks difficult margin identification and overcontour if poorly controlled
B	It is the preferred preparation for every bonded lithium-disilicate crown
C	It provides a thicker supported ceramic collar than a shoulder
D	It eliminates the need for adequate occlusal reduction and resistance form
E	It is always the least plaque-retentive margin regardless of laboratory execution

See answer and justification for Q572 →

Q573 INTERMEDIATE · Block H

A heavily restored mandibular molar has a short clinical crown and a proposed full-coverage restoration. Two clinicians propose different preparations: one with a large total occlusal convergence and polished short walls, the other with a more controlled taper and auxiliary grooves without sacrificing ferrule. Which principle most directly improves resistance to tipping?

A	Increasing cement thickness to create a hydraulic lock
B	Increasing taper to ease insertion despite short walls
C	Relying on occlusal porcelain alone to resist lateral displacement
D	Shortening axial walls to avoid a path of insertion
E	Preserving adequate axial wall height and controlled taper, with auxiliary resistance features when appropriate

See answer and justification for Q573 →

Q575 INTERMEDIATE · Block H

A two-wing resin-bonded bridge replacing a maxillary lateral incisor has recurrent unilateral retainer debonding; the other wing remains attached, and the occlusion causes differential movement of the abutment teeth. Both abutments are otherwise sound. Which redesign most directly addresses the failure pattern?

A	Remake both wings with greater metal thickness but retain the fixed–fixed physiological connection
B	Assess suitability for a single-retainer cantilever RBB, including adhesive substrate and dynamic occlusion
C	Increase the number of retainer wings but reduce the cement thickness on the frequently debonded side
D	Use a rigid additional connector without changing the independent mobility of the two abutments
E	Repeat the fixed–fixed design and increase surface roughness alone without investigating movement

See answer and justification for Q575 →

Q576 INTERMEDIATE · Block H

A single missing maxillary lateral incisor is bordered by intact enamel-rich teeth; occlusion permits a conservative wing and the patient avoids implant surgery. Which option is especially appropriate?

A	Reassess connector design where porcelain/metal fatigue occurs
B	Avoid an irreversible fixed solution until disease and tooth prognosis are stable
C	Discuss no replacement as a valid choice when function and patient preferences permit
D	Evaluate abutment periodontal prognosis and overload before choosing a long-span fixed bridge
E	Consider an adhesive resin-bonded single-retainer bridge in favourable enamel/occlusion

See answer and justification for Q576 →

Q577 INTERMEDIATE · Block H

A zirconia crown debonds despite good isolation after only hydrofluoric acid conditioning and no specified primer/abrasion. What is most likely?

A	Inadequate material-specific substrate conditioning
B	Inadequate isolation of adhesive surfaces
C	Compromised abutment prognosis from untreated disease
D	Inadequate resistance form in a short tapered preparation
E	Undersized ceramic thickness in a high-load area

See answer and justification for Q577 →

Q578 INTERMEDIATE · Block H

A vital upper lateral incisor is missing and both adjacent teeth are minimally restored, with sound enamel and a suitable occlusion. Which treatment is most conservative?

A	Defer definitive bridge while active caries risk is controlled
B	Discuss a removable temporary replacement while prognosis is uncertain
C	Refer for coordinated implant/orthodontic planning where space and roots are unsuitable
D	Avoid the design when occlusal clearance cannot be provided safely
E	Enamel-supported conservative single-retainer design

See answer and justification for Q578 →

Q579 INTERMEDIATE · Block H

An anterior porcelain-fused-to-metal crown repeatedly chips where veneering porcelain is excessively thick and unsupported. Which correction targets the failure mechanism?

A	Use a high-strength monolithic material when load and connector geometry dominate
B	Choose adhesive glass ceramic where enamel bonding and aesthetics dominate
C	Reassess occlusal reduction and restorative space before material change
D	Prefer repairable direct composite when a reversible additive solution is appropriate
E	Correct unsupported veneering porcelain/design before remaking a metal–ceramic crown

See answer and justification for Q579 →

Q580 INTERMEDIATE · Block H

A 26-year-old has intact enamel and a small labial aesthetic defect; excellent isolation is available and minimal preparation is desired. What material strategy best preserves the tooth?

A	Prefer repairable direct composite when a reversible additive solution is appropriate
B	Choose adhesive glass ceramic where enamel bonding and aesthetics dominate
C	Reassess occlusal reduction and restorative space before material change
D	Use a high-strength monolithic material when load and connector geometry dominate
E	Correct unsupported veneering porcelain/design before remaking a metal–ceramic crown

See answer and justification for Q580 →

Q581 INTERMEDIATE · Block H

A proposed posterior ceramic crown has only 0.4 mm occlusal clearance after preparation, below the chosen manufacturer’s minimum. What is the best next planning step?

A	Use a high-strength monolithic material when load and connector geometry dominate
B	Reassess occlusal reduction and restorative space before material change
C	Prefer repairable direct composite when a reversible additive solution is appropriate
D	Choose adhesive glass ceramic where enamel bonding and aesthetics dominate
E	Correct unsupported veneering porcelain/design before remaking a metal–ceramic crown

See answer and justification for Q581 →

Q582 INTERMEDIATE · Block H

A long-span anterior PFM bridge has repeatedly chipped veneering porcelain adjacent to a pontic, but the framework remains seated. The opposing guidance has steep, heavy anterior contacts and the laboratory identifies limited veneer support. What is the strongest combined explanation?

A	The bridge must have failed because all porcelain is chemically inert
B	Cement washout alone explains cohesive fracture in an otherwise stable bridge
C	High functional tensile stress acting on poorly supported veneer promotes crack initiation and propagation
D	A cervical knife-edge margin necessarily produces identical chipping at every pontic
E	Ceramic chipping cannot relate to occlusal scheme if the bridge is still retained

See answer and justification for Q582 →

Q583 INTERMEDIATE · Block H

A mandibular molar requires replacement of a subgingival crown with recurrent bleeding on probing beside a deep proximal margin. The margin lies very close to the alveolar crest radiographically, while plaque control has improved. Which plan best limits recurrence?

A	Place the replacement finish line at the same depth and use a stronger cement only
B	Move the new margin further apically to hide it better without assessing soft-tissue attachment
C	Assess whether the restoration impinges on the supracrestal tissue attachment and whether a sound cleansable margin can be established with appropriate periodontal/restorative planning
D	Use a knife-edge subgingival margin because it cannot violate the attachment zone
E	Treat the bleeding as unrelated to restoration contour once an antibacterial mouthwash has been prescribed

See answer and justification for Q583 →

Q584 INTERMEDIATE · Block H

A crown margin placed 1 mm from the bone crest has caused chronic gingival inflammation.
What has happened?

A	Allergy to the crown
B	Caries
C	Invasion of the supracrestal tissue attachment, causing chronic inflammation, recession or bone loss
D	Normal healing
E	Occlusal trauma only

See answer and justification for Q584 →

Q585 INTERMEDIATE · Block H

Aesthetic demands require porcelain on the facial surface of a maxillary premolar, while the palatal surface will be metal only. The patient has healthy, shallow sulci and good plaque control. Which finish-line design best balances ceramic support and conservation?

A	A knife-edge finish line circumferentially to maximise porcelain thickness
B	A deep circumferential shoulder irrespective of material coverage
C	An adequately supported facial shoulder or deep chamfer and a more conservative palatal chamfer for metal
D	A subgingival feather edge because all crown margins require sulcular placement
E	No visible finish line so the technician can shape the casting freely

See answer and justification for Q585 →

Q586 INTERMEDIATE · Block H

You need gingival retraction for 24 in a patient with poorly controlled hypertension.
Which retraction medicament should be avoided?

A	Aluminium chloride
B	Ferric sulphate
C	Plain cord
D	Retraction paste
E	Adrenaline-impregnated cord

See answer and justification for Q586 →

Q588 INTERMEDIATE · Block H

A long-span metal–ceramic bridge repeatedly fractures veneering porcelain at the connector, while abutments remain stable. What should be reassessed?

A	Evaluate abutment periodontal prognosis and overload before choosing a long-span fixed bridge
B	Discuss no replacement as a valid choice when function and patient preferences permit
C	Reassess connector design where porcelain/metal fatigue occurs
D	Avoid an irreversible fixed solution until disease and tooth prognosis are stable
E	Consider an adhesive resin-bonded single-retainer bridge in favourable enamel/occlusion

See answer and justification for Q588 →

Q589 INTERMEDIATE · Block H

A lithium disilicate crown will be resin-bonded next week.
Which temporary cement should be used now?

A	A eugenol-free temporary cement
B	Zinc oxide–eugenol
C	Zinc phosphate
D	Glass ionomer
E	Resin cement

See answer and justification for Q589 →

Q590 INTERMEDIATE · Block H

You are restoring a single crown on 36 in a patient with a stable, comfortable ICP.
Which occlusal approach is appropriate?

A	Reorganise to RCP
B	Increase the OVD
C	Conformative: restore to the existing ICP
D	Full-mouth equilibration
E	Canine guidance on all teeth

See answer and justification for Q590 →

Q591 INTERMEDIATE · Block H

Extensive restoration is planned with a change of OVD, and the ICP is unstable.
Which approach and level apply?

A	Conformative, Level 1
B	No approach needed
C	Reorganised (RCP), usually Level 3 when major
D	Conformative, Level 2
E	Hygienist level

See answer and justification for Q591 →

Q592 INTERMEDIATE · Block H

A vertical root fracture extends below the bone crest in a strategically poor abutment and the patient asks for a crown. What is the most appropriate discussion?

A	Consider orthodontic extrusion when biologically and strategically suitable
B	Consider crown lengthening only after evaluating crown-to-root and biological consequences
C	Discuss extraction/replacement when a deep fracture makes predictable restoration untenable
D	Avoid a post when adequate coronal structure and a core are already present
E	Reassess ferrule and remaining coronal dentine before post placement

See answer and justification for Q592 →

Q593 INTERMEDIATE · Block H

The palatal contact is extremely tight and there is no feasible wing space without significant irreversible occlusal or tooth reduction. What should be reconsidered?

A	Defer definitive bridge while active caries risk is controlled
B	Discuss a removable temporary replacement while prognosis is uncertain
C	Avoid the design when occlusal clearance cannot be provided safely
D	Refer for coordinated implant/orthodontic planning where space and roots are unsuitable
E	Enamel-supported conservative single-retainer design

See answer and justification for Q593 →

Q594 INTERMEDIATE · Block H

A colleague takes a dual-arch (triple) tray impression for a 4-unit bridge.
Why is this inappropriate?

A	Dual-arch trays suit single units with stable intercuspation only
B	They are too accurate
C	They cannot record margins
D	They are only for dentures
E	They need alginate

See answer and justification for Q594 →

Q595 INTERMEDIATE · Block H

The laboratory applies die spacer to the die.
What thickness is typical, and where does it stop?

A	200 µm, over the margin
B	1 mm, everywhere
C	None is needed
D	About 20–40 µm, stopping short of the margin
E	5 µm, on the margin only

See answer and justification for Q595 →

Q596 INTERMEDIATE · Block H

A lower first molar has had root canal treatment and has an MOD cavity.
What restoration improves survival?

A	An amalgam without cusp coverage
B	Cuspal coverage (an onlay or crown)
C	A composite without coverage
D	A veneer
E	No restoration

See answer and justification for Q596 →

Q597 INTERMEDIATE · Block H

An upper central incisor has had root canal treatment through a small access cavity, with little other tissue loss.
What restoration is appropriate?

A	A composite restoration of the access cavity
B	A post and crown
C	A full metal–ceramic crown
D	Extraction
E	A veneer and post

See answer and justification for Q597 →

Q598 INTERMEDIATE · Block H

A root-filled lower molar has a large pulp chamber and adequate remaining walls.
How can the core be retained without a post?

A	It cannot
B	With a threaded post only
C	A Nayyar core (amalgam or composite retained in the pulp chamber)
D	With pins only
E	With temporary cement

See answer and justification for Q598 →

Q600 INTERMEDIATE · Block H

A missing lateral space is narrow with root convergence after orthodontics and implant placement cannot be safely decided from available imaging. What action is appropriate?

A	Defer definitive bridge while active caries risk is controlled
B	Discuss a removable temporary replacement while prognosis is uncertain
C	Avoid the design when occlusal clearance cannot be provided safely
D	Refer for coordinated implant/orthodontic planning where space and roots are unsuitable
E	Enamel-supported conservative single-retainer design

See answer and justification for Q600 →

Q602 INTERMEDIATE · Block H

A 4-unit posterior bridge is proposed between two abutments with mobility, large restorations and reduced bone support. Which issue takes priority?

A	Consider an adhesive resin-bonded single-retainer bridge in favourable enamel/occlusion
B	Reassess connector design where porcelain/metal fatigue occurs
C	Evaluate abutment periodontal prognosis and overload before choosing a long-span fixed bridge
D	Avoid an irreversible fixed solution until disease and tooth prognosis are stable
E	Discuss no replacement as a valid choice when function and patient preferences permit

See answer and justification for Q602 →

Q605 INTERMEDIATE · Block H

You are choosing a single-rooted upper abutment for a cantilever.
Which single-rooted upper tooth has the largest root surface area?

A	Canine
B	Central incisor
C	Lateral incisor
D	First premolar
E	Second premolar

See answer and justification for Q605 →

Q606 INTERMEDIATE · Block H

The proposed abutment teeth have multiple active proximal caries lesions and high plaque indices. What should precede definitive bonding?

A	Enamel-supported conservative single-retainer design
B	Refer for coordinated implant/orthodontic planning where space and roots are unsuitable
C	Avoid the design when occlusal clearance cannot be provided safely
D	Discuss a removable temporary replacement while prognosis is uncertain
E	Defer definitive bridge while active caries risk is controlled

See answer and justification for Q606 →

Q607 INTERMEDIATE · Block H

A technician doubles the occlusogingival height of a bridge connector.
How does deflection change?

A	It doubles
B	It halves
C	It quadruples
D	It is unchanged
E	It falls to about one-eighth

See answer and justification for Q607 →

Q608 INTERMEDIATE · Block H

A bridge replaces all four upper incisors between the canines.
What extra mechanical problem does arch curvature create?

A	The pontics lie outside the line joining the abutments and act as a lever arm
B	The pontics are too small
C	The connectors become thicker
D	The canines are too long
E	None

See answer and justification for Q608 →

Q610 INTERMEDIATE · Block H

A young patient has lost 12; 13 is sound with a long root.
Which conventional design is suitable?

A	A fixed-fixed bridge from 11 to 13
B	A cantilever from 13 replacing 12
C	A distal cantilever from 11
D	A spring cantilever
E	A fixed-movable bridge from 14

See answer and justification for Q610 →

Q611 INTERMEDIATE · Block H

A single-rooted anterior tooth has a subgingival fracture and favourable root length; exposing sound tooth by controlled extrusion may preserve adjacent tissue levels. What is a reasonable option?

A	Avoid a post when adequate coronal structure and a core are already present
B	Consider orthodontic extrusion when biologically and strategically suitable
C	Reassess ferrule and remaining coronal dentine before post placement
D	Discuss extraction/replacement when a deep fracture makes predictable restoration untenable
E	Consider crown lengthening only after evaluating crown-to-root and biological consequences

See answer and justification for Q611 →

Q612 INTERMEDIATE · Block H

47 is tilted mesially and will be the distal abutment of a bridge replacing 46, with 45 as the mesial abutment. Which design avoids the need for parallel preparations?

A	A fixed-fixed bridge
B	A fixed-movable bridge
C	A cantilever from 47
D	A spring cantilever
E	A conventional fixed-fixed with deeper preparations

See answer and justification for Q612 →

Q613 INTERMEDIATE · Block H

An endodontically treated molar retains substantial sound cuspal dentine and a well-retained coronal core. A clinician plans a post solely because the tooth is root-filled. What should be advised?

A	Avoid a post when adequate coronal structure and a core are already present
B	Consider orthodontic extrusion when biologically and strategically suitable
C	Reassess ferrule and remaining coronal dentine before post placement
D	Discuss extraction/replacement when a deep fracture makes predictable restoration untenable
E	Consider crown lengthening only after evaluating crown-to-root and biological consequences

See answer and justification for Q613 →

Q614 INTERMEDIATE · Block H

A student describes an old bridge with a palatal bar from a premolar to a distant incisor pontic. What is it, and why is it rarely used?

A	A Maryland bridge; too strong
B	A fixed-movable bridge; too flexible
C	A resin-bonded cantilever; poor aesthetics
D	A spring cantilever
E	A hybrid bridge; too costly

See answer and justification for Q614 →

Q617 INTERMEDIATE · Block H

A root-filled premolar has little coronal dentine and a plan for a long metal post; there is no circumferential ferrule. What should be assessed before choosing the post?

A	Consider crown lengthening only after evaluating crown-to-root and biological consequences
B	Discuss extraction/replacement when a deep fracture makes predictable restoration untenable
C	Consider orthodontic extrusion when biologically and strategically suitable
D	Avoid a post when adequate coronal structure and a core are already present
E	Reassess ferrule and remaining coronal dentine before post placement

See answer and justification for Q617 →

Q620 INTERMEDIATE · Block H

A resin-bonded wing is planned on 23 but the occlusion is tight.
How much clearance does a Co-Cr wing need, and how can it be gained without preparation?

A	About 0.7 mm
B	About 2 mm; by extracting the opposing tooth
C	None
D	About 3 mm; by orthodontics
E	About 0.1 mm; by grinding the wing thin

See answer and justification for Q620 →

Q621 INTERMEDIATE · Block H

You are designing a resin-bonded cantilever bridge from 13 replacing 12.
Which design features improve its success?

A	A thin wing covering the minimum enamel
B	No rest
C	Two wings for safety
D	Maximum enamel coverage within aesthetic limits, about 180° wraparound, a cingulum rest and minimal preparation for a path
E	A labial wing

See answer and justification for Q621 →

Q624 INTERMEDIATE · Block H

A two-retainer resin-bonded bridge replacing a maxillary lateral incisor repeatedly debonds from one abutment only. There is no new caries, but adjacent abutments show different mobility under function. The patient wishes to avoid major preparation. What redesign should be evaluated?

A	Use a longer fixed-fixed retainer on two additional abutments without revisiting mobility
B	Use a single retainer only if occlusion forces it into tension
C	Increase only the resin-cement thickness
D	Consider an appropriately designed single-retainer cantilever RBB after assessing occlusion, support and available enamel
E	Place two separate metal wings that do not contact the pontic

See answer and justification for Q624 →

Q625 INTERMEDIATE · Block H

A pontic is needed for 46 in a patient with no aesthetic concern.
Which pontic design is most hygienic?

A	Saddle
B	Ovate
C	Sanitary (hygienic), with no ridge contact and at least a 3 mm gap
D	Ridge lap
E	Modified ridge lap

See answer and justification for Q625 →

Q626 INTERMEDIATE · Block H

A pontic replaces 14 in a patient who smiles broadly.
Which design balances appearance and hygiene?

A	Sanitary
B	Saddle
C	Conical
D	Ovate in a thin ridge
E	Modified ridge lap, contacting the labial side of the ridge only

See answer and justification for Q626 →

Q627 INTERMEDIATE · Block H

A patient with a thick ridge after socket preservation wants the most natural emergence for a 21 pontic.
Which pontic is indicated?

A	Sanitary
B	Saddle
C	Ovate, sitting in a concavity in the ridge
D	Conical
E	Hygienic

See answer and justification for Q627 →

Q629 INTERMEDIATE · Block H

An anterior tooth was lost to trauma and neighbouring teeth have uncertain vitality pending follow-up tests. A replacement is wanted in one week. What is safest?

A	Refer for coordinated implant/orthodontic planning where space and roots are unsuitable
B	Defer definitive bridge while active caries risk is controlled
C	Avoid the design when occlusal clearance cannot be provided safely
D	Enamel-supported conservative single-retainer design
E	Discuss a removable temporary replacement while prognosis is uncertain

See answer and justification for Q629 →

Q630 INTERMEDIATE · Block H

A posterior bridge is planned for a patient with heavy occlusal forces.
How can load on the pontic be reduced?

A	Widen the pontic
B	Use steep cusps
C	Increase the OVD
D	Narrow the buccolingual width of the pontic
E	Use porcelain occlusal surfaces

See answer and justification for Q630 →

Q631 INTERMEDIATE · Block H

A young patient has small peg lateral incisors, good enamel and a wax-up that demonstrates additive correction without preparation. What is the first restorative choice?

A	Choose adhesive glass ceramic where enamel bonding and aesthetics dominate
B	Prefer repairable direct composite when a reversible additive solution is appropriate
C	Reassess occlusal reduction and restorative space before material change
D	Use a high-strength monolithic material when load and connector geometry dominate
E	Correct unsupported veneering porcelain/design before remaking a metal–ceramic crown

See answer and justification for Q631 →

Q632 INTERMEDIATE · Block H

A lithium disilicate bridge is planned from 13 to 15 replacing 14.
What is the minimum connector cross-section, and how far back can such a bridge extend?

A	About 16 mm² in the premolar region
B	4 mm²; any length
C	9 mm²; to the second molar
D	25 mm²; five units
E	No minimum

See answer and justification for Q632 →

Q633 INTERMEDIATE · Block H

A three-unit posterior zirconia bridge is planned.
What approximate minimum connector cross-section is needed?

A	2 mm²
B	4 mm²
C	30 mm²
D	About 9 mm²
E	50 mm²

See answer and justification for Q633 →

Q634 INTERMEDIATE · Block H

A technician can enlarge a connector either in height or in width.
Which matters more, and why?

A	Width, because strength rises with the cube of width
B	Neither
C	Width, for aesthetics
D	Both equally
E	Height, because strength rises with the square of height and deflection falls with its cube

See answer and justification for Q634 →

Q635 INTERMEDIATE · Block H

A 30-year-old has tetracycline staining that has not responded to bleaching and wants better-looking teeth.
Which indication supports porcelain veneers?

A	Heavy bruxism
B	Class III incisors
C	Discolouration unresponsive to bleaching
D	Little enamel
E	Poor oral hygiene

See answer and justification for Q635 →

Q636 INTERMEDIATE · Block H

A bonded onlay loses retention rapidly after a procedure with uncontrolled saliva and bleeding, despite correct material-specific pretreatment. What factor is most implicated?

A	Undersized ceramic thickness in a high-load area
B	Compromised abutment prognosis from untreated disease
C	Inadequate isolation of adhesive surfaces
D	Inadequate resistance form in a short tapered preparation
E	Inadequate material-specific substrate conditioning

See answer and justification for Q636 →

Q637 INTERMEDIATE · Block H

You are choosing a preparation design for upper incisor veneers.
Which design is most common, for stress distribution and seating?

A	Window
B	Feathered incisal
C	Butt joint
D	Full crown
E	Incisal overlap with a palatal finish line

See answer and justification for Q637 →

Q639 INTERMEDIATE · Block H

A lower first premolar has an MOD restoration, and its buccal cusp is less than half as wide as it is high. What restoration is indicated?

A	An intracoronal inlay
B	A larger amalgam
C	Cusp coverage with an onlay
D	A veneer
E	No restoration

See answer and justification for Q639 →

Q640 INTERMEDIATE · Block H

A posterior fixed prosthesis needs high connector strength, occlusal forces are substantial, and a suitable monolithic design is feasible. Which broad material strategy is most defensible?

A	Prefer repairable direct composite when a reversible additive solution is appropriate
B	Use a high-strength monolithic material when load and connector geometry dominate
C	Reassess occlusal reduction and restorative space before material change
D	Correct unsupported veneering porcelain/design before remaking a metal–ceramic crown
E	Choose adhesive glass ceramic where enamel bonding and aesthetics dominate

See answer and justification for Q640 →

Q641 INTERMEDIATE · Block H

A posterior ceramic onlay fractures through a very thin functional cusp; adhesion remains intact on the tooth. What is the most likely contributing factor?

A	Inadequate material-specific substrate conditioning
B	Compromised abutment prognosis from untreated disease
C	Inadequate isolation of adhesive surfaces
D	Undersized ceramic thickness in a high-load area
E	Inadequate resistance form in a short tapered preparation

See answer and justification for Q641 →

Q642 INTERMEDIATE · Block H

A crown on a root-filled tooth repeatedly fails while new periapical symptoms and widening radiolucency have developed. What takes precedence?

A	Undersized ceramic thickness in a high-load area
B	Compromised abutment prognosis from untreated disease
C	Inadequate isolation of adhesive surfaces
D	Inadequate material-specific substrate conditioning
E	Inadequate resistance form in a short tapered preparation

See answer and justification for Q642 →

Q643 INTERMEDIATE · Block H

A dental technician asks whether she must register with a professional regulator.
Who regulates dental technicians and clinical dental technicians?

A	The GDC
B	The MHRA
C	The CQC
D	The BDA
E	No one

See answer and justification for Q643 →

Q644 INTERMEDIATE · Block H

An older adult has a stable shortened dental arch, good mastication, and no aesthetic concern about a missing terminal molar. What management is reasonable?

A	Consider an adhesive resin-bonded single-retainer bridge in favourable enamel/occlusion
B	Avoid an irreversible fixed solution until disease and tooth prognosis are stable
C	Reassess connector design where porcelain/metal fatigue occurs
D	Discuss no replacement as a valid choice when function and patient preferences permit
E	Evaluate abutment periodontal prognosis and overload before choosing a long-span fixed bridge

See answer and justification for Q644 →

Q646 INTERMEDIATE · Block H

A 25-year-old has a single darkened, root-filled upper incisor and asks for a crown.
What should be considered first?

A	A crown
B	Investigating the cause, then non-vital (internal) bleaching
C	A veneer
D	Extraction and an implant
E	An RBB

See answer and justification for Q646 →

Q649 INTERMEDIATE · Block H

A 30-year-old has crowded, rotated upper incisors and wants them "straightened" with crowns.
What should be offered first?

A	Crowns
B	Veneers
C	Extraction
D	Composite only
E	Orthodontics, before any elective tooth preparation

See answer and justification for Q649 →

Q650 INTERMEDIATE · Block H

A patient insists on porcelain veneers for her upper incisors.
What should be done before any preparation?

A	A diagnostic wax-up and mock-up, assessment of enamel, parafunction and overbite, and a discussion of irreversibility and replacement
B	Immediate preparation
C	A shade guide only
D	Whitening the day before bonding
E	Nothing

See answer and justification for Q650 →

Q651 INTERMEDIATE · Block H

A 28-year-old with sound, healthy upper incisors wants crowns because one is slightly darker.
Why are crowns inappropriate?

A	They are too weak
B	They are too cheap
C	They are less conservative than whitening, orthodontics or composite
D	They cannot match shade
E	Crowns are illegal for aesthetic reasons

See answer and justification for Q651 →

Q652 INTERMEDIATE · Block H

At a 5-year review, a bridge is still in place but has recurrent caries and recession.
How should its outcome be described?

A	A success
B	It has survived but is not a success
C	A failure, because it is present
D	A complete failure
E	Not assessable

See answer and justification for Q652 →

Q653 INTERMEDIATE · Block H

A patient wants elective replacement of a missing premolar, but there is active caries and untreated periodontal inflammation in proposed abutments. What is the first sequencing decision?

A	Discuss no replacement as a valid choice when function and patient preferences permit
B	Avoid an irreversible fixed solution until disease and tooth prognosis are stable
C	Reassess connector design where porcelain/metal fatigue occurs
D	Evaluate abutment periodontal prognosis and overload before choosing a long-span fixed bridge
E	Consider an adhesive resin-bonded single-retainer bridge in favourable enamel/occlusion

See answer and justification for Q653 →

Q654 INTERMEDIATE · Block H

A patient needs crowns or bridges on most teeth with an OVD increase.
At what level is this care?

A	Level 1
B	Level 2
C	Hygienist
D	Level 3 (referral)
E	Not provided

See answer and justification for Q654 →

Q655 INTERMEDIATE · Block H

A patient with uncontrolled hypertension needs gingival retraction before an impression.
Which retraction method should be avoided?

A	Adrenaline-impregnated cord
B	Aluminium chloride cord
C	Ferric sulphate
D	Retraction paste
E	Plain cord

See answer and justification for Q655 →

Q656 INTERMEDIATE · Block H

A short clinical molar crown debonds twice despite correct cement mixing and good isolation; the preparation has marked taper. What is the dominant problem?

A	Inadequate isolation of adhesive surfaces
B	Undersized ceramic thickness in a high-load area
C	Inadequate resistance form in a short tapered preparation
D	Inadequate material-specific substrate conditioning
E	Compromised abutment prognosis from untreated disease

See answer and justification for Q656 →

Q657 INTERMEDIATE · Block H

A colleague plans to scan a full arch digitally for a 10-unit bridge.
Which statement about intraoral scans is correct?

A	They suit short-span work best
B	They are most accurate for full arches
C	They need no calibration
D	They replace occlusal records
E	They avoid all errors

See answer and justification for Q657 →

Q658 INTERMEDIATE · Block H

A margin lies just subgingivally but periodontal support is adequate and surgical access might create a ferrule. What must be evaluated before choosing crown lengthening?

A	Consider orthodontic extrusion when biologically and strategically suitable
B	Discuss extraction/replacement when a deep fracture makes predictable restoration untenable
C	Reassess ferrule and remaining coronal dentine before post placement
D	Avoid a post when adequate coronal structure and a core are already present
E	Consider crown lengthening only after evaluating crown-to-root and biological consequences

See answer and justification for Q658 →

Q660 INTERMEDIATE · Block H

You are choosing between an inlay, an onlay and a crown for a premolar with a large cavity.
Which statement is correct?

A	A crown always preserves most tooth
B	An inlay protects weak cusps
C	An adhesive onlay preserves more tooth than a crown while covering the cusps
D	Onlays cannot be bonded
E	Crowns need no preparation

See answer and justification for Q660 →

Q661 INTERMEDIATE · Block H

You are assessing the lower anterior teeth as abutments.
Which lower tooth has the smallest root surface area?

A	Central incisor
B	Lateral incisor
C	Canine
D	First premolar
E	Second premolar

See answer and justification for Q661 →

Q662 INTERMEDIATE · Block H

46 is missing. 45 is upright, but 47 has tipped mesially by 25°.
Which approach is appropriate?

A	A fixed-fixed bridge with parallel preparations through 47
B	A distal cantilever from 45
C	Extract 47
D	A fixed-movable bridge, or uprighting 47 orthodontically
E	A spring cantilever

See answer and justification for Q662 →

Q666 INTERMEDIATE · Block H

A 16-year-old finishes orthodontics with small lateral incisors and spaces, and growth is ongoing.
What is the preferred restoration?

A	Porcelain veneers
B	Crowns
C	Direct composite build-ups
D	Implants
E	Conventional bridges

See answer and justification for Q666 →

Q667 INTERMEDIATE · Block H

A patient has just finished whitening and wants composite bonding the next day.
Why should you wait?

A	Whitening strengthens enamel
B	The shade may still change and residual oxygen reduces the resin bond for a time
C	Composite cannot be used after whitening
D	Whitening etches enamel
E	There is no reason to wait

See answer and justification for Q667 →

Q670 INTERMEDIATE · Block H

At review of a five-year-old fixed bridge, the patient is satisfied with appearance but the margins bleed on probing and radiographs show new distal caries on an abutment. What should the review prioritise?

A	Patient satisfaction alone as evidence of success
B	Shade matching before any biological investigation
C	Immediate recementation without removing deposits
D	Caries, tissue health, margins, contacts and occlusion before cosmetic changes
E	Repolishing the pontic while leaving the abutment unchanged

See answer and justification for Q670 →

BLOCK I · Occlusion and temporomandibular disorders (44)

Q671 INTERMEDIATE · Block I

A 42-year-old has intermittent preauricular clicking and myalgia but no locking, progressive limitation or alarming signs. A reversible appliance and self-care have helped. The patient asks for extensive occlusal equilibration to permanently stop clicking. Which advice is best supported?

A	Recommend full-mouth adjustment as a definitive proven cure for benign disc sounds
B	Provide occlusal adjustment immediately because the splint was effective
C	Extract the posterior interfering tooth to reduce joint loading
D	Explain the uncertain causal relationship and favour reversible care, monitoring and further assessment if symptoms progress
E	Offer prophylactic TMJ surgery while range of movement is preserved

See answer and justification for Q671 →

Q672 INTERMEDIATE · Block I

A periodontitis patient has a fremitus on an anterior tooth, increasing mobility and widened periodontal ligament space. Plaque-associated inflammation and deep pockets are also present. Which interpretation best guides the initial management plan?

A	Occlusal trauma alone proves the patient has plaque-independent attachment loss
B	The periodontal disease can be treated by selective grinding without controlling biofilm
C	Mobility means the tooth must be extracted before reassessing periodontal response
D	Mechanical overload is irrelevant whenever bleeding occurs
E	Control periodontal inflammation and reassess traumatic loading/mobility; occlusal trauma may modify support and function but does not replace the microbial diagnosis

See answer and justification for Q672 →

Q677 INTERMEDIATE · Block I

A new maxillary complete denture will oppose mandibular natural teeth with canine guidance and pronounced posterior cusps. Trial insertion reveals repeated unilateral dislodgement during lateral movements despite adequate suction and stable jaw relations. Which consideration is most directly relevant?

A	Restore bilateral posterior contacts at all excursive positions without assessing the existing natural guidance or stability
B	Accept the natural canine guidance as necessarily ideal for the complete denture
C	Shorten the maxillary buccal flanges rather than examine functional contacts
D	Increase the anterior vertical overlap to strengthen canine disclusion
E	Reassess the occlusal scheme and eliminate destabilising excursive interferences while considering denture support and opposing natural dentition

See answer and justification for Q677 →

Q678 INTERMEDIATE · Block I

A textbook describes an occlusion in which the anterior teeth protect the posteriors in excursions, and the posteriors protect the anteriors in ICP.
What is this?

A	Balanced occlusion
B	Group function
C	Lingualised occlusion
D	Mutually protected occlusion
E	Monoplane occlusion

See answer and justification for Q678 →

Q682 INTERMEDIATE · Block I

A person with diffuse TMD pain requests multiple full-coverage crowns as a definitive cure despite stable teeth and no structural indication. What should be advised?

A	Urgent specialist investigation for unexplained progressive neurosensory change
B	Differentiate disc displacement with reduction from persistent closed lock
C	Avoid irreversible occlusal reconstruction solely to treat nonspecific TMD pain
D	Recognition of acute mandibular dislocation requiring timely reduction/referral
E	Conservative reversible measures for myalgia without red flags

See answer and justification for Q682 →

Q684 INTERMEDIATE · Block I

A patient's maximum mouth opening is 34 mm.
How should this be interpreted?

A	Normal
B	Restricted (normal is about 40–55 mm)
C	Hypermobile
D	Normal for women
E	Excessive

See answer and justification for Q684 →

Q686 INTERMEDIATE · Block I

A patient opens with pure hinge rotation of the condyles before translation.
About how much incisal opening does pure rotation give?

A	5 mm
B	20–25 mm
C	40 mm
D	55 mm
E	70 mm

See answer and justification for Q686 →

Q687 INTERMEDIATE · Block I

A patient reports progressive unilateral facial numbness and worsening pain without a clear dental cause.
What takes priority?

A	Urgent specialist investigation for unexplained progressive neurosensory change
B	Differentiate disc displacement with reduction from persistent closed lock
C	Recognition of acute mandibular dislocation requiring timely reduction/referral
D	Avoid irreversible occlusal reconstruction solely to treat nonspecific TMD pain
E	Conservative reversible measures for myalgia without red flags

See answer and justification for Q687 →

Q688 INTERMEDIATE · Block I

A 30-year-old has aching cheeks, worse in the morning, and familiar pain when the masseters are palpated.
Joint movements are painless.
What is the DC/TMD diagnosis?

A	Myalgia
B	Arthralgia
C	Disc displacement with reduction
D	Degenerative joint disease
E	Subluxation

See answer and justification for Q688 →

Q689 INTERMEDIATE · Block I

A patient has a reproducible reciprocal click with normal maximal opening; there is no current locking. Which joint-mechanics interpretation is most likely?

A	Recognition of acute mandibular dislocation requiring timely reduction/referral
B	Avoid irreversible occlusal reconstruction solely to treat nonspecific TMD pain
C	Conservative reversible measures for myalgia without red flags
D	Urgent specialist investigation for unexplained progressive neurosensory change
E	Differentiate disc displacement with reduction from persistent closed lock

See answer and justification for Q689 →

Q690 INTERMEDIATE · Block I

A patient has a click on opening and another on closing (reciprocal clicking), without pain or locking. What is the DC/TMD diagnosis?

A	Disc displacement without reduction
B	Degenerative joint disease
C	Subluxation
D	Disc displacement with reduction
E	Myalgia

See answer and justification for Q690 →

Q691 INTERMEDIATE · Block I

A patient has reciprocal clicking and occasional brief episodes when the jaw catches and will not open fully. What is the DC/TMD diagnosis?

A	Disc displacement without reduction, with limited opening
B	Myalgia
C	Disc displacement with reduction, with intermittent locking
D	Subluxation
E	Degenerative joint disease

See answer and justification for Q691 →

Q692 INTERMEDIATE · Block I

A patient's click stopped suddenly a week ago. She now opens 28 mm, deviating to the right, with restricted left lateral movement.
What is the diagnosis?

A	Disc displacement with reduction
B	Subluxation
C	Myalgia
D	Disc displacement without reduction, with limited opening (right closed lock)
E	Degenerative joint disease

See answer and justification for Q692 →

Q694 INTERMEDIATE · Block I

A 68-year-old has crepitus in both TMJs, and a radiograph shows flattening and osteophytes on the condyles.
What is the diagnosis?

A	Degenerative joint disease
B	Disc displacement with reduction
C	Myalgia
D	Subluxation
E	Arthralgia only

See answer and justification for Q694 →

Q695 INTERMEDIATE · Block I

After a wide yawn, a patient cannot close her mouth. The condyles are palpable in front of the eminences.
What is the diagnosis?

A	Closed lock
B	Disc displacement with reduction
C	Myalgia
D	Degenerative joint disease
E	Dislocation (open lock)

See answer and justification for Q695 →

Q696 INTERMEDIATE · Block I

A patient has temporal headaches that worsen with chewing and are reproduced by palpating the temporalis.
What is the DC/TMD diagnosis?

A	Headache attributed to TMD
B	Migraine
C	Giant cell arteritis
D	Trigeminal neuralgia
E	Arthralgia

See answer and justification for Q696 →

Q698 INTERMEDIATE · Block I

You are taking the history of a patient with TMD.
In which group is TMD commonest?

A	Men over 60
B	Children under 10
C	Men aged 20–40
D	People over 80
E	Women aged about 20–40

See answer and justification for Q698 →

Q699 INTERMEDIATE · Block I

A 25-year-old has a painless click on opening, with no locking or limitation.
What is the appropriate management?

A	An anterior repositioning splint full-time
B	Occlusal equilibration
C	An MRI scan
D	Reassurance only
E	Referral for surgery

See answer and justification for Q699 →

Q700 INTERMEDIATE · Block I

A patient with masticatory myalgia asks what she can do herself.
Which self-care advice is appropriate?

A	Chew gum to exercise the muscles
B	A soft diet, avoiding wide opening and gum, warm compresses, jaw rest, and keeping "lips together, teeth apart"
C	Open as wide as possible several times a day
D	Avoid all analgesics
E	Sleep on the painful side

See answer and justification for Q700 →

Q701 INTERMEDIATE · Block I

A stabilisation splint is planned for a patient with myalgia.
Which design is correct?

A	Soft vacuum-formed, worn full-time
B	Anterior repositioning, worn full-time
C	Partial-coverage anterior bite plane worn continuously
D	A posterior-only splint
E	Full-coverage hard acrylic (often maxillary), with even contacts in RCP and canine guidance

See answer and justification for Q701 →

Q703 INTERMEDIATE · Block I

A patient has worn an anterior repositioning splint day and night for 9 months for a clicking joint.
What complication is likely?

A	Anterior open bite
B	Increased overbite
C	Canine wear
D	Posterior open bite
E	Gingival recession

See answer and justification for Q703 →

Q704 INTERMEDIATE · Block I

A patient has worn an anterior bite plane continuously for a year.
What complication may follow?

A	Anterior open bite
B	Increased freeway space
C	Over-eruption of the uncovered posterior teeth
D	Recession of the incisors
E	Root resorption of the canines

See answer and justification for Q704 →

Q705 INTERMEDIATE · Block I

A patient's left TMJ locked 2 days ago, with sudden loss of clicking and restricted opening.
What might be tried first?

A	Manipulation to try to recapture the disc, with reassurance, a splint and exercises
B	Immediate surgery
C	Occlusal grinding
D	Extraction of the third molars
E	Nothing

See answer and justification for Q705 →

Q706 INTERMEDIATE · Block I

A patient's jaw dislocates during an extraction. The condyles are anterior to the eminences.
How is it reduced?

A	Pull the chin forwards
B	Push the chin upwards
C	Wait for spontaneous reduction
D	Press the mandible downwards and backwards, with the thumbs on the lower molars or external oblique ridges
E	Ask the patient to bite hard

See answer and justification for Q706 →

Q707 INTERMEDIATE · Block I

A 72-year-old has new jaw pain on chewing that eases with rest, temporal headaches and a tender scalp.
What is the most appropriate action?

A	Same-day medical referral for suspected giant cell arteritis (urgent ESR or CRP)
B	A stabilisation splint
C	Ibuprofen and review in a month
D	Occlusal adjustment
E	TMJ MRI

See answer and justification for Q707 →

Q708 INTERMEDIATE · Block I

A 58-year-old has progressive trismus, numbness of the lower lip, unexplained weight loss and earache with a normal ear examination.
What must be excluded?

A	Myalgia
B	Malignancy (for example nasopharyngeal, parotid or infratemporal)
C	Disc displacement with reduction
D	Bruxism
E	Dental caries

See answer and justification for Q708 →

Q709 INTERMEDIATE · Block I

A 64-year-old has jaw ache that comes on when climbing stairs and settles with rest.
What must be considered?

A	Myalgia
B	Pulpitis
C	Disc displacement
D	Sinusitis
E	Cardiac pain referred to the jaw

See answer and justification for Q709 →

Q710 INTERMEDIATE · Block I

A patient has brief, electric-shock pains in the right cheek triggered by light touch when shaving.
What is the likely diagnosis and treatment?

A	Myalgia; splint
B	Arthralgia; NSAIDs
C	Trigeminal neuralgia
D	Pulpitis; RCT
E	Sinusitis; antibiotics

See answer and justification for Q710 →

Q712 INTERMEDIATE · Block I

A patient with widespread painful masticatory myalgia but no locking or structural joint pathology asks for a minimally invasive starting treatment. Which approach is most appropriate?

A	Immediate arthrocentesis despite no joint indication
B	Full-mouth occlusal reconstruction as first-line therapy
C	Botulinum toxin as routine first-line treatment
D	Education, self-care, conservative analgesia or reversible therapy with review
E	A long-term opioid prescription without reassessment

See answer and justification for Q712 →

Q713 INTERMEDIATE · Block I

A patient's chronic TMD pain persists despite conservative care.
Who would usually start low-dose tricyclic medication?

A	The GMP or specialist
B	The dental nurse
C	The technician
D	The patient alone
E	The pharmacist without prescription

See answer and justification for Q713 →

Q714 INTERMEDIATE · Block I

You examine a patient who says she grinds her teeth.
Which signs suggest bruxism?

A	Erosion cupping only
B	Abrasion notches
C	Root caries
D	Attrition facets, fractured restorations and cusps, masseteric hypertrophy, a scalloped tongue and linea alba
E	Gingival recession only

See answer and justification for Q714 →

Q715 INTERMEDIATE · Block I

A known bruxist needs new posterior crowns.
How should the new work be protected?

A	Layered feldspathic porcelain
B	No protection is needed
C	A hard night-time splint, with metal or monolithic zirconia occlusal surfaces
D	A soft daytime splint
E	Temporary crowns only

See answer and justification for Q715 →

Q716 INTERMEDIATE · Block I

Multiple posterior crowns are planned.
What should precede them?

A	Occlusal grinding
B	A soft splint
C	Examination on mounted casts to find interferences, the slide and the guidance
D	Extraction of the third molars
E	Nothing

See answer and justification for Q716 →

Q717 INTERMEDIATE · Block I

A patient needs restorations on most teeth with a planned OVD increase, and has an unstable ICP.
Which approach is appropriate?

A	Conform to ICP
B	Restore one tooth at a time without a plan
C	Ignore the occlusion
D	Reorganise to RCP
E	Splint therapy only

See answer and justification for Q717 →

Q718 INTERMEDIATE · Block I

A colleague plans occlusal equilibration.
When is it justified?

A	Only to remove specific interferences for restorative reasons
B	As a routine TMD cure
C	In all new patients
D	To prevent caries
E	Before any splint

See answer and justification for Q718 →

Q719 INTERMEDIATE · Block I

A 28-year-old has typical masticatory myalgia with no red flags.
Is routine imaging needed?

A	Yes, MRI in all cases
B	No; imaging is targeted to specific questions such as bone pathology or an alternative diagnosis
C	Yes, a panoramic radiograph always
D	Yes, CBCT always
E	Yes, a skull radiograph

See answer and justification for Q719 →

Q721 INTERMEDIATE · Block I

A heavy bruxist with sound canines is having new posterior crowns.
Why is canine guidance favoured?

A	Canine guidance discludes the posterior teeth, reducing lateral load on the new crowns
B	It increases posterior contact
C	It improves appearance
D	It reduces OVD
E	It prevents caries

See answer and justification for Q721 →

Q722 INTERMEDIATE · Block I

After yawning widely, a patient cannot close and the condyles are displaced anterior to the eminence. What is the appropriate diagnosis/action pathway?

A	Urgent specialist investigation for unexplained progressive neurosensory change
B	Avoid irreversible occlusal reconstruction solely to treat nonspecific TMD pain
C	Differentiate disc displacement with reduction from persistent closed lock
D	Conservative reversible measures for myalgia without red flags
E	Recognition of acute mandibular dislocation requiring timely reduction/referral

See answer and justification for Q722 →

Q724 INTERMEDIATE · Block I

A patient has an acute left disc displacement without reduction.
Which way does the mandible deviate on opening?

A	Right
B	Left (towards the affected side)
C	No deviation
D	Forwards only
E	It alternates

See answer and justification for Q724 →

Q725 INTERMEDIATE · Block I

A patient has bilateral masseter tenderness, stress-related clenching, normal opening and no constitutional symptoms. What initial approach is best?

A	Urgent specialist investigation for unexplained progressive neurosensory change
B	Differentiate disc displacement with reduction from persistent closed lock
C	Avoid irreversible occlusal reconstruction solely to treat nonspecific TMD pain
D	Recognition of acute mandibular dislocation requiring timely reduction/referral
E	Conservative reversible measures for myalgia without red flags

See answer and justification for Q725 →

Q727 INTERMEDIATE · Block I

You are examining a new TMD patient.
What should be documented?

A	Only the bite
B	Only radiographs
C	Only the shade of the teeth
D	Pain location, movement limitation, joint noises, locking and functional impact
E	Nothing until imaging

See answer and justification for Q727 →

Q728 INTERMEDIATE · Block I

A patient's upper canine has reduced periodontal support, so it cannot take all lateral load alone.
Which guidance is preferable when restoring?

A	Canine guidance on that canine
B	A non-working contact
C	Group function, sharing the working-side load
D	Balanced occlusion
E	No guidance

See answer and justification for Q728 →

Q729 INTERMEDIATE · Block I

A patient has a stable, comfortable ICP and needs a single onlay.
Which approach is correct?

A	Reorganise to RCP
B	Full-mouth equilibration
C	Increase the OVD
D	Splint first
E	Conform to the existing ICP

See answer and justification for Q729 →

Q730 INTERMEDIATE · Block I

A patient denies grinding but has a marked ridge on the buccal mucosa and indentations along the tongue. What do these signs suggest?

A	Clenching or bruxism (linea alba and a scalloped tongue)
B	Candida
C	Lichen planus
D	Iron deficiency
E	Normal anatomy only

See answer and justification for Q730 →

BLOCK J · Tooth wear, the failing dentition, interfaces and trauma (57)

Q731 INTERMEDIATE · Block J

A 49-year-old reports a sharp pain on release when biting a hard granule on a heavily restored mandibular second molar. Cold produces brief, non-lingering pain; percussion is normal and no periapical lesion is seen. Which diagnosis is most consistent with the pattern?

A	Symptomatic apical periodontitis from a necrotic pulp
B	Primary myofascial pain from TMD without a dental component
C	A cracked tooth with a vital pulp and probable reversible pulpal inflammation
D	A vertical root fracture proved by a normal periapical radiograph
E	Persistent idiopathic dentoalveolar pain requiring extraction

See answer and justification for Q731 →

Q732 INTERMEDIATE · Block J

A 45-year-old has symptomatic localised anterior wear, no interocclusal space, intact posterior support and wishes to avoid crowns. Which restorative approach is particularly suitable?

A	Review bruxism and restorative material/load before replacing repeated fractured restorations
B	Use monitoring/prevention when wear is localised, asymptomatic and stable
C	Refer for multidisciplinary assessment of severe generalised wear with complex occlusal changes
D	Consider additive anterior Dahl restoration where space is limited but posterior support stable
E	Control intrinsic or dietary acid challenge before definitive restoration

See answer and justification for Q732 →

Q733 INTERMEDIATE · Block J

A root-filled mandibular molar shows a narrow isolated 10-mm periodontal pocket adjacent to one root, intermittent sinus drainage and a J-shaped radiolucency. General periodontal probing elsewhere is shallow. What is the most appropriate next step before assigning a prognosis or preparing a replacement bridge?

A	Prioritise investigations for possible vertical root fracture, integrating isolated probing, sinus tracing, imaging and history
B	Treat as localised Stage III periodontal disease on the strength of the 10-mm pocket alone
C	Interpret the radiographic halo as conclusive persistent apical infection and repeat root treatment immediately
D	Treat the isolated lesion as occlusal trauma and splint the tooth before testing its structural integrity
E	Begin definitive bridge preparation because a previous root filling rules out a structural root problem

See answer and justification for Q733 →

Q734 INTERMEDIATE · Block J

A patient has severe generalised attrition/erosion, unstable posterior support, altered jaw relations and multiple failed restorations. What is the most appropriate overall approach?

A	Refer for multidisciplinary assessment of severe generalised wear with complex occlusal changes
B	Use monitoring/prevention when wear is localised, asymptomatic and stable
C	Consider additive anterior Dahl restoration where space is limited but posterior support stable
D	Review bruxism and restorative material/load before replacing repeated fractured restorations
E	Control intrinsic or dietary acid challenge before definitive restoration

See answer and justification for Q734 →

Q736 INTERMEDIATE · Block J

A mandibular first molar has an isolated narrow probing depth of 9 mm on the buccal aspect. The tooth is tender to percussion and does not respond to sensibility tests, while neighbouring teeth have shallow pockets and there is no significant plaque-associated attachment loss. A periapical lesion tracks toward the furcation. What is the preferred initial disease-directed approach?

A	Perform definitive subgingival instrumentation first without pulp assessment because the tract reaches the gingival sulcus
B	Assess and control a probable pulpal infection through an endodontic pathway, then reassess periodontal healing
C	Treat both lesions surgically at the same visit without establishing the likely primary source
D	Classify the lesion by the periodontal pocket depth alone and begin regenerative furcation therapy
E	Schedule periodontal recall alone because a narrow defect indicates transient trauma rather than infection

See answer and justification for Q736 →

Q737 INTERMEDIATE · Block J

A patient has long-standing generalised attachment loss, interproximal radiographic bone loss and a heavily involved molar furcation. The tooth previously responded normally to cold but now has no pulpal response and a new periapical radiolucency. What clinical reasoning is strongest?

A	A primary pulpal lesion is proven because the tooth currently fails cold testing
B	A vertical root fracture is established by the new apical shadow alone
C	The denture design is the direct cause of pulp necrosis
D	Long-standing periodontal disease may have progressed to secondary pulpal involvement, requiring integrated endodontic and periodontal assessment
E	The periodontal component should be ignored until a new crown is placed

See answer and justification for Q737 →

Q738 INTERMEDIATE · Block J

A 19-year-old suffers trauma to a mature maxillary central incisor. A periapical radiograph taken at two projections shows a horizontal root fracture in the cervical third, and the coronal fragment is displaced and mobile. Which initial management principle is most appropriate?

A	Assume that immediate extraction is required for every cervical third fracture
B	Leave the displaced segment un-repositioned because pulp tests are initially unreliable
C	Perform endodontic treatment of both segments before repositioning
D	Provide a definitive ceramic crown immediately to splint the tooth
E	Promptly reposition the coronal segment, assess stabilisation options and arrange trauma-guideline follow-up/referral

See answer and justification for Q738 →

Q739 INTERMEDIATE · Block J

A young adult with active bulimia has worsening palatal erosion and requests definitive ceramic veneers. What takes priority?

A	Modify parafunctional loading and restore selectively after risk assessment
B	Staged provisional increase in OVD for complex generalised rehabilitation
C	Localised additive composite using the Dahl principle
D	Control erosive drivers and defer irreversible restorations
E	Preventive non-restorative approach with monitoring

See answer and justification for Q739 →

Q740 INTERMEDIATE · Block J

A 17-year-old has generalised thin, pitted enamel affecting both primary and permanent teeth, with exposed yellow dentine, rapid posteruptive wear and several similarly affected relatives. Radiographs show reduced enamel thickness but identifiable normal-density enamel where present. Which explanation best fits the findings?

A	Hypomaturation AI characterised primarily by normal enamel thickness with decreased hardness
B	Molar-incisor hypomineralisation restricted to first permanent molars and incisors
C	Hypoplastic amelogenesis imperfecta with insufficient enamel matrix formation
D	Dental fluorosis causing solely symmetrical opaque surface areas without loss of thickness
E	Dentinogenesis imperfecta with bulbous crowns and obliterated pulp chambers as the defining signs

See answer and justification for Q740 →

Q742 INTERMEDIATE · Block J

A teenager has blue-grey opalescent teeth, severe crown wear and a family history of similar teeth. Imaging shows bulbous crowns with cervical constriction and markedly obliterated pulp chambers. What is the strongest diagnosis to investigate?

A	Dentinogenesis imperfecta
B	Isolated fluorosis
C	Hypoplastic amelogenesis imperfecta
D	Tetracycline staining without structural abnormality
E	Molar-incisor hypomineralisation

See answer and justification for Q742 →

Q743 INTERMEDIATE · Block J

A bruxist with good periodontal health breaks repeated small resin restorations in heavy excursive contacts. What is the best next step?

A	Control erosive drivers and defer irreversible restorations
B	Staged provisional increase in OVD for complex generalised rehabilitation
C	Preventive non-restorative approach with monitoring
D	Localised additive composite using the Dahl principle
E	Modify parafunctional loading and restore selectively after risk assessment

See answer and justification for Q743 →

Q744 INTERMEDIATE · Block J

A 44-year-old presents with deep proximal caries on 46. Cold produces severe pain that persists well after the stimulus is removed; spontaneous episodes wake the patient, but percussion is not tender and the periapical film shows no definite apical change. Which clinical diagnosis best guides definitive care?

A	Dentine hypersensitivity that can be treated with a desensitising toothpaste alone
B	Symptomatic irreversible pulpitis in a vital tooth, requiring a discussion of appropriate definitive pulpal treatment or extraction
C	Pulp necrosis because spontaneous pain excludes any vital pulp
D	Symptomatic apical periodontitis established by the absence of periapical change
E	Temporomandibular myalgia because the tooth is percussion negative

See answer and justification for Q744 →

Q746 INTERMEDIATE · Block J

A 23-year-old presents with an avulsed permanent maxillary central incisor after a fall. The tooth has been dry for 80 minutes and is visibly contaminated, but there is no major facial fracture. Which feature is most important when explaining replantation prognosis and subsequent monitoring?

A	Dry time does not affect periodontal ligament viability if the root looks clean
B	Replantation becomes a guarantee of normal ligament healing once the socket is irrigated
C	A mature apex rules out ankylosis after delayed replantation
D	Prolonged dry extraoral time markedly worsens periodontal ligament prognosis, requiring guideline-based replantation decisions and monitoring for replacement resorption/ankylosis
E	The sole long-term concern is tooth colour rather than root resorption

See answer and justification for Q746 →

Q747 INTERMEDIATE · Block J

A 12-year-old presents with a fractured permanent maxillary incisor after sports injury. Dentine is exposed across one incisal angle, the tooth responds to cold and there is no visible pulp exposure, abnormal mobility or displacement. What is the most appropriate immediate restorative goal?

A	Protect exposed dentine and restore the fractured crown, after assessing for associated luxation or root injury and arranging follow-up
B	Begin elective root canal therapy because every dentine fracture compromises pulpal viability
C	Extract the tooth because the patient is still growing
D	Bleach the tooth first to standardise the colour before protecting exposed dentine
E	Delay any dentine coverage until pulpal necrosis is evident

See answer and justification for Q747 →

Q748 INTERMEDIATE · Block J

You consider the Dahl approach for a patient with localised anterior wear.
Which is a relative contraindication?

A	Good compliance
B	Stable periodontal health
C	Long-term bisphosphonate therapy
D	Unworn posterior teeth
E	A deep overbite

See answer and justification for Q748 →

Q749 INTERMEDIATE · Block J

You place Dahl composites today.
What should the patient be warned about?

A	Nothing
B	Immediate perfect chewing
C	Permanent speech change
D	Pulp death
E	Temporary speech change, tenderness and eating difficulty, with posterior contact taking months

See answer and justification for Q749 →

Q750 INTERMEDIATE · Block J

A patient has generalised severe wear and needs an OVD increase on most teeth.
How is this planned?

A	Direct composite on all teeth in one visit without testing
B	A diagnostic wax-up and a provisional trial at the new OVD
C	Extraction of all teeth
D	Crowns at the existing OVD
E	Dahl on posterior teeth only

See answer and justification for Q750 →

Q751 INTERMEDIATE · Block J

You list ways of creating space for restorations.
Which method should be avoided?

A	The Dahl approach
B	Increasing the OVD
C	Orthodontic intrusion or extrusion
D	Minor adjustment of opposing teeth
E	Elective root canal treatment to gain space

See answer and justification for Q751 →

Q753 INTERMEDIATE · Block J

A colleague assumes a patient's OVD is reduced because the teeth are worn.
Why is this unsafe?

A	Worn teeth always mean reduced OVD
B	Dentoalveolar compensation often maintains OVD, so it should not be inferred from wear alone
C	OVD cannot be measured
D	OVD is irrelevant
E	Wear always increases OVD

See answer and justification for Q753 →

Q754 INTERMEDIATE · Block J

A patient with active, untreated reflux has severe palatal erosion.
Why should crowns not be planned yet?

A	Crowns are too expensive
B	Erosion stops naturally
C	Reflux does not affect crowns
D	The cause must be stabilised and the remaining tissue reviewed first
E	Crowns cure reflux

See answer and justification for Q754 →

Q755 INTERMEDIATE · Block J

A patient has erosion, a bruxism habit and toothbrush abrasion.
How should the wear be approached?

A	Recognise that mixed mechanisms are common and address all relevant risk factors
B	Treat only the erosion
C	Treat only the bruxism
D	Assume one cause
E	Ignore the brushing

See answer and justification for Q755 →

Q756 INTERMEDIATE · Block J

You are deciding which teeth to keep in a failing dentition.
What gives a tooth strategic value?

A	Its colour
B	Its abutment potential, occlusal stops and role in canine guidance
C	Its size
D	Being the last molar
E	Previous treatment cost

See answer and justification for Q756 →

Q757 INTERMEDIATE · Block J

A 70-year-old has sound premolars and anteriors, and compromised lower molars needing complex treatment.
What approach is reasonable?

A	Heroic molar treatment
B	Full clearance
C	Implants on all molars
D	Crowns on all teeth
E	A shortened dental arch, avoiding heroic treatment of the molars when function is adequate

See answer and justification for Q757 →

Q758 INTERMEDIATE · Block J

A 60-year-old will gradually lose several teeth of poor prognosis over the next few years.
How should the transition to edentulism be managed?

A	Transitional acrylic RPDs to which teeth can be added, with overdenture abutments where possible
B	Immediate total clearance
C	No prosthesis
D	Implants in every socket
E	Co-Cr frameworks immediately

See answer and justification for Q758 →

Q759 INTERMEDIATE · Block J

A root is to be retained under a complete denture.
What must be true of it?

A	It can be carious
B	It must be crowned
C	It must be root-filled or asymptomatic and free of caries
D	It must be mobile
E	It must be above the ridge by 5 mm

See answer and justification for Q759 →

Q761 INTERMEDIATE · Block J

A patient finishing periodontal therapy wants a bridge.
When is the periodontium stable enough?

A	Any pocket depth
B	6 mm pockets that bleed
C	PPD 4 mm or less with no bleeding at those sites
D	BOP at 50%
E	Mobility grade 3

See answer and justification for Q761 →

Q763 INTERMEDIATE · Block J

You restore a patient with a history of periodontitis.
Which features protect the periodontium?

A	Subgingival margins everywhere
B	Overcontoured crowns
C	Closed embrasures
D	Saddle pontics
E	Supragingival margins, correct contour, and cleansable embrasures and pontics

See answer and justification for Q763 →

Q764 INTERMEDIATE · Block J

A partial denture is needed for periodontally involved abutments.
Which design is appropriate?

A	Acrylic mucosa-borne with gingival coverage
B	Tooth-supported Co-Cr with rests, an open design away from the gingivae, guide planes and minimal clasp torque
C	Rigid clasps in deep undercuts
D	A horseshoe with no rests
E	A flexible nylon denture

See answer and justification for Q764 →

Q765 INTERMEDIATE · Block J

A patient's lower incisors are mobile, and the mobility now makes biting uncomfortable. When is splinting indicated, and which type is preferred?

A	Always; with wire
B	Never
C	Only for aesthetics
D	When increasing mobility interferes with function or comfort; adhesive splints
E	When mobility is grade 0

See answer and justification for Q765 →

Q767 INTERMEDIATE · Block J

Two patients have problems around implants: one has bleeding without bone loss; the other has bone loss. Which is correct?

A	Both are Level 1
B	Both are Level 3
C	Peri-implant mucositis (reversible, no bone loss) is Level 2
D	Both need explantation
E	Peri-implantitis is reversible

See answer and justification for Q767 →

Q768 INTERMEDIATE · Block J

A 49-year-old has sharp pain on release of biting on a heavily restored premolar. Cold sensitivity lasts for about two seconds and the tooth is not spontaneously painful; a periapical radiograph is normal. A visible crack does not extend onto the root surface. What is the best next step?

A	Complete crack assessment, test the segments and pulpal status, stabilise the cusps and plan appropriate coverage if the tooth is restorable
B	Start root canal treatment solely because pain is released on biting
C	Extract immediately because every crack propagates below the bone
D	Adjust the occlusion and stop all monitoring
E	Diagnose neuralgia because the radiograph is normal

See answer and justification for Q768 →

Q769 INTERMEDIATE · Block J

A patient has shallow stable tooth wear for years, no discomfort and no aesthetic complaint. Which management is most justified?

A	Modify parafunctional loading and restore selectively after risk assessment
B	Staged provisional increase in OVD for complex generalised rehabilitation
C	Preventive non-restorative approach with monitoring
D	Control erosive drivers and defer irreversible restorations
E	Localised additive composite using the Dahl principle

See answer and justification for Q769 →

Q770 INTERMEDIATE · Block J

A strategic upper canine has a combined endodontic–periodontal lesion and is the key abutment of a planned bridge.
At which level is this managed?

A	Level 1
B	Level 3
C	Level 2
D	Hygienist
E	Not treatable

See answer and justification for Q770 →

Q771 INTERMEDIATE · Block J

A patient has stable posterior stops and symptomatic localised anterior erosion with no restorative space; they want a repairable conservative method. What is most appropriate?

A	Localised additive composite using the Dahl principle
B	Control erosive drivers and defer irreversible restorations
C	Preventive non-restorative approach with monitoring
D	Staged provisional increase in OVD for complex generalised rehabilitation
E	Modify parafunctional loading and restore selectively after risk assessment

See answer and justification for Q771 →

Q772 INTERMEDIATE · Block J

A patient asks about an implant to replace 21.
Which disadvantages of implants should be discussed?

A	None
B	Only cost
C	Implants damage adjacent teeth
D	Surgery, cost, time, bone requirements, peri-implant disease and limited NHS access
E	Implants never fail

See answer and justification for Q772 →

Q773 INTERMEDIATE · Block J

A patient needs routine conforming fixed and removable restorations with healthy tissues.
At what NHS complexity level is this?

A	Level 3
B	Level 1
C	Level 2
D	Not provided
E	Hygienist only

See answer and justification for Q773 →

Q774 INTERMEDIATE · Block J

A patient needs extensive reconstruction after longstanding tooth wear, missing posterior support and uncertain tolerance of altered bite height. What is the prudent strategy?

A	Staged provisional increase in OVD for complex generalised rehabilitation
B	Control erosive drivers and defer irreversible restorations
C	Preventive non-restorative approach with monitoring
D	Modify parafunctional loading and restore selectively after risk assessment
E	Localised additive composite using the Dahl principle

See answer and justification for Q774 →

Q775 INTERMEDIATE · Block J

A 20-year-old with hypodontia needs multiple replacements.
At which level is this care?

A	Level 1
B	Level 2
C	Level 3, along with cleft, amelogenesis imperfecta and head and neck cancer rehabilitation
D	Hygienist
E	Not provided

See answer and justification for Q775 →

Q776 INTERMEDIATE · Block J

A patient's severe gag reflex has prevented every attempt at a denture.
At which level does care sit?

A	Level 1
B	Level 2
C	Hygienist
D	Level 3
E	Not provided

See answer and justification for Q776 →

Q777 INTERMEDIATE · Block J

A patient has undiagnosed facial pain and wants new crowns to cure it.
At which level is this managed?

A	Level 1
B	Level 3
C	Level 2
D	Not provided
E	Hygienist

See answer and justification for Q777 →

Q778 INTERMEDIATE · Block J

A bruxist repeatedly fractures posterior direct composites despite good moisture control and margins. What should be reassessed before routine replacement?

A	Refer for multidisciplinary assessment of severe generalised wear with complex occlusal changes
B	Use monitoring/prevention when wear is localised, asymptomatic and stable
C	Review bruxism and restorative material/load before replacing repeated fractured restorations
D	Control intrinsic or dietary acid challenge before definitive restoration
E	Consider additive anterior Dahl restoration where space is limited but posterior support stable

See answer and justification for Q778 →

Q780 INTERMEDIATE · Block J

An avulsed permanent tooth cannot be replanted at the scene.
What should be done?

A	Wrap it in tissue
B	Scrub it clean
C	Store it in an appropriate medium and arrange urgent expert care
D	Leave it dry
E	Discard it

See answer and justification for Q780 →

Q781 INTERMEDIATE · Block J

A replanted tooth is under review.
Which long-term complications are monitored?

A	Caries only
B	Pulpal necrosis, inflammatory or replacement resorption, ankylosis, colour change and root fracture healing
C	Gingival recession only
D	Allergy
E	None

See answer and justification for Q781 →

Q783 INTERMEDIATE · Block J

The 13-year-old in the previous question needs a replacement until growth finishes.
Which interim options preserve future choices?

A	Implants
B	Conventional bridges
C	Crowns on adjacent teeth
D	A resin-bonded bridge, a removable partial denture or orthodontic space management
E	No replacement

See answer and justification for Q783 →

Q784 INTERMEDIATE · Block J

A replanted incisor in a 12-year-old has become progressively infra-occluded.
What is the likely cause?

A	Caries
B	Pulpitis
C	Ankylosis (replacement resorption)
D	Bruxism
E	Normal eruption

See answer and justification for Q784 →

Q785 INTERMEDIATE · Block J

A 15-year-old presents two hours after trauma to a permanent maxillary central incisor with a small vital pulp exposure. The tooth has a mature root, is not displaced, and there is no root fracture on appropriately angled radiographs. What is the most appropriate principle of care?

A	Extract the tooth immediately because pulp exposure is present
B	Aim to preserve pulp vitality with appropriate vital-pulp therapy after assessment and provide a sealed restoration
C	Observe without covering the exposed pulp until symptoms occur
D	Perform elective root canal treatment in every mature permanent tooth with a traumatic exposure
E	Use a flexible partial denture as first-line management of the fractured crown

See answer and justification for Q785 →

Q786 INTERMEDIATE · Block J

A 10-year-old has complex trauma to several incisors.
How should care be planned?

A	With multidisciplinary advice
B	Immediate implants
C	Crowns on all teeth
D	Extraction of all affected teeth
E	No follow-up

See answer and justification for Q786 →

Q787 INTERMEDIATE · Block J

A 24-year-old with bulimia has palatal erosion and wants veneers.
What is the most appropriate approach?

A	Immediate veneers
B	Full-mouth crowns
C	Liaise with the GMP, prevent and monitor, and delay definitive restoration until the condition is stable
D	Extraction
E	Ignore the cause

See answer and justification for Q787 →

Q788 INTERMEDIATE · Block J

A bruxist has flat, shiny facets and recently fractured cusps.
Which preventive measure is appropriate?

A	An acid-free diet only
B	A night-time hard splint
C	Whitening
D	Orthodontics
E	Extraction of canines

See answer and justification for Q788 →

Q790 INTERMEDIATE · Block J

A patient has minor cupped wear facets documented as unchanged over three years, no sensitivity and no dissatisfaction. What is the best management principle?

A	Control intrinsic or dietary acid challenge before definitive restoration
B	Use monitoring/prevention when wear is localised, asymptomatic and stable
C	Consider additive anterior Dahl restoration where space is limited but posterior support stable
D	Refer for multidisciplinary assessment of severe generalised wear with complex occlusal changes
E	Review bruxism and restorative material/load before replacing repeated fractured restorations

See answer and justification for Q790 →

Q791 INTERMEDIATE · Block J

A 17-year-old has generalised thin, pitted enamel affecting most teeth in both dentitions. Her father describes a similar pattern; fluoride exposure is unremarkable. Radiographs show reduced enamel thickness but relatively preserved enamel–dentine contrast. Which diagnosis is best supported?

A	Dental fluorosis from childhood fluoride excess
B	Molar-incisor hypomineralisation affecting only first molars and incisors
C	Hypoplastic amelogenesis imperfecta with a familial presentation
D	Dentinogenesis imperfecta with pulpal obliteration
E	Acquired erosion causing generalised enamel hypoplasia from eruption

See answer and justification for Q791 →

Q793 INTERMEDIATE · Block J

Conventional bridge preparation is planned on vital abutments.
What should be done before preparation?

A	Nothing
B	Elective root canal treatment of both
C	Assess pulp status pre-operatively and record it
D	Extract one abutment
E	Prescribe antibiotics

See answer and justification for Q793 →

Q794 INTERMEDIATE · Block J

You plan the transition to complete dentures in a 70-year-old with a failing dentition.
Which teeth are most worth keeping as overdenture abutments?

A	Mobile incisors
B	Canines, with their long roots and strategic position
C	Third molars
D	Teeth with Grade III furcation
E	Any carious tooth

See answer and justification for Q794 →

Q795 INTERMEDIATE · Block J

A patient with several failing teeth asks for them all to be removed at once.
What is generally preferred?

A	Immediate total clearance always
B	No prosthesis
C	Dental implants on the same day
D	A planned, staged transition, avoiding sudden total clearance where possible
E	Leaving all teeth untreated

See answer and justification for Q795 →

Q796 INTERMEDIATE · Block J

An upper lateral incisor has had root canal treatment through a small access with little other tissue loss.
What restoration is appropriate?

A	A post crown
B	A full ceramic crown
C	A conservative composite restoration
D	A veneer with post
E	Extraction

See answer and justification for Q796 →

Q797 INTERMEDIATE · Block J

A lingual plate is proposed to splint mobile lower incisors.
When is this appropriate?

A	Always
B	Never
C	Only with an acrylic base
D	When hygiene is poor
E	Only if the patient's oral hygiene allows

See answer and justification for Q797 →

Q799 INTERMEDIATE · Block J

A 34-year-old has severe palatal erosion, admits nocturnal vomiting and asks for immediate porcelain veneers. What is the first step?

A	Use monitoring/prevention when wear is localised, asymptomatic and stable
B	Review bruxism and restorative material/load before replacing repeated fractured restorations
C	Refer for multidisciplinary assessment of severe generalised wear with complex occlusal changes
D	Control intrinsic or dietary acid challenge before definitive restoration
E	Consider additive anterior Dahl restoration where space is limited but posterior support stable

See answer and justification for Q799 →

BLOCK K · Implants and digital prosthodontics (47)

Q801 INTERMEDIATE · Block K

An implant shows increasing crestal bone loss over sequential radiographs with bleeding and suppuration. What management threshold has been reached?

A	Evaluate abutment screw loosening, fit and occlusal forces before retightening
B	Investigate peri-implant inflammation and hygiene access before simply polishing the crown
C	Provide conservative shared planning where implant maintenance capacity is poor
D	Escalate progressive radiographic bone loss with bleeding/suppuration for specialist evaluation
E	Assess prosthesis design and keratinised-tissue comfort without assuming implant failure

See answer and justification for Q801 →

Q802 INTERMEDIATE · Block K

A patient with a mandibular two-implant overdenture has bleeding on gentle peri-implant probing and erythema. Compared with the reliable prosthesis-delivery baseline radiographs there is no additional bone loss, no suppuration and the implants are stable. Which interpretation and initial management are best?

A	Diagnose early peri-implantitis from bleeding alone even when sequential bone levels are unchanged
B	Diagnose an abutment microgap as the sole cause and avoid assessment of plaque-retentive design
C	Treat peri-implant redness as a physiological response to metal attachments without monitoring
D	Manage peri-implant mucositis with plaque control, accessible debridement and tissue reassessment
E	Plan surgical decontamination as the only durable treatment for an inflamed site without bone loss

See answer and justification for Q802 →

Q803 INTERMEDIATE · Block K

An implant is stable and clean but a patient reports pain when brushing around a narrow band of keratinised tissue. What should be assessed?

A	Assess prosthesis design and keratinised-tissue comfort without assuming implant failure
B	Evaluate abutment screw loosening, fit and occlusal forces before retightening
C	Investigate peri-implant inflammation and hygiene access before simply polishing the crown
D	Escalate progressive radiographic bone loss with bleeding/suppuration for specialist evaluation
E	Provide conservative shared planning where implant maintenance capacity is poor

See answer and justification for Q803 →

Q804 INTERMEDIATE · Block K

A smoking patient with poorly controlled diabetes returns with bleeding, suppuration and increasing probing depth around an implant supporting a lower overdenture. Sequential radiographs show progressive crestal bone loss since loading. At Level 2, which step is the best?

A	First replace worn attachments and defer diagnosis until the patient can retain the overdenture well
B	Recognise probable peri-implantitis, control accessible biofilm/risk factors and refer progressive bone loss appropriately
C	Treat the suppuration as mechanical overload and carry out occlusal adjustment as the sole therapy
D	Use systemic antimicrobials as definitive therapy without a mechanical or prosthetic maintenance plan
E	Reinforce personal cleaning only and reassess after a year despite serial radiographic progression

See answer and justification for Q804 →

Q805 INTERMEDIATE · Block K

A medically suitable edentulous patient requests a mandibular implant-retained overdenture opposed by a stable maxillary complete denture. Planning shows limited prosthetic vertical space and substantial mandibular ridge resorption. The clinician is choosing between solitary attachments and a splinted bar. Which approach is best?

A	Use a bar without measuring space because the metal can always be made thinner
B	Recommend fixed full-arch prosthesis because a two-implant overdenture cannot function
C	Select an attachment by colour and stated retention alone
D	Make the lower overdenture as thin as possible without assessing fracture resistance
E	Measure the restorative envelope for components, hygiene access, acrylic thickness and attachment maintenance before selecting the design

See answer and justification for Q805 →

Q806 INTERMEDIATE · Block K

A maxillary complete denture is opposed by a well-retained mandibular two-implant overdenture. Six months after insertion the patient reports repeated maxillary denture dislodgement during strong chewing, with wear facets and anterior loading. Which intervention is most appropriate before replacing the upper denture?

A	Review the entire occlusion, attachment retention, tooth positions, ridge support and denture stability under function, then modify the prosthetic scheme as indicated
B	Increase mandibular attachment retention to the maximum possible without inspecting the upper denture
C	Replace the upper denture immediately because two lower implants inevitably destabilise it
D	Reduce the upper posterior occlusal contacts until only incisors contact
E	Shorten the maxillary posterior border to permit a greater range of movement

See answer and justification for Q806 →

Q807 INTERMEDIATE · Block K

A patient with a bar-retained mandibular overdenture has plaque retention beneath the bar, bleeding and progressive crestal bone loss at one implant. A colleague suggests that a single cultured periodontal pathogen would be sufficient to diagnose peri-implantitis. Which interpretation is more appropriate?

A	Peri-implantitis is caused exclusively by one named species, which proves the diagnosis when cultured
B	A negative culture rules out inflammatory peri-implant disease despite serial bone loss
C	Clinical inflammation and progressive radiographic bone loss are central to diagnosis; the microbial biofilm is polymicrobial and cultures alone are not sufficient
D	Peri-implant disease is sterile and caused only by occlusal load
E	Attachment wear must be the underlying cause because overdentures cannot retain bacterial biofilm

See answer and justification for Q807 →

Q808 INTERMEDIATE · Block K

A 76-year-old with limited hand dexterity and an otherwise maintainable mandibular two-implant overdenture needs replacement attachments. A bar could provide splinting but would create a narrow space beneath the connector that she cannot clean reliably. Which factor should dominate the decision?

A	Select the attachment with highest manufacturer-stated retention regardless of hygiene
B	Use a bar to remove the need for implant maintenance
C	Choose any attachment based only on the initial component cost
D	Select the deepest possible attachment to increase vertical load transmission
E	Balance retentive needs with available vertical space, patient dexterity, cleaning access and realistic maintenance

See answer and justification for Q808 →

Q809 INTERMEDIATE · Block K

A temporary crown is placed on an implant 3 days after surgery.
Which ITI loading protocol is this?

A	Conventional
B	Early
C	Immediate (within 1 week)
D	Delayed
E	Submerged

See answer and justification for Q809 →

Q810 INTERMEDIATE · Block K

An implant is restored 6 weeks after placement.
Which ITI loading protocol is this?

A	Immediate
B	Early (1 week to 2 months)
C	Conventional
D	Delayed
E	Late

See answer and justification for Q810 →

Q811 INTERMEDIATE · Block K

A patient's older implant crown keeps loosening.
Which implant–abutment connection is associated with more screw loosening?

A	Conical (Morse taper)
B	External hex
C	Internal octagon
D	Internal conical with platform switching
E	One-piece zirconia

See answer and justification for Q811 →

Q812 INTERMEDIATE · Block K

A colleague prefers a particular implant connection for its seal and stability.
Which connection offers these?

A	External hex
B	No connection
C	Butt joint
D	Conical (Morse taper)
E	Slip fit

See answer and justification for Q812 →

Q813 INTERMEDIATE · Block K

A restored implant has increasing bleeding on probing with plaque around a bulky emergence profile but no mobility. What is the first clinical review priority?

A	Provide conservative shared planning where implant maintenance capacity is poor
B	Assess prosthesis design and keratinised-tissue comfort without assuming implant failure
C	Evaluate abutment screw loosening, fit and occlusal forces before retightening
D	Escalate progressive radiographic bone loss with bleeding/suppuration for specialist evaluation
E	Investigate peri-implant inflammation and hygiene access before simply polishing the crown

See answer and justification for Q813 →

Q814 INTERMEDIATE · Block K

A patient asks whether an implant is suitable for her.
Which history most increases the risk of peri-implantitis?

A	A history of periodontitis, especially with smoking
B	Good oral hygiene
C	Controlled hypertension
D	A penicillin allergy
E	Previous orthodontics

See answer and justification for Q814 →

Q815 INTERMEDIATE · Block K

An edentulous mandible is planned for a two-implant overdenture. The patient has a limited interarch restorative envelope, uneven implant angulation and difficulty cleaning complex bars. A bar-and-clip design is proposed simply because it is presumed most retentive. What should guide attachment selection?

A	Use a bar in all cases regardless of vertical space
B	Use the same unsplinted attachments even when implant divergence exceeds the system limits
C	Choose a bar solely from ridge height without considering opposing occlusion
D	Prescribe fixed implant-supported prosthesis because two implants always support it
E	Evaluate available restorative height, implant position/divergence, retention requirement, opposing arch and hygiene to select the most maintainable system

See answer and justification for Q815 →

Q816 INTERMEDIATE · Block K

Unsplinted ball or locator attachments are chosen for a two-implant mandibular overdenture.
What are their characteristics?

A	Easier hygiene and maintenance, less restorative space needed
B	They need the most vertical space
C	They splint the implants rigidly
D	They never wear
E	They are hardest to clean

See answer and justification for Q816 →

Q817 INTERMEDIATE · Block K

Implants for an overdenture are markedly divergent.
Which attachment system suits them?

A	A bar
B	Single unsplinted balls
C	Magnets only
D	No attachment
E	Denture adhesive

See answer and justification for Q817 →

Q819 INTERMEDIATE · Block K

A single implant crown on 36 is planned and the patient wants it to be retrievable.
Which retention type is preferred?

A	Cement-retained with resin cement
B	Cement-retained with glass ionomer
C	Friction-retained
D	Screw-retained: retrievable, with no subgingival cement
E	Adhesive bridge

See answer and justification for Q819 →

Q820 INTERMEDIATE · Block K

A cement-retained implant crown has bleeding and suppuration around it a year after fitting.
What is a common cause?

A	Occlusal overload only
B	Shade mismatch
C	Residual subgingival cement
D	Platform switching
E	A bar attachment

See answer and justification for Q820 →

Q821 INTERMEDIATE · Block K

A full-arch fixed implant bridge is planned.
Which statement is correct?

A	It is Level 1
B	Long distal cantilevers are preferred
C	Passive fit is unimportant
D	It is Level 3
E	It needs no maintenance

See answer and justification for Q821 →

Q822 INTERMEDIATE · Block K

A patient with a two-implant mandibular overdenture complains it is loose after 18 months.
What is the commonest cause?

A	Wear of the attachment inserts, needing replacement
B	Implant fracture
C	Peri-implantitis
D	Mandibular fracture
E	Allergy

See answer and justification for Q822 →

Q824 INTERMEDIATE · Block K

You tighten the abutment screw of an implant crown.
How should the screw be torqued?

A	By hand until tight
B	With pliers
C	With a calibrated torque driver to the manufacturer's value
D	As tight as possible
E	It does not matter

See answer and justification for Q824 →

Q825 INTERMEDIATE · Block K

A colleague plans a bridge joining a natural premolar to an implant.
Why should this be avoided?

A	Implants are too weak
B	Implants have no periodontal ligament, so differential mobility causes problems
C	Teeth cannot be crowned
D	It is illegal
E	It is always successful

See answer and justification for Q825 →

Q826 INTERMEDIATE · Block K

You review a patient with an implant crown.
What defines peri-implant health?

A	Bleeding on probing
B	Progressive bone loss
C	No erythema, bleeding or suppuration, and no bone loss beyond initial remodelling
D	A 6 mm pocket
E	Suppuration

See answer and justification for Q826 →

Q827 INTERMEDIATE · Block K

A patient with a two-implant mandibular overdenture has bleeding on gentle probing around one implant. The probing depth is 4 mm and unchanged from baseline, and serial comparable radiographs show no progressive bone loss. Plaque accumulates around the attachment. What is the most appropriate Level 2 interpretation and response?

A	Diagnose peri-implantitis and prescribe systemic antibiotics routinely
B	Interpret peri-implant health because bleeding is common around attachments
C	Assume implant failure and remove the attachment immediately
D	Diagnose peri-implant mucositis, address plaque control and prosthetic cleansability, and reassess
E	Arrange exploratory surgery without non-surgical care or reference to baseline data

See answer and justification for Q827 →

Q828 INTERMEDIATE · Block K

No baseline radiograph exists for an implant with bleeding.
What findings indicate peri-implantitis?

A	Any bleeding
B	A 4 mm pocket alone
C	Mucosal redness alone
D	Bone 3 mm or more apical to the implant platform, with PPD 6 mm or more and bleeding
E	No bone loss

See answer and justification for Q828 →

Q829 INTERMEDIATE · Block K

You fit a new implant-retained crown.
What should be recorded at fitting?

A	A baseline radiograph and probing depths
B	The shade only
C	Nothing
D	A CBCT
E	A photograph only

See answer and justification for Q829 →

Q830 INTERMEDIATE · Block K

A patient with an implant bar overdenture asks how to clean it.
Which maintenance is appropriate?

A	No special cleaning
B	Abrasive toothpaste on the abutments
C	Metal scalers on the implants
D	Chlorhexidine only
E	Interdental brushes and floss under the bar, professional debridement with instruments that will not scratch titanium, and annual review

See answer and justification for Q830 →

Q831 INTERMEDIATE · Block K

A hygienist is about to debride around a titanium implant abutment.
Which instruments should be used?

A	Instruments that will not scratch titanium (such as plastic, titanium or carbon-fibre tips)
B	Stainless-steel scalers
C	Diamond burs
D	Coarse prophylaxis paste
E	None; implants need no cleaning

See answer and justification for Q831 →

Q833 INTERMEDIATE · Block K

An implant is still in place after 10 years but has 5 mm of bone loss and recurrent suppuration.
How should its outcome be described?

A	A success
B	It has survived, but is not a success
C	A failure because it is present
D	Fully healthy
E	Not assessable

See answer and justification for Q833 →

Q834 INTERMEDIATE · Block K

A patient asks for an implant to replace 25.
What should the assessment start with?

A	Implant brand
B	Bone graft options
C	The CBCT
D	The fee
E	The prosthesis: the missing-tooth problem, desired restoration, smile line, restorative space, occlusion, cleansability and the patient's priorities

See answer and justification for Q834 →

Q836 INTERMEDIATE · Block K

A colleague plans to scan a full arch digitally for a cross-arch bridge.
Which statement about intraoral scanners is correct?

A	They are most accurate over full arches
B	They are excellent for short spans
C	They are unaffected by saliva
D	They record functional borders
E	They record mucodisplaced edentulous tissue well

See answer and justification for Q836 →

Q839 INTERMEDIATE · Block K

A clinician wants a digital model that shows gingival colour for shade communication.
Which file formats can carry colour and texture?

A	STL
B	DICOM
C	PDF
D	PLY or OBJ
E	JPG only

See answer and justification for Q839 →

Q840 INTERMEDIATE · Block K

A surgical guide is planned for an implant at 36.
Which data are merged?

A	CBCT (DICOM) with surface scans (STL)
B	Two STL files only
C	Photographs only
D	A panoramic radiograph and an alginate cast
E	A facebow record only

See answer and justification for Q840 →

Q841 INTERMEDIATE · Block K

A manufacturer quotes the trueness and precision of its scanner.
What do these terms mean?

A	Trueness is speed; precision is colour
B	Both mean resolution
C	Trueness is repeatability; precision is closeness to the reference
D	Both mean file size
E	Trueness is closeness to the reference; precision is repeatability

See answer and justification for Q841 →

Q842 INTERMEDIATE · Block K

A patient has limited ability to perform implant hygiene and unreliable recall attendance but seeks a complex full-arch fixed prosthesis. What planning principle is most important?

A	Provide conservative shared planning where implant maintenance capacity is poor
B	Assess prosthesis design and keratinised-tissue comfort without assuming implant failure
C	Escalate progressive radiographic bone loss with bleeding/suppuration for specialist evaluation
D	Evaluate abutment screw loosening, fit and occlusal forces before retightening
E	Investigate peri-implant inflammation and hygiene access before simply polishing the crown

See answer and justification for Q842 →

Q843 INTERMEDIATE · Block K

A laboratory 3D-prints a model and a Co-Cr framework.
Which processes are additive?

A	Milling
B	Casting
C	Grinding
D	Stereolithography (SLA), digital light processing (DLP) and selective laser melting (for metals)
E	Turning

See answer and justification for Q843 →

Q846 INTERMEDIATE · Block K

A patient's smile has been designed digitally.
What is still useful before treatment?

A	Nothing
B	Immediate preparation
C	Only the digital image
D	A physical mock-up from a wax-up for the patient's approval
E	A CBCT

See answer and justification for Q846 →

Q847 INTERMEDIATE · Block K

A clinician wants to record a patient's dynamic occlusion digitally.
Which technology does this?

A	An STL scan
B	A panoramic radiograph
C	Jaw motion tracking
D	A shade photograph
E	An alginate impression

See answer and justification for Q847 →

Q848 INTERMEDIATE · Block K

A practice designs crowns digitally.
How should the records be managed?

A	Save source files securely and document revisions
B	Delete them after milling
C	Keep only the final file
D	Share them publicly
E	No records are needed

See answer and justification for Q848 →

Q849 INTERMEDIATE · Block K

A printed provisional crown is weaker than expected.
What affects the properties of printed restorations?

A	Nothing
B	Only the colour
C	Only the printer brand
D	The patient's age
E	Post-curing and using a material indicated for the purpose

See answer and justification for Q849 →

Q851 INTERMEDIATE · Block K

An implant crown on 46 has loosened three times.
What should be checked?

A	Shade
B	Nothing; just retighten
C	Screw torque, occlusal loading (including bruxism) and the connection
D	The opposing tooth's caries
E	The patient's diet only

See answer and justification for Q851 →

Q852 INTERMEDIATE · Block K

A patient has a two-implant mandibular overdenture opposing an upper complete denture.
What complication of the upper denture is common?

A	Fracture of the upper denture
B	Implant failure in the maxilla
C	Upper denture loosening is impossible
D	Lower implant fracture
E	None

See answer and justification for Q852 →

Q855 INTERMEDIATE · Block K

A screw-retained posterior implant crown loosens twice over six months but the tissues remain healthy. What must be assessed before another torque application?

A	Investigate peri-implant inflammation and hygiene access before simply polishing the crown
B	Evaluate abutment screw loosening, fit and occlusal forces before retightening
C	Provide conservative shared planning where implant maintenance capacity is poor
D	Assess prosthesis design and keratinised-tissue comfort without assuming implant failure
E	Escalate progressive radiographic bone loss with bleeding/suppuration for specialist evaluation

See answer and justification for Q855 →

Q857 INTERMEDIATE · Block K

A Level 2 dentist manages peri-implant mucositis on a bar overdenture.
What does management include?

A	Implant removal
B	Bone grafting
C	Surgical access
D	Mechanical debridement, OHI and a check of the prosthesis design for cleansability
E	Systemic antibiotics only

See answer and justification for Q857 →

Q859 INTERMEDIATE · Block K

A patient's locator-retained overdenture has lost its retention.
What is the usual first step?

A	Replace the worn matrix (insert)
B	Remove the implants
C	Remake the denture
D	Add denture adhesive permanently
E	Reline with a soft liner

See answer and justification for Q859 →

Q860 INTERMEDIATE · Block K

A clinician plans aesthetic rehabilitation using a facial scan and a virtual articulator.
What do these add?

A	They replace consent
B	They guarantee results
C	They integrate facial references into the digital plan
D	They replace clinical examination
E	They remove the need for records

See answer and justification for Q860 →

BLOCK L · Complications, failures and maintenance (23)

Q861 INTERMEDIATE · Block L

An implant-supported crown looks intact but is difficult to clean and bleeds on probing during maintenance.
What is the most important review?

A	Check restoration thickness and load for ceramic chipping
B	Investigate cementation contamination and surface treatment after early adhesive debond
C	Assess peri-implant tissue health and prosthesis cleansability at review
D	Identify repetitive flexure and support deficiency before denture-base repair
E	Investigate occlusion and jaw relation when prosthesis tips during contact

See answer and justification for Q861 →

Q862 INTERMEDIATE · Block L

An 80-year-old returns two days after fitting a mandibular complete denture. A painful ulcer lies over a sharp mylohyoid ridge, while articulating marks reveal one dominant balancing-side interference. Which sequence is most appropriate?

A	Relieve all of the lingual flange without investigating whether the ulcer is associated with the bony prominence
B	Identify the true fitting-surface pressure and the destabilising occlusal contact, correct each conservatively and review healing
C	Remake both dentures immediately without assessing selective correction
D	Prescribe systemic antifungals because ulcers beneath new dentures are necessarily candidal
E	Increase OVD to distribute more force to the painful ridge

See answer and justification for Q862 →

Q863 INTERMEDIATE · Block L

A resin-bonded bridge debonds six days after placement, and operative records document saliva contamination during bonding. What should be investigated?

A	Investigate occlusion and jaw relation when prosthesis tips during contact
B	Assess peri-implant tissue health and prosthesis cleansability at review
C	Identify repetitive flexure and support deficiency before denture-base repair
D	Check restoration thickness and load for ceramic chipping
E	Investigate cementation contamination and surface treatment after early adhesive debond

See answer and justification for Q863 →

Q864 INTERMEDIATE · Block L

A repair holds at rest but a sharp crack opens when a patient closes into a heavy anterior contact. What is the next diagnostic focus?

A	Discuss design reinforcement or alternative base material if risk persists
B	Review fit and tissue support over the resorbed ridge
C	Check occlusal contacts and opposing dentition for overload
D	Evaluate palatal vault/section thickness and flexural fatigue
E	Inspect porosity and repair-interface quality

See answer and justification for Q864 →

Q865 INTERMEDIATE · Block L

A cast Co–Cr clasp fractures near its origin after several chairside tightening adjustments. The abutment has now drifted and the RPD no longer follows its original surveyed path. What is the best management strategy?

A	Solder the broken clasp and preserve the original path without assessing the drifted abutment
B	Replace the clasp with a deeper engaging tip because this will compensate for drift
C	Reassess the dentition, surveyed path and support, then decide whether repair, redesign or replacement can achieve a reliable long-term fit
D	Remove the reciprocal arm to decrease insertion forces on the new clasp
E	Add a soft liner to the saddle as the sole treatment for clasp fatigue

See answer and justification for Q865 →

Q866 INTERMEDIATE · Block L

An upper denture repeatedly cracks across a narrow palatal midline while the lower arch has sound natural teeth; the vault is high. What design/biomechanical factor should be examined first?

A	Check occlusal contacts and opposing dentition for overload
B	Discuss design reinforcement or alternative base material if risk persists
C	Evaluate palatal vault/section thickness and flexural fatigue
D	Inspect porosity and repair-interface quality
E	Review fit and tissue support over the resorbed ridge

See answer and justification for Q866 →

Q868 INTERMEDIATE · Block L

Bleeding and swelling develop around one newly fitted crown.
What should be suspected first?

A	Allergy
B	Caries
C	Root fracture
D	Residual excess cement, until proven otherwise
E	Normal healing

See answer and justification for Q868 →

Q869 INTERMEDIATE · Block L

A patient complains of food packing between a new crown on 36 and 37.
What is the likely cause and remedy?

A	An open or weak contact
B	Caries; restore 37
C	Gingivitis; chlorhexidine
D	Occlusal overload; splint
E	Normal; reassure

See answer and justification for Q869 →

Q870 INTERMEDIATE · Block L

A posterior glass-ceramic onlay chips twice at a thin unsupported cusp under heavy contact. What design variable needs reassessment?

A	Investigate occlusion and jaw relation when prosthesis tips during contact
B	Assess peri-implant tissue health and prosthesis cleansability at review
C	Investigate cementation contamination and surface treatment after early adhesive debond
D	Check restoration thickness and load for ceramic chipping
E	Identify repetitive flexure and support deficiency before denture-base repair

See answer and justification for Q870 →

Q871 INTERMEDIATE · Block L

You audit failures of conventional bridges in your practice.
What are the leading causes of failure in the literature?

A	Caries and loss of retention
B	Pontic fracture and allergy
C	Shade mismatch
D	Gingival recession only
E	Opposing tooth wear

See answer and justification for Q871 →

Q872 INTERMEDIATE · Block L

You consent a patient for crowns on vital teeth.
Which common biological complication should be mentioned?

A	Allergy to gold
B	Root resorption
C	Tooth discolouration only
D	The later need for endodontic treatment
E	TMD

See answer and justification for Q872 →

Q873 INTERMEDIATE · Block L

A Co-Cr framework will not seat at try-in. Disclosing medium shows widespread contacts, with no single interference.
What is the most likely cause and action?

A	Normal; force it
B	Grind every contact
C	A distorted impression or cast; re-impression and remake
D	Too much retention; open the clasps
E	Reline it

See answer and justification for Q873 →

Q874 INTERMEDIATE · Block L

A once well-fitting upper denture has lost tissue support after years of ridge resorption and now develops a midline fracture without being dropped. What should be checked?

A	Inspect porosity and repair-interface quality
B	Check occlusal contacts and opposing dentition for overload
C	Review fit and tissue support over the resorbed ridge
D	Evaluate palatal vault/section thickness and flexural fatigue
E	Discuss design reinforcement or alternative base material if risk persists

See answer and justification for Q874 →

Q875 INTERMEDIATE · Block L

A Co-Cr clasp fractures at the shoulder after several adjustments.
What is the cause?

A	Corrosion
B	Fatigue, aggravated by repeated adjustment of low-ductility Co-Cr or casting porosity
C	Too little undercut
D	Acid erosion
E	A loose rest

See answer and justification for Q875 →

Q876 INTERMEDIATE · Block L

An RPD abutment has become painful and mobile.
Which causes should be considered?

A	Shade
B	Saliva
C	Denture adhesive
D	Occlusal overload, an unseated rest, free-end saddle rotation or a clasp active at rest
E	Night removal

See answer and justification for Q876 →

Q878 INTERMEDIATE · Block L

A patient has had repeated fracture despite corrected fit and occlusal loading; a new prosthesis is being designed. What additional measure is reasonable?

A	Check occlusal contacts and opposing dentition for overload
B	Discuss design reinforcement or alternative base material if risk persists
C	Inspect porosity and repair-interface quality
D	Review fit and tissue support over the resorbed ridge
E	Evaluate palatal vault/section thickness and flexural fatigue

See answer and justification for Q878 →

Q880 INTERMEDIATE · Block L

A patient's dentures are loose, but the appearance, OVD and occlusion are good.
What is the appropriate treatment?

A	Remake
B	Reline or rebase
C	Adhesive only
D	Implants
E	No treatment

See answer and justification for Q880 →

Q881 INTERMEDIATE · Block L

A denture breaks at the margin of a previous large cold-cure patch and pores are evident at the repair line.
What should be evaluated?

A	Check occlusal contacts and opposing dentition for overload
B	Inspect porosity and repair-interface quality
C	Discuss design reinforcement or alternative base material if risk persists
D	Review fit and tissue support over the resorbed ridge
E	Evaluate palatal vault/section thickness and flexural fatigue

See answer and justification for Q881 →

Q883 INTERMEDIATE · Block L

Newly processed complete dentures are too tall at fitting, though the wax try-in was correct.
Which laboratory error is likely?

A	Over-polished fitting surface
B	Abraded dies
C	A wrong shade
D	Raised OVD from flash during processing
E	Distorted impression

See answer and justification for Q883 →

Q885 INTERMEDIATE · Block L

A new complete denture tips at closure although border extension is within functional limits and tissue fit is acceptable. What should be checked?

A	Investigate occlusion and jaw relation when prosthesis tips during contact
B	Assess peri-implant tissue health and prosthesis cleansability at review
C	Check restoration thickness and load for ceramic chipping
D	Identify repetitive flexure and support deficiency before denture-base repair
E	Investigate cementation contamination and surface treatment after early adhesive debond

See answer and justification for Q885 →

Q887 INTERMEDIATE · Block L

An upper denture has been repaired three times at the same midline location; cracks appear without dropping. What is the first principle?

A	Investigate cementation contamination and surface treatment after early adhesive debond
B	Check restoration thickness and load for ceramic chipping
C	Investigate occlusion and jaw relation when prosthesis tips during contact
D	Assess peri-implant tissue health and prosthesis cleansability at review
E	Identify repetitive flexure and support deficiency before denture-base repair

See answer and justification for Q887 →

Q889 INTERMEDIATE · Block L

A practice plans maintenance appointments for its patients.
Which prosthesis needs frequent maintenance?

A	A single gold crown
B	A composite filling
C	A metal–ceramic bridge
D	A veneer
E	Implant overdenture attachments

See answer and justification for Q889 →

Q890 INTERMEDIATE · Block L

A well-fitting acrylic maxillary denture has a fine crack starting at the anterior labial frenal notch, and the patient reports repeated high bite forces. It has not yet separated and occlusal contacts are uneven. Which approach is most likely to prevent propagation?

A	Assess stress concentration, occlusion, extension and base thickness, then plan an appropriate laboratory repair or replacement with reinforcement where indicated
B	Polish the crackline and assume the acrylic has regained its original strength
C	Apply more pressure to the high contact to keep both halves in position
D	Ignore the crack until complete separation occurs
E	Use a stronger denture cleanser because it changes acrylic fracture toughness

See answer and justification for Q890 →

BLOCK M · Governance, law and communication (45)

Q891 INTERMEDIATE · Block M

A patient signed a consent form for a bridge, but the risks were never discussed and she felt pressured. What makes consent valid?

A	A signature alone
B	It is informed, voluntary, given by a person with capacity, specific to the treatment, documented, and can be withdrawn
C	Verbal agreement only
D	The dentist's judgement
E	Payment

See answer and justification for Q891 →

Q892 INTERMEDIATE · Block M

You are consenting a professional violinist for a lower molar extraction near the lingual nerve. Which case set the standard for disclosure of risks?

A	Bolam (1957)
B	Bolitho (1997)
C	Gillick (1985)
D	Caparo (1990)
E	Montgomery v Lanarkshire (2015): material risks to a reasonable patient, or to this patient, and reasonable alternatives

See answer and justification for Q892 →

Q893 INTERMEDIATE · Block M

A clinician's crown preparation technique is challenged in a negligence claim.
Which test applies to the standard of treatment?

A	Bolam, qualified by Bolitho (opinion must withstand logical analysis)
B	Montgomery
C	Gillick
D	The Mental Capacity Act
E	The Equality Act

See answer and justification for Q893 →

Q894 INTERMEDIATE · Block M

A patient alleges negligence after a crown.
What must be proved?

A	Only that harm occurred
B	Only breach of duty
C	Duty of care, breach of that duty, causation, and resulting harm
D	Only lack of consent
E	Only that the patient is unhappy

See answer and justification for Q894 →

Q895 INTERMEDIATE · Block M

An associate's error injures a patient. The practice owner is also sued.
What principle allows this?

A	Duty of candour
B	Montgomery
C	Caldicott
D	Vicarious liability: an employer is liable for employees' acts in the course of employment
E	Gillick

See answer and justification for Q895 →

Q896 INTERMEDIATE · Block M

A patient with capacity declines a recommended bridge and chooses no replacement.
Which Mental Capacity Act principle applies?

A	Assume incapacity in older patients
B	Override the decision if unwise
C	Decide in the clinician's best interests
D	An unwise decision does not mean the person lacks capacity
E	Refer to court

See answer and justification for Q896 →

Q897 INTERMEDIATE · Block M

A patient has limited English and a relative is answering most questions about irreversible reconstruction.
What is a prudent step?

A	Use a qualified interpreter for important complex consent discussions
B	Audit an identified pattern of clinical failure and implement an improvement cycle
C	Obtain renewed informed consent when the agreed plan materially changes
D	Disclose and manage significant errors transparently with appropriate documentation
E	Discuss reasonable alternatives, benefits, risks and maintenance in language the patient understands

See answer and justification for Q897 →

Q898 INTERMEDIATE · Block M

A patient with fluctuating dementia consented last week but seems confused today.
Which principle about capacity applies?

A	Capacity once given lasts
B	Capacity is global
C	Consent from last week stands
D	The carer decides
E	Capacity is decision-specific and time-specific

See answer and justification for Q898 →

Q900 INTERMEDIATE · Block M

A 14-year-old requests treatment without her parents knowing.
When may she consent?

A	Never
B	Only with parental consent
C	If she is Gillick competent
D	At age 16 only
E	If she pays

See answer and justification for Q900 →

Q901 INTERMEDIATE · Block M

A Scottish dentist treats an adult lacking capacity.
Which legislation applies?

A	Mental Capacity Act 2005
B	Children Act 1989
C	Equality Act 2010
D	Data Protection Act 2018
E	Adults with Incapacity (Scotland) Act 2000

See answer and justification for Q901 →

Q902 INTERMEDIATE · Block M

You use a decision aid with a patient choosing between a bridge and an implant.
What does NICE NG197 (shared decision-making) recommend discussing?

A	Only the recommended option
B	Only cost
C	Benefits, risks, alternatives and doing nothing (BRAN)
D	Only risks over 10%
E	Nothing in writing

See answer and justification for Q902 →

Q903 INTERMEDIATE · Block M

A student lists the GDC's nine principles.
Which is NOT one of them?

A	Put patients' interests first
B	Obtain valid consent
C	Raise concerns if patients are at risk
D	Maximise the practice's income
E	Have a clear and effective complaints procedure

See answer and justification for Q903 →

Q906 INTERMEDIATE · Block M

A dentist extracts a tooth and discovers it was the wrong one.
Who has a professional duty of candour?

A	All registrants
B	Only the practice owner
C	Only the CQC
D	No one
E	Only the nurse

See answer and justification for Q906 →

Q908 INTERMEDIATE · Block M

A CQC-registered practice has a notifiable safety incident.
Which duty applies to the provider?

A	None
B	A voluntary apology
C	The statutory duty of candour
D	Only internal discussion
E	Reporting to the police

See answer and justification for Q908 →

Q909 INTERMEDIATE · Block M

A patient develops a serious adverse reaction to a prescribed antibiotic.
How should it be reported?

A	To the GDC
B	To RIDDOR
C	To the CQC only
D	Through the MHRA Yellow Card scheme
E	Not at all

See answer and justification for Q909 →

Q910 INTERMEDIATE · Block M

A batch of defective impression trays causes repeated distortions.
Where are medical device incidents reported?

A	The GDC
B	The MHRA
C	The HSE
D	The police
E	The BDA

See answer and justification for Q910 →

Q911 INTERMEDIATE · Block M

A nurse suffers a serious injury from a workplace accident.
Under which regulations, and to whom, is it reported?

A	IR(ME)R, to the CQC
B	GDPR, to the ICO
C	RIDDOR, to the HSE
D	The Equality Act, to the court
E	Montgomery, to the GDC

See answer and justification for Q911 →

Q912 INTERMEDIATE · Block M

A healthy lower molar is extracted instead of its carious neighbour.
How is this classified?

A	A never event (wrong-site surgery)
B	A minor incident
C	A complaint only
D	Acceptable error
E	A near miss

See answer and justification for Q912 →

Q913 INTERMEDIATE · Block M

An NHS patient complains about her dentures.
What is the acknowledgement timescale and time limit for complaints?

A	Acknowledge within 1 month; no time limit
B	Acknowledge within 3 working days
C	Within 6 months; 3 years
D	No acknowledgement needed
E	Within a week; 6 months

See answer and justification for Q913 →

Q914 INTERMEDIATE · Block M

A complaint is not resolved locally.
Where does it escalate?

A	NHS: the GDC; private: the CQC
B	Always to court
C	To the BDA
D	Both to the police
E	NHS: the Parliamentary and Health Service Ombudsman

See answer and justification for Q914 →

Q915 INTERMEDIATE · Block M

A nurse raises concerns about unsafe practice by her employer.
Which law protects her?

A	The Equality Act 2010
B	The Mental Capacity Act 2005
C	Data Protection Act 2018
D	The Public Interest Disclosure Act 1998
E	The Children Act 1989

See answer and justification for Q915 →

Q916 INTERMEDIATE · Block M

A dentist realises an entry in yesterday's notes is wrong.
How should it be corrected?

A	Delete and rewrite it
B	Add a dated, signed note of the correction, leaving the original visible
C	Use correction fluid
D	Ignore it
E	Ask the nurse to change it

See answer and justification for Q916 →

Q917 INTERMEDIATE · Block M

A practice is clearing old records.
How long must adult and child records be kept (NHS England guidance)?

A	Adults 3 years; children 5 years
B	Adults 7 years; children 18 years
C	Indefinitely
D	Adults at least 11 years after last treatment
E	Until the patient dies

See answer and justification for Q917 →

Q919 INTERMEDIATE · Block M

Police ask for a patient's records without a court order or consent, in a minor matter.
When may confidentiality be breached?

A	Whenever police ask
B	Never
C	With consent, when required by law (such as a court order), or in the public interest to prevent serious harm, sharing the minimum necessary
D	If the patient owes money
E	At the dentist's discretion

See answer and justification for Q919 →

Q920 INTERMEDIATE · Block M

A practice reviews how it shares patient information.
Which principles govern confidential information sharing?

A	Bolam principles
B	Montgomery principles
C	Gillick principles
D	Hanau principles
E	Caldicott principles

See answer and justification for Q920 →

Q923 INTERMEDIATE · Block M

You review who does what under IR(ME)R 2017.
Who are the duty holders?

A	Patient and nurse
B	HSE and CQC
C	GDC and BDA
D	Only the dentist
E	Employer, referrer, practitioner and operator

See answer and justification for Q923 →

Q924 INTERMEDIATE · Block M

An audit shows radiographs taken with round collimation and no film holders.
Which optimisation measures should be introduced?

A	A higher exposure
B	More repeat films
C	Rectangular collimation, film holders, the fastest suitable receptor, appropriate exposure settings, and image-quality audit
D	Panoramic views instead
E	No change

See answer and justification for Q924 →

Q926 INTERMEDIATE · Block M

A practice will sterilise wrapped handpieces and hollow instruments.
Which steriliser is needed?

A	A type N (non-vacuum) steriliser
B	A boiling water bath
C	An ultrasonic bath
D	A hot air oven
E	A type B (vacuum) steriliser

See answer and justification for Q926 →

Q929 INTERMEDIATE · Block M

A patient requests fixed implant rehabilitation but has not been told that a simpler removable option would meet the same functional goals. What should the clinician do?

A	Audit an identified pattern of clinical failure and implement an improvement cycle
B	Obtain renewed informed consent when the agreed plan materially changes
C	Disclose and manage significant errors transparently with appropriate documentation
D	Use a qualified interpreter for important complex consent discussions
E	Discuss reasonable alternatives, benefits, risks and maintenance in language the patient understands

See answer and justification for Q929 →

Q930 INTERMEDIATE · Block M

You review dental unit waterline management.
What flushing is recommended?

A	None
B	Once a week
C	Only at the end of the day
D	About 2 minutes at the start of the day and 20–30 seconds between patients, with a biocide
E	10 minutes between patients

See answer and justification for Q930 →

Q931 INTERMEDIATE · Block M

A nurse sustains a needlestick from a used syringe.
What is the immediate management?

A	Encourage bleeding, wash with soap and water without scrubbing, cover, report, and contact occupational health urgently (HIV PEP ideally within 1 hour)
B	Suck the wound
C	Scrub with alcohol
D	Ignore it if the patient seems healthy
E	Wait until the end of the day

See answer and justification for Q931 →

Q933 INTERMEDIATE · Block M

A 6-year-old has unexplained facial bruising and multiple untreated carious teeth.
What is the dentist's role?

A	Investigate the family
B	Confront the parents
C	Recognise, record and refer
D	Ignore it
E	Treat and say nothing

See answer and justification for Q933 →

Q934 INTERMEDIATE · Block M

A 15-year-old discloses female genital mutilation.
What must a regulated professional in England and Wales do?

A	Nothing
B	Report the known case to the police
C	Tell only her parents
D	Record it only
E	Refer to a dermatologist

See answer and justification for Q934 →

Q936 INTERMEDIATE · Block M

A practice manager lists the pillars of clinical governance.
Which list is correct?

A	Profit, marketing and sales
B	Only audit
C	Clinical effectiveness and audit, risk management, education and training, patient involvement, information management, staff management and research
D	Only complaints
E	Only CPD

See answer and justification for Q936 →

Q938 INTERMEDIATE · Block M

An audit found poor recording of RPD designs, and changes have been made.
What closes the audit loop?

A	Publishing the audit
B	Discarding the data
C	Writing a guideline
D	Re-audit
E	Disciplining staff

See answer and justification for Q938 →

Q939 INTERMEDIATE · Block M

A practice uses small, repeated cycles to test a new recall system.
What is this quality-improvement method?

A	Root cause analysis
B	A randomised trial
C	Audit only
D	Significant event analysis
E	Plan–Do–Study–Act (PDSA) cycles

See answer and justification for Q939 →

Q940 INTERMEDIATE · Block M

You are appraising evidence for a new bonding protocol.
Which evidence is at the top of the hierarchy?

A	A systematic review and meta-analysis of RCTs
B	A single RCT
C	A cohort study
D	A case series
E	Expert opinion

See answer and justification for Q940 →

Q941 INTERMEDIATE · Block M

A patient seeks a complex implant-supported full-arch prosthesis after failing multiple treatments. There is active Grade C periodontitis, uncontrolled diabetes and unclear bone volume. The dentist has experience with simple restorative care but not complex full-arch surgery. What is the strongest Level 2 response?

A	Proceed if the patient signs a broad disclaimer
B	Arrange multiple implants before periodontal stabilisation to secure the available bone
C	Assume existing knowledge is sufficient because the prosthetic phase is familiar
D	Stabilise active disease, explain the increased risks, obtain appropriate investigations and refer for multidisciplinary specialist planning
E	Refuse to offer any interim functional or preventive care

See answer and justification for Q941 →

Q942 INTERMEDIATE · Block M

An abutment thought to be restorable fractures during preparation; the clinician now recommends extraction instead of the agreed crown. What is the appropriate next step?

A	Disclose and manage significant errors transparently with appropriate documentation
B	Audit an identified pattern of clinical failure and implement an improvement cycle
C	Use a qualified interpreter for important complex consent discussions
D	Discuss reasonable alternatives, benefits, risks and maintenance in language the patient understands
E	Obtain renewed informed consent when the agreed plan materially changes

See answer and justification for Q942 →

Q943 INTERMEDIATE · Block M

The same test is used in a population with very low caries prevalence.
What happens to its positive predictive value?

A	It falls
B	It rises
C	It is unchanged
D	It equals sensitivity
E	It equals specificity

See answer and justification for Q943 →

Q945 INTERMEDIATE · Block M

The practice has had repeated denture fractures after a laboratory switches to a faster curing protocol. What quality action is most appropriate?

A	Use a qualified interpreter for important complex consent discussions
B	Disclose and manage significant errors transparently with appropriate documentation
C	Discuss reasonable alternatives, benefits, risks and maintenance in language the patient understands
D	Obtain renewed informed consent when the agreed plan materially changes
E	Audit an identified pattern of clinical failure and implement an improvement cycle

See answer and justification for Q945 →

Q946 INTERMEDIATE · Block M

A dentist moves from England to Wales.
Which bodies regulate services in each of the four UK nations?

A	CQC (England), Healthcare Improvement Scotland, Healthcare Inspectorate Wales, RQIA (Northern Ireland)
B	The GDC everywhere
C	The BDA everywhere
D	The NHS everywhere
E	The MHRA everywhere

See answer and justification for Q946 →

Q948 INTERMEDIATE · Block M

You must tell a patient that her failing teeth cannot be saved.
Which framework structures breaking bad news?

A	BRAN
B	PDSA
C	SPIKES: setting, perception, invitation, knowledge, emotions, strategy and summary
D	GRADE
E	ABCDE

See answer and justification for Q948 →

Q949 INTERMEDIATE · Block M

A crown is cemented on the wrong tooth and recognised immediately. What professional response is required?

A	Obtain renewed informed consent when the agreed plan materially changes
B	Use a qualified interpreter for important complex consent discussions
C	Disclose and manage significant errors transparently with appropriate documentation
D	Discuss reasonable alternatives, benefits, risks and maintenance in language the patient understands
E	Audit an identified pattern of clinical failure and implement an improvement cycle

See answer and justification for Q949 →

Q950 INTERMEDIATE · Block M

A clinician recommends a fixed bridge for one missing premolar, but the patient is comfortable with the space and declines replacement after discussing possible drift, aesthetics, maintenance and costs. The remaining dentition is stable. What is the most appropriate documentation and plan?

A	Record refusal only as a signed form without recording clinical discussion
B	Prescribe a bridge because no replacement is never acceptable
C	Document the patient's informed choice, alternatives including no treatment, expected consequences, and a risk-based review plan
D	Ask the patient to find another dentist before discussing no treatment
E	Fit a temporary prosthesis without consent

See answer and justification for Q950 →

BLOCK N · Integrated cases (40)

Q951 INTERMEDIATE · Block N

A 72-year-old has an upper complete denture and a lower Kennedy Class I RPD. 33 is broken down to the gingival margin but its root is sound. She takes apixaban for atrial fibrillation and has taken alendronic acid for 7 years.
What is the most appropriate management of 33?

A	Stop apixaban for 3 days and extract it
B	Extract it under antibiotic cover after a 3-month drug holiday
C	Root-treat and retain the root as an overdenture abutment, if feasible, rather than extracting
D	Leave it untreated under the denture
E	Crown it to act as a clasped abutment

See answer and justification for Q951 →

Q952 INTERMEDIATE · Block N

A 28-year-old has palatal erosion of the upper incisors and reflux symptoms. There is no space to restore the teeth, and she asks for porcelain veneers.
What is the most appropriate first step?

A	Liaise with her GMP about the reflux, give preventive advice, and monitor before any restoration
B	Prepare the teeth for veneers now
C	Place Dahl composites today
D	Crown lengthening
E	Extract and replace with implants

See answer and justification for Q952 →

Q953 INTERMEDIATE · Block N

An 83-year-old care-home resident with moderate dementia has lost both dentures. Staff report that she cannot eat solid food.
What is the most appropriate approach?

A	Implant overdentures
B	Do nothing, as dementia contraindicates dentures
C	Copy dentures from the lost originals
D	An immediate denture
E	Assess capacity, act in her best interests with carer involvement, and provide simple conventional dentures with labelling

See answer and justification for Q953 →

Q954 INTERMEDIATE · Block N

A patient with a known latex allergy develops urticaria, wheeze and hypotension minutes after an impression was taken by a nurse wearing latex gloves.
What is the immediate management?

A	Chlorphenamine orally and observation
B	Remove the allergen, call for help, and give adrenaline 1:1000, 500 micrograms IM
C	Salbutamol and discharge
D	Hydrocortisone only
E	Glucose gel

See answer and justification for Q954 →

Q955 INTERMEDIATE · Block N

A cement-retained implant crown on 21 was fitted 6 months ago. There is bleeding on probing at a 4 mm pocket, and the radiograph shows a small radiopaque fleck at the margin with no bone loss.
What is the diagnosis and management?

A	Peri-implantitis; refer for surgery
B	Implant failure; explant
C	Normal remodelling; no action
D	Peri-implant mucositis from residual cement: remove the cement, debride, give OHI and review (Level 2)
E	Occlusal overload; adjust the opposing tooth

See answer and justification for Q955 →

Q956 INTERMEDIATE · Block N

A 64-year-old had radiotherapy to the mandible 3 years ago. A lower molar under his RPD has become non-restorable.
What is the most appropriate action?

A	Extract it in primary care with antibiotics
B	Refer for specialist assessment because of the lifelong osteoradionecrosis risk
C	Leave it and reline the RPD
D	Extract with a drug holiday
E	Place an implant at the same visit

See answer and justification for Q956 →

Q957 INTERMEDIATE · Block N

A 19-year-old has palatal erosion, parotid enlargement and calluses on her knuckles. She asks for crowns and does not want her parents to know.
What is the most appropriate approach?

A	Tell her parents
B	Refuse treatment
C	Place crowns immediately
D	Report to social services
E	Approach with empathy, seek her consent to liaise with her GMP, and focus on prevention and monitoring before restoration

See answer and justification for Q957 →

Q958 INTERMEDIATE · Block N

A 77-year-old with an acrylic upper denture has diffuse palatal erythema. She takes warfarin and simvastatin and wears the denture day and night.
What is the most appropriate management?

A	Denture hygiene, night removal, soaking, correcting the fit, and nystatin if it persists
B	Miconazole oral gel
C	Fluconazole 50 mg daily
D	Biopsy of the palate
E	Stop warfarin

See answer and justification for Q958 →

Q959 INTERMEDIATE · Block N

A resin-bonded bridge replacing 21 has debonded three times. The patient is a bruxist with a deep overbite and heavy contact on the wing.
What is the most appropriate next step?

A	Rebond again with the same design
B	Use temporary cement
C	Add a second wing to 22
D	Review the occlusion and design, and discuss alternatives such as a conventional bridge, an implant or an RPD
E	Extract 11

See answer and justification for Q959 →

Q960 INTERMEDIATE · Block N

A 55-year-old has generalised severe wear, dentine exposure on most teeth and a planned OVD increase of 4 mm.
What is the most appropriate plan?

A	Crowns on all teeth at the current OVD
B	Dahl composites on the posterior teeth only
C	A diagnostic wax-up and provisional trial at the new OVD, with referral for Level 3 planning
D	Extraction and complete dentures
E	Monitor only, despite symptoms

See answer and justification for Q960 →

Q961 INTERMEDIATE · Block N

A 58-year-old with treated, stable periodontitis and high caries risk needs a lower Kennedy Class II RPD. The lingual sulcus is 9 mm deep.
Which design is most appropriate?

A	An acrylic mucosa-borne plate covering the gingivae
B	A flexible nylon denture
C	Co-Cr with a lingual plate and distal rests
D	Co-Cr, tooth-supported, with a mesial rest and RPI on the terminal abutment, a lingual bar, an open design and maximum saddle extension
E	A swing-lock

See answer and justification for Q961 →

Q962 INTERMEDIATE · Block N

A new upper complete denture drops on wide opening, and the patient gags. The posterior border is thick, and the distobuccal flange is long.
What is the most appropriate correction?

A	Add adhesive and increase the OVD
B	Remake with porcelain teeth
C	Trim the distobuccal flange and thin the posterior border, preserving the post-dam seal
D	Remove the post-dam
E	Reline with a soft liner

See answer and justification for Q962 →

Q963 INTERMEDIATE · Block N

A 14-year-old's replanted upper central incisor is now infra-occluded and ankylosed. He is still growing.
What is the most appropriate approach?

A	Monitor and seek multidisciplinary specialist advice
B	Place an implant now
C	Crown the tooth to level it
D	Extract it and leave the space
E	Orthodontic extrusion of the ankylosed tooth

See answer and justification for Q963 →

Q964 INTERMEDIATE · Block N

A 70-year-old on IV zoledronic acid for myeloma wants implants to stabilise her lower complete denture.
What is the most appropriate advice?

A	Implants are routine
B	Stop the zoledronic acid and place implants
C	Place implants with antibiotic cover
D	Refuse any denture
E	A well-fitting conventional denture with relief over bony prominences

See answer and justification for Q964 →

Q965 INTERMEDIATE · Block N

A 52-year-old with an HbA1c of 80 mmol/mol, generalised Stage III periodontitis and missing 46 wants a bridge.
What is the most appropriate sequence?

A	Prepare the bridge now
B	Periodontal therapy and referral to the GMP for diabetic control first
C	Extract all molars
D	Place an implant
E	Prescribe antibiotics and proceed

See answer and justification for Q965 →

Q966 INTERMEDIATE · Block N

A heavy bruxist had root canal treatment on 46 two weeks ago. The tooth has an MOD cavity with thin cusps and a temporary restoration.
What is the most appropriate restoration?

A	Prompt cuspal coverage (onlay or crown) in metal or monolithic zirconia, plus a night guard
B	A large composite without coverage
C	Layered feldspathic porcelain onlay
D	Leave the temporary for a year
E	A post and amalgam core only

See answer and justification for Q966 →

Q967 INTERMEDIATE · Block N

You realise after the patient has left that you extracted 15 instead of 14.
What is the most appropriate action?

A	Say nothing unless she notices
B	Alter the notes to read 15
C	Wait for a complaint
D	Contact her, explain, apologise, offer a remedy, report it as a never event and document everything
E	Ask a colleague to see her next time

See answer and justification for Q967 →

Q969 INTERMEDIATE · Block N

A patient with an implanted cardiac defibrillator and a high smile line needs a crown on 11 with a slightly subgingival margin.
What retraction method and margin position are appropriate?

A	Electrosurgery and a deep margin at the bone crest
B	Adrenaline cord and a 3 mm subgingival margin
C	Laser troughing to the crest
D	No retraction and a supragingival margin regardless of aesthetics
E	Plain or aluminium chloride cord, with the margin just subgingival and about 3 mm from the bone crest

See answer and justification for Q969 →

Q970 INTERMEDIATE · Block N

An upper RPD is needed for anterior and posterior saddles. The patient has a large torus and refuses surgery.
Which major connector is most appropriate?

A	Full palatal plate
B	Anteroposterior strap avoiding the torus
C	Horseshoe
D	Narrow mid-palatal bar over the torus
E	Lingual bar

See answer and justification for Q970 →

Q971 INTERMEDIATE · Block N

An 85-year-old attending for a denture reline has fingertip bruising on both arms. Her carer answers all questions for her.
What is the most appropriate action?

A	Ignore it
B	Ask the carer to explain
C	Photograph the bruising and investigate
D	Report directly to the police without discussion
E	Record your findings and refer through the practice safeguarding lead to adult social care

See answer and justification for Q971 →

Q972 INTERMEDIATE · Block N

A 20-year-old with hypodontia (missing 12, 22, 35 and 45) wants fixed replacements.
What is the most appropriate pathway?

A	Implants in general practice immediately
B	Interim RBBs or an RPD, with referral for multidisciplinary Level 3 care
C	Conventional bridges on all sites
D	No treatment
E	Crowns on all adjacent teeth

See answer and justification for Q972 →

Q973 INTERMEDIATE · Block N

During try-in, a crown slips from your fingers to the back of the patient's mouth. He is coughing forcefully and can speak.
What is the most appropriate immediate action?

A	Start back blows at once
B	Abdominal thrusts
C	Lay him flat
D	Encourage coughing
E	Ignore it

See answer and justification for Q973 →

Q974 INTERMEDIATE · Block N

An implant supporting a crown on 36 shows suppuration, a 7 mm pocket and 4 mm of radiographic bone loss since the baseline radiograph.
What is the most appropriate action?

A	Debride and review in a year
B	Retighten the abutment screw
C	Refer as peri-implantitis (Level 3)
D	Remove the crown permanently
E	Prescribe antibiotics only

See answer and justification for Q974 →

Q975 INTERMEDIATE · Block N

A patient with masticatory myalgia insists on "having her bite adjusted" before her crowns are made.
What is the most appropriate approach?

A	Explain that reversible care comes first (self-care and a stabilisation splint), and do not equilibrate to treat TMD
B	Equilibrate the whole dentition
C	Prepare the crowns at a new OVD
D	Prescribe long-term opioids
E	Refer for TMJ surgery

See answer and justification for Q975 →

Q976 INTERMEDIATE · Block N

A 79-year-old with a severely resorbed mandible has burning and numbness of the lower lip whenever she wears her new lower denture. The symptoms resolve when it is out.
What is the most likely cause and action?

A	Candida; antifungal
B	Allergy; patch test
C	Pressure on the mental nerve
D	Excessive OVD; remake
E	Malignancy; urgent referral

See answer and justification for Q976 →

Q977 INTERMEDIATE · Block N

A patient with lower canine overdenture abutments has new root caries on both.
What is the most appropriate preventive measure?

A	Fluoride gel in the fitting surface of the denture, with meticulous hygiene and a diet review
B	Extract both roots
C	Wear the denture day and night
D	Crown both roots
E	Antibiotics

See answer and justification for Q977 →

Q978 INTERMEDIATE · Block N

A patient has a two-implant lower overdenture opposing an upper complete denture. The upper anterior ridge has become flabby.
What is the most appropriate approach to a new upper denture?

A	A compressive putty impression and heavy anterior contacts
B	A selective-pressure (windowed) impression, with balanced occlusion avoiding heavy anterior contact
C	Extraction of the lower implants
D	Excision of the flabby ridge first, routinely
E	A soft liner only

See answer and justification for Q978 →

Q979 INTERMEDIATE · Block N

A 60-year-old with Sjögren's syndrome has a Co-Cr RPD and multiple new root caries lesions.
What is the most appropriate integrated plan?

A	Remove the RPD permanently
B	Use acidic saliva substitutes
C	Extract all carious teeth
D	High-fluoride toothpaste, a neutral saliva substitute, an open RPD design and a short recall
E	Antibiotics

See answer and justification for Q979 →

Q984 INTERMEDIATE · Block N

An 81-year-old with dementia whose capacity fluctuates needs a painful, infected tooth removed. On the day, she cannot weigh the information.
What is the most appropriate approach?

A	Proceed because she agreed last month
B	Ask a neighbour to sign
C	Act in her best interests, consulting those close to her and choosing the least restrictive option, unless she regains capacity
D	Never treat her
E	Wait for a court order in every case

See answer and justification for Q984 →

Q985 INTERMEDIATE · Block N

A crown arrives from a new laboratory with no statement of manufacture.
What should you do?

A	Fit it anyway
B	Throw it away
C	Accept it if it fits well
D	Ask the patient to sign a waiver
E	Request the statement and confirm the laboratory is MHRA-registered before fitting

See answer and justification for Q985 →

Q987 INTERMEDIATE · Block N

A rigid five-unit bridge from 13 to 17, with a pier at 15, has debonded at 13. There is caries under the 13 retainer.
What is the most appropriate management?

A	Recement it
B	Cut the bridge and leave 13 untreated
C	Remove it, treat the caries, and redesign with a non-rigid connector at the pier or as separate bridges
D	Add a sixth unit
E	Use temporary cement

See answer and justification for Q987 →

Q988 INTERMEDIATE · Block N

A crown on 11 looks too light. The shade was taken after preparation under the operating light.
What explains the mismatch?

A	Wrong cement
B	Wrong material
C	Laboratory error only
D	Dehydration and poor lighting
E	The patient's diet

See answer and justification for Q988 →

Q989 INTERMEDIATE · Block N

A patient with moderate haemophilia A needs several extractions before an immediate upper denture.
What is the most appropriate management?

A	Proceed in practice with local measures only
B	Liaise with the haemophilia centre, avoiding regional blocks without factor cover
C	Give aspirin for pain
D	Avoid dentures
E	Extract one tooth per year

See answer and justification for Q989 →

Q990 INTERMEDIATE · Block N

A 68-year-old has taken oral alendronate for 2 years for osteoporosis, without steroids. A tooth under her RPD needs extraction.
What is the most appropriate approach?

A	Proceed with standard care and no drug holiday; review healing
B	Stop alendronate for 6 months first
C	Refuse extraction
D	Antibiotics for a month first
E	Refer every case to hospital

See answer and justification for Q990 →

Q993 INTERMEDIATE · Block N

A 30-year-old has lost 12. 13 is sound, unrestored and enamel-rich, and the occlusion is light.
What is the most appropriate first-choice replacement?

A	A conventional fixed-fixed bridge from 11 to 13
B	An immediate implant without assessment
C	A single-wing resin-bonded cantilever from 13
D	An acrylic RPD
E	A spring cantilever

See answer and justification for Q993 →

Q994 INTERMEDIATE · Block N

A patient with an upper complete denture and only lower anterior teeth has a flabby premaxilla and enlarged tuberosities.
What is the most appropriate management?

A	Provide posterior support with a lower RPD, and use a selective-pressure impression for the flabby ridge
B	Extract the lower anteriors immediately
C	Increase anterior loading
D	Excise all flabby tissue routinely
E	No change

See answer and justification for Q994 →

Q995 INTERMEDIATE · Block N

A 74-year-old attends for a denture adjustment and mentions new pain in the jaw when chewing, a headache and scalp tenderness.
What is the most appropriate action?

A	Adjust the denture and review in a month
B	A stabilisation splint
C	Ibuprofen
D	Same-day medical referral for suspected giant cell arteritis
E	TMJ radiograph

See answer and justification for Q995 →

Q996 INTERMEDIATE · Block N

A patient had a myocardial infarction 3 weeks ago and asks to start elective bridgework now.
What is the most appropriate approach?

A	Proceed today with sedation
B	Refuse care permanently
C	Give GTN and proceed
D	Liaise with the physician and defer elective treatment until he is medically stable
E	Proceed without local anaesthetic

See answer and justification for Q996 →

Q998 INTERMEDIATE · Block N

A 66-year-old smoker has an ulcer under her lower denture flange that persists 4 weeks after the flange was adjusted.
What is the most appropriate action?

A	Adjust further and review in 3 months
B	A soft liner
C	Chlorhexidine
D	Antifungal
E	Urgent referral on the suspected cancer pathway

See answer and justification for Q998 →

Q1000 INTERMEDIATE · Block N

A patient has completed treatment with an implant overdenture, a Co-Cr RPD and several crowns.
What is the most appropriate long-term plan?

A	No further review
B	Review only when there is pain
C	Risk-based recall with supportive periodontal and peri-implant care, prosthesis maintenance, and recording of patient-reported outcomes
D	Annual CBCT
E	Replace all prostheses every 3 years

See answer and justification for Q1000 →

Advanced questions · 500 items

BLOCK A · Applied anatomy (21)

Q1088 ADVANCED · Block A

A 47-year-old with a mandibular second molar infection develops a rapidly rising floor of mouth, muffled voice, drooling and bilateral submandibular swelling. What is the immediate priority?

A	Use an intraoral drain and routine review in one week
B	Provide routine denture relief and review
C	Seek urgent assessment for a progressive cranial neuropathy
D	Manage localised swelling after appropriate assessment and source control
E	Refer immediately for urgent airway and maxillofacial management

See answer and justification for Q1088 →

Q1089 ADVANCED · Block A

A 43-year-old has a fluctuant localised vestibular swelling adjacent to a non-restorable premolar. There is no fever, dysphagia, trismus or other spread, and the airway is normal. Which management principle fits?

A	Use an intraoral drain and routine review in one week
B	Seek urgent assessment for a progressive cranial neuropathy
C	Refer immediately for urgent airway and maxillofacial management
D	Manage localised swelling after appropriate assessment and source control
E	Provide routine denture relief and review

See answer and justification for Q1089 →

Q1090 ADVANCED · Block A

A 71-year-old with an adequate denture and no evidence of local pressure develops persistent unilateral lower lip numbness and progressive facial sensory changes. What should happen?

A	Refer immediately for urgent airway and maxillofacial management
B	Manage localised swelling after appropriate assessment and source control
C	Provide routine denture relief and review
D	Seek urgent assessment for a progressive cranial neuropathy
E	Use an intraoral drain and routine review in one week

See answer and justification for Q1090 →

Q1091 ADVANCED · Block A

A mandibular complete denture rises when the patient elevates the tongue and swallows. The lingual flange in the molar region appears deep and sharply inclined. What should be assessed?

A	Assess the anterior labial contour and modiolus activity
B	Assess lingual flange extension against mylohyoid and tongue movements
C	Assess posterior maxillary flange interference from the coronoid process
D	Assess the distobuccal flange against masseter action
E	Modify occlusal contacts without checking flange movement

See answer and justification for Q1091 →

Q1092 ADVANCED · Block A

A mandibular complete denture dislodges only when the patient clenches strongly. Pressure is visible at the distobuccal flange near the ramus. Which functional structure is relevant?

A	Assess posterior maxillary flange interference from the coronoid process
B	Modify occlusal contacts without checking flange movement
C	Assess lingual flange extension against mylohyoid and tongue movements
D	Assess the distobuccal flange against masseter action
E	Assess the anterior labial contour and modiolus activity

See answer and justification for Q1092 →

Q1093 ADVANCED · Block A

A maxillary complete denture remains seated in quiet closure but loosens when the patient opens very widely. The distobuccal border is thick. Which structure should be checked?

A	Assess the anterior labial contour and modiolus activity
B	Assess the distobuccal flange against masseter action
C	Assess posterior maxillary flange interference from the coronoid process
D	Modify occlusal contacts without checking flange movement
E	Assess lingual flange extension against mylohyoid and tongue movements

See answer and justification for Q1093 →

Q1157 ADVANCED · Block A

An edentulous patient has lower lip tingling immediately after inserting a new mandibular denture. The ridge is severely resorbed and the symptom stops when it is removed. Which anatomical risk should be checked?

A	Use functional border moulding to register a masseteric notch
B	Ignore a torus if the denture appears retentive on the cast
C	Identify the mental foramen before relieving a severely resorbed lower denture
D	Assess the hamular notches and vibrating lines for a posterior palatal seal
E	Extend the maxillary post-dam without assessing palate movement

See answer and justification for Q1157 →

Q1158 ADVANCED · Block A

A lower denture is stable at rest but lifts when the patient contracts the masseters strongly. What functional impression feature is most relevant?

A	Use functional border moulding to register a masseteric notch
B	Assess the hamular notches and vibrating lines for a posterior palatal seal
C	Extend the maxillary post-dam without assessing palate movement
D	Ignore a torus if the denture appears retentive on the cast
E	Identify the mental foramen before relieving a severely resorbed lower denture

See answer and justification for Q1158 →

Q1159 ADVANCED · Block A

An upper complete denture has no apparent posterior seal and falls during speech. The existing posterior extension was guessed from the cast without functional examination. What must be reassessed?

A	Assess the hamular notches and vibrating lines for a posterior palatal seal
B	Use functional border moulding to register a masseteric notch
C	Ignore a torus if the denture appears retentive on the cast
D	Identify the mental foramen before relieving a severely resorbed lower denture
E	Extend the maxillary post-dam without assessing palate movement

See answer and justification for Q1159 →

Q1181 ADVANCED · Block A

A mandibular complete denture wearer reports focal tingling only during chewing. The ridge is very resorbed; the new base presses directly over a palpable mental foramen. Sensation is normal when the denture is removed.
Which is the most appropriate next step?

A	Assess a distobuccal flange against coronoid interference
B	Investigate the lingual flange against dynamic tongue movement and adjust it
C	Relieve the denture over the mental foramen and review symptoms promptly
D	Arrange an urgent assessment of unexplained persistent sensory change
E	Reassess posterior palatal seal limits with functional soft-palate movements

See answer and justification for Q1181 →

Q1182 ADVANCED · Block A

A complete mandibular denture fits at rest but lifts on speech and tongue protrusion. A functional impression confirms inadequate clearance of the anterior lingual sulcus without an occlusal interference.
Which is the most appropriate next step?

A	Reassess posterior palatal seal limits with functional soft-palate movements
B	Relieve the denture over the mental foramen and review symptoms promptly
C	Arrange an urgent assessment of unexplained persistent sensory change
D	Investigate the lingual flange against dynamic tongue movement and adjust it
E	Assess a distobuccal flange against coronoid interference

See answer and justification for Q1182 →

Q1183 ADVANCED · Block A

A 74-year-old has unilateral lip numbness for six weeks despite stopping an old denture. There is no flange tenderness, and examination reveals a firm asymmetry of the mandibular body.
Which is the most appropriate next step?

A	Arrange an urgent assessment of unexplained persistent sensory change
B	Relieve the denture over the mental foramen and review symptoms promptly
C	Reassess posterior palatal seal limits with functional soft-palate movements
D	Investigate the lingual flange against dynamic tongue movement and adjust it
E	Assess a distobuccal flange against coronoid interference

See answer and justification for Q1183 →

Q1184 ADVANCED · Block A

A maxillary denture repeatedly dislodges on wide opening and lateral excursions, although retention is good at rest. Its distal buccal flange is bulky immediately anterior to the tuberosity.
Which is the most appropriate next step?

A	Assess a distobuccal flange against coronoid interference
B	Reassess posterior palatal seal limits with functional soft-palate movements
C	Relieve the denture over the mental foramen and review symptoms promptly
D	Investigate the lingual flange against dynamic tongue movement and adjust it
E	Arrange an urgent assessment of unexplained persistent sensory change

See answer and justification for Q1184 →

Q1185 ADVANCED · Block A

A patient with a lower second-molar infection has a raised tongue, bilateral submandibular swelling, drooling, muffled voice and difficulty swallowing. Respiratory observations are deteriorating.
What is the priority management?

A	Arrange review after adjustment of a traumatic denture flange
B	Obtain targeted imaging and refer for suspected deep-space infection
C	Assess whether an isolated sublingual swelling needs urgent maxillofacial evaluation
D	Drain a localised fluctuant abscess and address the cause if appropriate within competence
E	Arrange emergency admission for airway assessment and infection control

See answer and justification for Q1185 →

Q1186 ADVANCED · Block A

A systemically well adult has a small, fluctuant buccal swelling next to an obviously non-restorable molar. Mouth opening is unrestricted and there are no deep-space signs.
What is the priority management?

A	Drain a localised fluctuant abscess and address the cause if appropriate within competence
B	Arrange emergency admission for airway assessment and infection control
C	Arrange review after adjustment of a traumatic denture flange
D	Assess whether an isolated sublingual swelling needs urgent maxillofacial evaluation
E	Obtain targeted imaging and refer for suspected deep-space infection

See answer and justification for Q1186 →

Q1187 ADVANCED · Block A

After an infected mandibular molar, a patient develops increasing trismus and swelling around the angle but no current respiratory symptoms. The suspected infection is deep to the masseter.
What is the priority management?

A	Drain a localised fluctuant abscess and address the cause if appropriate within competence
B	Arrange review after adjustment of a traumatic denture flange
C	Assess whether an isolated sublingual swelling needs urgent maxillofacial evaluation
D	Obtain targeted imaging and refer for suspected deep-space infection
E	Arrange emergency admission for airway assessment and infection control

See answer and justification for Q1187 →

Q1188 ADVANCED · Block A

A stable denture wearer has a small superficial ulcer at the labial flange, appearing after a recent denture extension. There is no facial swelling, fever, altered sensation or persistent mass.
What is the priority management?

A	Drain a localised fluctuant abscess and address the cause if appropriate within competence
B	Arrange emergency admission for airway assessment and infection control
C	Obtain targeted imaging and refer for suspected deep-space infection
D	Arrange review after adjustment of a traumatic denture flange
E	Assess whether an isolated sublingual swelling needs urgent maxillofacial evaluation

See answer and justification for Q1188 →

Q1189 ADVANCED · Block A

A mandibular denture repeatedly lifts along its posterior lingual border when the patient swallows. The impression shows a flat, under-recorded retromylohyoid area; the labial and buccal borders are satisfactory.
Which approach best explains and manages the instability?

A	Re-establish the posterior palatal seal after locating the vibrating line
B	Assess and correct over-extension against the coronoid process
C	Relieve a flange interfering with the masseteric notch during clenching
D	Record the retromylohyoid space during functional moulding
E	Reassess neutral-zone and polished-surface contour

See answer and justification for Q1189 →

Q1190 ADVANCED · Block A

The lower denture is stable on talking but the left posterior buccal corner is sore each time the patient clenches. The lateral border extends into the contracted masseter.
Which approach best explains and manages the instability?

A	Assess and correct over-extension against the coronoid process
B	Re-establish the posterior palatal seal after locating the vibrating line
C	Reassess neutral-zone and polished-surface contour
D	Relieve a flange interfering with the masseteric notch during clenching
E	Record the retromylohyoid space during functional moulding

See answer and justification for Q1190 →

Q1191 ADVANCED · Block A

The upper denture has excellent retention at rest, but dislodges every time the patient opens widely. The distobuccal flange is unusually thick and extends around the tuberosity.
Which approach best explains and manages the instability?

A	Assess and correct over-extension against the coronoid process
B	Relieve a flange interfering with the masseteric notch during clenching
C	Reassess neutral-zone and polished-surface contour
D	Record the retromylohyoid space during functional moulding
E	Re-establish the posterior palatal seal after locating the vibrating line

See answer and justification for Q1191 →

Q1192 ADVANCED · Block A

A severely resorbed mandibular denture remains loose despite accurate functional borders and acceptable occlusion. It is displaced by cheek and tongue movements; wax trial contours project beyond the neutral zone.
Which approach best explains and manages the instability?

A	Record the retromylohyoid space during functional moulding
B	Re-establish the posterior palatal seal after locating the vibrating line
C	Relieve a flange interfering with the masseteric notch during clenching
D	Assess and correct over-extension against the coronoid process
E	Reassess neutral-zone and polished-surface contour

See answer and justification for Q1192 →

BLOCK B · Physiology and pathology (25)

Q1013 ADVANCED · Block B

A 61-year-old taking several new antimuscarinic medicines has resting saliva of 0.07 mL/min, rapid new root caries and an RPD. What should be addressed first?

A	Investigate the medicines with the prescriber and institute saliva and caries prevention
B	Replace sound restorations with full coverage crowns to prevent caries
C	Investigate for systemic causes and manage oral disease while awaiting results
D	Provide a hard reline as the only intervention
E	Recommend acidic sweets to stimulate saliva

See answer and justification for Q1013 →

Q1014 ADVANCED · Block B

A 70-year-old reports new widespread dry mouth, dry eyes and fatigue, with no recent medication changes and unexplained mucosal discomfort. What is most appropriate?

A	Investigate for systemic causes and manage oral disease while awaiting results
B	Recommend acidic sweets to stimulate saliva
C	Provide a hard reline as the only intervention
D	Investigate the medicines with the prescriber and institute saliva and caries prevention
E	Replace sound restorations with full coverage crowns to prevent caries

See answer and justification for Q1014 →

Q1015 ADVANCED · Block B

A 74-year-old requests a reline because a well-fitting upper denture loses retention after radiotherapy. Mucosa is dry and the film of saliva is scant. What is the best management priority?

A	Recommend acidic sweets to stimulate saliva
B	Investigate the medicines with the prescriber and institute saliva and caries prevention
C	Investigate for systemic causes and manage oral disease while awaiting results
D	Provide a hard reline as the only intervention
E	Replace sound restorations with full coverage crowns to prevent caries

See answer and justification for Q1015 →

Q1016 ADVANCED · Block B

A 28-year-old has palatal erosive wear of upper anterior teeth, parotid enlargement and reports frequent unexplained nausea. Flat matching facets are absent. What diagnostic direction is most appropriate?

A	Assume abfraction alone and place cervical crowns
B	Identify attrition linked to tooth contact and assess function
C	Identify extrinsic acid erosion and counsel on beverage frequency
D	Suspect intrinsic acid exposure and address the medical cause sensitively
E	Investigate combined chemical and mechanical wear rather than assign one cause

See answer and justification for Q1016 →

Q1017 ADVANCED · Block B

A 52-year-old has matching polished facets on opposing canine teeth and fractured posterior cusps. The teeth have no silky enamel or cupped occlusal defects. What most likely predominates?

A	Investigate combined chemical and mechanical wear rather than assign one cause
B	Identify extrinsic acid erosion and counsel on beverage frequency
C	Identify attrition linked to tooth contact and assess function
D	Suspect intrinsic acid exposure and address the medical cause sensitively
E	Assume abfraction alone and place cervical crowns

See answer and justification for Q1017 →

Q1018 ADVANCED · Block B

A 42-year-old has widespread cupped cusps and smooth enamel, drinks acidic sports drinks through the working day, and has no reflux symptoms. What should be prioritised?

A	Suspect intrinsic acid exposure and address the medical cause sensitively
B	Assume abfraction alone and place cervical crowns
C	Identify extrinsic acid erosion and counsel on beverage frequency
D	Identify attrition linked to tooth contact and assess function
E	Investigate combined chemical and mechanical wear rather than assign one cause

See answer and justification for Q1018 →

Q1094 ADVANCED · Block B

A 57-year-old with periodontitis, multiple recurrent infections and poorly controlled type 2 diabetes requests immediate full-mouth implants. What is most appropriate?

A	Assume osteoporosis alone makes conventional dentures contraindicated
B	Coordinate assessment of glycaemic control before complex elective care
C	Prioritise an adaptable low-maintenance prosthetic approach
D	Investigate haematinic deficiency alongside oral findings
E	Treat periodontal inflammation without accounting for systemic risk

See answer and justification for Q1094 →

Q1095 ADVANCED · Block B

A 68-year-old denture wearer reports painful tongue, recurrent angular cheilitis and fatigue. Examination shows glossitis but a satisfactory prosthesis. What should be investigated?

A	Coordinate assessment of glycaemic control before complex elective care
B	Prioritise an adaptable low-maintenance prosthetic approach
C	Treat periodontal inflammation without accounting for systemic risk
D	Assume osteoporosis alone makes conventional dentures contraindicated
E	Investigate haematinic deficiency alongside oral findings

See answer and justification for Q1095 →

Q1096 ADVANCED · Block B

An 85-year-old with tremor, frailty and frequent carer support wants replacement of a failing partial denture. Hygiene and attendance are unreliable. What should guide the design?

A	Prioritise an adaptable low-maintenance prosthetic approach
B	Treat periodontal inflammation without accounting for systemic risk
C	Investigate haematinic deficiency alongside oral findings
D	Assume osteoporosis alone makes conventional dentures contraindicated
E	Coordinate assessment of glycaemic control before complex elective care

See answer and justification for Q1096 →

Q1193 ADVANCED · Block B

A 65-year-old recently started two anticholinergic medicines and now has a dry mouth, multiple new root lesions and reduced salivary flow. An RPD is otherwise serviceable.
What is the best initial comprehensive plan?

A	Manage suspected intrinsic acid exposure through sensitive medical liaison and prevention
B	Apply evidence-based prevention and monitor stable tooth wear before intervening
C	Urgently investigate a persistent suspicious mucosal lesion
D	Coordinate medication review and intensive caries/xerostomia prevention
E	Restore a symptomatic localised cervical defect conservatively after controlling risk

See answer and justification for Q1193 →

Q1194 ADVANCED · Block B

A young adult has rapid palatal upper-incisor wear, intermittent hoarseness and enamel thinning despite low acidic-drink intake. They privately describe nocturnal regurgitation.
What is the best initial comprehensive plan?

A	Urgently investigate a persistent suspicious mucosal lesion
B	Coordinate medication review and intensive caries/xerostomia prevention
C	Apply evidence-based prevention and monitor stable tooth wear before intervening
D	Restore a symptomatic localised cervical defect conservatively after controlling risk
E	Manage suspected intrinsic acid exposure through sensitive medical liaison and prevention

See answer and justification for Q1194 →

Q1195 ADVANCED · Block B

A 44-year-old has slight, longstanding wear facets, healthy teeth and no symptoms. Serial records over three years show no progression, and the patient requests preventive advice.
What is the best initial comprehensive plan?

A	Manage suspected intrinsic acid exposure through sensitive medical liaison and prevention
B	Apply evidence-based prevention and monitor stable tooth wear before intervening
C	Urgently investigate a persistent suspicious mucosal lesion
D	Coordinate medication review and intensive caries/xerostomia prevention
E	Restore a symptomatic localised cervical defect conservatively after controlling risk

See answer and justification for Q1195 →

Q1196 ADVANCED · Block B

A 59-year-old has a persistent indurated red-and-white lesion on the lateral tongue, first noticed four weeks ago during a denture review. It remains after removing a suspected mechanical cause.
What is the best initial comprehensive plan?

A	Restore a symptomatic localised cervical defect conservatively after controlling risk
B	Apply evidence-based prevention and monitor stable tooth wear before intervening
C	Urgently investigate a persistent suspicious mucosal lesion
D	Coordinate medication review and intensive caries/xerostomia prevention
E	Manage suspected intrinsic acid exposure through sensitive medical liaison and prevention

See answer and justification for Q1196 →

Q1197 ADVANCED · Block B

An older adult with exposed roots sips sweetened tea every 30 minutes. New cervical lesions develop despite occasional fluoride varnish and good technical restorations.
Which intervention most directly addresses the dominant risk?

A	Reduce between-meal sugar exposure and strengthen fluoride-based caries control
B	Modify frequent dietary acid exposure and postpone elective bonding until stabilised
C	Introduce a personalised denture hygiene and night-removal programme
D	Assess a possible eating disorder sensitively and coordinate appropriate support
E	Address poor plaque control and active periodontal inflammation before fixed treatment

See answer and justification for Q1197 →

Q1198 ADVANCED · Block B

A runner regularly holds sports drinks in the mouth during training. Serial photographs show progressing palatal cupping and no evidence of intrinsic acid exposure.
Which intervention most directly addresses the dominant risk?

A	Assess a possible eating disorder sensitively and coordinate appropriate support
B	Introduce a personalised denture hygiene and night-removal programme
C	Modify frequent dietary acid exposure and postpone elective bonding until stabilised
D	Address poor plaque control and active periodontal inflammation before fixed treatment
E	Reduce between-meal sugar exposure and strengthen fluoride-based caries control

See answer and justification for Q1198 →

Q1199 ADVANCED · Block B

An upper acrylic denture has diffuse fitting-surface plaque and palatal erythema. The patient sleeps in it and cleans it by rinsing under water.
Which intervention most directly addresses the dominant risk?

A	Modify frequent dietary acid exposure and postpone elective bonding until stabilised
B	Address poor plaque control and active periodontal inflammation before fixed treatment
C	Introduce a personalised denture hygiene and night-removal programme
D	Reduce between-meal sugar exposure and strengthen fluoride-based caries control
E	Assess a possible eating disorder sensitively and coordinate appropriate support

See answer and justification for Q1199 →

Q1200 ADVANCED · Block B

A 19-year-old has palatal erosion, enlarged parotids and subtle knuckle calluses, but declines to discuss possible vomiting when their parent is present.
Which intervention most directly addresses the dominant risk?

A	Address poor plaque control and active periodontal inflammation before fixed treatment
B	Assess a possible eating disorder sensitively and coordinate appropriate support
C	Reduce between-meal sugar exposure and strengthen fluoride-based caries control
D	Introduce a personalised denture hygiene and night-removal programme
E	Modify frequent dietary acid exposure and postpone elective bonding until stabilised

See answer and justification for Q1200 →

Q1201 ADVANCED · Block B

An asymptomatic denture wearer has generalised erythema confined to mucosa covered by a maxillary denture, which is worn overnight. The surrounding palate is normal.
What is the most appropriate interpretation or response?

A	Provide a non-compressive impression approach for a mobile ridge
B	Diagnose probable denture stomatitis and address hygiene and fit
C	Investigate burning-mouth symptoms after excluding local and systemic causes
D	Refer an unexplained persistent ulcer for urgent assessment
E	Assess haematinic and systemic factors contributing to angular cheilitis

See answer and justification for Q1201 →

Q1202 ADVANCED · Block B

A 75-year-old has bilateral mouth-corner fissures with a reduced occlusal vertical dimension and fatigue. Diet has deteriorated after a recent bereavement.
What is the most appropriate interpretation or response?

A	Provide a non-compressive impression approach for a mobile ridge
B	Investigate burning-mouth symptoms after excluding local and systemic causes
C	Assess haematinic and systemic factors contributing to angular cheilitis
D	Refer an unexplained persistent ulcer for urgent assessment
E	Diagnose probable denture stomatitis and address hygiene and fit

See answer and justification for Q1202 →

Q1203 ADVANCED · Block B

A sore mandibular denture flange was adjusted three weeks ago. The same ulcer is now indurated and enlarging despite no further contact.
What is the most appropriate interpretation or response?

A	Investigate burning-mouth symptoms after excluding local and systemic causes
B	Provide a non-compressive impression approach for a mobile ridge
C	Refer an unexplained persistent ulcer for urgent assessment
D	Diagnose probable denture stomatitis and address hygiene and fit
E	Assess haematinic and systemic factors contributing to angular cheilitis

See answer and justification for Q1203 →

Q1204 ADVANCED · Block B

The anterior edentulous maxillary ridge is flabby and displaces during preliminary impressions. The lower anterior natural teeth are retained, and a conventional compressive impression distorts the tissue.
What is the most appropriate interpretation or response?

A	Refer an unexplained persistent ulcer for urgent assessment
B	Diagnose probable denture stomatitis and address hygiene and fit
C	Assess haematinic and systemic factors contributing to angular cheilitis
D	Investigate burning-mouth symptoms after excluding local and systemic causes
E	Provide a non-compressive impression approach for a mobile ridge

See answer and justification for Q1204 →

Q1205 ADVANCED · Block B

A 77-year-old retains sound anterior and premolar contacts bilaterally, reports satisfactory chewing and requests no further intervention. Only the second molars are missing.
Which plan best fits the patient’s functional needs?

A	Use a carefully adapted copy-denture approach to preserve familiar contours
B	Evaluate a removable partial denture for an actual chewing or appearance deficit
C	Accept a shortened dental arch and monitor function without replacing every molar
D	Assess implant-retained overdenture suitability when conventional mandibular retention is inadequate
E	Treat active denture-bearing mucosal disease before replacing the prosthesis

See answer and justification for Q1205 →

Q1206 ADVANCED · Block B

A frail patient with Parkinsonism has worn a familiar denture comfortably for years. The base is worn but the patient finds new contours disorientating.
Which plan best fits the patient’s functional needs?

A	Assess implant-retained overdenture suitability when conventional mandibular retention is inadequate
B	Evaluate a removable partial denture for an actual chewing or appearance deficit
C	Use a carefully adapted copy-denture approach to preserve familiar contours
D	Accept a shortened dental arch and monitor function without replacing every molar
E	Treat active denture-bearing mucosal disease before replacing the prosthesis

See answer and justification for Q1206 →

Q1207 ADVANCED · Block B

A medically stable edentulous patient cannot control a repeatedly remade lower denture despite competent impressions and occlusion. They can maintain hygiene and seek better retention.
Which plan best fits the patient's functional needs?

A	Accept a shortened dental arch and monitor function without replacing every molar
B	Assess implant-retained overdenture suitability when conventional mandibular retention is inadequate
C	Treat active denture-bearing mucosal disease before replacing the prosthesis
D	Evaluate a removable partial denture for an actual chewing or appearance deficit
E	Use a carefully adapted copy-denture approach to preserve familiar contours

See answer and justification for Q1207 →

Q1208 ADVANCED · Block B

A 62-year-old with shortened arches has recently lost one functioning premolar pair and now struggles to chew. The remaining teeth have good support but the patient declines implants.
Which plan best fits the patient's functional needs?

A	Evaluate a removable partial denture for an actual chewing or appearance deficit
B	Treat active denture-bearing mucosal disease before replacing the prosthesis
C	Assess implant-retained overdenture suitability when conventional mandibular retention is inadequate
D	Accept a shortened dental arch and monitor function without replacing every molar
E	Use a carefully adapted copy-denture approach to preserve familiar contours

See answer and justification for Q1208 →

BLOCK C · Dental materials science (46)

Q1019 ADVANCED · Block C

A minimal-preparation lithium disilicate veneer debonds with an intact internal ceramic surface; adequate enamel and isolation remain. What surface strategy is most appropriate for rebonding?

A	Treat all ceramics identically with hydrofluoric acid
B	Reassess isolation and ferrule before selecting any bonding protocol
C	Use conventional glass ionomer alone for any thin indirect veneer
D	Etch a silica-based ceramic with an appropriate HF protocol and silanise before resin bonding
E	Air-abrade zirconia and use an appropriate phosphate-monomer primer or cement

See answer and justification for Q1019 →

Q1020 ADVANCED · Block C

A monolithic zirconia posterior onlay is being bonded in a case with limited mechanical retention and otherwise good isolation. Which pretreatment is appropriate?

A	Reassess isolation and ferrule before selecting any bonding protocol
B	Etch a silica-based ceramic with an appropriate HF protocol and silanise before resin bonding
C	Use conventional glass ionomer alone for any thin indirect veneer
D	Air-abrade zirconia and use an appropriate phosphate-monomer primer or cement
E	Treat all ceramics identically with hydrofluoric acid

See answer and justification for Q1020 →

Q1021 ADVANCED · Block C

A root-filled premolar has virtually no sound coronal dentine, a margin deep beneath the gingiva and impossible dry-field isolation. The patient requests an adhesive onlay. What is the next step?

A	Treat all ceramics identically with hydrofluoric acid
B	Air-abrade zirconia and use an appropriate phosphate-monomer primer or cement
C	Reassess isolation and ferrule before selecting any bonding protocol
D	Etch a silica-based ceramic with an appropriate HF protocol and silanise before resin bonding
E	Use conventional glass ionomer alone for any thin indirect veneer

See answer and justification for Q1021 →

Q1022 ADVANCED · Block C

A crown fits its working cast but has an open margin clinically. The addition-silicone impression tray was visibly flexible and the wash was thin and unsupported. What should be changed?

A	Use a highly hydrophilic impression material with appropriate moisture control
B	Use a selective-pressure technique to avoid displacing the flabby tissue
C	Repeat the impression using a rigid tray and correct material support
D	Disinfect by prolonged immersion in any solution available
E	Fit the inaccurate restoration and adjust the occlusion extensively

See answer and justification for Q1022 →

Q1023 ADVANCED · Block C

A deep subgingival margin remains wet with sulcular fluid despite suitable tissue management. A precise impression is required and material selection is being reviewed. Which principle is most useful?

A	Disinfect by prolonged immersion in any solution available
B	Fit the inaccurate restoration and adjust the occlusion extensively
C	Use a highly hydrophilic impression material with appropriate moisture control
D	Use a selective-pressure technique to avoid displacing the flabby tissue
E	Repeat the impression using a rigid tray and correct material support

See answer and justification for Q1023 →

Q1024 ADVANCED · Block C

An edentulous maxilla has a highly mobile anterior fibrous ridge opposing lower natural incisors. Conventional impression pressure has displaced it. What is most appropriate?

A	Repeat the impression using a rigid tray and correct material support
B	Use a highly hydrophilic impression material with appropriate moisture control
C	Disinfect by prolonged immersion in any solution available
D	Use a selective-pressure technique to avoid displacing the flabby tissue
E	Fit the inaccurate restoration and adjust the occlusion extensively

See answer and justification for Q1024 →

Q1025 ADVANCED · Block C

An upper complete denture develops a midline crack gradually after years of use; there has been no fall and repeated flexure occurs in function. What is the dominant failure mechanism?

A	Identify impact fracture and consider a high-impact base
B	Investigate loss of bonding and contamination at the interface
C	Identify investment wax distortion and make a fresh pattern promptly
D	Select the restoration solely by maximum elastic modulus
E	Identify fatigue from repeated flexure and redesign support or reinforcement

See answer and justification for Q1025 →

Q1026 ADVANCED · Block C

A patient with a tremor drops a new acrylic denture onto a hard tiled floor and it fractures immediately through the base. Which material property is most relevant to prevention?

A	Identify fatigue from repeated flexure and redesign support or reinforcement
B	Identify impact fracture and consider a high-impact base
C	Investigate loss of bonding and contamination at the interface
D	Select the restoration solely by maximum elastic modulus
E	Identify investment wax distortion and make a fresh pattern promptly

See answer and justification for Q1026 →

Q1027 ADVANCED · Block C

A wax pattern for an inlay was stored for two days before investing, then the casting distorted despite controlled alloy shrinkage. What should be addressed?

A	Investigate loss of bonding and contamination at the interface
B	Identify fatigue from repeated flexure and redesign support or reinforcement
C	Identify impact fracture and consider a high-impact base
D	Identify investment wax distortion and make a fresh pattern promptly
E	Select the restoration solely by maximum elastic modulus

See answer and justification for Q1027 →

Q1097 ADVANCED · Block C

A full-coverage cast crown has loosened three times from a very short, over-tapered preparation. Isolation is satisfactory and the cement has always remained on the tooth. What problem is primary?

A Treat the margin caries before recementing the existing crown

B Use MDP-compatible bonding for suitably conditioned zirconia

C Confirm whether the problem is inadequate retention form rather than cement strength

D Use a validated silane-based protocol for a glass ceramic

E Attribute every debond to an intrinsic defect in the cement

See answer and justification for Q1097 →

Q1098 ADVANCED · Block C

A zirconia restoration with limited macromechanical retention has separated cleanly from a suitably prepared tooth. After checking substrates, which bonding principle is relevant?

A Attribute every debond to an intrinsic defect in the cement

B Confirm whether the problem is inadequate retention form rather than cement strength

C Treat the margin caries before recementing the existing crown

D Use MDP-compatible bonding for suitably conditioned zirconia

E Use a validated silane-based protocol for a glass ceramic

See answer and justification for Q1098 →

Q1099 ADVANCED · Block C

A crown has become loose and clinical examination shows softened, discoloured margin dentine with radiographic evidence of recurrent caries. What comes before recementation?

A Confirm whether the problem is inadequate retention form rather than cement strength

B Attribute every debond to an intrinsic defect in the cement

C Use a validated silane-based protocol for a glass ceramic

D Treat the margin caries before recementing the existing crown

E Use MDP-compatible bonding for suitably conditioned zirconia

See answer and justification for Q1099 →

Q1100 ADVANCED · Block C

A working cast chips at fine marginal details; the technician admits adding excess water to dental stone for easier pouring. What is the likely material consequence?

A Review delayed pouring of dimensionally unstable alginate

B Review excessive water-to-powder ratio as a cause of low strength

C Attribute every inaccurate crown solely to the technician

D Review insufficient strength and expansion control in die material selection

E Assess the accuracy of impression, disinfection, cast and lab stages systematically

See answer and justification for Q1100 →

Q1101 ADVANCED · Block C

A series of base-metal castings are tight despite careful waxing, and the investment expansion used is intended for lower-shrinkage noble alloys. What should be reviewed?

A	Review excessive water-to-powder ratio as a cause of low strength
B	Attribute every inaccurate crown solely to the technician
C	Review delayed pouring of dimensionally unstable alginate
D	Review insufficient strength and expansion control in die material selection
E	Assess the accuracy of impression, disinfection, cast and lab stages systematically

See answer and justification for Q1101 →

Q1102 ADVANCED · Block C

An alginate preliminary impression is stored overnight with visible shrinkage and a distorted tray rim before the master cast is poured. What is the dominant preventable source?

A	Assess the accuracy of impression, disinfection, cast and lab stages systematically
B	Review insufficient strength and expansion control in die material selection
C	Attribute every inaccurate crown solely to the technician
D	Review delayed pouring of dimensionally unstable alginate
E	Review excessive water-to-powder ratio as a cause of low strength

See answer and justification for Q1102 →

Q1160 ADVANCED · Block C

During adhesive cementation of a glass-ceramic onlay, gingival bleeding contaminates the prepared enamel just before bonding. What should happen?

A	Re-evaluate rubber-dam isolation and re-treat contaminated enamel before bonding
B	Use a compatible zirconia primer after controlled surface preparation
C	Leave a known contaminated bonding interface without correction
D	Select a different ceramic solely to overcome uncontrolled haemorrhage
E	Use silane on a correctly etched silica-based surface

See answer and justification for Q1160 →

Q1161 ADVANCED · Block C

A zirconia restoration has been trial-fitted and salivary contamination occurred. The fit is acceptable and controlled air-abrasion procedures are available. What chemistry is relevant?

A	Use silane on a correctly etched silica-based surface
B	Use a compatible zirconia primer after controlled surface preparation
C	Re-evaluate rubber-dam isolation and re-treat contaminated enamel before bonding
D	Select a different ceramic solely to overcome uncontrolled haemorrhage
E	Leave a known contaminated bonding interface without correction

See answer and justification for Q1161 →

Q1162 ADVANCED · Block C

A feldspathic porcelain veneer is prepared under the correct glass-ceramic etching protocol with excellent isolation and sound enamel. Which coupling step is appropriate?

A	Re-evaluate rubber-dam isolation and re-treat contaminated enamel before bonding
B	Select a different ceramic solely to overcome uncontrolled haemorrhage
C	Leave a known contaminated bonding interface without correction
D	Use a compatible zirconia primer after controlled surface preparation
E	Use silane on a correctly etched silica-based surface

See answer and justification for Q1162 →

Q1209 ADVANCED · Block C

A vital upper central incisor has a moderate labial enamel defect, broad enamel margins and high aesthetic demand. Occlusion is favourable and remaining tissue supports bonding.
Which material or strategy is most appropriate?

A	Select monolithic zirconia where high strength and limited clearance dominate
B	Use a metal-ceramic crown where adequate clearance and predictable retention exist
C	Delay definitive indirect treatment until the high-risk disease state is stabilised
D	Use a bonded direct composite for repairable, minimally invasive restoration
E	Select a glass ceramic with enamel bonding after conservative preparation

See answer and justification for Q1209 →

Q1210 ADVANCED · Block C

A root-filled posterior molar needs cuspal coverage. Interocclusal clearance is constrained and parafunctional load is high; adhesive enamel margins are limited.
Which material or strategy is most appropriate?

A	Use a metal-ceramic crown where adequate clearance and predictable retention exist
B	Use a bonded direct composite for repairable, minimally invasive restoration
C	Select monolithic zirconia where high strength and limited clearance dominate
D	Delay definitive indirect treatment until the high-risk disease state is stabilised
E	Select a glass ceramic with enamel bonding after conservative preparation

See answer and justification for Q1210 →

Q1211 ADVANCED · Block C

A vital premolar has a small-to-moderate chipped cusp, largely intact tooth structure, and the patient values repairability and cost. Moisture control is satisfactory.
Which material or strategy is most appropriate?

A	Select monolithic zirconia where high strength and limited clearance dominate
B	Delay definitive indirect treatment until the high-risk disease state is stabilised
C	Use a bonded direct composite for repairable, minimally invasive restoration
D	Use a metal-ceramic crown where adequate clearance and predictable retention exist
E	Select a glass ceramic with enamel bonding after conservative preparation

See answer and justification for Q1211 →

Q1212 ADVANCED · Block C

A medically stable patient has a heavily restored posterior abutment, adequate axial height, suitable periodontal support, and enough space for a retentive conventional restoration.
Which material or strategy is most appropriate?

A	Use a metal-ceramic crown where adequate clearance and predictable retention exist
B	Select a glass ceramic with enamel bonding after conservative preparation
C	Delay definitive indirect treatment until the high-risk disease state is stabilised
D	Select monolithic zirconia where high strength and limited clearance dominate
E	Use a bonded direct composite for repairable, minimally invasive restoration

See answer and justification for Q1212 →

Q1213 ADVANCED · Block C

Several full-arch elastomeric impressions are reproducibly short across the arch. The thin plastic tray flexes when seated and released, while mixing and setting times were correct.
What is the most relevant cause or management change?

A	Investigate latex-associated inhibition of addition-silicone setting
B	Correct inadequate wash relief before repeating a two-stage PVS impression
C	Avoid prolonged immersion of a polyether impression and follow the manufacturer’s disinfection protocol
D	Use a mucostatic or selective-pressure technique for mobile ridge tissue
E	Repeat the impression promptly after correcting tray flexure and material support

See answer and justification for Q1213 →

Q1214 ADVANCED · Block C

A two-stage putty-wash record shows inaccurate crown preparations. The set putty was reseated without relief and visibly displaced almost all wash material.
What is the most relevant cause or management change?

A	Use a mucostatic or selective-pressure technique for mobile ridge tissue
B	Investigate latex-associated inhibition of addition-silicone setting
C	Repeat the impression promptly after correcting tray flexure and material support
D	Avoid prolonged immersion of a polyether impression and follow the manufacturer’s disinfection protocol
E	Correct inadequate wash relief before repeating a two-stage PVS impression

See answer and justification for Q1214 →

Q1215 ADVANCED · Block C

A polyether impression is immersed in disinfectant for hours before casting, and the completed die is dimensionally inaccurate.
What is the most relevant cause or management change?

A	Correct inadequate wash relief before repeating a two-stage PVS impression
B	Use a mucostatic or selective-pressure technique for mobile ridge tissue
C	Repeat the impression promptly after correcting tray flexure and material support
D	Avoid prolonged immersion of a polyether impression and follow the manufacturer’s disinfection protocol
E	Investigate latex-associated inhibition of addition-silicone setting

See answer and justification for Q1215 →

Q1216 ADVANCED · Block C

An addition-silicone wash remains tacky only at a margin manipulated with latex gloves, whereas the remainder sets normally.
What is the most relevant cause or management change?

A	Investigate latex-associated inhibition of addition-silicone setting
B	Use a mucostatic or selective-pressure technique for mobile ridge tissue
C	Avoid prolonged immersion of a polyether impression and follow the manufacturer’s disinfection protocol
D	Repeat the impression promptly after correcting tray flexure and material support
E	Correct inadequate wash relief before repeating a two-stage PVS impression

See answer and justification for Q1216 →

Q1217 ADVANCED · Block C

A thin lithium-disilicate veneer will be adhesively cemented to an enamel-rich anterior preparation. The fitting surface is a silica-based glass ceramic.
Which bonding pathway is best supported?

A	Control moisture and preparation form before selecting an adhesive protocol
B	A suitable metal primer and surface conditioning for a base-metal restoration
C	Air abrasion and phosphate-monomer chemistry for zirconia
D	Hydrofluoric acid conditioning and silane for suitable silica-based glass ceramic
E	Selective enamel etching with a suitable adhesive for direct resin composite

See answer and justification for Q1217 →

Q1218 ADVANCED · Block C

A monolithic zirconia crown has modest retention and requires resin bonding after extraoral surface preparation. No glass phase is available for conventional HF etching.
Which bonding pathway is best supported?

A	A suitable metal primer and surface conditioning for a base-metal restoration
B	Hydrofluoric acid conditioning and silane for suitable silica-based glass ceramic
C	Air abrasion and phosphate-monomer chemistry for zirconia
D	Selective enamel etching with a suitable adhesive for direct resin composite
E	Control moisture and preparation form before selecting an adhesive protocol

See answer and justification for Q1218 →

Q1219 ADVANCED · Block C

A Class IV direct composite restoration has broad sound enamel margins and good rubber-dam isolation. The preparation is conservative.
Which bonding pathway is best supported?

A	Selective enamel etching with a suitable adhesive for direct resin composite
B	Air abrasion and phosphate-monomer chemistry for zirconia
C	Hydrofluoric acid conditioning and silane for suitable silica-based glass ceramic
D	A suitable metal primer and surface conditioning for a base-metal restoration
E	Control moisture and preparation form before selecting an adhesive protocol

See answer and justification for Q1219 →

Q1220 ADVANCED · Block C

A base-metal resin-bonded retainer has debonded; the design and occlusion are sound. Laboratory evaluation confirms a contaminated alloy fitting surface.
Which bonding pathway is best supported?

A	Control moisture and preparation form before selecting an adhesive protocol
B	Air abrasion and phosphate-monomer chemistry for zirconia
C	Hydrofluoric acid conditioning and silane for suitable silica-based glass ceramic
D	A suitable metal primer and surface conditioning for a base-metal restoration
E	Selective enamel etching with a suitable adhesive for direct resin composite

See answer and justification for Q1220 →

Q1221 ADVANCED · Block C

Study casts are friable and abrade easily after a technician added excessive water to make stone flow better. The impression is otherwise accurate.
What should be changed first in the fabrication workflow?

A	Select investment expansion matched to the casting alloy and technique
B	Review digital scan margins, processing and milling limits before remaking
C	Invest wax patterns promptly to minimise time-dependent distortion
D	Correct the water-to-powder ratio and controlled spatulation for gypsum
E	Use a higher-strength, appropriate die stone for precision fixed restorations

See answer and justification for Q1221 →

Q1222 ADVANCED · Block C

A die for a thin-margin crown repeatedly fractures when finishing the wax pattern. Ordinary mounting plaster was used.
What should be changed first in the fabrication workflow?

A	Use a higher-strength, appropriate die stone for precision fixed restorations
B	Select investment expansion matched to the casting alloy and technique
C	Correct the water-to-powder ratio and controlled spatulation for gypsum
D	Review digital scan margins, processing and milling limits before remaking
E	Invest wax patterns promptly to minimise time-dependent distortion

See answer and justification for Q1222 →

Q1223 ADVANCED · Block C

A series of cobalt-chromium cast restorations are consistently too tight. The phosphate-bonded investment was selected from a protocol designed for low-shrinkage noble alloys.
What should be changed first in the fabrication workflow?

A	Invest wax patterns promptly to minimise time-dependent distortion
B	Use a higher-strength, appropriate die stone for precision fixed restorations
C	Select investment expansion matched to the casting alloy and technique
D	Correct the water-to-powder ratio and controlled spatulation for gypsum
E	Review digital scan margins, processing and milling limits before remaking

See answer and justification for Q1223 →

Q1224 ADVANCED · Block C

An inlay wax pattern is shaped accurately on Monday but not invested until Friday. Finished castings no longer seat despite accurate dies.
What should be changed first in the fabrication workflow?

A	Select investment expansion matched to the casting alloy and technique
B	Use a higher-strength, appropriate die stone for precision fixed restorations
C	Review digital scan margins, processing and milling limits before remaking
D	Correct the water-to-powder ratio and controlled spatulation for gypsum
E	Invest wax patterns promptly to minimise time-dependent distortion

See answer and justification for Q1224 →

Q1225 ADVANCED · Block C

An upper acrylic complete denture fractures along its midline after four years of normal use. It has never been dropped, and there is a prominent anterior contact on the opposing dentition.
What is the most likely failure mechanism or best correction?

A	Correct porosity associated with inadequate processing control
B	Eliminate stress concentration from an excessively deep frenum notch
C	Investigate dimensional distortion caused by hot-water cleaning
D	Diagnose fatigue from recurrent flexure and assess occlusal support
E	Diagnose impact fracture and consider a tougher denture base

See answer and justification for Q1225 →

Q1226 ADVANCED · Block C

A patient with tremor repeatedly drops a well-fitting acrylic denture. Fractures occur at different sites immediately after falls.
What is the most likely failure mechanism or best correction?

A	Investigate dimensional distortion caused by hot-water cleaning
B	Eliminate stress concentration from an excessively deep frenum notch
C	Diagnose fatigue from recurrent flexure and assess occlusal support
D	Correct porosity associated with inadequate processing control
E	Diagnose impact fracture and consider a tougher denture base

See answer and justification for Q1226 →

Q1227 ADVANCED · Block C

An acrylic denture base has voids concentrated in its thickest sections after a shortened, high-temperature curing cycle.
What is the most likely failure mechanism or best correction?

A	Correct porosity associated with inadequate processing control
B	Investigate dimensional distortion caused by hot-water cleaning
C	Diagnose fatigue from recurrent flexure and assess occlusal support
D	Diagnose impact fracture and consider a tougher denture base
E	Eliminate stress concentration from an excessively deep frenum notch

See answer and justification for Q1227 →

Q1228 ADVANCED · Block C

A new denture repeatedly cracks at a very deep, sharp labial frenum notch. The base elsewhere is appropriately thick and the bite is acceptable.
What is the most likely failure mechanism or best correction?

A	Correct porosity associated with inadequate processing control
B	Eliminate stress concentration from an excessively deep frenum notch
C	Diagnose fatigue from recurrent flexure and assess occlusal support
D	Investigate dimensional distortion caused by hot-water cleaning
E	Diagnose impact fracture and consider a tougher denture base

See answer and justification for Q1228 →

Q1229 ADVANCED · Block C

A periodontal patient has deep interproximal embrasures and narrow finish-line recesses. Several impression washes tear on removal despite sound mixing and tray support.
Which material choice or handling principle is decisive?

A	Evaluate scan accessibility and margin capture rather than assuming digital accuracy
B	Choose a suitable rigid mucostatic material only when undercuts do not preclude removal
C	Use an elastomer with adequate tear resistance and controlled bulk across deep undercuts
D	Avoid a condensation material if dimensional delay before pouring cannot be controlled
E	Select a more hydrophilic impression approach when moist surfaces cannot be completely dried

See answer and justification for Q1229 →

Q1230 ADVANCED · Block C

A patient has a difficult-to-isolate moist sulcus despite appropriate retraction. Bubbles recur in a highly hydrophobic wash even after operator technique has improved.
Which material choice or handling principle is decisive?

A	Select a more hydrophilic impression approach when moist surfaces cannot be completely dried
B	Choose a suitable rigid mucostatic material only when undercuts do not preclude removal
C	Use an elastomer with adequate tear resistance and controlled bulk across deep undercuts
D	Avoid a condensation material if dimensional delay before pouring cannot be controlled
E	Evaluate scan accessibility and margin capture rather than assuming digital accuracy

See answer and justification for Q1230 →

Q1231 ADVANCED · Block C

An edentulous maxillary ridge is displaceable and has no clinically relevant undercuts. A windowed technique is planned for a mucostatic record.
Which material choice or handling principle is decisive?

A	Choose a suitable rigid mucostatic material only when undercuts do not preclude removal
B	Avoid a condensation material if dimensional delay before pouring cannot be controlled
C	Select a more hydrophilic impression approach when moist surfaces cannot be completely dried
D	Evaluate scan accessibility and margin capture rather than assuming digital accuracy
E	Use an elastomer with adequate tear resistance and controlled bulk across deep undercuts

See answer and justification for Q1231 →

Q1232 ADVANCED · Block C

A long-span impression will travel to a distant laboratory; it cannot be cast for several days. The clinician can choose between elastomers with different by-products and storage stability.
Which material choice or handling principle is decisive?

A	Evaluate scan accessibility and margin capture rather than assuming digital accuracy
B	Avoid a condensation material if dimensional delay before pouring cannot be controlled
C	Choose a suitable rigid mucostatic material only when undercuts do not preclude removal
D	Use an elastomer with adequate tear resistance and controlled bulk across deep undercuts
E	Select a more hydrophilic impression approach when moist surfaces cannot be completely dried

See answer and justification for Q1232 →

Q1233 ADVANCED · Block C

A new elastomeric impression of several undercut teeth is removed extremely slowly, and the cast shows subtle distortion not present in earlier trials.
Which clinical implication best follows from the material behaviour?

A	Provide sufficient ceramic thickness or choose a suitable high-strength design
B	Select repairable additive restoration when frequent maintenance is expected
C	Check modulus, length, diameter and support before changing an RPD clasp alloy
D	Use a compatible coefficient of thermal expansion for metal-ceramic systems
E	Prioritise elastic recovery and predictable removal from undercuts

See answer and justification for Q1233 →

Q1234 ADVANCED · Block C

An attractive posterior glass-ceramic crown has chipped twice in a heavy occlusion; preparation clearance was insufficient and the patient prefers a conservative redesign.
Which clinical implication best follows from the material behaviour?

A	Use a compatible coefficient of thermal expansion for metal-ceramic systems
B	Check modulus, length, diameter and support before changing an RPD clasp alloy
C	Select repairable additive restoration when frequent maintenance is expected
D	Prioritise elastic recovery and predictable removal from undercuts
E	Provide sufficient ceramic thickness or choose a suitable high-strength design

See answer and justification for Q1234 →

Q1235 ADVANCED · Block C

A cobalt-chromium clasp fractures after repeated bending by the patient. The tooth undercut was large and the clasp arm unusually short.
Which clinical implication best follows from the material behaviour?

A	Check modulus, length, diameter and support before changing an RPD clasp alloy
B	Provide sufficient ceramic thickness or choose a suitable high-strength design
C	Prioritise elastic recovery and predictable removal from undercuts
D	Select repairable additive restoration when frequent maintenance is expected
E	Use a compatible coefficient of thermal expansion for metal-ceramic systems

See answer and justification for Q1235 →

Q1236 ADVANCED · Block C

Porcelain repeatedly cracks off a metal substructure after laboratory cooling, with no evidence of trauma or marginal misfit.
Which clinical implication best follows from the material behaviour?

A	Select repairable additive restoration when frequent maintenance is expected
B	Provide sufficient ceramic thickness or choose a suitable high-strength design
C	Prioritise elastic recovery and predictable removal from undercuts
D	Use a compatible coefficient of thermal expansion for metal-ceramic systems
E	Check modulus, length, diameter and support before changing an RPD clasp alloy

See answer and justification for Q1236 →

BLOCK D · Pharmacology, pain control, emergencies and the medically compromised patient (42)

Q1028 ADVANCED · Block D

During crown preparation a patient develops new central crushing chest pain, pallor and sweating, not resolving with their usual medication. What action takes priority?

A	Arrange a routine restorative review in six months
B	Stop elective treatment and seek emergency medical assessment
C	Prescribe empirical antibiotics without examination
D	Consult the medical team and plan care once systemic disease is stabilised
E	Continue treatment after reassuring the patient

See answer and justification for Q1028 →

Q1029 ADVANCED · Block D

A person with advanced uncontrolled heart failure and recent hospital admission requests elective multi-unit fixed prosthodontics. They are currently comfortable but medically unstable. What should happen?

A	Arrange a routine restorative review in six months
B	Stop elective treatment and seek emergency medical assessment
C	Consult the medical team and plan care once systemic disease is stabilised
D	Continue treatment after reassuring the patient
E	Prescribe empirical antibiotics without examination

See answer and justification for Q1029 →

Q1030 ADVANCED · Block D

A 68-year-old with an unchanged history of well-controlled hypertension and good functional capacity attends for simple denture adjustment and is asymptomatic. What is most proportionate?

A	Continue treatment after reassuring the patient
B	Prescribe empirical antibiotics without examination
C	Consult the medical team and plan care once systemic disease is stabilised
D	Arrange a routine restorative review in six months
E	Stop elective treatment and seek emergency medical assessment

See answer and justification for Q1030 →

Q1031 ADVANCED · Block D

A patient taking apixaban is scheduled for a non-invasive digital impression and try-in of a crown. What is the best approach?

A	Choose a non-surgical prosthetic option after shared decision-making
B	Assess procedure-specific bleeding risk and follow current anticoagulant guidance
C	Seek medical advice for a high-risk procedure when the indication and drug plan are uncertain
D	Proceed to extensive elective surgery without a medicines history
E	Stop all antiplatelets or anticoagulants for one week without medical input

See answer and justification for Q1031 →

Q1032 ADVANCED · Block D

A patient on dual antiplatelet therapy after recent coronary stenting requests elective alveolar surgery solely to improve an otherwise functional denture. What option deserves priority?

A	Seek medical advice for a high-risk procedure when the indication and drug plan are uncertain
B	Choose a non-surgical prosthetic option after shared decision-making
C	Assess procedure-specific bleeding risk and follow current anticoagulant guidance
D	Proceed to extensive elective surgery without a medicines history
E	Stop all antiplatelets or anticoagulants for one week without medical input

See answer and justification for Q1032 →

Q1033 ADVANCED · Block D

A patient on a DOAC needs several invasive procedures and has impaired renal function. You cannot determine the bleeding category or appropriate timing from the available history. What is the safest next step?

A	Proceed to extensive elective surgery without a medicines history
B	Assess procedure-specific bleeding risk and follow current anticoagulant guidance
C	Stop all antiplatelets or anticoagulants for one week without medical input
D	Seek medical advice for a high-risk procedure when the indication and drug plan are uncertain
E	Choose a non-surgical prosthetic option after shared decision-making

See answer and justification for Q1033 →

Q1034 ADVANCED · Block D

A 69-year-old reports “bone injections” but cannot identify the drug or its indication. She asks for removal of roots and immediate implants. What should happen before a definitive plan?

A	Assume all antiresorptive users have the same MRONJ risk
B	Confirm medication, indication, dose and exposure before assigning MRONJ risk
C	Arrange specialist input for a complex, high-risk oncology-related case
D	Prefer non-surgical prosthetic treatment where this meets the patient’s aims
E	Plan an elective surgical implant as a routine procedure without risk discussion

See answer and justification for Q1034 →

Q1035 ADVANCED · Block D

A patient receiving high-dose intravenous antiresorptive treatment for metastatic cancer has severe ridge atrophy and requests implant surgery. What is the most appropriate pathway?

A	Prefer non-surgical prosthetic treatment where this meets the patient’s aims
B	Confirm medication, indication, dose and exposure before assigning MRONJ risk
C	Assume all antiresorptive users have the same MRONJ risk
D	Arrange specialist input for a complex, high-risk oncology-related case
E	Plan an elective surgical implant as a routine procedure without risk discussion

See answer and justification for Q1035 →

Q1036 ADVANCED · Block D

An asymptomatic older patient on long-term antiresorptive therapy has a loose complete denture. A careful impression and reline will restore comfort. What is preferable initially?

A	Plan an elective surgical implant as a routine procedure without risk discussion
B	Prefer non-surgical prosthetic treatment where this meets the patient’s aims
C	Confirm medication, indication, dose and exposure before assigning MRONJ risk
D	Assume all antiresorptive users have the same MRONJ risk
E	Arrange specialist input for a complex, high-risk oncology-related case

See answer and justification for Q1036 →

Q1103 ADVANCED · Block D

Immediately after latex exposure, a patient develops generalised urticaria, wheeze, hypotension and facial swelling. What is the priority?

A	Prescribe oral antibiotics after discharge
B	Stop the procedure and initiate the anaphylaxis emergency algorithm
C	Assess positional syncope and manage the airway, positioning and vital signs
D	Continue treatment because the patient may be anxious
E	Assess a local anaesthetic systemic toxicity scenario and initiate emergency support

See answer and justification for Q1103 →

Q1104 ADVANCED · Block D

A patient becomes pale and light-headed while the chair is upright during a prolonged impression. They briefly lose consciousness but rapidly recover when reclined; there is no rash, wheeze or focal deficit. What is most likely?

A	Prescribe oral antibiotics after discharge
B	Stop the procedure and initiate the anaphylaxis emergency algorithm
C	Continue treatment because the patient may be anxious
D	Assess positional syncope and manage the airway, positioning and vital signs
E	Assess a local anaesthetic systemic toxicity scenario and initiate emergency support

See answer and justification for Q1104 →

Q1105 ADVANCED · Block D

Shortly after an inadvertently rapid large local anaesthetic injection, a patient develops tinnitus, circumoral tingling, agitation and a seizure. What syndrome needs recognition?

A	Assess a local anaesthetic systemic toxicity scenario and initiate emergency support
B	Prescribe oral antibiotics after discharge
C	Continue treatment because the patient may be anxious
D	Stop the procedure and initiate the anaphylaxis emergency algorithm
E	Assess positional syncope and manage the airway, positioning and vital signs

See answer and justification for Q1105 →

Q1106 ADVANCED · Block D

A 54-year-old is to start head-and-neck radiotherapy in three weeks. Several molars are unrestorable and likely to require extraction. What is the dental priority?

A	Delay cancer therapy indefinitely until prosthodontic rehabilitation is complete
B	Plan routine removable repair without concern when no invasive treatment is needed
C	Coordinate pre-radiotherapy dental assessment with the oncology team
D	Seek specialist guidance where post-radiotherapy extractions could threaten bone health
E	Prioritise full-mouth crowns despite a rapidly approaching oncological deadline

See answer and justification for Q1106 →

Q1107 ADVANCED · Block D

A patient irradiated to the mandible three years ago has a grossly carious lower molar and is requesting extraction. The radiotherapy dose map is unavailable. What is the most appropriate next step?

A	Coordinate pre-radiotherapy dental assessment with the oncology team
B	Plan routine removable repair without concern when no invasive treatment is needed
C	Seek specialist guidance where post-radiotherapy extractions could threaten bone health
D	Prioritise full-mouth crowns despite a rapidly approaching oncological deadline
E	Delay cancer therapy indefinitely until prosthodontic rehabilitation is complete

See answer and justification for Q1107 →

Q1108 ADVANCED · Block D

A patient receiving chemotherapy has an otherwise intact complete denture with one chipped acrylic incisal edge; there is no mucosal injury or active infection. What is reasonable?

A	Seek specialist guidance where post-radiotherapy extractions could threaten bone health
B	Plan routine removable repair without concern when no invasive treatment is needed
C	Delay cancer therapy indefinitely until prosthodontic rehabilitation is complete
D	Coordinate pre-radiotherapy dental assessment with the oncology team
E	Prioritise full-mouth crowns despite a rapidly approaching oncological deadline

See answer and justification for Q1108 →

Q1163 ADVANCED · Block D

During a lengthy denture impression a patient becomes unresponsive, cyanosed and breathes abnormally. What is the immediate priority?

A	Arrange urgent assessment for a suspicious malignant lesion
B	Activate emergency escalation for rapidly spreading infection and airway symptoms
C	Stop treatment and assess airway, breathing and circulation
D	Provide symptom-led conservative dental care after excluding red flags
E	Continue cementing and arrange a later medical appointment

See answer and justification for Q1163 →

Q1164 ADVANCED · Block D

A patient with lower molar sepsis develops increasing trismus, drooling and difficulty swallowing over several hours. What should be done?

A	Activate emergency escalation for rapidly spreading infection and airway symptoms
B	Stop treatment and assess airway, breathing and circulation
C	Arrange urgent assessment for a suspicious malignant lesion
D	Continue cementing and arrange a later medical appointment
E	Provide symptom-led conservative dental care after excluding red flags

See answer and justification for Q1164 →

Q1165 ADVANCED · Block D

An asymptomatic patient has mild transient denture sore spots which resolve after removing an obvious overextended flange. No ulcer persists and no systemic symptoms are present. What is reasonable?

A	Activate emergency escalation for rapidly spreading infection and airway symptoms
B	Arrange urgent assessment for a suspicious malignant lesion
C	Provide symptom-led conservative dental care after excluding red flags
D	Stop treatment and assess airway, breathing and circulation
E	Continue cementing and arrange a later medical appointment

See answer and justification for Q1165 →

Q1237 ADVANCED · Block D

During preparation, an anxious patient becomes pale, sweaty and briefly loses consciousness after the injection. Pulse is slow and they regain consciousness on laying supine.
What is the safest immediate action?

A	Treat a suspected hypoglycaemic event promptly according to consciousness and local emergency protocol
B	Arrange immediate hospital assessment for suspected acute coronary syndrome
C	Activate the emergency response for suspected anaphylaxis and give first-line emergency treatment
D	Stop treatment, place the patient appropriately, assess and manage a probable vasovagal episode
E	Stop the procedure and assess possible local anaesthetic toxicity with emergency support

See answer and justification for Q1237 →

Q1238 ADVANCED · Block D

A diabetic patient who skipped lunch becomes sweaty, confused and tremulous during a long appointment; capillary glucose is markedly low and they can swallow.
What is the safest immediate action?

A	Arrange immediate hospital assessment for suspected acute coronary syndrome
B	Stop the procedure and assess possible local anaesthetic toxicity with emergency support
C	Stop treatment, place the patient appropriately, assess and manage a probable vasovagal episode
D	Treat a suspected hypoglycaemic event promptly according to consciousness and local emergency protocol
E	Activate the emergency response for suspected anaphylaxis and give first-line emergency treatment

See answer and justification for Q1238 →

Q1239 ADVANCED · Block D

Minutes after a new drug is administered, a patient develops wheeze, tongue swelling, generalized urticaria and falling blood pressure.
What is the safest immediate action?

A	Stop the procedure and assess possible local anaesthetic toxicity with emergency support
B	Arrange immediate hospital assessment for suspected acute coronary syndrome
C	Stop treatment, place the patient appropriately, assess and manage a probable vasovagal episode
D	Treat a suspected hypoglycaemic event promptly according to consciousness and local emergency protocol
E	Activate the emergency response for suspected anaphylaxis and give first-line emergency treatment

See answer and justification for Q1239 →

Q1240 ADVANCED · Block D

A 62-year-old develops crushing central chest pain radiating to the arm, nausea and breathlessness during treatment; symptoms persist despite stopping.
What is the safest immediate action?

A	Stop the procedure and assess possible local anaesthetic toxicity with emergency support
B	Stop treatment, place the patient appropriately, assess and manage a probable vasovagal episode
C	Activate the emergency response for suspected anaphylaxis and give first-line emergency treatment
D	Arrange immediate hospital assessment for suspected acute coronary syndrome
E	Treat a suspected hypoglycaemic event promptly according to consciousness and local emergency protocol

See answer and justification for Q1240 →

Q1241 ADVANCED · Block D

A patient with coronary stents takes dual antiplatelet therapy and needs a straightforward extraction of a hopeless premolar. They have no other bleeding disorder.
Which medication-related strategy is most appropriate?

A	Liaise with the relevant prescriber about significant drug interactions before treatment
B	Identify an interacting antifungal and avoid unsafe concomitant use
C	Assess bleeding risk and use local haemostasis rather than routinely stopping antiplatelet treatment
D	Obtain relevant INR information before invasive care in a warfarin-treated patient
E	Consult the current DOAC dental guidance and tailor timing to procedure bleeding risk

See answer and justification for Q1241 →

Q1242 ADVANCED · Block D

A patient takes apixaban and needs a procedure judged at higher bleeding risk than routine subgingival debridement. Renal and medical history are available.
Which medication-related strategy is most appropriate?

A	Identify an interacting antifungal and avoid unsafe concomitant use
B	Consult the current DOAC dental guidance and tailor timing to procedure bleeding risk
C	Obtain relevant INR information before invasive care in a warfarin-treated patient
D	Assess bleeding risk and use local haemostasis rather than routinely stopping antiplatelet treatment
E	Liaise with the relevant prescriber about significant drug interactions before treatment

See answer and justification for Q1242 →

Q1243 ADVANCED · Block D

A warfarin patient has persistent denture-associated candidosis and requests miconazole oral gel. Their anticoagulation is currently stable.
Which medication-related strategy is most appropriate?

A	Identify an interacting antifungal and avoid unsafe concomitant use
B	Obtain relevant INR information before invasive care in a warfarin-treated patient
C	Liaise with the relevant prescriber about significant drug interactions before treatment
D	Assess bleeding risk and use local haemostasis rather than routinely stopping antiplatelet treatment
E	Consult the current DOAC dental guidance and tailor timing to procedure bleeding risk

See answer and justification for Q1243 →

Q1244 ADVANCED · Block D

A warfarin-treated patient needs extraction of a non-restorable tooth, but their anticoagulation status has not been checked and the last INR is months old.
Which medication-related strategy is most appropriate?

A	Liaise with the relevant prescriber about significant drug interactions before treatment
B	Consult the current DOAC dental guidance and tailor timing to procedure bleeding risk
C	Identify an interacting antifungal and avoid unsafe concomitant use
D	Obtain relevant INR information before invasive care in a warfarin-treated patient
E	Assess bleeding risk and use local haemostasis rather than routinely stopping antiplatelet treatment

See answer and justification for Q1244 →

Q1245 ADVANCED · Block D

A patient receiving high-dose intravenous antiresorptive therapy for metastatic cancer has a symptomatic but potentially restorable molar. Extraction is not the only option.
How should antiresorptive-related risk influence care?

A	Explain a comparatively low but non-zero risk in a stable osteoporosis-dose patient
B	Provide clear preventive advice and regular review without unnecessary dental intervention
C	Investigate exposed necrotic bone and coordinate appropriate specialist assessment
D	Coordinate dental stabilisation and medical liaison before initiating high-risk therapy
E	Prioritise retention and non-surgical management where feasible in high-risk oncology therapy

See answer and justification for Q1245 →

Q1246 ADVANCED · Block D

A patient is due to start oncology antiresorptive therapy in several weeks and has several septic non-restorable roots and active periodontitis.
How should antiresorptive-related risk influence care?

A	Explain a comparatively low but non-zero risk in a stable osteoporosis-dose patient
B	Prioritise retention and non-surgical management where feasible in high-risk oncology therapy
C	Coordinate dental stabilisation and medical liaison before initiating high-risk therapy
D	Provide clear preventive advice and regular review without unnecessary dental intervention
E	Investigate exposed necrotic bone and coordinate appropriate specialist assessment

See answer and justification for Q1246 →

Q1247 ADVANCED · Block D

A woman taking oral alendronate for osteoporosis needs a clinically indicated extraction and has no other major risk factors.
How should antiresorptive-related risk influence care?

A	Investigate exposed necrotic bone and coordinate appropriate specialist assessment
B	Provide clear preventive advice and regular review without unnecessary dental intervention
C	Prioritise retention and non-surgical management where feasible in high-risk oncology therapy
D	Explain a comparatively low but non-zero risk in a stable osteoporosis-dose patient
E	Coordinate dental stabilisation and medical liaison before initiating high-risk therapy

See answer and justification for Q1247 →

Q1248 ADVANCED · Block D

A patient on antiresorptives presents with exposed mandibular bone for more than eight weeks and associated discomfort.
How should antiresorptive-related risk influence care?

A	Investigate exposed necrotic bone and coordinate appropriate specialist assessment
B	Explain a comparatively low but non-zero risk in a stable osteoporosis-dose patient
C	Coordinate dental stabilisation and medical liaison before initiating high-risk therapy
D	Provide clear preventive advice and regular review without unnecessary dental intervention
E	Prioritise retention and non-surgical management where feasible in high-risk oncology therapy

See answer and justification for Q1248 →

Q1249 ADVANCED · Block D

A patient requests four implant crowns but reports recent unstable angina, increasing exertional chest symptoms and no medical review.
Which modification to prosthodontic treatment is most justified?

A	Provide enhanced xerostomia prevention and review medicines with the prescriber
B	Defer elective extensive care until unstable cardiovascular or systemic disease has been assessed
C	Develop a shared-care pathway for patients with complex oncological sequelae
D	Adapt appointment structure and maintenance for cognitive and physical limitations
E	Coordinate periodontal control and glycaemic-risk assessment before invasive rehabilitation

See answer and justification for Q1249 →

Q1250 ADVANCED · Block D

An older adult with mild dementia can tolerate short visits, forgets cleaning instructions and relies on a carer for denture handling.
Which modification to prosthodontic treatment is most justified?

A	Provide enhanced xerostomia prevention and review medicines with the prescriber
B	Defer elective extensive care until unstable cardiovascular or systemic disease has been assessed
C	Develop a shared-care pathway for patients with complex oncological sequelae
D	Adapt appointment structure and maintenance for cognitive and physical limitations
E	Coordinate periodontal control and glycaemic-risk assessment before invasive rehabilitation

See answer and justification for Q1250 →

Q1251 ADVANCED · Block D

A 61-year-old takes several xerogenic medicines and has rapidly worsening root caries around crown margins while planning a new bridge.
Which modification to prosthodontic treatment is most justified?

A	Develop a shared-care pathway for patients with complex oncological sequelae
B	Adapt appointment structure and maintenance for cognitive and physical limitations
C	Provide enhanced xerostomia prevention and review medicines with the prescriber
D	Defer elective extensive care until unstable cardiovascular or systemic disease has been assessed
E	Coordinate periodontal control and glycaemic-risk assessment before invasive rehabilitation

See answer and justification for Q1251 →

Q1252 ADVANCED · Block D

A patient with poorly controlled diabetes and active periodontal suppuration requests definitive implant-supported full-arch work.
Which modification to prosthodontic treatment is most justified?

A	Adapt appointment structure and maintenance for cognitive and physical limitations
B	Provide enhanced xerostomia prevention and review medicines with the prescriber
C	Develop a shared-care pathway for patients with complex oncological sequelae
D	Defer elective extensive care until unstable cardiovascular or systemic disease has been assessed
E	Coordinate periodontal control and glycaemic-risk assessment before invasive rehabilitation

See answer and justification for Q1252 →

Q1253 ADVANCED · Block D

A patient develops a generalised tonic-clonic seizure during a long visit. They have known epilepsy, and the team is tempted to restrain the limbs.
What is the most appropriate response during the incident?

A	Treat a severe asthma exacerbation with emergency inhaled therapy and escalation
B	Respond to suspected stroke with immediate emergency activation and time documentation
C	Stop treatment and manage suspected myocardial ischaemia with emergency services
D	Manage a seizure safely, protect from injury and activate help according to duration or recovery
E	Assess the airway and provide emergency support after aspiration of an instrument

See answer and justification for Q1253 →

Q1254 ADVANCED · Block D

A patient develops wheezing, cough and marked breathlessness shortly after sitting back. Their own reliever inhaler has given little relief.
What is the most appropriate response during the incident?

A	Treat a severe asthma exacerbation with emergency inhaled therapy and escalation
B	Stop treatment and manage suspected myocardial ischaemia with emergency services
C	Assess the airway and provide emergency support after aspiration of an instrument
D	Manage a seizure safely, protect from injury and activate help according to duration or recovery
E	Respond to suspected stroke with immediate emergency activation and time documentation

See answer and justification for Q1254 →

Q1255 ADVANCED · Block D

During a crown preparation a patient develops crushing chest pain, cold sweats and dyspnoea, with symptoms not relieved by stopping work.
What is the most appropriate response during the incident?

A	Respond to suspected stroke with immediate emergency activation and time documentation
B	Treat a severe asthma exacerbation with emergency inhaled therapy and escalation
C	Manage a seizure safely, protect from injury and activate help according to duration or recovery
D	Stop treatment and manage suspected myocardial ischaemia with emergency services
E	Assess the airway and provide emergency support after aspiration of an instrument

See answer and justification for Q1255 →

Q1256 ADVANCED · Block D

The patient suddenly has facial drooping, unilateral arm weakness and dysphasia during the impression appointment. Onset was observed five minutes ago.
What is the most appropriate response during the incident?

A	Stop treatment and manage suspected myocardial ischaemia with emergency services
B	Assess the airway and provide emergency support after aspiration of an instrument
C	Treat a severe asthma exacerbation with emergency inhaled therapy and escalation
D	Manage a seizure safely, protect from injury and activate help according to duration or recovery
E	Respond to suspected stroke with immediate emergency activation and time documentation

See answer and justification for Q1256 →

Q1257 ADVANCED · Block D

A patient with early dementia can explain their denture options but struggles to retain information unless it is presented slowly with written prompts.
Which consent or treatment-safety action is most appropriate?

A	Respect a capacitous patient’s informed refusal and document alternatives
B	Reassess anxiety and offer graded or supported treatment before elective care
C	Offer accessible communication and the option of a suitable interpreter
D	Arrange best-interests decision making when the patient lacks decision-specific capacity
E	Establish decision-specific capacity and support understanding before proceeding

See answer and justification for Q1257 →

Q1258 ADVANCED · Block D

A patient speaks limited English and nods through a complex implant-consent discussion but cannot explain the risks back to the clinician.
Which consent or treatment-safety action is most appropriate?

A	Respect a capacitous patient’s informed refusal and document alternatives
B	Offer accessible communication and the option of a suitable interpreter
C	Establish decision-specific capacity and support understanding before proceeding
D	Reassess anxiety and offer graded or supported treatment before elective care
E	Arrange best-interests decision making when the patient lacks decision-specific capacity

See answer and justification for Q1258 →

Q1259 ADVANCED · Block D

A competent adult declines a clinically indicated bridge after understanding benefits, risks, alternatives and the consequences of no treatment.
Which consent or treatment-safety action is most appropriate?

A	Respect a capacitous patient’s informed refusal and document alternatives
B	Establish decision-specific capacity and support understanding before proceeding
C	Offer accessible communication and the option of a suitable interpreter
D	Arrange best-interests decision making when the patient lacks decision-specific capacity
E	Reassess anxiety and offer graded or supported treatment before elective care

See answer and justification for Q1259 →

Q1260 ADVANCED · Block D

An adult with advanced cognitive impairment cannot understand or weigh the decision about replacing a severely painful prosthesis despite maximal support.
Which consent or treatment-safety action is most appropriate?

A	Reassess anxiety and offer graded or supported treatment before elective care
B	Respect a capacitous patient’s informed refusal and document alternatives
C	Establish decision-specific capacity and support understanding before proceeding
D	Arrange best-interests decision making when the patient lacks decision-specific capacity
E	Offer accessible communication and the option of a suitable interpreter

See answer and justification for Q1260 →

BLOCK E · Assessment, diagnosis, imaging, classifications and planning (54)

Q1001 ADVANCED · Block E

A root-treated mandibular molar has persistent symptoms despite conventional assessment. Two diagnostic periapicals at different angles remain inconclusive about a suspected untreated canal; the result would change retreatment planning. What is the best next step?

A	Refer urgently on a suspected malignancy pathway
B	Obtain a limited-field CBCT after confirming its diagnostic benefit and justification
C	Request a panoramic radiograph to replace the intraoral examination
D	Repeat a high-quality periapical radiograph using a second horizontal angulation
E	Begin irreversible treatment without additional investigation

See answer and justification for Q1001 →

Q1002 ADVANCED · Block E

Before an indirect restoration, a premolar has an equivocal periapical radiolucency on a blurred, elongated film. No comparison view has been taken and there are no red flags. What is most appropriate?

A	Obtain a limited-field CBCT after confirming its diagnostic benefit and justification
B	Repeat a high-quality periapical radiograph using a second horizontal angulation
C	Request a panoramic radiograph to replace the intraoral examination
D	Refer urgently on a suspected malignancy pathway
E	Begin irreversible treatment without additional investigation

See answer and justification for Q1002 →

Q1003 ADVANCED · Block E

A 67-year-old with a newly discovered persistent indurated lateral tongue ulcer and unilateral cervical lymphadenopathy asks for a CBCT before the denture is remade. What is the priority?

A	Repeat a high-quality periapical radiograph using a second horizontal angulation
B	Refer urgently on a suspected malignancy pathway
C	Obtain a limited-field CBCT after confirming its diagnostic benefit and justification
D	Begin irreversible treatment without additional investigation
E	Request a panoramic radiograph to replace the intraoral examination

See answer and justification for Q1003 →

Q1004 ADVANCED · Block E

A root-treated mandibular molar with a crown has isolated 9-mm narrow probing on one root, pain on release, and a J-shaped radiolucency. The tooth repeatedly develops a sinus. What is the most appropriate next step?

A	Investigate for a vertical root fracture and discuss extraction if confirmed
B	Investigate periodontal and endodontic causes before assigning prognosis
C	Place a post-and-core and full crown immediately
D	Restore the tooth without addressing a persistent active periodontal problem
E	Provide an indirect restoration after managing the limited coronal defect

See answer and justification for Q1004 →

Q1005 ADVANCED · Block E

An asymptomatic vital premolar has a fractured marginal ridge with sound peripheral enamel, no deep caries, stable periodontium, and a favourable occlusion. Which approach is most proportionate?

A	Provide an indirect restoration after managing the limited coronal defect
B	Investigate for a vertical root fracture and discuss extraction if confirmed
C	Restore the tooth without addressing a persistent active periodontal problem
D	Place a post-and-core and full crown immediately
E	Investigate periodontal and endodontic causes before assigning prognosis

See answer and justification for Q1005 →

Q1006 ADVANCED · Block E

A heavily restored first molar has a 6-mm pocket at its distal surface, bleeding, furcation involvement and an uncertain periapical change; the patient requests a new crown immediately. What should happen first?

A	Restore the tooth without addressing a persistent active periodontal problem
B	Provide an indirect restoration after managing the limited coronal defect
C	Place a post-and-core and full crown immediately
D	Investigate periodontal and endodontic causes before assigning prognosis
E	Investigate for a vertical root fracture and discuss extraction if confirmed

See answer and justification for Q1006 →

Q1007 ADVANCED · Block E

A healthy 72-year-old with sound anterior and premolar occlusion is missing both mandibular second molars, has no symptoms and is satisfied with chewing. What is the best recommendation?

A	Accept a shortened dental arch and monitor function
B	Plan multiple implants immediately without disease stabilisation
C	Offer a resin-bonded adhesive bridge after occlusal assessment
D	Use a simple hygienic removable partial denture
E	Extract remaining strategic teeth and provide complete dentures

See answer and justification for Q1007 →

Q1008 ADVANCED · Block E

An 81-year-old with reduced hand dexterity and a few missing posterior teeth reports difficulty retaining food but has stable, maintainable abutments and requests a removable appliance. What option fits best?

A	Offer a resin-bonded adhesive bridge after occlusal assessment
B	Accept a shortened dental arch and monitor function
C	Use a simple hygienic removable partial denture
D	Extract remaining strategic teeth and provide complete dentures
E	Plan multiple implants immediately without disease stabilisation

See answer and justification for Q1008 →

Q1009 ADVANCED · Block E

A 24-year-old has a single missing maxillary lateral incisor, sound adjacent unrestored teeth, good hygiene, and is keen to avoid adjacent tooth preparation or surgery. Orthodontic space is stable. What should be explored?

A	Offer a resin-bonded adhesive bridge after occlusal assessment
B	Plan multiple implants immediately without disease stabilisation
C	Use a simple hygienic removable partial denture
D	Extract remaining strategic teeth and provide complete dentures
E	Accept a shortened dental arch and monitor function

See answer and justification for Q1009 →

Q1010 ADVANCED · Block E

A 49-year-old requests an extensive fixed bridge. Active generalised periodontitis, plaque scores above 60% and recurrent bleeding remain despite initial advice. Abutment prognosis is uncertain. What is the priority?

A	Stabilise periodontal disease and reassess prognosis before final prosthodontics
B	Arrange specialist multidisciplinary input before attempting full-mouth reconstruction
C	Proceed with definitive rehabilitation now because the patient has paid a deposit
D	Extract every mobile tooth without further assessment
E	Provide provisional, repairable restorations while medical and behavioural risks are addressed

See answer and justification for Q1010 →

Q1011 ADVANCED · Block E

A 37-year-old with severe generalised tooth wear, an unstable occlusion, multiple missing posterior units and suspected altered vertical dimension requests full-mouth crowns. What is the most appropriate planning step?

A	Proceed with definitive rehabilitation now because the patient has paid a deposit
B	Stabilise periodontal disease and reassess prognosis before final prosthodontics
C	Extract every mobile tooth without further assessment
D	Arrange specialist multidisciplinary input before attempting full-mouth reconstruction
E	Provide provisional, repairable restorations while medical and behavioural risks are addressed

See answer and justification for Q1011 →

Q1012 ADVANCED · Block E

A 58-year-old has severe xerostomia after medication changes and several active root caries lesions. Cosmetic replacement of anterior restorations is requested. What is the best initial strategy?

A	Provide provisional, repairable restorations while medical and behavioural risks are addressed
B	Arrange specialist multidisciplinary input before attempting full-mouth reconstruction
C	Extract every mobile tooth without further assessment
D	Proceed with definitive rehabilitation now because the patient has paid a deposit
E	Stabilise periodontal disease and reassess prognosis before final prosthodontics

See answer and justification for Q1012 →

Q1109 ADVANCED · Block E

A recently traumatised vital-looking incisor has no response to a single cold test at its first post-injury visit. There is no discoloration, sinus or radiographic pathology. What is the best interpretation?

A	Use a second sensibility test and diagnostic imaging before assigning pulpal diagnosis
B	Assess periodontal charting and maintenance history before assigning abutment prognosis
C	Seek urgent investigation of a persistent unusual soft-tissue lesion
D	Investigate cracks and occlusal loading with appropriate tests
E	Assume a single negative cold test proves necrosis

See answer and justification for Q1109 →

Q1110 ADVANCED · Block E

A crowned molar has sharp pain on release from biting, normal periapical imaging and no deep pocket. What investigation best tests a likely cause?

A	Investigate cracks and occlusal loading with appropriate tests
B	Seek urgent investigation of a persistent unusual soft-tissue lesion
C	Assess periodontal charting and maintenance history before assigning abutment prognosis
D	Assume a single negative cold test proves necrosis
E	Use a second sensibility test and diagnostic imaging before assigning pulpal diagnosis

See answer and justification for Q1110 →

Q1111 ADVANCED · Block E

A new patient has unexplained persistent ulceration on the ventral tongue with induration; teeth and prosthesis have no source of trauma. What is the priority?

A	Investigate cracks and occlusal loading with appropriate tests
B	Assume a single negative cold test proves necrosis
C	Use a second sensibility test and diagnostic imaging before assigning pulpal diagnosis
D	Assess periodontal charting and maintenance history before assigning abutment prognosis
E	Seek urgent investigation of a persistent unusual soft-tissue lesion

See answer and justification for Q1111 →

Q1166 ADVANCED · Block E

A heavily restored molar proposed as a bridge abutment has deep tenderness to percussion and an uncertain endodontic history. No current intraoral image exists. What is appropriate?

A	Repeat CBCT serially to monitor a stable uncomplicated bridge
B	Request CBCT only when a defined 3-D clinical question will alter management
C	Manage without imaging if there is no new clinical indication
D	Use a justified periapical radiograph to evaluate a specific abutment concern
E	Use a panoramic image as a substitute for all detailed periapical assessment

See answer and justification for Q1166 →

Q1167 ADVANCED · Block E

A proposed implant will be close to the mandibular canal, and existing images cannot define the cross-sectional bone envelope. How should further imaging be chosen?

A	Request CBCT only when a defined 3-D clinical question will alter management
B	Manage without imaging if there is no new clinical indication
C	Repeat CBCT serially to monitor a stable uncomplicated bridge
D	Use a panoramic image as a substitute for all detailed periapical assessment
E	Use a justified periapical radiograph to evaluate a specific abutment concern

See answer and justification for Q1167 →

Q1168 ADVANCED · Block E

A patient with a well-fitting bridge, stable periodontal health and no pain asks for another CT solely to “check everything.” The last imaging is adequate and recent. What is best?

A	Repeat CBCT serially to monitor a stable uncomplicated bridge
B	Manage without imaging if there is no new clinical indication
C	Use a justified periapical radiograph to evaluate a specific abutment concern
D	Request CBCT only when a defined 3-D clinical question will alter management
E	Use a panoramic image as a substitute for all detailed periapical assessment

See answer and justification for Q1168 →

Q1261 ADVANCED · Block E

An asymptomatic heavily restored premolar has an equivocal root-filling quality and a localised apical sign on examination. No recent periapical image exists.
Which investigation is best justified at this stage?

A	Use focused periapical imaging after assessing the clinical question
B	Consider limited-field CBCT only after 2-D imaging fails to answer a decision-critical 3-D question
C	Compare existing diagnostic photographs, mounted casts and occlusal records
D	Use a panoramic survey to assess the distribution of a partially dentate arch where justified
E	Arrange appropriate specialist evaluation without delaying for routine prosthodontic images

See answer and justification for Q1261 →

Q1262 ADVANCED · Block E

A patient presents with multiple missing teeth and complex patterns of retained roots in both arches. Clinical examination suggests a need for overview imaging in planning.
Which investigation is best justified at this stage?

A	Arrange appropriate specialist evaluation without delaying for routine prosthodontic images
B	Consider limited-field CBCT only after 2-D imaging fails to answer a decision-critical 3-D question
C	Use a panoramic survey to assess the distribution of a partially dentate arch where justified
D	Compare existing diagnostic photographs, mounted casts and occlusal records
E	Use focused periapical imaging after assessing the clinical question

See answer and justification for Q1262 →

Q1263 ADVANCED · Block E

A previously root-treated molar remains symptomatic. Multiple technically adequate parallax periapicals do not identify a suspected missed canal that would alter referral and retreatment.
Which investigation is best justified at this stage?

A	Compare existing diagnostic photographs, mounted casts and occlusal records
B	Use a panoramic survey to assess the distribution of a partially dentate arch where justified
C	Consider limited-field CBCT only after 2-D imaging fails to answer a decision-critical 3-D question
D	Use focused periapical imaging after assessing the clinical question
E	Arrange appropriate specialist evaluation without delaying for routine prosthodontic images

See answer and justification for Q1263 →

Q1264 ADVANCED · Block E

A patient develops unilateral altered sensation, an enlarging submucosal mass and unexplained weight loss. They ask for an OPG before referral.
Which investigation is best justified at this stage?

A	Arrange appropriate specialist evaluation without delaying for routine prosthodontic images
B	Use focused periapical imaging after assessing the clinical question
C	Use a panoramic survey to assess the distribution of a partially dentate arch where justified
D	Compare existing diagnostic photographs, mounted casts and occlusal records
E	Consider limited-field CBCT only after 2-D imaging fails to answer a decision-critical 3-D question

See answer and justification for Q1264 →

Q1265 ADVANCED · Block E

A molar has pain on biting a cotton roll that improves on release. Routine radiographs look normal and there is no isolated deep pocket.
Which diagnostic interpretation is most appropriate?

A	Assess periodontal activity and overall support before bridge planning
B	Evaluate occlusal load and parafunction as contributors to restoration fracture
C	Establish pulpal and periapical status before assigning a restorative prognosis
D	Assess possible non-odontogenic pain and refer when findings do not fit a tooth cause
E	Investigate a possible cracked tooth with targeted tests before choosing a restoration

See answer and justification for Q1265 →

Q1266 ADVANCED · Block E

A crown-treated premolar has intermittent spontaneous pain and an equivocal cold test; there is a large deep restoration but no reliable pulpal diagnosis.
Which diagnostic interpretation is most appropriate?

A	Assess periodontal activity and overall support before bridge planning
B	Evaluate occlusal load and parafunction as contributors to restoration fracture
C	Establish pulpal and periapical status before assigning a restorative prognosis
D	Assess possible non-odontogenic pain and refer when findings do not fit a tooth cause
E	Investigate a possible cracked tooth with targeted tests before choosing a restoration

See answer and justification for Q1266 →

Q1267 ADVANCED · Block E

A planned bridge abutment has mobility, bleeding on probing and angular bone loss, although the proposed bridge has adequate space.
Which diagnostic interpretation is most appropriate?

A	Assess periodontal activity and overall support before bridge planning
B	Evaluate occlusal load and parafunction as contributors to restoration fracture
C	Establish pulpal and periapical status before assigning a restorative prognosis
D	Assess possible non-odontogenic pain and refer when findings do not fit a tooth cause
E	Investigate a possible cracked tooth with targeted tests before choosing a restoration

See answer and justification for Q1267 →

Q1268 ADVANCED · Block E

A patient reports constant diffuse facial burning that is not provoked by chewing, has persisted after two root treatments and shows no corresponding dental lesion.
Which diagnostic interpretation is most appropriate?

A	Assess periodontal activity and overall support before bridge planning
B	Evaluate occlusal load and parafunction as contributors to restoration fracture
C	Establish pulpal and periapical status before assigning a restorative prognosis
D	Assess possible non-odontogenic pain and refer when findings do not fit a tooth cause
E	Investigate a possible cracked tooth with targeted tests before choosing a restoration

See answer and justification for Q1268 →

Q1269 ADVANCED · Block E

A root-treated premolar has extensive coronal breakdown, but caries removal reveals 2 mm of sound supragingival tooth structure around most of the circumference. Periodontal support is good.
Which factor most strongly changes the tooth’s prognosis?

A	A likely vertical root fracture despite otherwise acceptable crown margins
B	Demonstrable sound circumferential ferrule with adequate retention after caries removal
C	A favourable root form and periodontium despite significant but restorable coronal loss
D	Uncontrolled periodontal support loss and mobility affecting strategic abutment value
E	Inability to maintain plaque control around a proposed complex restoration

See answer and justification for Q1269 →

Q1270 ADVANCED · Block E

A crowned molar has a recurrent narrow isolated 10-mm pocket, a J-shaped radiolucency and pain on release after years of apparently satisfactory crown margins.
Which factor most strongly changes the tooth’s prognosis?

A	Demonstrable sound circumferential ferrule with adequate retention after caries removal
B	A likely vertical root fracture despite otherwise acceptable crown margins
C	Uncontrolled periodontal support loss and mobility affecting strategic abutment value
D	A favourable root form and periodontium despite significant but restorable coronal loss
E	Inability to maintain plaque control around a proposed complex restoration

See answer and justification for Q1270 →

Q1271 ADVANCED · Block E

An otherwise sound premolar is proposed as the terminal abutment for a long fixed span, but has progressive stage III periodontitis with marked mobility and poor attendance.
Which factor most strongly changes the tooth’s prognosis?

A	Demonstrable sound circumferential ferrule with adequate retention after caries removal
B	Inability to maintain plaque control around a proposed complex restoration
C	A likely vertical root fracture despite otherwise acceptable crown margins
D	Uncontrolled periodontal support loss and mobility affecting strategic abutment value
E	A favourable root form and periodontium despite significant but restorable coronal loss

See answer and justification for Q1271 →

Q1272 ADVANCED · Block E

A partially dentate patient requests a sophisticated fixed solution but repeatedly declines interdental cleaning or supportive periodontal appointments and has recurrent plaque-related disease.
Which factor most strongly changes the tooth’s prognosis?

A	A favourable root form and periodontium despite significant but restorable coronal loss
B	Demonstrable sound circumferential ferrule with adequate retention after caries removal
C	Uncontrolled periodontal support loss and mobility affecting strategic abutment value
D	A likely vertical root fracture despite otherwise acceptable crown margins
E	Inability to maintain plaque control around a proposed complex restoration

See answer and justification for Q1272 →

Q1273 ADVANCED · Block E

A patient requests an expensive upper bridge but has several active root caries lesions, generalised bleeding on probing and no established preventive routine.
What is the most appropriate sequence of treatment?

A	Refer for multidisciplinary planning of a complex full-mouth rehabilitation
B	Stabilise caries and periodontal disease, then reassess definitive restorative options
C	Perform a diagnostic wax-up/trial to assess the desired occlusal changes before irreversible work
D	Maintain function without restoration while monitoring a stable situation
E	Provide an interim removable prosthesis while establishing prognosis and patient priorities

See answer and justification for Q1273 →

Q1274 ADVANCED · Block E

A patient has multiple doubtful-prognosis strategic teeth after trauma but urgently needs an acceptable appearance for work. Prognosis cannot be finalised for several weeks.
What is the most appropriate sequence of treatment?

A	Provide an interim removable prosthesis while establishing prognosis and patient priorities
B	Stabilise caries and periodontal disease, then reassess definitive restorative options
C	Refer for multidisciplinary planning of a complex full-mouth rehabilitation
D	Perform a diagnostic wax-up/trial to assess the desired occlusal changes before irreversible work
E	Maintain function without restoration while monitoring a stable situation

See answer and justification for Q1274 →

Q1275 ADVANCED · Block E

A patient seeks a substantial increase in vertical dimension for widespread tooth wear, but there is uncertainty about aesthetics, speech and adaptation.
What is the most appropriate sequence of treatment?

A	Perform a diagnostic wax-up/trial to assess the desired occlusal changes before irreversible work
B	Stabilise caries and periodontal disease, then reassess definitive restorative options
C	Provide an interim removable prosthesis while establishing prognosis and patient priorities
D	Refer for multidisciplinary planning of a complex full-mouth rehabilitation
E	Maintain function without restoration while monitoring a stable situation

See answer and justification for Q1275 →

Q1276 ADVANCED · Block E

A patient has severe hypodontia, complex skeletal discrepancies and several congenitally absent teeth requiring interdisciplinary restorative and orthodontic decisions.
What is the most appropriate sequence of treatment?

A	Provide an interim removable prosthesis while establishing prognosis and patient priorities
B	Stabilise caries and periodontal disease, then reassess definitive restorative options
C	Maintain function without restoration while monitoring a stable situation
D	Perform a diagnostic wax-up/trial to assess the desired occlusal changes before irreversible work
E	Refer for multidisciplinary planning of a complex full-mouth rehabilitation

See answer and justification for Q1276 →

Q1277 ADVANCED · Block E

A patient with a functional shortened arch asks whether missing molars must be replaced and worries about cost. Their chewing is satisfactory and they do not want implants.
Which response best reconciles clinical need and patient preference?

A	Provide balanced options, including doing nothing, and elicit the patient’s priorities
B	Adapt the plan to affordability while retaining safety and minimum maintenance standards
C	Defer elective irreversible treatment if expectations remain demonstrably unrealistic
D	Seek an independent specialist opinion for a highly complex elective reconstruction
E	Use a provisional aesthetic mock-up to resolve a realistic but uncertain visual objective

See answer and justification for Q1277 →

Q1278 ADVANCED · Block E

A patient wants the maxillary incisors slightly longer but cannot visualise the proposal and worries about phonetics. Teeth are healthy and a conservative additive approach is feasible.
Which response best reconciles clinical need and patient preference?

A	Defer elective irreversible treatment if expectations remain demonstrably unrealistic
B	Use a provisional aesthetic mock-up to resolve a realistic but uncertain visual objective
C	Seek an independent specialist opinion for a highly complex elective reconstruction
D	Provide balanced options, including doing nothing, and elicit the patient’s priorities
E	Adapt the plan to affordability while retaining safety and minimum maintenance standards

See answer and justification for Q1278 →

Q1279 ADVANCED · Block E

A patient with healthy natural teeth demands extensive extraction and full-mouth fixed implants to achieve a photographically perfect smile. They reject discussion of limitations.
Which response best reconciles clinical need and patient preference?

A	Defer elective irreversible treatment if expectations remain demonstrably unrealistic
B	Adapt the plan to affordability while retaining safety and minimum maintenance standards
C	Provide balanced options, including doing nothing, and elicit the patient’s priorities
D	Seek an independent specialist opinion for a highly complex elective reconstruction
E	Use a provisional aesthetic mock-up to resolve a realistic but uncertain visual objective

See answer and justification for Q1279 →

Q1280 ADVANCED · Block E

A patient cannot afford implants, has stable periodontal health, and prioritises a comfortable, cleanable removable solution. They can attend follow-up reliably.
Which response best reconciles clinical need and patient preference?

A	Seek an independent specialist opinion for a highly complex elective reconstruction
B	Defer elective irreversible treatment if expectations remain demonstrably unrealistic
C	Adapt the plan to affordability while retaining safety and minimum maintenance standards
D	Use a provisional aesthetic mock-up to resolve a realistic but uncertain visual objective
E	Provide balanced options, including doing nothing, and elicit the patient’s priorities

See answer and justification for Q1280 →

Q1281 ADVANCED · Block E

A young adult has slowly progressing smooth palatal enamel loss and frequent acidic drinks, with stable posterior contacts and no bite change.
Which interpretation guides the next management decision?

A	Investigate a discrepancy between retruded contact and habitual intercuspation
B	Assess whether symptoms reflect muscle-related TMD before considering occlusal alteration
C	Record extent and progression of erosive wear and establish risk control before restoration
D	Assess reduced restorative space versus true loss of vertical dimension
E	Use diagnostic records to assess occlusal stability after loss of posterior support

See answer and justification for Q1281 →

Q1282 ADVANCED · Block E

An adult has severe local anterior wear but posterior teeth meet consistently and face height appears maintained. There is no free space for incisal addition.
Which interpretation guides the next management decision?

A	Record extent and progression of erosive wear and establish risk control before restoration
B	Assess reduced restorative space versus true loss of vertical dimension
C	Assess whether symptoms reflect muscle-related TMD before considering occlusal alteration
D	Investigate a discrepancy between retruded contact and habitual intercuspation
E	Use diagnostic records to assess occlusal stability after loss of posterior support

See answer and justification for Q1282 →

Q1283 ADVANCED · Block E

The first tooth contact on gentle guided closure is on an angled cusp, followed by a noticeable lateral slide into maximal intercuspation. The patient has no joint pain.
Which interpretation guides the next management decision?

A	Assess reduced restorative space versus true loss of vertical dimension
B	Assess whether symptoms reflect muscle-related TMD before considering occlusal alteration
C	Investigate a discrepancy between retruded contact and habitual intercuspation
D	Record extent and progression of erosive wear and establish risk control before restoration
E	Use diagnostic records to assess occlusal stability after loss of posterior support

See answer and justification for Q1283 →

Q1284 ADVANCED · Block E

A patient has diffuse masseter tenderness, morning jaw ache and normal radiographs; they request full-mouth bite reorganisation to cure presumed TMD.
Which interpretation guides the next management decision?

A	Use diagnostic records to assess occlusal stability after loss of posterior support
B	Investigate a discrepancy between retruded contact and habitual intercuspation
C	Assess whether symptoms reflect muscle-related TMD before considering occlusal alteration
D	Record extent and progression of erosive wear and establish risk control before restoration
E	Assess reduced restorative space versus true loss of vertical dimension

See answer and justification for Q1284 →

Q1285 ADVANCED · Block E

A long-span bridge is being planned, but baseline periodontal charting is absent. The proposed distal abutment bleeds and has mobility that may change the plan.
What is the most appropriate next diagnostic step?

A	Obtain an adequate clinical periodontal chart and targeted radiographs
B	Refer an unexplained non-healing oral lesion urgently for assessment
C	Use pulpal sensibility and percussion tests alongside imaging
D	Request appropriate blood tests or medical liaison for suspected systemic disease
E	Examine fit, border extension and occlusion of the existing prosthesis dynamically

See answer and justification for Q1285 →

Q1286 ADVANCED · Block E

A restored premolar has bite sensitivity and intermittent thermal symptoms with an uncertain pulp response. A periapical image shows no definite lesion.
What is the most appropriate next diagnostic step?

A	Refer an unexplained non-healing oral lesion urgently for assessment
B	Use pulpal sensibility and percussion tests alongside imaging
C	Examine fit, border extension and occlusion of the existing prosthesis dynamically
D	Request appropriate blood tests or medical liaison for suspected systemic disease
E	Obtain an adequate clinical periodontal chart and targeted radiographs

See answer and justification for Q1286 →

Q1287 ADVANCED · Block E

A denture wearer has recurrent angular cheilitis, glossitis, fatigue and poor diet after significant weight loss, without an obvious local prosthesis cause.
What is the most appropriate next diagnostic step?

A	Refer an unexplained non-healing oral lesion urgently for assessment
B	Examine fit, border extension and occlusion of the existing prosthesis dynamically
C	Request appropriate blood tests or medical liaison for suspected systemic disease
D	Use pulpal sensibility and percussion tests alongside imaging
E	Obtain an adequate clinical periodontal chart and targeted radiographs

See answer and justification for Q1287 →

Q1288 ADVANCED · Block E

An elderly patient has a firm red-and-white patch on the lateral tongue that persists after removal of the adjacent rough acrylic edge.
What is the most appropriate next diagnostic step?

A	Refer an unexplained non-healing oral lesion urgently for assessment
B	Request appropriate blood tests or medical liaison for suspected systemic disease
C	Use pulpal sensibility and percussion tests alongside imaging
D	Examine fit, border extension and occlusion of the existing prosthesis dynamically
E	Obtain an adequate clinical periodontal chart and targeted radiographs

See answer and justification for Q1288 →

Q1289 ADVANCED · Block E

An active older adult with stable premolars has lost a molar and now describes a clear new chewing deficit on that side, despite unchanged oral hygiene.
Which factor should dominate the decision to replace missing teeth?

A	Plaque control and periodontal maintenance capacity around a proposed bridge
B	Aesthetic concern in a visible region despite acceptable mastication
C	Occlusal space and opposing tooth position after long-standing edentulism
D	Demonstrated chewing impairment and the patient’s preference for replacement
E	Overall medical fitness and the likely burden of follow-up

See answer and justification for Q1289 →

Q1290 ADVANCED · Block E

A patient wants a long fixed bridge but has uncontrolled periodontitis, poor dexterity and repeatedly misses maintenance appointments.
Which factor should dominate the decision to replace missing teeth?

A	Overall medical fitness and the likely burden of follow-up
B	Occlusal space and opposing tooth position after long-standing edentulism
C	Aesthetic concern in a visible region despite acceptable mastication
D	Demonstrated chewing impairment and the patient’s preference for replacement
E	Plaque control and periodontal maintenance capacity around a proposed bridge

See answer and justification for Q1290 →

Q1291 ADVANCED · Block E

A patient with an intact posterior bite lost a lateral incisor and is distressed by the visible space; they otherwise eat normally.
Which factor should dominate the decision to replace missing teeth?

A	Overall medical fitness and the likely burden of follow-up
B	Occlusal space and opposing tooth position after long-standing edentulism
C	Demonstrated chewing impairment and the patient’s preference for replacement
D	Aesthetic concern in a visible region despite acceptable mastication
E	Plaque control and periodontal maintenance capacity around a proposed bridge

See answer and justification for Q1291 →

Q1292 ADVANCED · Block E

A first molar was lost years ago and the opposing molar has over-erupted into the space. The patient requests a direct implant crown without occlusal treatment.
Which factor should dominate the decision to replace missing teeth?

A	Plaque control and periodontal maintenance capacity around a proposed bridge
B	Occlusal space and opposing tooth position after long-standing edentulism
C	Overall medical fitness and the likely burden of follow-up
D	Demonstrated chewing impairment and the patient’s preference for replacement
E	Aesthetic concern in a visible region despite acceptable mastication

See answer and justification for Q1292 →

Q1293 ADVANCED · Block E

An anterior composite has chipped once after trauma. Enamel is sound, isolation is straightforward and occlusal examination identifies a discrete edge-to-edge contact that can be managed.
Which plan is the most proportionate?

A	Reconsider prosthesis design and strategic abutments before repeating a failed long-span bridge
B	Repair the local restoration after addressing the specific failure mechanism
C	Replace a repeatedly failing restoration with a conservative indirect option when tissue permits
D	Refer a multi-factorial full-mouth case for interdisciplinary assessment
E	Stabilise a high-risk oral environment before undertaking definitive rehabilitation

See answer and justification for Q1293 →

Q1294 ADVANCED · Block E

A large anterior direct composite has fractured cohesively several times despite controlled occlusion, good isolation and abundant surrounding enamel. The defect is mainly labio-incisal.
Which plan is the most proportionate?

A	Stabilise a high-risk oral environment before undertaking definitive rehabilitation
B	Reconsider prosthesis design and strategic abutments before repeating a failed long-span bridge
C	Refer a multi-factorial full-mouth case for interdisciplinary assessment
D	Replace a repeatedly failing restoration with a conservative indirect option when tissue permits
E	Repair the local restoration after addressing the specific failure mechanism

See answer and justification for Q1294 →

Q1295 ADVANCED · Block E

A long fixed bridge has debonded twice. One abutment has marginal bone loss, the span is extended and the occlusal scheme loads the cantilever heavily.
Which plan is the most proportionate?

A	Refer a multi-factorial full-mouth case for interdisciplinary assessment
B	Stabilise a high-risk oral environment before undertaking definitive rehabilitation
C	Repair the local restoration after addressing the specific failure mechanism
D	Replace a repeatedly failing restoration with a conservative indirect option when tissue permits
E	Reconsider prosthesis design and strategic abutments before repeating a failed long-span bridge

See answer and justification for Q1295 →

Q1296 ADVANCED · Block E

A patient seeks many new crowns but currently has high root-caries activity, irregular attendance, frequent acidic intake and gingival inflammation.
Which plan is the most proportionate?

A	Repair the local restoration after addressing the specific failure mechanism
B	Replace a repeatedly failing restoration with a conservative indirect option when tissue permits
C	Reconsider prosthesis design and strategic abutments before repeating a failed long-span bridge
D	Stabilise a high-risk oral environment before undertaking definitive rehabilitation
E	Refer a multi-factorial full-mouth case for interdisciplinary assessment

See answer and justification for Q1296 →

BLOCK F · Removable partial dentures (54)

Q1037 ADVANCED · Block F

A lower Kennedy Class I RPD is planned on healthy premolar abutments. Distal-extension saddles are expected to rotate toward the mucosa under load. Which design principle best protects abutments?

A	Rely on clasp retention without occlusal support
B	Use distal rests with rigid circumferential clasps to maximise torque
C	Extract healthy premolars to convert the saddle to a complete denture
D	Reconsider the design after surveying and evaluating abutment contours
E	Use mesial rests and an appropriate stress-releasing retainer assembly

See answer and justification for Q1037 →

Q1038 ADVANCED · Block F

A proposed distal-extension framework uses very short guiding surfaces and an I-bar that would cross a high soft-tissue undercut. What is the correct next step?

A	Rely on clasp retention without occlusal support
B	Extract healthy premolars to convert the saddle to a complete denture
C	Use mesial rests and an appropriate stress-releasing retainer assembly
D	Reconsider the design after surveying and evaluating abutment contours
E	Use distal rests with rigid circumferential clasps to maximise torque

See answer and justification for Q1038 →

Q1039 ADVANCED · Block F

A tooth-supported bounded saddle is planned with a sound distal abutment and favourable contours. What feature should guide its support?

A	Extract healthy premolars to convert the saddle to a complete denture
B	Reconsider the design after surveying and evaluating abutment contours
C	Use mesial rests and an appropriate stress-releasing retainer assembly
D	Use distal rests with rigid circumferential clasps to maximise torque
E	Rely on clasp retention without occlusal support

See answer and justification for Q1039 →

Q1040 ADVANCED · Block F

A partially dentate mandible has a 9-mm functional distance between gingival margins and the highest position of the floor of mouth, with healthy periodontal tissues. Which connector is normally most suitable?

A	Use a lingual plate when clearance is inadequate and appropriate coverage is possible
B	Use a lingual bar when functional clearance is adequate
C	Consider a labial bar or surgical correction where lingual tori preclude conventional designs
D	Use an unrelieved bar pressing on the mobile floor of mouth
E	Omit a major connector to reduce plaque

See answer and justification for Q1040 →

Q1041 ADVANCED · Block F

A mandibular RPD is planned in a shallow lingual sulcus with 5-mm functional clearance; remaining incisors require bracing and future addition may be needed. What connector is preferable?

A	Use an unrelieved bar pressing on the mobile floor of mouth
B	Use a lingual plate when clearance is inadequate and appropriate coverage is possible
C	Omit a major connector to reduce plaque
D	Consider a labial bar or surgical correction where lingual tori preclude conventional designs
E	Use a lingual bar when functional clearance is adequate

See answer and justification for Q1041 →

Q1042 ADVANCED · Block F

Large bilateral mandibular lingual tori prevent insertion of a lingual bar and would prevent a lingual plate from being appropriately relieved. What is the design issue?

A	Use a lingual plate when clearance is inadequate and appropriate coverage is possible
B	Consider a labial bar or surgical correction where lingual tori preclude conventional designs
C	Use an unrelieved bar pressing on the mobile floor of mouth
D	Use a lingual bar when functional clearance is adequate
E	Omit a major connector to reduce plaque

See answer and justification for Q1042 →

Q1043 ADVANCED · Block F

A Co-Cr RPD formerly stable has gradually become loose. The clasp tip is distorted and has been repeatedly bent by the patient; rests and base fit are satisfactory. What should be assessed?

A	Remake the entire denture before examining it
B	Identify a missing or worn guide plane/rest and redesign the deficient component
C	Adjust the clasp only after surveying its support, position and fatigue history
D	Provide adhesive as the sole long-term solution
E	Investigate loss of occlusal support and tissue adaptation before clasp tightening

See answer and justification for Q1043 →

Q1044 ADVANCED · Block F

A distal-extension RPD rocks under chewing, though its clasps retain when the patient closes gently. The residual ridge has resorbed and the base no longer contacts fully. What matters first?

A	Adjust the clasp only after surveying its support, position and fatigue history
B	Investigate loss of occlusal support and tissue adaptation before clasp tightening
C	Remake the entire denture before examining it
D	Identify a missing or worn guide plane/rest and redesign the deficient component
E	Provide adhesive as the sole long-term solution

See answer and justification for Q1044 →

Q1045 ADVANCED · Block F

An RPD repeatedly dislodges despite adequately retentive clasp tips; the design lacks an intended guiding plane and the only occlusal rest seats incompletely. What is most helpful?

A	Investigate loss of occlusal support and tissue adaptation before clasp tightening
B	Identify a missing or worn guide plane/rest and redesign the deficient component
C	Adjust the clasp only after surveying its support, position and fatigue history
D	Provide adhesive as the sole long-term solution
E	Remake the entire denture before examining it

See answer and justification for Q1045 →

Q1112 ADVANCED · Block F

A maxillary Kennedy Class I framework rocks away from tissues at the distal saddles when sticky foods are removed. What design principle most directly counters rotational dislodgement?

A	Assess the need for altered-cast or functional impression techniques in distal extension
B	Position indirect retention as far as feasible from the fulcrum line
C	Position all occlusal rests distal to the abutments
D	Provide rigid tooth-borne support for a bounded saddle
E	Rely on direct clasps alone to resist base rotation

See answer and justification for Q1112 →

Q1113 ADVANCED · Block F

A Kennedy Class III space has healthy abutments at each end and good periodontal support. Which framework principle is paramount?

A	Rely on direct clasps alone to resist base rotation
B	Assess the need for altered-cast or functional impression techniques in distal extension
C	Provide rigid tooth-borne support for a bounded saddle
D	Position indirect retention as far as feasible from the fulcrum line
E	Position all occlusal rests distal to the abutments

See answer and justification for Q1113 →

Q1114 ADVANCED · Block F

A lower bilateral distal-extension RPD consistently sinks into compressible mucosa despite accurately fitting casts and sound abutment rests. What should be evaluated?

A	Assess the need for altered-cast or functional impression techniques in distal extension
B	Position all occlusal rests distal to the abutments
C	Position indirect retention as far as feasible from the fulcrum line
D	Provide rigid tooth-borne support for a bounded saddle
E	Rely on direct clasps alone to resist base rotation

See answer and justification for Q1114 →

Q1115 ADVANCED · Block F

A mandibular RPD framework will engage opposing undercuts that prevent simultaneous insertion. What should the clinician do before design approval?

A	Place clasps where most visible because access is easiest
B	Modify tooth contours conservatively where necessary for appropriate guide planes
C	Prefer broad tissue coverage independent of plaque risk
D	Avoid all rest seats to preserve enamel
E	Survey path of insertion and evaluate guide planes and retentive undercuts

See answer and justification for Q1115 →

Q1116 ADVANCED · Block F

A long, tilted premolar in an RPD design has no suitable guiding plane, but enamel is sufficient for conservative reshaping. What may improve the path?

A	Modify tooth contours conservatively where necessary for appropriate guide planes
B	Place clasps where most visible because access is easiest
C	Avoid all rest seats to preserve enamel
D	Prefer broad tissue coverage independent of plaque risk
E	Survey path of insertion and evaluate guide planes and retentive undercuts

See answer and justification for Q1116 →

Q1117 ADVANCED · Block F

A heavily restored posterior abutment has an ideal retentive undercut but no stable occlusal rest seat. What must be prioritised?

A	Modify tooth contours conservatively where necessary for appropriate guide planes
B	Avoid all rest seats to preserve enamel
C	Survey path of insertion and evaluate guide planes and retentive undercuts
D	Prefer broad tissue coverage independent of plaque risk
E	Place clasps where most visible because access is easiest

See answer and justification for Q1117 →

Q1169 ADVANCED · Block F

A complex RPD with tilted abutments fails to seat uniformly and requires force over the surveyed survey lines. What should be reconsidered first?

A	Evaluate a wrought-wire retainer where flexibility or adjustment is needed
B	Use the surveyor to locate a single feasible path of insertion
C	Select an I-bar irrespective of vestibular tissue undercuts
D	Increase clasp engagement until the patient reports forceful retention
E	Choose a rigid cast clasp where geometry and support permit predictable deflection

See answer and justification for Q1169 →

Q1170 ADVANCED · Block F

An RPD requires a retainer on a weakened abutment with unfavourable severe tissue undercut for an I-bar, and adjustment is likely. Which option may be evaluated?

A	Use the surveyor to locate a single feasible path of insertion
B	Evaluate a wrought-wire retainer where flexibility or adjustment is needed
C	Select an I-bar irrespective of vestibular tissue undercuts
D	Choose a rigid cast clasp where geometry and support permit predictable deflection
E	Increase clasp engagement until the patient reports forceful retention

See answer and justification for Q1170 →

Q1171 ADVANCED · Block F

A well-supported bounded saddle has a favourable tooth undercut, adequate height of contour, and strong abutment support. Which clasp strategy could be satisfactory?

A	Use the surveyor to locate a single feasible path of insertion
B	Select an I-bar irrespective of vestibular tissue undercuts
C	Choose a rigid cast clasp where geometry and support permit predictable deflection
D	Increase clasp engagement until the patient reports forceful retention
E	Evaluate a wrought-wire retainer where flexibility or adjustment is needed

See answer and justification for Q1171 →

Q1297 ADVANCED · Block F

A mandibular Kennedy Class I RPD is planned on periodontally stable premolars. The clinician wants to reduce harmful distal torque on the terminal abutments as the saddle settles.
Which component or design choice best addresses the problem?

A	Survey and establish an appropriate path of insertion before designing clasps
B	Place an indirect retainer anterior to the fulcrum line on a suitable abutment
C	Use a mesial occlusal rest on a distal-extension terminal abutment in a suitable stress-releasing design
D	Use a rigid reciprocal element opposite the retentive clasp arm
E	Provide broad denture-base coverage to improve support of a free-end saddle

See answer and justification for Q1297 →

Q1298 ADVANCED · Block F

A unilateral distal-extension RPD repeatedly rotates away from the ridge during sticky-food removal. The direct retainers and saddles are otherwise satisfactory.
Which component or design choice best addresses the problem?

A	Use a rigid reciprocal element opposite the retentive clasp arm
B	Use a mesial occlusal rest on a distal-extension terminal abutment in a suitable stress-releasing design
C	Survey and establish an appropriate path of insertion before designing clasps
D	Provide broad denture-base coverage to improve support of a free-end saddle
E	Place an indirect retainer anterior to the fulcrum line on a suitable abutment

See answer and justification for Q1298 →

Q1299 ADVANCED · Block F

A free-end saddle RPD sinks into the mucosa under chewing despite sound occlusal rests. The denture base is very small and offers poor mucosal support.
Which component or design choice best addresses the problem?

A	Use a rigid reciprocal element opposite the retentive clasp arm
B	Place an indirect retainer anterior to the fulcrum line on a suitable abutment
C	Provide broad denture-base coverage to improve support of a free-end saddle
D	Survey and establish an appropriate path of insertion before designing clasps
E	Use a mesial occlusal rest on a distal-extension terminal abutment in a suitable stress-releasing design

See answer and justification for Q1299 →

Q1300 ADVANCED · Block F

A new cobalt-chromium RPD repeatedly binds on insertion. The master cast was not surveyed and clasp tips were chosen solely from a visual estimate.
Which component or design choice best addresses the problem?

A	Survey and establish an appropriate path of insertion before designing clasps
B	Use a mesial occlusal rest on a distal-extension terminal abutment in a suitable stress-releasing design
C	Use a rigid reciprocal element opposite the retentive clasp arm
D	Place an indirect retainer anterior to the fulcrum line on a suitable abutment
E	Provide broad denture-base coverage to improve support of a free-end saddle

See answer and justification for Q1300 →

Q1301 ADVANCED · Block F

A mandibular Kennedy Class III case has a normal floor-of-mouth depth and ample clearance below the gingival margins, with healthy lingual surfaces.
Which major connector best fits the anatomy and hygiene?

A	Consider a broad palatal plate for poor support in extensive maxillary distal extensions
B	Consider a lingual plate where clearance is inadequate and tooth contours permit
C	Choose a lingual bar where functional depth and gingival clearance are adequate
D	Choose a palatal strap with adequate width and rigidity for a bounded saddle
E	Consider a labial connector where obstructive lingual tori cannot otherwise be accommodated

See answer and justification for Q1301 →

Q1302 ADVANCED · Block F

A lower partial denture is needed where the functional lingual sulcus is shallow and there is insufficient space for a conventional bar. The remaining incisors can support a plate.
Which major connector best fits the anatomy and hygiene?

A	Consider a broad palatal plate for poor support in extensive maxillary distal extensions
B	Consider a lingual plate where clearance is inadequate and tooth contours permit
C	Consider a labial connector where obstructive lingual tori cannot otherwise be accommodated
D	Choose a palatal strap with adequate width and rigidity for a bounded saddle
E	Choose a lingual bar where functional depth and gingival clearance are adequate

See answer and justification for Q1302 →

Q1303 ADVANCED · Block F

A maxillary bounded edentulous span requires a rigid connector; palatal support is good and there is no need for extensive palatal coverage.
Which major connector best fits the anatomy and hygiene?

A	Consider a lingual plate where clearance is inadequate and tooth contours permit
B	Consider a broad palatal plate for poor support in extensive maxillary distal extensions
C	Choose a lingual bar where functional depth and gingival clearance are adequate
D	Consider a labial connector where obstructive lingual tori cannot otherwise be accommodated
E	Choose a palatal strap with adequate width and rigidity for a bounded saddle

See answer and justification for Q1303 →

Q1304 ADVANCED · Block F

An extensively edentulous maxilla has bilateral distal extensions, compromised residual ridges and few remaining abutments requiring broad tissue support.
Which major connector best fits the anatomy and hygiene?

A	Choose a lingual bar where functional depth and gingival clearance are adequate
B	Consider a labial connector where obstructive lingual tori cannot otherwise be accommodated
C	Consider a broad palatal plate for poor support in extensive maxillary distal extensions
D	Consider a lingual plate where clearance is inadequate and tooth contours permit
E	Choose a palatal strap with adequate width and rigidity for a bounded saddle

See answer and justification for Q1304 →

Q1305 ADVANCED · Block F

A terminal premolar in a Kennedy Class I RPD has a suitable buccal undercut and tissue anatomy for an I-bar. The design aims to minimise torque as the free-end base moves.
Which clasp strategy is most appropriate?

A	Use a wrought-wire retentive arm where increased flexibility is needed
B	Use a cast circumferential clasp for a suitable bounded saddle abutment
C	Use an appropriately designed RPI-type assembly for a suitable distal extension
D	Adjust the survey line or guide plane through conservative recontouring where safe
E	Select a non-clasp retention approach when visible metal is a primary aesthetic concern

See answer and justification for Q1305 →

Q1306 ADVANCED · Block F

A terminal abutment has a deep undercut but good support. A cast cobalt-chromium clasp fractured after excessive repeated flexure.
Which clasp strategy is most appropriate?

A	Select a non-clasp retention approach when visible metal is a primary aesthetic concern
B	Use a wrought-wire retentive arm where increased flexibility is needed
C	Use a cast circumferential clasp for a suitable bounded saddle abutment
D	Use an appropriately designed RPI-type assembly for a suitable distal extension
E	Adjust the survey line or guide plane through conservative recontouring where safe

See answer and justification for Q1306 →

Q1307 ADVANCED · Block F

A bounded Kennedy Class III space has a sound tooth abutment, adequate survey line and no rotational free-end saddle movement.
Which clasp strategy is most appropriate?

A	Use an appropriately designed RPI-type assembly for a suitable distal extension
B	Adjust the survey line or guide plane through conservative recontouring where safe
C	Use a wrought-wire retentive arm where increased flexibility is needed
D	Select a non-clasp retention approach when visible metal is a primary aesthetic concern
E	Use a cast circumferential clasp for a suitable bounded saddle abutment

See answer and justification for Q1307 →

Q1308 ADVANCED · Block F

A proposed clasp engages an unfavourable undercut caused by an old over-contoured crown, and the insertion path is obstructed despite good alternative tooth surfaces.
Which clasp strategy is most appropriate?

A	Adjust the survey line or guide plane through conservative recontouring where safe
B	Use a cast circumferential clasp for a suitable bounded saddle abutment
C	Select a non-clasp retention approach when visible metal is a primary aesthetic concern
D	Use a wrought-wire retentive arm where increased flexibility is needed
E	Use an appropriately designed RPI-type assembly for a suitable distal extension

See answer and justification for Q1308 →

Q1309 ADVANCED · Block F

An RPD sinks under biting even though the saddle is large. Examination shows the planned terminal occlusal rest seats were omitted and the framework contacts only the proximal enamel.
What should be corrected first when the RPD is unstable?

A	Adjust an occlusal interference causing denture rocking in function
B	Manage ongoing periodontal inflammation before remaking a framework
C	Correct absent or ineffective occlusal rests on the abutments
D	Reassess clasp reciprocity and framework fit before tightening retention
E	Record appropriate functional support for a distal-extension base

See answer and justification for Q1309 →

Q1310 ADVANCED · Block F

A Kennedy Class I RPD seats on teeth but the free-end acrylic rocks under compressive loading. The master cast was made without assessing the differential tissue support.
What should be corrected first when the RPD is unstable?

A	Reassess clasp reciprocity and framework fit before tightening retention
B	Correct absent or ineffective occlusal rests on the abutments
C	Adjust an occlusal interference causing denture rocking in function
D	Manage ongoing periodontal inflammation before remaking a framework
E	Record appropriate functional support for a distal-extension base

See answer and justification for Q1310 →

Q1311 ADVANCED · Block F

The RPD visibly moves during insertion and the clasp tips alone contact before the framework seats; an ineffective reciprocal arm allows the abutment to be pushed laterally.
What should be corrected first when the RPD is unstable?

A	Adjust an occlusal interference causing denture rocking in function
B	Record appropriate functional support for a distal-extension base
C	Manage ongoing periodontal inflammation before remaking a framework
D	Correct absent or ineffective occlusal rests on the abutments
E	Reassess clasp reciprocity and framework fit before tightening retention

See answer and justification for Q1311 →

Q1312 ADVANCED · Block F

A lower RPD rocks only during lateral excursions; at rest it seats accurately. Articulating paper shows an isolated high contact on a distal artificial molar.
What should be corrected first when the RPD is unstable?

A	Reassess clasp reciprocity and framework fit before tightening retention
B	Record appropriate functional support for a distal-extension base
C	Correct absent or ineffective occlusal rests on the abutments
D	Adjust an occlusal interference causing denture rocking in function
E	Manage ongoing periodontal inflammation before remaking a framework

See answer and justification for Q1312 →

Q1313 ADVANCED · Block F

A mandibular free-end RPD includes two artificial molars extending far beyond the useful retromolar support. The patient reports recurring tipping and painful distal pressure.
What design decision best protects the abutments?

A	Reduce an excessive distal cantilever of artificial teeth on a free-end saddle
B	Reduce plaque traps and improve cleansability around clasp and connector surfaces
C	Use appropriate indirect retention to control rotational displacement
D	Reassess abutment prognosis and consider alternative support where periodontal disease is active
E	Use an appropriate guide-plane and rest configuration to control insertion forces

See answer and justification for Q1313 →

Q1314 ADVANCED · Block F

A new framework repeatedly pushes the terminal premolar laterally during insertion. Examination shows a long, angled, unplanned guide plane and poorly aligned rests.
What design decision best protects the abutments?

A	Reduce an excessive distal cantilever of artificial teeth on a free-end saddle
B	Use an appropriate guide-plane and rest configuration to control insertion forces
C	Reduce plaque traps and improve cleansability around clasp and connector surfaces
D	Use appropriate indirect retention to control rotational displacement
E	Reassess abutment prognosis and consider alternative support where periodontal disease is active

See answer and justification for Q1314 →

Q1315 ADVANCED · Block F

A distal-extension RPD is planned on a mobile terminal premolar with deep bleeding pockets and recent progressive bone loss.
What design decision best protects the abutments?

A	Reassess abutment prognosis and consider alternative support where periodontal disease is active
B	Reduce plaque traps and improve cleansability around clasp and connector surfaces
C	Reduce an excessive distal cantilever of artificial teeth on a free-end saddle
D	Use appropriate indirect retention to control rotational displacement
E	Use an appropriate guide-plane and rest configuration to control insertion forces

See answer and justification for Q1315 →

Q1316 ADVANCED · Block F

An RPD with large acrylic gingival coverage has repeatedly inflamed papillae adjacent to several stable abutments, with plaque accumulation under connectors.
What design decision best protects the abutments?

A	Use an appropriate guide-plane and rest configuration to control insertion forces
B	Reduce an excessive distal cantilever of artificial teeth on a free-end saddle
C	Reduce plaque traps and improve cleansability around clasp and connector surfaces
D	Reassess abutment prognosis and consider alternative support where periodontal disease is active
E	Use appropriate indirect retention to control rotational displacement

See answer and justification for Q1316 →

Q1317 ADVANCED · Block F

An RPD distal-extension saddle has repeated mucosal soreness, and the artificial distal molar carries a heavy contact while anterior contacts are light.
How should the occlusal/restorative plan be modified?

A	Reassess the occlusal plane before arranging artificial teeth in a missing-tooth space
B	Avoid replacing every missing molar when an adequate shortened dental arch exists
C	Design an appropriate indirect retainer for rotation away from tissues
D	Reduce excessive occlusal loading on a distal-extension RPD
E	Investigate a compromised periodontal abutment before using it for support

See answer and justification for Q1317 →

Q1318 ADVANCED · Block F

A free-end saddle lifts during chewing sticky food, although functional ridge support is adequate and the framework seats precisely.
How should the occlusal/restorative plan be modified?

A	Investigate a compromised periodontal abutment before using it for support
B	Design an appropriate indirect retainer for rotation away from tissues
C	Reduce excessive occlusal loading on a distal-extension RPD
D	Reassess the occlusal plane before arranging artificial teeth in a missing-tooth space
E	Avoid replacing every missing molar when an adequate shortened dental arch exists

See answer and justification for Q1318 →

Q1319 ADVANCED · Block F

A premolar designated for an RPD rest has a deep angular bone defect, suppuration and rapidly changing mobility.
How should the occlusal/restorative plan be modified?

A	Design an appropriate indirect retainer for rotation away from tissues
B	Reassess the occlusal plane before arranging artificial teeth in a missing-tooth space
C	Investigate a compromised periodontal abutment before using it for support
D	Reduce excessive occlusal loading on a distal-extension RPD
E	Avoid replacing every missing molar when an adequate shortened dental arch exists

See answer and justification for Q1319 →

Q1320 ADVANCED · Block F

A patient lost a lower molar years ago. Its upper antagonist has over-erupted 3 mm, leaving insufficient occlusal clearance for a proposed denture tooth.
How should the occlusal/restorative plan be modified?

A	Avoid replacing every missing molar when an adequate shortened dental arch exists
B	Reduce excessive occlusal loading on a distal-extension RPD
C	Reassess the occlusal plane before arranging artificial teeth in a missing-tooth space
D	Investigate a compromised periodontal abutment before using it for support
E	Design an appropriate indirect retainer for rotation away from tissues

See answer and justification for Q1320 →

Q1321 ADVANCED · Block F

A 72-year-old has isolated bounded missing spaces and several stable premolar abutments. Oral hygiene is good, cost excludes implants, and the patient wants a removable option.
Which management best suits this partially dentate patient?

A	Arrange specialist assessment for a very complex combined periodontal and restorative case
B	Use an interim acrylic RPD where rapid changes or additions are expected
C	Consider a shortened dental arch without distal tooth replacement
D	Preserve sound strategically important teeth and plan a cleanable tooth-supported RPD
E	Use a distal-extension RPD with realistic patient counselling about movement and maintenance

See answer and justification for Q1321 →

Q1322 ADVANCED · Block F

A patient has bilateral distal-extension spaces, good remaining anterior teeth and a strong preference against implants. They understand that the saddle will move relative to the teeth.
Which management best suits this partially dentate patient?

A	Preserve sound strategically important teeth and plan a cleanable tooth-supported RPD
B	Use an interim acrylic RPD where rapid changes or additions are expected
C	Use a distal-extension RPD with realistic patient counselling about movement and maintenance
D	Consider a shortened dental arch without distal tooth replacement
E	Arrange specialist assessment for a very complex combined periodontal and restorative case

See answer and justification for Q1322 →

Q1323 ADVANCED · Block F

An 81-year-old has stable bilateral premolar contacts and no functional or aesthetic complaint after loss of second molars. They prefer no appliance.
Which management best suits this partially dentate patient?

A	Consider a shortened dental arch without distal tooth replacement
B	Use a distal-extension RPD with realistic patient counselling about movement and maintenance
C	Arrange specialist assessment for a very complex combined periodontal and restorative case
D	Preserve sound strategically important teeth and plan a cleanable tooth-supported RPD
E	Use an interim acrylic RPD where rapid changes or additions are expected

See answer and justification for Q1323 →

Q1324 ADVANCED · Block F

An older adult is undergoing staged extractions and healing; definitive abutment prognosis will be reviewed in several months, but appearance is a concern.
Which management best suits this partially dentate patient?

A	Consider a shortened dental arch without distal tooth replacement
B	Preserve sound strategically important teeth and plan a cleanable tooth-supported RPD
C	Arrange specialist assessment for a very complex combined periodontal and restorative case
D	Use an interim acrylic RPD where rapid changes or additions are expected
E	Use a distal-extension RPD with realistic patient counselling about movement and maintenance

See answer and justification for Q1324 →

Q1325 ADVANCED · Block F

A three-year-old RPD has begun to rock at rest. The terminal rest no longer seats after a new crown was placed on its abutment.
Which cause best explains the new complaint?

A	Clasp deformation from inappropriate patient adjustment
B	Excessive posterior flange extension in a distal-extension saddle
C	A high artificial-tooth contact creating a functional pivot
D	Loss of rest support due to a poorly fitting framework
E	Inflammation caused by plaque-retentive denture contours

See answer and justification for Q1325 →

Q1326 ADVANCED · Block F

A recently relined mandibular RPD causes soreness in the retromolar region only when the mouth opens widely. The distal flange visibly impinges during movement.
Which cause best explains the new complaint?

A	Inflammation caused by plaque-retentive denture contours
B	A high artificial-tooth contact creating a functional pivot
C	Excessive posterior flange extension in a distal-extension saddle
D	Clasp deformation from inappropriate patient adjustment
E	Loss of rest support due to a poorly fitting framework

See answer and justification for Q1326 →

Q1327 ADVANCED · Block F

After months of increasing looseness, a patient repeatedly bends the RPD clasp at home. The arm now no longer engages correctly and has visible distortion.
Which cause best explains the new complaint?

A	Clasp deformation from inappropriate patient adjustment
B	Excessive posterior flange extension in a distal-extension saddle
C	A high artificial-tooth contact creating a functional pivot
D	Inflammation caused by plaque-retentive denture contours
E	Loss of rest support due to a poorly fitting framework

See answer and justification for Q1327 →

Q1328 ADVANCED · Block F

A stable RPD wearer develops recurrent gingival bleeding under a broad acrylic connector and around poorly cleaned clasps; pressure spots are absent.
Which cause best explains the new complaint?

A	Excessive posterior flange extension in a distal-extension saddle
B	A high artificial-tooth contact creating a functional pivot
C	Loss of rest support due to a poorly fitting framework
D	Clasp deformation from inappropriate patient adjustment
E	Inflammation caused by plaque-retentive denture contours

See answer and justification for Q1328 →

Q1329 ADVANCED · Block F

A Kennedy Class I lower RPD with distal rests repeatedly torques otherwise healthy premolar abutments during function. Tissue support is poorer than tooth support.
Which design revision is most appropriate?

A	Provide a guiding plane and reciprocation rather than relying on frictional clasping
B	Reduce an over-extended labial/buccal flange that conflicts with muscles
C	Use a flexible clasp assembly where a deep undercut demands it
D	Shift the fulcrum strategy by appropriately located mesial rests and stress release
E	Investigate an aesthetic alternative to visible clasps where clinically sound

See answer and justification for Q1329 →

Q1330 ADVANCED · Block F

A retained lower canine has a pronounced undercut and poor path for a short stiff cast clasp. A more flexible design is feasible without compromising reciprocation.
Which design revision is most appropriate?

A	Investigate an aesthetic alternative to visible clasps where clinically sound
B	Provide a guiding plane and reciprocation rather than relying on frictional clasping
C	Use a flexible clasp assembly where a deep undercut demands it
D	Shift the fulcrum strategy by appropriately located mesial rests and stress release
E	Reduce an over-extended labial/buccal flange that conflicts with muscles

See answer and justification for Q1330 →

Q1331 ADVANCED · Block F

An RPD inserts through an irregular path and flexes an abutment because only a retentive arm was designed, with no reciprocal contact.
Which design revision is most appropriate?

A	Provide a guiding plane and reciprocation rather than relying on frictional clasping
B	Shift the fulcrum strategy by appropriately located mesial rests and stress release
C	Reduce an over-extended labial/buccal flange that conflicts with muscles
D	Use a flexible clasp assembly where a deep undercut demands it
E	Investigate an aesthetic alternative to visible clasps where clinically sound

See answer and justification for Q1331 →

Q1332 ADVANCED · Block F

An anterior mandibular acrylic RPD causes blanching when smiling, but tooth support and occlusion are satisfactory. The labial flange clearly displaces mobile mucosa.
Which design revision is most appropriate?

A	Reduce an over-extended labial/buccal flange that conflicts with muscles
B	Use a flexible clasp assembly where a deep undercut demands it
C	Shift the fulcrum strategy by appropriately located mesial rests and stress release
D	Provide a guiding plane and reciprocation rather than relying on frictional clasping
E	Investigate an aesthetic alternative to visible clasps where clinically sound

See answer and justification for Q1332 →

BLOCK G · Complete dentures (50)

Q1046 ADVANCED · Block G

A new lower complete denture is stable when seated but lifts whenever the patient protrudes the tongue. Pressure indicator paste shows a prominent lingual flange. What is the likely first correction?

A	Apply adhesive and discharge without examination
B	Investigate occlusal errors and centric discrepancy
C	Check functional peripheral border extension and polished surface contours
D	Increase occlusal vertical dimension without assessment
E	Manage xerostomia and tissue condition before assuming a new denture is required

See answer and justification for Q1046 →

Q1047 ADVANCED · Block G

Both dentures resist manual removal but the maxillary appliance rocks at every attempt at bilateral closure. Clinical remount reveals prematurities from jaw-relation error. What should be addressed?

A	Apply adhesive and discharge without examination
B	Manage xerostomia and tissue condition before assuming a new denture is required
C	Check functional peripheral border extension and polished surface contours
D	Investigate occlusal errors and centric discrepancy
E	Increase occlusal vertical dimension without assessment

See answer and justification for Q1047 →

Q1048 ADVANCED · Block G

An older patient whose upper denture fit well now has marked dryness after new medication and complains of intermittent loss of suction. Borders and occlusion are satisfactory. What should be investigated?

A	Manage xerostomia and tissue condition before assuming a new denture is required
B	Check functional peripheral border extension and polished surface contours
C	Investigate occlusal errors and centric discrepancy
D	Increase occlusal vertical dimension without assessment
E	Apply adhesive and discharge without examination

See answer and justification for Q1048 →

Q1049 ADVANCED · Block G

A new complete denture wearer reports tired elevator muscles, clicking teeth during speech and loss of comfortable interocclusal rest space. The clinician suspects excessive OVD. What should occur first?

A	Increase OVD to the maximum available space to improve retention
B	Remount and correct occlusion while preserving an otherwise acceptable OVD
C	Reduce OVD without measuring or testing
D	Assess freeway space, speech, aesthetics and existing vertical dimension before adjustment
E	Use a copy-denture approach retaining tolerated relationships

See answer and justification for Q1049 →

Q1050 ADVANCED · Block G

An 84-year-old with Parkinsonism is comfortable in very worn dentures; the jaw relationship is tolerated but the bases are degraded. The patient has previously failed to adapt to a radically new set. What is best?

A	Assess freeway space, speech, aesthetics and existing vertical dimension before adjustment
B	Use a copy-denture approach retaining tolerated relationships
C	Increase OVD to the maximum available space to improve retention
D	Remount and correct occlusion while preserving an otherwise acceptable OVD
E	Reduce OVD without measuring or testing

See answer and justification for Q1050 →

Q1051 ADVANCED · Block G

A complete denture patient reports intermittent dislodgement during chewing; aesthetics, freeway space and phonetics are good, but a remount shows a unilateral deflective contact. What is best?

A	Reduce OVD without measuring or testing
B	Use a copy-denture approach retaining tolerated relationships
C	Increase OVD to the maximum available space to improve retention
D	Remount and correct occlusion while preserving an otherwise acceptable OVD
E	Assess freeway space, speech, aesthetics and existing vertical dimension before adjustment

See answer and justification for Q1051 →

Q1052 ADVANCED · Block G

An edentulous patient has mobile anterior maxillary tissue opposing lower natural incisors. Repeated conventional secondary impressions produce a denture that dislodges after the tissue rebounds. What technique is indicated?

A	Investigate underlying pathology before assuming the ridge is harmless
B	Use a close-fitting thermoplastic compound impression to compress all tissues
C	Use a window or selective-pressure impression technique
D	Provide a conventional reline without evaluating tissue displacement
E	Use mucostatic registration of the mobile anterior tissues with controlled support elsewhere

See answer and justification for Q1052 →

Q1053 ADVANCED · Block G

A stable mandibular denture is planned for a severely atrophic ridge with a freely mobile fibrous crest but firm buccal shelves. What impression principle is appropriate?

A	Provide a conventional reline without evaluating tissue displacement
B	Use a close-fitting thermoplastic compound impression to compress all tissues
C	Use a window or selective-pressure impression technique
D	Investigate underlying pathology before assuming the ridge is harmless
E	Use mucostatic registration of the mobile anterior tissues with controlled support elsewhere

See answer and justification for Q1053 →

Q1054 ADVANCED · Block G

A unilateral enlarging displaceable ridge is associated with persistent ulceration and altered sensation, unlike a longstanding symmetric flabby ridge. What comes first?

A	Use a close-fitting thermoplastic compound impression to compress all tissues
B	Use mucostatic registration of the mobile anterior tissues with controlled support elsewhere
C	Use a window or selective-pressure impression technique
D	Investigate underlying pathology before assuming the ridge is harmless
E	Provide a conventional reline without evaluating tissue displacement

See answer and justification for Q1054 →

Q1118 ADVANCED · Block G

A new denture produces a localised ulcer over the left mylohyoid ridge only after swallowing; the fitting surface is otherwise satisfactory. What is best?

A	Refer a persistent unexplained mucosal ulcer for investigation
B	Treat confirmed denture stomatitis with hygiene, fit correction and appropriate therapy
C	Assess occlusal deflective contacts and correct them after remount
D	Use pressure-indicator paste and functional movement to locate overextension
E	Reduce the entire fitting surface indiscriminately

See answer and justification for Q1118 →

Q1119 ADVANCED · Block G

A new complete denture repeatedly causes bruising on one ridge crest after chewing; pressure-indicator paste is satisfactory but a remount identifies an isolated early posterior contact. What is the first correction?

A	Use pressure-indicator paste and functional movement to locate overextension
B	Assess occlusal deflective contacts and correct them after remount
C	Reduce the entire fitting surface indiscriminately
D	Treat confirmed denture stomatitis with hygiene, fit correction and appropriate therapy
E	Refer a persistent unexplained mucosal ulcer for investigation

See answer and justification for Q1119 →

Q1120 ADVANCED · Block G

A chronic ulcer persists for three weeks after denture relief and removal of the suspected cause; it has a firm edge. What is the next step?

A	Reduce the entire fitting surface indiscriminately
B	Refer a persistent unexplained mucosal ulcer for investigation
C	Assess occlusal deflective contacts and correct them after remount
D	Use pressure-indicator paste and functional movement to locate overextension
E	Treat confirmed denture stomatitis with hygiene, fit correction and appropriate therapy

See answer and justification for Q1120 →

Q1121 ADVANCED · Block G

A 76-year-old with a newly made lower complete denture reports instability. Border moulding was incomplete and obvious occlusal errors remain. What should occur first?

A	Offer full-mouth extractions to create symmetry
B	Optimise conventional mandibular denture design and tissue condition before escalating
C	Assess whether patient dexterity and cleaning capacity support attachment maintenance
D	Refer for complex implant planning where anatomy or systemic risks exceed Level 2
E	Discuss two-implant overdenture retention as a potential option after assessment

See answer and justification for Q1121 →

Q1122 ADVANCED · Block G

A fit independent patient has persistent instability of multiple well-designed mandibular complete dentures and requests improved retention. Bone and systemic assessment are favourable. What replacement concept merits discussion?

A	Assess whether patient dexterity and cleaning capacity support attachment maintenance
B	Refer for complex implant planning where anatomy or systemic risks exceed Level 2
C	Offer full-mouth extractions to create symmetry
D	Discuss two-implant overdenture retention as a potential option after assessment
E	Optimise conventional mandibular denture design and tissue condition before escalating

See answer and justification for Q1122 →

Q1123 ADVANCED · Block G

A patient with severe mandibular atrophy and an altered inferior alveolar nerve course requests implant overdentures. The proposed positions are close to vital anatomy. What is appropriate?

A	Refer for complex implant planning where anatomy or systemic risks exceed Level 2
B	Offer full-mouth extractions to create symmetry
C	Assess whether patient dexterity and cleaning capacity support attachment maintenance
D	Optimise conventional mandibular denture design and tissue condition before escalating
E	Discuss two-implant overdenture retention as a potential option after assessment

See answer and justification for Q1123 →

Q1172 ADVANCED · Block G

A first-time complete denture wearer complains of a lisp and mild soreness at the first review. Borders and occlusion appear satisfactory. What is the best initial approach?

A	Modify the denture to improve cleansability where stomatitis is recurrent
B	Assess the specific cause of the patient’s complaint before proposing a remake
C	Use a copy-denture technique when established contours aid adaptation
D	Explain adaptation, agree targeted adjustments and set review criteria
E	Remake the denture for every complaint about speech

See answer and justification for Q1172 →

Q1173 ADVANCED · Block G

An 83-year-old with neuromuscular disease strongly prefers the familiar contours of worn dentures and has failed to adapt to newly designed sets. What technique is appropriate?

A	Use a copy-denture technique when established contours aid adaptation
B	Modify the denture to improve cleansability where stomatitis is recurrent
C	Assess the specific cause of the patient’s complaint before proposing a remake
D	Explain adaptation, agree targeted adjustments and set review criteria
E	Remake the denture for every complaint about speech

See answer and justification for Q1173 →

Q1174 ADVANCED · Block G

An upper denture has persistent erythematous mucosa beneath it, plaque accumulation and nightly wear. Its fitting surface is rough. What should be addressed?

A	Modify the denture to improve cleansability where stomatitis is recurrent
B	Remake the denture for every complaint about speech
C	Assess the specific cause of the patient’s complaint before proposing a remake
D	Use a copy-denture technique when established contours aid adaptation
E	Explain adaptation, agree targeted adjustments and set review criteria

See answer and justification for Q1174 →

Q1333 ADVANCED · Block G

A complete upper denture drops when speaking. The posterior border ends anterior to the vibrating line and seal is absent; ridge form and occlusal contacts are satisfactory.
Which intervention best addresses the failure?

A	Correct high occlusal contacts causing denture displacement
B	Re-establish the mandibular neutral-zone contour and tongue space
C	Reassess maxillary posterior palatal seal position and function
D	Reduce overextension in the mandibular masseteric notch
E	Improve retention by accurately recording functional border depth

See answer and justification for Q1333 →

Q1334 ADVANCED · Block G

A mandibular denture is comfortable at rest but repeatedly lifts and abrades the distobuccal sulcus when the patient clenches.
Which intervention best addresses the failure?

A	Reassess maxillary posterior palatal seal position and function
B	Reduce overextension in the mandibular masseteric notch
C	Improve retention by accurately recording functional border depth
D	Re-establish the mandibular neutral-zone contour and tongue space
E	Correct high occlusal contacts causing denture displacement

See answer and justification for Q1334 →

Q1335 ADVANCED · Block G

A patient’s lower denture is continuously destabilised by the tongue despite adequate ridge adaptation and functional border moulding. Its lingual polished surface is markedly convex.
Which intervention best addresses the failure?

A	Improve retention by accurately recording functional border depth
B	Correct high occlusal contacts causing denture displacement
C	Re-establish the mandibular neutral-zone contour and tongue space
D	Reduce overextension in the mandibular masseteric notch
E	Reassess maxillary posterior palatal seal position and function

See answer and justification for Q1335 →

Q1336 ADVANCED · Block G

A new lower denture has poor retention in all movements and the lingual and labial borders are demonstrably short of their functional sulci; the occlusion is even.
Which intervention best addresses the failure?

A	Reassess maxillary posterior palatal seal position and function
B	Reduce overextension in the mandibular masseteric notch
C	Re-establish the mandibular neutral-zone contour and tongue space
D	Correct high occlusal contacts causing denture displacement
E	Improve retention by accurately recording functional border depth

See answer and justification for Q1336 →

Q1337 ADVANCED · Block G

Rest vertical dimension varies between two visits by several millimetres as the patient becomes more relaxed. A single freeway-space reading has been used to prescribe OVD.
How should the jaw-relation finding influence management?

A	Use a reversible trial when the patient has limited adaptive capacity
B	Correct an excessive OVD causing clicking, muscular strain and limited freeway space
C	Investigate a reduced OVD associated with poor support and aesthetics
D	Reassess OVD using several clinical criteria rather than a single number
E	Record a reproducible horizontal jaw relation before setting posterior teeth

See answer and justification for Q1337 →

Q1338 ADVANCED · Block G

A patient has repeatedly displaced mandibular rims on closing. Centric records vary by 4 mm and artificial teeth have not been set.
How should the jaw-relation finding influence management?

A	Investigate a reduced OVD associated with poor support and aesthetics
B	Use a reversible trial when the patient has limited adaptive capacity
C	Reassess OVD using several clinical criteria rather than a single number
D	Correct an excessive OVD causing clicking, muscular strain and limited freeway space
E	Record a reproducible horizontal jaw relation before setting posterior teeth

See answer and justification for Q1338 →

Q1339 ADVANCED · Block G

New complete dentures produce tooth clicking, muscle fatigue and difficulty closing lips comfortably. The interocclusal rest space appears virtually absent.
How should the jaw-relation finding influence management?

A	Use a reversible trial when the patient has limited adaptive capacity
B	Correct an excessive OVD causing clicking, muscular strain and limited freeway space
C	Reassess OVD using several clinical criteria rather than a single number
D	Record a reproducible horizontal jaw relation before setting posterior teeth
E	Investigate a reduced OVD associated with poor support and aesthetics

See answer and justification for Q1339 →

Q1340 ADVANCED · Block G

A long-term denture wearer has worn occlusal teeth, reduced lower facial support and angular commissural folds. Clinical assessment suggests an altered bite relationship.
How should the jaw-relation finding influence management?

A	Reassess OVD using several clinical criteria rather than a single number
B	Use a reversible trial when the patient has limited adaptive capacity
C	Investigate a reduced OVD associated with poor support and aesthetics
D	Correct an excessive OVD causing clicking, muscular strain and limited freeway space
E	Record a reproducible horizontal jaw relation before setting posterior teeth

See answer and justification for Q1340 →

Q1341 ADVANCED · Block G

A long-term denture wearer has a very mobile anterior maxillary ridge opposing natural mandibular incisors. Compressing the ridge during trial impression changes the recorded position.
What is the most appropriate impression strategy?

A	Provide targeted relief over prominent pressure-sensitive bony anatomy
B	Control the inflamed denture-bearing mucosa before taking the definitive impression
C	Use functional border moulding to record a resorbed mandibular lingual sulcus
D	Use a copy-denture approach where adaptation and established contours are key
E	Use a selective-pressure technique for a displaceable anterior maxillary ridge

See answer and justification for Q1341 →

Q1342 ADVANCED · Block G

An edentulous patient has an atrophic lower ridge and a deep retromylohyoid fossa; the old denture fails to engage the functional sulcus.
What is the most appropriate impression strategy?

A	Use a selective-pressure technique for a displaceable anterior maxillary ridge
B	Use functional border moulding to record a resorbed mandibular lingual sulcus
C	Provide targeted relief over prominent pressure-sensitive bony anatomy
D	Use a copy-denture approach where adaptation and established contours are key
E	Control the inflamed denture-bearing mucosa before taking the definitive impression

See answer and justification for Q1342 →

Q1343 ADVANCED · Block G

A patient with Parkinsonism likes the arrangement and polished surfaces of a very worn denture and has difficulty adapting to unfamiliar contours.
What is the most appropriate impression strategy?

A	Use functional border moulding to record a resorbed mandibular lingual sulcus
B	Use a copy-denture approach where adaptation and established contours are key
C	Control the inflamed denture-bearing mucosa before taking the definitive impression
D	Use a selective-pressure technique for a displaceable anterior maxillary ridge
E	Provide targeted relief over prominent pressure-sensitive bony anatomy

See answer and justification for Q1343 →

Q1344 ADVANCED · Block G

An upper denture repeatedly ulcerates over a sharp torus palatinus. The rest of the supporting palate and ridge are comfortable.
What is the most appropriate impression strategy?

A	Use a selective-pressure technique for a displaceable anterior maxillary ridge
B	Provide targeted relief over prominent pressure-sensitive bony anatomy
C	Use a copy-denture approach where adaptation and established contours are key
D	Control the inflamed denture-bearing mucosa before taking the definitive impression
E	Use functional border moulding to record a resorbed mandibular lingual sulcus

See answer and justification for Q1344 →

Q1345 ADVANCED · Block G

A maxillary denture dislodges when the patient speaks; seating pressure is good and there is no eccentric tipping. Air leaks at the posterior border on function.
Which factor most directly accounts for the reported complaint?

A	A reduced functional sulcus depth due to flange overextension
B	Reduced salivary film from xerostomia
C	Inadequate upper posterior palatal seal or border adaptation
D	An incorrect neutral-zone contour in a highly resorbed lower arch
E	Excessive posterior occlusal contact causing tipping

See answer and justification for Q1345 →

Q1346 ADVANCED · Block G

A new lower complete denture has stable retention until the first bite; articulating contacts are concentrated at the distal artificial molars.
Which factor most directly accounts for the reported complaint?

A	An incorrect neutral-zone contour in a highly resorbed lower arch
B	Inadequate upper posterior palatal seal or border adaptation
C	Excessive posterior occlusal contact causing tipping
D	Reduced salivary film from xerostomia
E	A reduced functional sulcus depth due to flange overextension

See answer and justification for Q1346 →

Q1347 ADVANCED · Block G

An edentulous patient has severely resorbed lower ridges. The new lower denture rocks on speech because its polished surfaces bulge into the tongue and cheek space.
Which factor most directly accounts for the reported complaint?

A	Reduced salivary film from xerostomia
B	Inadequate upper posterior palatal seal or border adaptation
C	A reduced functional sulcus depth due to flange overextension
D	An incorrect neutral-zone contour in a highly resorbed lower arch
E	Excessive posterior occlusal contact causing tipping

See answer and justification for Q1347 →

Q1348 ADVANCED · Block G

After head and neck radiotherapy the patient reports dry, tacky mucosa and a previously serviceable upper denture has gradually lost retention without a major change in fit.
Which factor most directly accounts for the reported complaint?

A	Excessive posterior occlusal contact causing tipping
B	A reduced functional sulcus depth due to flange overextension
C	An incorrect neutral-zone contour in a highly resorbed lower arch
D	Inadequate upper posterior palatal seal or border adaptation
E	Reduced salivary film from xerostomia

See answer and justification for Q1348 →

Q1349 ADVANCED · Block G

A healthy edentulous patient has adequate ridges, good neuromuscular control and wishes to avoid surgery; conventional dentures are acceptable after a realistic discussion.
Which rehabilitation route best matches these circumstances?

A	Assess a mandibular implant-retained overdenture with ongoing maintenance needs
B	Refer for a complex surgical-prosthetic plan when standard anatomy is profoundly altered
C	Consider a copy denture where established tooth position and acceptance matter
D	Use an interim immediate denture while post-extraction tissues change
E	Use conventional complete dentures with explicit discussion of achievable limitations

See answer and justification for Q1349 →

Q1350 ADVANCED · Block G

An edentulous person cannot retain a well-made conventional mandibular denture after multiple attempts and is willing to consider surgery and lifelong maintenance.
Which rehabilitation route best matches these circumstances?

A	Use an interim immediate denture while post-extraction tissues change
B	Use conventional complete dentures with explicit discussion of achievable limitations
C	Refer for a complex surgical-prosthetic plan when standard anatomy is profoundly altered
D	Assess a mandibular implant-retained overdenture with ongoing maintenance needs
E	Consider a copy denture where established tooth position and acceptance matter

See answer and justification for Q1350 →

Q1351 ADVANCED · Block G

A patient will have final remaining teeth extracted next week and needs acceptable appearance at work immediately afterward. They understand that rapid ridge changes will follow.
Which rehabilitation route best matches these circumstances?

A	Refer for a complex surgical-prosthetic plan when standard anatomy is profoundly altered
B	Consider a copy denture where established tooth position and acceptance matter
C	Assess a mandibular implant-retained overdenture with ongoing maintenance needs
D	Use conventional complete dentures with explicit discussion of achievable limitations
E	Use an interim immediate denture while post-extraction tissues change

See answer and justification for Q1351 →

Q1352 ADVANCED · Block G

A patient has extensive post-oncological tissue and bony defects affecting swallowing and prosthesis support, requiring coordinated surgical and restorative expertise.
Which rehabilitation route best matches these circumstances?

A	Consider a copy denture where established tooth position and acceptance matter
B	Assess a mandibular implant-retained overdenture with ongoing maintenance needs
C	Use conventional complete dentures with explicit discussion of achievable limitations
D	Use an interim immediate denture while post-extraction tissues change
E	Refer for a complex surgical-prosthetic plan when standard anatomy is profoundly altered

See answer and justification for Q1352 →

Q1353 ADVANCED · Block G

A new lower complete denture causes a discrete painful spot at a bony ridge 24 hours after insertion; pressure-indicating paste identifies a localised fitting-surface contact.
Which corrective action is most appropriate?

A	Reassess altered denture-bearing tissues before undertaking a hard reline
B	Escalate a non-healing indurated lesion instead of repeating adjustments
C	Investigate recurrent midline fracture and redesign for the underlying load mechanism
D	Re-mount and correct demonstrable occlusal errors after processing
E	Adjust a localised traumatic pressure area and review healing promptly

See answer and justification for Q1353 →

Q1354 ADVANCED · Block G

New dentures fit and retain well but produce asymmetric early contacts and a slide on closure after processing; mounting records were consistent before processing.
Which corrective action is most appropriate?

A	Reassess altered denture-bearing tissues before undertaking a hard reline
B	Adjust a localised traumatic pressure area and review healing promptly
C	Re-mount and correct demonstrable occlusal errors after processing
D	Escalate a non-healing indurated lesion instead of repeating adjustments
E	Investigate recurrent midline fracture and redesign for the underlying load mechanism

See answer and justification for Q1354 →

Q1355 ADVANCED · Block G

An upper denture fractures repeatedly along the midline despite technically sound repair. An opposing anterior natural dentition creates high cyclic loading.
Which corrective action is most appropriate?

A	Reassess altered denture-bearing tissues before undertaking a hard reline
B	Re-mount and correct demonstrable occlusal errors after processing
C	Adjust a localised traumatic pressure area and review healing promptly
D	Escalate a non-healing indurated lesion instead of repeating adjustments
E	Investigate recurrent midline fracture and redesign for the underlying load mechanism

See answer and justification for Q1355 →

Q1356 ADVANCED · Block G

A ten-year-old denture is increasingly loose because of progressive ridge resorption. Teeth and occlusion remain satisfactory; tissues are healthy.
Which corrective action is most appropriate?

A	Re-mount and correct demonstrable occlusal errors after processing
B	Escalate a non-healing indurated lesion instead of repeating adjustments
C	Adjust a localised traumatic pressure area and review healing promptly
D	Investigate recurrent midline fracture and redesign for the underlying load mechanism
E	Reassess altered denture-bearing tissues before undertaking a hard reline

See answer and justification for Q1356 →

Q1357 ADVANCED · Block G

A patient reports that wax-trial upper incisors make the upper lip look unsupported, despite a stable base and comfortable bite.
How should the patient’s complaint guide tooth arrangement?

A	Assess lip support and anterior tooth display using try-in and photographs
B	Evaluate bilateral contacts and occlusal plane to improve functional stability
C	Assess phonetics and incisal edge position using speech tests
D	Explore patient-specific expectations and trial tooth setup before processing
E	Adjust tongue space and lower posterior tooth position within the neutral zone

See answer and justification for Q1357 →

Q1358 ADVANCED · Block G

The patient can swallow comfortably with wax rims but cannot control the trial lower denture with the tongue; its posterior teeth project into tongue space.
How should the patient’s complaint guide tooth arrangement?

A	Assess lip support and anterior tooth display using try-in and photographs
B	Explore patient-specific expectations and trial tooth setup before processing
C	Assess phonetics and incisal edge position using speech tests
D	Adjust tongue space and lower posterior tooth position within the neutral zone
E	Evaluate bilateral contacts and occlusal plane to improve functional stability

See answer and justification for Q1358 →

Q1359 ADVANCED · Block G

At try-in, the denture teeth meet heavily on one side and the lower trial base repeatedly tips during lateral movement.
How should the patient’s complaint guide tooth arrangement?

A	Evaluate bilateral contacts and occlusal plane to improve functional stability
B	Assess phonetics and incisal edge position using speech tests
C	Assess lip support and anterior tooth display using try-in and photographs
D	Explore patient-specific expectations and trial tooth setup before processing
E	Adjust tongue space and lower posterior tooth position within the neutral zone

See answer and justification for Q1359 →

Q1360 ADVANCED · Block G

A patient lisps specifically on “s” and “f” sounds at wax try-in while the vertical relation and posterior contacts are satisfactory.
How should the patient’s complaint guide tooth arrangement?

A	Assess phonetics and incisal edge position using speech tests
B	Assess lip support and anterior tooth display using try-in and photographs
C	Evaluate bilateral contacts and occlusal plane to improve functional stability
D	Explore patient-specific expectations and trial tooth setup before processing
E	Adjust tongue space and lower posterior tooth position within the neutral zone

See answer and justification for Q1360 →

Q1361 ADVANCED · Block G

A healthy denture wearer has no complaint but has not attended for five years. The denture looks intact and the patient asks if reviews are unnecessary.
Which long-term action is most appropriate?

A	Arrange modification or reline for loss of tissue adaptation despite stable teeth
B	Investigate a persistent unexplained sore area for potentially serious disease
C	Re-evaluate a change in medical or salivary status affecting comfort
D	Reassess prosthesis suitability when manual dexterity declines markedly
E	Review tissue health, fit, occlusion and hygiene at planned intervals

See answer and justification for Q1361 →

Q1362 ADVANCED · Block G

A denture gradually becomes loose while the patient remains happy with tooth positions and aesthetics; residual-ridge change is apparent and the mucosa is healthy.
Which long-term action is most appropriate?

A	Arrange modification or reline for loss of tissue adaptation despite stable teeth
B	Investigate a persistent unexplained sore area for potentially serious disease
C	Reassess prosthesis suitability when manual dexterity declines markedly
D	Review tissue health, fit, occlusion and hygiene at planned intervals
E	Re-evaluate a change in medical or salivary status affecting comfort

See answer and justification for Q1362 →

Q1363 ADVANCED · Block G

A previously comfortable denture suddenly becomes difficult to wear after a new antidepressant is started. Mucosa is dry and there is new root caries around remaining teeth.
Which long-term action is most appropriate?

A	Reassess prosthesis suitability when manual dexterity declines markedly
B	Investigate a persistent unexplained sore area for potentially serious disease
C	Arrange modification or reline for loss of tissue adaptation despite stable teeth
D	Review tissue health, fit, occlusion and hygiene at planned intervals
E	Re-evaluate a change in medical or salivary status affecting comfort

See answer and justification for Q1363 →

Q1364 ADVANCED · Block G

An older denture wearer recently developed severe hand tremor and can no longer insert an RPD accurately or clean it, despite a technically excellent framework.
Which long-term action is most appropriate?

A	Re-evaluate a change in medical or salivary status affecting comfort
B	Review tissue health, fit, occlusion and hygiene at planned intervals
C	Reassess prosthesis suitability when manual dexterity declines markedly
D	Arrange modification or reline for loss of tissue adaptation despite stable teeth
E	Investigate a persistent unexplained sore area for potentially serious disease

See answer and justification for Q1364 →

BLOCK H · Fixed prosthodontics (61)

Q1055 ADVANCED · Block H

A vital mandibular first molar has a large MOD restoration and fractured one functional cusp. Most remaining cuspal structure and margins are sound, caries is controlled, and rubber dam is possible. What is most suitable?

A	Place a full-coverage crown routinely on all fractured teeth
B	Re-evaluate restorability and ferrule before preparing the tooth
C	Use a provisional restoration to assess occlusion and prognosis before a complex definitive plan
D	Select a resin-bonded bridge rather than full-coverage abutment preparations
E	Use a conservative bonded indirect restoration where substrate and isolation permit

See answer and justification for Q1055 →

Q1056 ADVANCED · Block H

A root-filled maxillary premolar has little supragingival dentine on one side and a deep margin, and has repeatedly lost post cores. Before another indirect restoration, what must be decided?

A	Use a conservative bonded indirect restoration where substrate and isolation permit
B	Re-evaluate restorability and ferrule before preparing the tooth
C	Select a resin-bonded bridge rather than full-coverage abutment preparations
D	Use a provisional restoration to assess occlusion and prognosis before a complex definitive plan
E	Place a full-coverage crown routinely on all fractured teeth

See answer and justification for Q1056 →

Q1057 ADVANCED · Block H

A 30-year-old lacks UL2 and has intact vital UL1 and UL3, a satisfactory occlusion, and wishes to minimise damage to adjacent teeth. What fixed option fits?

A	Select a resin-bonded bridge rather than full-coverage abutment preparations
B	Use a conservative bonded indirect restoration where substrate and isolation permit
C	Place a full-coverage crown routinely on all fractured teeth
D	Use a provisional restoration to assess occlusion and prognosis before a complex definitive plan
E	Re-evaluate restorability and ferrule before preparing the tooth

See answer and justification for Q1057 →

Q1058 ADVANCED · Block H

A proposed four-unit bridge would use abutments with deep bleeding pockets and progressive attachment loss. The patient is willing to undergo periodontal therapy. What happens first?

A	Choose implant replacement after appropriate planning rather than overloading weak abutments
B	Proceed because the abutments have sufficient combined root surface area
C	Accept no replacement where stable function and patient preference permit
D	Investigate and stabilise active periodontal disease before definitive bridgework
E	Consider cantilevered adhesive replacement with strict case selection

See answer and justification for Q1058 →

Q1059 ADVANCED · Block H

An otherwise healthy adult has a small bounded anterior space, sound enamel for bonding, favourable occlusion and no parafunction. The patient strongly wishes to avoid surgery and destructive preparation. What might be reasonable?

A	Proceed because the abutments have sufficient combined root surface area
B	Consider cantilevered adhesive replacement with strict case selection
C	Choose implant replacement after appropriate planning rather than overloading weak abutments
D	Accept no replacement where stable function and patient preference permit
E	Investigate and stabilise active periodontal disease before definitive bridgework

See answer and justification for Q1059 →

Q1060 ADVANCED · Block H

A lower posterior bounded span is long; the only proposed terminal abutment is a periodontally compromised premolar with uncertain long-term prognosis. The patient desires fixed replacement and is open to surgery. What alternative merits investigation?

A	Investigate and stabilise active periodontal disease before definitive bridgework
B	Consider cantilevered adhesive replacement with strict case selection
C	Choose implant replacement after appropriate planning rather than overloading weak abutments
D	Accept no replacement where stable function and patient preference permit
E	Proceed because the abutments have sufficient combined root surface area

See answer and justification for Q1060 →

Q1061 ADVANCED · Block H

A molar crown has repeatedly decemented intact; there is no caries or pain. The preparation is short with excessive taper and almost no grooves. What is the main corrective assessment?

A	Investigate secondary caries or fracture before making a new core
B	Use the strongest permanent cement available without further diagnosis
C	Investigate occlusion and parafunction contributing to loss of retention
D	Assess preparation geometry, height and taper before remaking the crown
E	Assess substrate contamination and bonding strategy for an adhesive restoration

See answer and justification for Q1061 →

Q1062 ADVANCED · Block H

A crown on an adequate preparation repeatedly loosens only after recurrent lateral grinding, with polished non-axial contact marks. What additional factor should be assessed?

A	Investigate secondary caries or fracture before making a new core
B	Investigate occlusion and parafunction contributing to loss of retention
C	Use the strongest permanent cement available without further diagnosis
D	Assess preparation geometry, height and taper before remaking the crown
E	Assess substrate contamination and bonding strategy for an adhesive restoration

See answer and justification for Q1062 →

Q1063 ADVANCED · Block H

A crown repeatedly loosens and the abutment shows new softness, tenderness to biting and a radiographic marginal defect. What should be established first?

A	Assess substrate contamination and bonding strategy for an adhesive restoration
B	Investigate occlusion and parafunction contributing to loss of retention
C	Use the strongest permanent cement available without further diagnosis
D	Assess preparation geometry, height and taper before remaking the crown
E	Investigate secondary caries or fracture before making a new core

See answer and justification for Q1063 →

Q1064 ADVANCED · Block H

A missing UL2 is bordered by sound unrestored teeth and ample enamel. The canine and central incisor have different physiological mobility. Which fixed adhesive design is usually preferable?

A	Choose a full-coverage conventional bridge with both adjacent teeth prepared
B	Assess available enamel, occlusal clearance and abutment movement before design
C	Use a two-wing rigid adhesive bridge across differently mobile abutments
D	Investigate implant-supported replacement if the adhesive substrate is inadequate
E	Consider a single-retainer cantilever resin-bonded bridge

See answer and justification for Q1064 →

Q1065 ADVANCED · Block H

A dentist proposes a palatal wing for an anterior space but has not recorded overbite, guidance or enamel area. What should precede the design?

A	Assess available enamel, occlusal clearance and abutment movement before design
B	Choose a full-coverage conventional bridge with both adjacent teeth prepared
C	Investigate implant-supported replacement if the adhesive substrate is inadequate
D	Use a two-wing rigid adhesive bridge across differently mobile abutments
E	Consider a single-retainer cantilever resin-bonded bridge

See answer and justification for Q1065 →

Q1066 ADVANCED · Block H

An upper lateral space has adjacent teeth heavily restored in dentine with little enamel for bonding, and a failed adhesive bridge. The patient is medically suitable and requests fixed replacement. What should be explored?

A	Investigate implant-supported replacement if the adhesive substrate is inadequate
B	Use a two-wing rigid adhesive bridge across differently mobile abutments
C	Consider a single-retainer cantilever resin-bonded bridge
D	Choose a full-coverage conventional bridge with both adjacent teeth prepared
E	Assess available enamel, occlusal clearance and abutment movement before design

See answer and justification for Q1066 →

Q1124 ADVANCED · Block H

A posterior tooth has a sound supragingival margin, low aesthetic demand and a healthy periodontium. Where should the new crown margin ideally be placed?

A	Use a crown to splint a mobile tooth without periodontal diagnosis
B	Assess ferrule and biologic tissues before modifying a deep margin
C	Prefer accessible supragingival margins when clinically feasible
D	Complete periodontal treatment and plaque control before definitive crownwork
E	Extend all crown margins deep subgingivally for aesthetics

See answer and justification for Q1124 →

Q1125 ADVANCED · Block H

A short root-filled premolar has a deep proximal carious margin close to bone and no circumferential ferrule. The patient asks for a crown. What is essential?

A	Complete periodontal treatment and plaque control before definitive crownwork
B	Extend all crown margins deep subgingivally for aesthetics
C	Assess ferrule and biologic tissues before modifying a deep margin
D	Prefer accessible supragingival margins when clinically feasible
E	Use a crown to splint a mobile tooth without periodontal diagnosis

See answer and justification for Q1125 →

Q1126 ADVANCED · Block H

A heavily restored molar planned for a crown has recurrent bleeding and active periodontal pockets. The patient wants the definitive crown immediately. What is the priority?

A	Extend all crown margins deep subgingivally for aesthetics
B	Use a crown to splint a mobile tooth without periodontal diagnosis
C	Prefer accessible supragingival margins when clinically feasible
D	Complete periodontal treatment and plaque control before definitive crownwork
E	Assess ferrule and biologic tissues before modifying a deep margin

See answer and justification for Q1126 →

Q1127 ADVANCED · Block H

A root-filled molar under a crown has recurrent sinus formation, a narrow deep probing tract and a halo radiolucency extending along one root. What should be suspected first?

A	Repair a small accessible marginal ceramic chip when the rest of the crown is sound
B	Investigate suspected vertical root fracture before committing to further restoration
C	Replace an extensive defective restoration after confirming restorability
D	Refer a complex root fracture or uncertain prognosis for specialist evaluation
E	Assume that pain following root treatment always requires retreatment

See answer and justification for Q1127 →

Q1128 ADVANCED · Block H

A zirconia crown has a small polished buccal porcelain chip away from the functional contact, sound margins and no recurrent caries. The patient is otherwise satisfied. What is proportionate?

A	Assume that pain following root treatment always requires retreatment
B	Investigate suspected vertical root fracture before committing to further restoration
C	Refer a complex root fracture or uncertain prognosis for specialist evaluation
D	Replace an extensive defective restoration after confirming restorability
E	Repair a small accessible marginal ceramic chip when the rest of the crown is sound

See answer and justification for Q1128 →

Q1129 ADVANCED · Block H

An upper first molar with complex endodontic history shows contradictory symptoms and imaging, with uncertain restorability after cusp fracture. What is reasonable?

A	Replace an extensive defective restoration after confirming restorability
B	Assume that pain following root treatment always requires retreatment
C	Repair a small accessible marginal ceramic chip when the rest of the crown is sound
D	Investigate suspected vertical root fracture before committing to further restoration
E	Refer a complex root fracture or uncertain prognosis for specialist evaluation

See answer and justification for Q1129 →

Q1175 ADVANCED · Block H

A 16-year-old with peg-shaped vital laterals, fully intact enamel and continuing facial growth has a no-preparation wax-up. The family values repairability and conservatism. What is best?

A	Select a conservative minimal-preparation ceramic veneer where enamel is plentiful and prior composites have failed
B	Place a full crown regardless of remaining enamel
C	Use a conventional full-coverage fixed bridge in preference to adhesive options for an isolated gap
D	Use an additive direct composite where no-preparation reshaping is feasible
E	Reassess the occlusal and parafunctional cause before another definitive restoration

See answer and justification for Q1175 →

Q1176 ADVANCED · Block H

A vital central incisor has had two cohesive fractures of a large labio-incisal composite despite good isolation. There is sound palatal and abundant labial enamel, no parafunction and good occlusion. What may be more durable while preserving tissue?

A	Place a full crown regardless of remaining enamel
B	Use an additive direct composite where no-preparation reshaping is feasible
C	Select a conservative minimal-preparation ceramic veneer where enamel is plentiful and prior composites have failed
D	Reassess the occlusal and parafunctional cause before another definitive restoration
E	Use a conventional full-coverage fixed bridge in preference to adhesive options for an isolated gap

See answer and justification for Q1176 →

Q1177 ADVANCED · Block H

A ceramic veneer has fractured twice in a patient with severe nocturnal bruxism, an unrecognised protrusive interference and poor appliance use. What precedes a new definitive restoration?

A	Use an additive direct composite where no-preparation reshaping is feasible
B	Reassess the occlusal and parafunctional cause before another definitive restoration
C	Use a conventional full-coverage fixed bridge in preference to adhesive options for an isolated gap
D	Select a conservative minimal-preparation ceramic veneer where enamel is plentiful and prior composites have failed
E	Place a full crown regardless of remaining enamel

See answer and justification for Q1177 →

Q1365 ADVANCED · Block H

An upper central incisor has a small first-time incisal chip after minor trauma. It is vital, the enamel is plentiful and there are no significant occlusal interferences.
Which definitive restoration is most appropriate?

A	Adhesively bonded labio-incisal glass-ceramic veneer
B	Full-coverage crown where remaining geometry supports retentive preparation
C	Cuspal-coverage bonded onlay for a heavily undermined posterior cusp
D	No definitive restoration until caries or periodontal activity is controlled
E	Direct composite with a palatal index and conservative enamel margin

See answer and justification for Q1365 →

Q1366 ADVANCED · Block H

A vital upper incisor has a large labio-incisal defect that has repeatedly failed by cohesive composite fracture despite good isolation and no parafunction. Palatal enamel is intact.
Which definitive restoration is most appropriate?

A	Full-coverage crown where remaining geometry supports retentive preparation
B	No definitive restoration until caries or periodontal activity is controlled
C	Adhesively bonded labio-incisal glass-ceramic veneer
D	Cuspal-coverage bonded onlay for a heavily undermined posterior cusp
E	Direct composite with a palatal index and conservative enamel margin

See answer and justification for Q1366 →

Q1367 ADVANCED · Block H

A lower molar has a cracked non-functional cusp and large MOD restoration, but adequate remaining enamel and a sound pulp. The patient wants to conserve structure.
Which definitive restoration is most appropriate?

A	Direct composite with a palatal index and conservative enamel margin
B	Full-coverage crown where remaining geometry supports retentive preparation
C	Adhesively bonded labio-incisal glass-ceramic veneer
D	No definitive restoration until caries or periodontal activity is controlled
E	Cuspal-coverage bonded onlay for a heavily undermined posterior cusp

See answer and justification for Q1367 →

Q1368 ADVANCED · Block H

An endodontically treated heavily restored premolar has several missing axial walls, but ferrule and sound root support can be achieved. An adhesive partial coverage is mechanically unsuitable.
Which definitive restoration is most appropriate?

A	Adhesively bonded labio-incisal glass-ceramic veneer
B	Direct composite with a palatal index and conservative enamel margin
C	No definitive restoration until caries or periodontal activity is controlled
D	Cuspal-coverage bonded onlay for a heavily undermined posterior cusp
E	Full-coverage crown where remaining geometry supports retentive preparation

See answer and justification for Q1368 →

Q1369 ADVANCED · Block H

A short clinical crown repeatedly loses a conventional full-coverage restoration. Its margins are well fitting but the axial walls are low and preparation convergence is excessive.
Which principle is most important for long-term crown retention?

A	Choose an adhesive restoration when enamel and isolation support bonding
B	Preserve or develop an adequate ferrule rather than rely on a post alone
C	Increase sound axial wall height and improve preparation resistance form where feasible
D	Reconsider restorability if sound cervical tooth structure cannot be obtained
E	Correct excessive occlusal loading and crown geometry before recementing

See answer and justification for Q1369 →

Q1370 ADVANCED · Block H

A root-treated maxillary premolar needs substantial build-up. The proposed cast post is long, but there is little remaining coronal dentine around the cervical circumference.
Which principle is most important for long-term crown retention?

A	Choose an adhesive restoration when enamel and isolation support bonding
B	Preserve or develop an adequate ferrule rather than rely on a post alone
C	Reconsider restorability if sound cervical tooth structure cannot be obtained
D	Increase sound axial wall height and improve preparation resistance form where feasible
E	Correct excessive occlusal loading and crown geometry before recementing

See answer and justification for Q1370 →

Q1371 ADVANCED · Block H

A vital premolar has an isolated occlusal defect and abundant enamel. The patient prioritises conservation and suitable dry-field bonding can be achieved.
Which principle is most important for long-term crown retention?

A	Increase sound axial wall height and improve preparation resistance form where feasible
B	Preserve or develop an adequate ferrule rather than rely on a post alone
C	Reconsider restorability if sound cervical tooth structure cannot be obtained
D	Choose an adhesive restoration when enamel and isolation support bonding
E	Correct excessive occlusal loading and crown geometry before recementing

See answer and justification for Q1371 →

Q1372 ADVANCED · Block H

A root-treated tooth has caries extending well below the alveolar crest, and proposed crown-lengthening would compromise the neighbouring periodontal support.
Which principle is most important for long-term crown retention?

A	Preserve or develop an adequate ferrule rather than rely on a post alone
B	Choose an adhesive restoration when enamel and isolation support bonding
C	Correct excessive occlusal loading and crown geometry before recementing
D	Reconsider restorability if sound cervical tooth structure cannot be obtained
E	Increase sound axial wall height and improve preparation resistance form where feasible

See answer and justification for Q1372 →

Q1373 ADVANCED · Block H

A vital maxillary lateral incisor needs a minimal labial veneer. Broad enamel margins remain and reliable isolation is possible.
Which factor should drive the restoration or cement selection?

A	Preparation retention and material requirements for a conventional posterior crown
B	Reliable enamel bonding and conservative preparation for an aesthetic indirect restoration
C	Moisture isolation and suitable surface treatment for resin-bonded retention
D	Occlusal load and available thickness for posterior material selection
E	Caries activity and cleanability before definitive indirect replacement

See answer and justification for Q1373 →

Q1374 ADVANCED · Block H

A posterior crown with excellent preparation height and taper is being placed on a low-risk tooth; a conventional luting strategy is supported by the chosen restoration.
Which factor should drive the restoration or cement selection?

A	Moisture isolation and suitable surface treatment for resin-bonded retention
B	Occlusal load and available thickness for posterior material selection
C	Preparation retention and material requirements for a conventional posterior crown
D	Reliable enamel bonding and conservative preparation for an aesthetic indirect restoration
E	Caries activity and cleanability before definitive indirect replacement

See answer and justification for Q1374 →

Q1375 ADVANCED · Block H

A resin-bonded retainer has failed after salivary contamination during bonding, though the abutment enamel and framework design are adequate.
Which factor should drive the restoration or cement selection?

A	Moisture isolation and suitable surface treatment for resin-bonded retention
B	Occlusal load and available thickness for posterior material selection
C	Reliable enamel bonding and conservative preparation for an aesthetic indirect restoration
D	Caries activity and cleanability before definitive indirect replacement
E	Preparation retention and material requirements for a conventional posterior crown

See answer and justification for Q1375 →

Q1376 ADVANCED · Block H

A patient has recurrent cervical caries at several crown margins, frequent sugar exposures and poor interdental cleaning. Multiple new definitive crowns are planned.
Which factor should drive the restoration or cement selection?

A	Moisture isolation and suitable surface treatment for resin-bonded retention
B	Occlusal load and available thickness for posterior material selection
C	Reliable enamel bonding and conservative preparation for an aesthetic indirect restoration
D	Preparation retention and material requirements for a conventional posterior crown
E	Caries activity and cleanability before definitive indirect replacement

See answer and justification for Q1376 →

Q1377 ADVANCED · Block H

A young adult lacks an upper lateral incisor. The adjacent canine has broad enamel, intact periodontal support and no major occlusal interference; they prefer not to prepare the tooth.
Which replacement strategy is most proportionate?

A	Specialist multi-disciplinary planning when defect complexity exceeds routine care
B	A single-wing resin-bonded bridge on a sound enamel-rich abutment
C	A conventional fixed bridge supported by suitable teeth with adequate periodontal reserve
D	A removable replacement when hygiene, cost or future flexibility dominates
E	No replacement with monitoring when function and preferences support it

See answer and justification for Q1377 →

Q1378 ADVANCED · Block H

An adult lacks a premolar with adjacent teeth already requiring full-coverage restorations. Abutment support and span geometry are good, and hygiene is reliable.
Which replacement strategy is most proportionate?

A	Specialist multi-disciplinary planning when defect complexity exceeds routine care
B	No replacement with monitoring when function and preferences support it
C	A removable replacement when hygiene, cost or future flexibility dominates
D	A single-wing resin-bonded bridge on a sound enamel-rich abutment
E	A conventional fixed bridge supported by suitable teeth with adequate periodontal reserve

See answer and justification for Q1378 →

Q1379 ADVANCED · Block H

A patient needs interim tooth replacement while the long-term prognosis of two neighbouring abutments remains uncertain. Costs and cleaning simplicity matter.
Which replacement strategy is most proportionate?

A	A removable replacement when hygiene, cost or future flexibility dominates
B	A conventional fixed bridge supported by suitable teeth with adequate periodontal reserve
C	Specialist multi-disciplinary planning when defect complexity exceeds routine care
D	No replacement with monitoring when function and preferences support it
E	A single-wing resin-bonded bridge on a sound enamel-rich abutment

See answer and justification for Q1379 →

Q1380 ADVANCED · Block H

An older adult lacks a second molar, has a stable functional occlusion and no chewing or aesthetic complaint. They prefer not to undergo more treatment.
Which replacement strategy is most proportionate?

A	A single-wing resin-bonded bridge on a sound enamel-rich abutment
B	A removable replacement when hygiene, cost or future flexibility dominates
C	A conventional fixed bridge supported by suitable teeth with adequate periodontal reserve
D	Specialist multi-disciplinary planning when defect complexity exceeds routine care
E	No replacement with monitoring when function and preferences support it

See answer and justification for Q1380 →

Q1381 ADVANCED · Block H

A proposed fixed bridge would replace three posterior teeth with two small abutments of reduced periodontal support. The span is long, and the patient has bruxism.
Which bridge-design decision best addresses the risk?

A	Avoid a long unsupported span when flexure and abutment reserve are inadequate
B	Reduce an excessive cantilever moment arm and occlusal loading
C	Reassess the connector or fixed-movable design to accommodate differential abutment movement
D	Abandon a fixed plan when hygiene and periodontal support make it unsuitable
E	Use a minimal single-retainer adhesive design where anatomy supports it

See answer and justification for Q1381 →

Q1382 ADVANCED · Block H

A complex bridge is proposed on abutments with markedly different mobility and path of insertion, and a rigid one-piece design would compromise seating.
Which bridge-design decision best addresses the risk?

A	Reduce an excessive cantilever moment arm and occlusal loading
B	Abandon a fixed plan when hygiene and periodontal support make it unsuitable
C	Use a minimal single-retainer adhesive design where anatomy supports it
D	Reassess the connector or fixed-movable design to accommodate differential abutment movement
E	Avoid a long unsupported span when flexure and abutment reserve are inadequate

See answer and justification for Q1382 →

Q1383 ADVANCED · Block H

A distal cantilever pontic fractures repeatedly under heavy occlusion despite adequate crown margins and otherwise stable abutments.
Which bridge-design decision best addresses the risk?

A	Abandon a fixed plan when hygiene and periodontal support make it unsuitable
B	Reduce an excessive cantilever moment arm and occlusal loading
C	Use a minimal single-retainer adhesive design where anatomy supports it
D	Reassess the connector or fixed-movable design to accommodate differential abutment movement
E	Avoid a long unsupported span when flexure and abutment reserve are inadequate

See answer and justification for Q1383 →

Q1384 ADVANCED · Block H

A young patient requires replacement of one anterior tooth between intact neighbours. Enamel support is good and an appropriate occlusal scheme allows a single-retainer adhesive design.
Which bridge-design decision best addresses the risk?

A	Avoid a long unsupported span when flexure and abutment reserve are inadequate
B	Abandon a fixed plan when hygiene and periodontal support make it unsuitable
C	Reduce an excessive cantilever moment arm and occlusal loading
D	Reassess the connector or fixed-movable design to accommodate differential abutment movement
E	Use a minimal single-retainer adhesive design where anatomy supports it

See answer and justification for Q1384 →

Q1385 ADVANCED · Block H

A resin-bonded anterior bridge debonds intact, leaving a clean tooth surface. The patient recalls poor isolation during cementation and bites directly on the pontic.
How should failure be investigated before treatment?

A	Check for vertical root fracture in a repeatedly symptomatic root-filled abutment
B	Examine abutment mobility and periodontal support after a bridge failure
C	Inspect a retainer’s bonding failure mechanism and verify isolation and occlusion
D	Assess marginal leakage, recurrent caries and pulpal status beneath a failed crown
E	Assess parafunction and unsupported ceramic thickness after chipping

See answer and justification for Q1385 →

Q1386 ADVANCED · Block H

A conventional crown is loose and the underlying dentine is soft at the margin. The tooth has intermittent spontaneous pain.
How should failure be investigated before treatment?

A	Assess parafunction and unsupported ceramic thickness after chipping
B	Examine abutment mobility and periodontal support after a bridge failure
C	Inspect a retainer’s bonding failure mechanism and verify isolation and occlusion
D	Check for vertical root fracture in a repeatedly symptomatic root-filled abutment
E	Assess marginal leakage, recurrent caries and pulpal status beneath a failed crown

See answer and justification for Q1386 →

Q1387 ADVANCED · Block H

A bridge fractures at its connector and one abutment is newly mobile with deep bleeding pockets. Maintenance has lapsed.
How should failure be investigated before treatment?

A	Examine abutment mobility and periodontal support after a bridge failure
B	Inspect a retainer’s bonding failure mechanism and verify isolation and occlusion
C	Assess marginal leakage, recurrent caries and pulpal status beneath a failed crown
D	Assess parafunction and unsupported ceramic thickness after chipping
E	Check for vertical root fracture in a repeatedly symptomatic root-filled abutment

See answer and justification for Q1387 →

Q1388 ADVANCED · Block H

A posterior monolithic restoration repeatedly chips at a thin opposing cusp after heavy lateral grinding. Margins and retention are good.
How should failure be investigated before treatment?

A	Inspect a retainer’s bonding failure mechanism and verify isolation and occlusion
B	Assess marginal leakage, recurrent caries and pulpal status beneath a failed crown
C	Assess parafunction and unsupported ceramic thickness after chipping
D	Examine abutment mobility and periodontal support after a bridge failure
E	Check for vertical root fracture in a repeatedly symptomatic root-filled abutment

See answer and justification for Q1388 →

Q1389 ADVANCED · Block H

An upper lateral incisor has a developmental labial defect with broad enamel, stable occlusion and a highly visible smile. A conservative external restoration is requested.
Which material treatment or restoration best suits the defect?

A	Select a suitable cast/onlay restoration for extensive posterior cuspal compromise
B	Choose a robust posterior monolithic restoration where load and clearance dominate
C	Stabilise active disease before elective indirect work
D	Use a conservative direct composite for a small, repairable lesion
E	Use bonded ceramic for enamel-rich anterior aesthetic restoration

See answer and justification for Q1389 →

Q1390 ADVANCED · Block H

A bruxist needs a posterior crown, has modest restorative clearance and retains adequate preparation height. The patient values durability over maximal translucency.
Which material treatment or restoration best suits the defect?

A	Stabilise active disease before elective indirect work
B	Choose a robust posterior monolithic restoration where load and clearance dominate
C	Use a conservative direct composite for a small, repairable lesion
D	Use bonded ceramic for enamel-rich anterior aesthetic restoration
E	Select a suitable cast/onlay restoration for extensive posterior cuspal compromise

See answer and justification for Q1390 →

Q1391 ADVANCED · Block H

A vital premolar has a small non-carious cervical lesion with manageable sensitivity. Tooth structure is otherwise sound and the patient requests a simple repair.
Which material treatment or restoration best suits the defect?

A	Choose a robust posterior monolithic restoration where load and clearance dominate
B	Use a conservative direct composite for a small, repairable lesion
C	Use bonded ceramic for enamel-rich anterior aesthetic restoration
D	Select a suitable cast/onlay restoration for extensive posterior cuspal compromise
E	Stabilise active disease before elective indirect work

See answer and justification for Q1391 →

Q1392 ADVANCED · Block H

A molar has a very large MOD restoration with cracked cusps and repeated direct composite fracture, but adequate remaining tooth and conservative cuspal coverage are feasible.
Which material treatment or restoration best suits the defect?

A	Use bonded ceramic for enamel-rich anterior aesthetic restoration
B	Select a suitable cast/onlay restoration for extensive posterior cuspal compromise
C	Choose a robust posterior monolithic restoration where load and clearance dominate
D	Use a conservative direct composite for a small, repairable lesion
E	Stabilise active disease before elective indirect work

See answer and justification for Q1392 →

Q1393 ADVANCED · Block H

A patient with generalised active periodontitis asks for a three-unit bridge to replace a lower premolar. Plaque control and bleeding have not yet improved.
Which treatment-planning factor has greatest priority?

A	Coordinate specialist care for severe developmental abnormalities
B	Perform a diagnostic trial for substantial aesthetic or occlusal alteration
C	Consider an adhesive single-retainer design to conserve intact adjacent teeth
D	Stabilise periodontal and caries risk before evaluating bridge retainers
E	Avoid fixed replacement when the patient cannot maintain complex contours

See answer and justification for Q1393 →

Q1394 ADVANCED · Block H

A patient wants extensive changes to crown height, smile line and OVD with no prior diagnostic records; they have difficulty expressing the aesthetic goal.
Which treatment-planning factor has greatest priority?

A	Coordinate specialist care for severe developmental abnormalities
B	Avoid fixed replacement when the patient cannot maintain complex contours
C	Consider an adhesive single-retainer design to conserve intact adjacent teeth
D	Perform a diagnostic trial for substantial aesthetic or occlusal alteration
E	Stabilise periodontal and caries risk before evaluating bridge retainers

See answer and justification for Q1394 →

Q1395 ADVANCED · Block H

A healthy 26-year-old lost a lateral incisor. Both potential abutments are untouched and sufficiently enamel-rich; the bite is favourable.
Which treatment-planning factor has greatest priority?

A	Stabilise periodontal and caries risk before evaluating bridge retainers
B	Coordinate specialist care for severe developmental abnormalities
C	Perform a diagnostic trial for substantial aesthetic or occlusal alteration
D	Avoid fixed replacement when the patient cannot maintain complex contours
E	Consider an adhesive single-retainer design to conserve intact adjacent teeth

See answer and justification for Q1395 →

Q1396 ADVANCED · Block H

An older patient has limited dexterity and cannot clean under fixed pontics even with assistance, yet requests a long-span fixed reconstruction.
Which treatment-planning factor has greatest priority?

A	Stabilise periodontal and caries risk before evaluating bridge retainers
B	Perform a diagnostic trial for substantial aesthetic or occlusal alteration
C	Coordinate specialist care for severe developmental abnormalities
D	Consider an adhesive single-retainer design to conserve intact adjacent teeth
E	Avoid fixed replacement when the patient cannot maintain complex contours

See answer and justification for Q1396 →

Q1397 ADVANCED · Block H

A vital anterior tooth chips for the first time after minor trauma. There is a broad enamel margin, and the defect is confined to a small incisal corner.
Which management strategy is most evidence-consistent?

A	Consider minimal bonded ceramic after repeated cohesive direct-composite fracture
B	Use cuspal coverage to prevent propagation of a significant posterior cusp crack
C	Refer for interdisciplinary planning of a severely compromised dentition
D	Reassess ferrule, root condition and prognosis before crown on a root-filled tooth
E	Use direct bonded composite to repair a localised conservative anterior defect

See answer and justification for Q1397 →

Q1398 ADVANCED · Block H

A vital upper incisor has a labio-incisal direct composite that has failed three times cohesively despite good isolation and controlled occlusion. Palatal enamel remains strong.
Which management strategy is most evidence-consistent?

A	Use direct bonded composite to repair a localised conservative anterior defect
B	Use cuspal coverage to prevent propagation of a significant posterior cusp crack
C	Consider minimal bonded ceramic after repeated cohesive direct-composite fracture
D	Reassess ferrule, root condition and prognosis before crown on a root-filled tooth
E	Refer for interdisciplinary planning of a severely compromised dentition

See answer and justification for Q1398 →

Q1399 ADVANCED · Block H

A posterior molar has reproducible pain on release from a wedged cusp, a large MOD restoration and a vital responsive pulp; there is no deep periodontal pocket.
Which management strategy is most evidence-consistent?

A	Refer for interdisciplinary planning of a severely compromised dentition
B	Reassess ferrule, root condition and prognosis before crown on a root-filled tooth
C	Use cuspal coverage to prevent propagation of a significant posterior cusp crack
D	Use direct bonded composite to repair a localised conservative anterior defect
E	Consider minimal bonded ceramic after repeated cohesive direct-composite fracture

See answer and justification for Q1399 →

Q1400 ADVANCED · Block H

A root-filled premolar lacks most coronal walls. A proposal for a post and crown has been made without confirming ferrule or root integrity.
Which management strategy is most evidence-consistent?

A	Refer for interdisciplinary planning of a severely compromised dentition
B	Consider minimal bonded ceramic after repeated cohesive direct-composite fracture
C	Reassess ferrule, root condition and prognosis before crown on a root-filled tooth
D	Use direct bonded composite to repair a localised conservative anterior defect
E	Use cuspal coverage to prevent propagation of a significant posterior cusp crack

See answer and justification for Q1400 →

Q1401 ADVANCED · Block H

A proposed bridge has clinically long abutment crowns but marked horizontal bone loss and mobility on several retainers.
Which principle best determines whether a bridge is appropriate?

A	Use sound enamel and appropriate case selection for adhesive retainer design
B	Evaluate span length, occlusion and likely flexure of the planned prosthesis
C	Evaluate hygiene access and expected maintenance burden
D	Do not replace a missing tooth without meaningful patient benefit
E	Evaluate total abutment periodontal support, not merely crown length

See answer and justification for Q1401 →

Q1402 ADVANCED · Block H

A bridge is proposed across three posterior missing units in a heavy grinder. The abutments are healthy, but the unbroken span would be unusually long.
Which principle best determines whether a bridge is appropriate?

A	Evaluate span length, occlusion and likely flexure of the planned prosthesis
B	Evaluate hygiene access and expected maintenance burden
C	Evaluate total abutment periodontal support, not merely crown length
D	Do not replace a missing tooth without meaningful patient benefit
E	Use sound enamel and appropriate case selection for adhesive retainer design

See answer and justification for Q1402 →

Q1403 ADVANCED · Block H

A patient has limited manual dexterity and very high plaque accumulation under an existing bridge; maintenance for a longer span is unrealistic.
Which principle best determines whether a bridge is appropriate?

A	Evaluate hygiene access and expected maintenance burden
B	Do not replace a missing tooth without meaningful patient benefit
C	Evaluate total abutment periodontal support, not merely crown length
D	Use sound enamel and appropriate case selection for adhesive retainer design
E	Evaluate span length, occlusion and likely flexure of the planned prosthesis

See answer and justification for Q1403 →

Q1404 ADVANCED · Block H

A single missing lateral incisor is bounded by intact enamel and stable periodontal teeth. A minimally invasive solution is a priority.
Which principle best determines whether a bridge is appropriate?

A	Evaluate total abutment periodontal support, not merely crown length
B	Do not replace a missing tooth without meaningful patient benefit
C	Use sound enamel and appropriate case selection for adhesive retainer design
D	Evaluate span length, occlusion and likely flexure of the planned prosthesis
E	Evaluate hygiene access and expected maintenance burden

See answer and justification for Q1404 →

BLOCK I · Occlusion and temporomandibular disorders (22)

Q1067 ADVANCED · Block I

A 32-year-old with intermittent bilateral masseter pain has tenderness on palpation, normal opening and no red flags. They request an occlusal adjustment to “cure the bite.” What is the best initial management?

A	Replace all posterior restorations to achieve a new occlusal scheme
B	Assess for red flags and refer if there are concerning systemic or neurological findings
C	Adjust intact teeth irreversibly to remove all non-working contacts
D	Consider a reversible stabilisation appliance where indicated
E	Begin with explanation, self-care, and reversible management

See answer and justification for Q1067 →

Q1068 ADVANCED · Block I

A 43-year-old with morning jaw soreness, cracked restorations and sleep-related clenching has not improved with advice alone. Examination identifies no inflammatory or neurological red flags. What can be considered?

A	Begin with explanation, self-care, and reversible management
B	Consider a reversible stabilisation appliance where indicated
C	Adjust intact teeth irreversibly to remove all non-working contacts
D	Replace all posterior restorations to achieve a new occlusal scheme
E	Assess for red flags and refer if there are concerning systemic or neurological findings

See answer and justification for Q1068 →

Q1069 ADVANCED · Block I

A 65-year-old reports new severe unilateral jaw pain, headache, scalp tenderness and transient visual disturbance. What is the appropriate response?

A	Replace all posterior restorations to achieve a new occlusal scheme
B	Consider a reversible stabilisation appliance where indicated
C	Begin with explanation, self-care, and reversible management
D	Assess for red flags and refer if there are concerning systemic or neurological findings
E	Adjust intact teeth irreversibly to remove all non-working contacts

See answer and justification for Q1069 →

Q1130 ADVANCED · Block I

A patient needs extensive posterior rehabilitation and has a shifting habitual closure after loss of several occlusal units. The recorded intercuspal position is not repeatable. What is the priority?

A	Make reversible assessment of mandibular position and occlusal stability before extensive reconstruction
B	Seek specialist advice where severe occlusal instability makes rehabilitation high complexity
C	Use a deprogramming or trial provisional phase when jaw position is uncertain
D	Accept a harmless non-working contact without irreversible intervention
E	Increase OVD solely because restorative space is limited

See answer and justification for Q1130 →

Q1131 ADVANCED · Block I

A symptomatic patient with severe wear and large discrepancy between recorded jaw relationships is planned for irreversible full-mouth crowns. What intermediate stage may help?

A	Seek specialist advice where severe occlusal instability makes rehabilitation high complexity
B	Accept a harmless non-working contact without irreversible intervention
C	Increase OVD solely because restorative space is limited
D	Make reversible assessment of mandibular position and occlusal stability before extensive reconstruction
E	Use a deprogramming or trial provisional phase when jaw position is uncertain

See answer and justification for Q1131 →

Q1132 ADVANCED · Block I

A well-functioning patient has an isolated non-working contact on an intact molar. There are no symptoms, mobility, wear progression or prosthetic plans. What is most appropriate?

A	Use a deprogramming or trial provisional phase when jaw position is uncertain
B	Increase OVD solely because restorative space is limited
C	Seek specialist advice where severe occlusal instability makes rehabilitation high complexity
D	Accept a harmless non-working contact without irreversible intervention
E	Make reversible assessment of mandibular position and occlusal stability before extensive reconstruction

See answer and justification for Q1132 →

Q1405 ADVANCED · Block I

A patient has bilateral masseter tenderness, morning jaw stiffness and pain with chewing. Imaging and oral examination reveal no serious pathology; they ask for a bite change to cure pain.
Which action is most appropriate?

A	Investigate a suspected disc-related locking problem and refer if persistent or severe
B	Avoid irreversible occlusal reconstruction when symptoms are non-specific
C	Assess parafunctional loading and protective strategies for repeated restoration failures
D	Begin reversible education, self-care and symptom-directed TMD management
E	Check contacts and selective adjustment of a new restoration that clearly creates an acute interference

See answer and justification for Q1405 →

Q1406 ADVANCED · Block I

A patient developed a single deflective contact immediately after a new onlay. A tooth now feels high and painful on biting, while symptoms were absent before cementation.
Which action is most appropriate?

A	Avoid irreversible occlusal reconstruction when symptoms are non-specific
B	Check contacts and selective adjustment of a new restoration that clearly creates an acute interference
C	Investigate a suspected disc-related locking problem and refer if persistent or severe
D	Assess parafunctional loading and protective strategies for repeated restoration failures
E	Begin reversible education, self-care and symptom-directed TMD management

See answer and justification for Q1406 →

Q1407 ADVANCED · Block I

A patient reports a painful jaw that suddenly cannot open beyond 24 mm after prior intermittent clicking. There is no history of infection or trauma.
Which action is most appropriate?

A	Investigate a suspected disc-related locking problem and refer if persistent or severe
B	Begin reversible education, self-care and symptom-directed TMD management
C	Check contacts and selective adjustment of a new restoration that clearly creates an acute interference
D	Avoid irreversible occlusal reconstruction when symptoms are non-specific
E	Assess parafunctional loading and protective strategies for repeated restoration failures

See answer and justification for Q1407 →

Q1408 ADVANCED · Block I

A patient repeatedly fractures posterior ceramics and has large matching wear facets, masseter hypertrophy and a history of awake clenching.
Which action is most appropriate?

A	Begin reversible education, self-care and symptom-directed TMD management
B	Assess parafunctional loading and protective strategies for repeated restoration failures
C	Avoid irreversible occlusal reconstruction when symptoms are non-specific
D	Check contacts and selective adjustment of a new restoration that clearly creates an acute interference
E	Investigate a suspected disc-related locking problem and refer if persistent or severe

See answer and justification for Q1408 →

Q1409 ADVANCED · Block I

An asymptomatic adult has a 1.5-mm slide between guided closure and habitual intercuspation. No tooth wear or joint complaints are progressing.
Which conclusion is most defensible?

A	Conservative TMD care is appropriate before irreversible bite modification
B	A reproducible closure slide is a finding to document, not automatic grounds for equilibration
C	New symptomatic occlusal contacts should be identified and corrected when causally linked
D	A stable anterior guidance scheme may need protection during new posterior restoration design
E	A change in OVD requires evaluation of function, speech, comfort and adaptation

See answer and justification for Q1409 →

Q1410 ADVANCED · Block I

A patient with recurrent jaw muscle ache asks for extensive equilibration after reading that all TMD is caused by occlusal imbalance.
Which conclusion is most defensible?

A	A reproducible closure slide is a finding to document, not automatic grounds for equilibration
B	New symptomatic occlusal contacts should be identified and corrected when causally linked
C	Conservative TMD care is appropriate before irreversible bite modification
D	A change in OVD requires evaluation of function, speech, comfort and adaptation
E	A stable anterior guidance scheme may need protection during new posterior restoration design

See answer and justification for Q1410 →

Q1411 ADVANCED · Block I

After a new posterior crown, a patient has immediate discomfort on one tooth only during excursion. The crown has a clear new eccentric interference.
Which conclusion is most defensible?

A	Conservative TMD care is appropriate before irreversible bite modification
B	A stable anterior guidance scheme may need protection during new posterior restoration design
C	New symptomatic occlusal contacts should be identified and corrected when causally linked
D	A change in OVD requires evaluation of function, speech, comfort and adaptation
E	A reproducible closure slide is a finding to document, not automatic grounds for equilibration

See answer and justification for Q1411 →

Q1412 ADVANCED · Block I

A patient undergoing extensive anterior wear rehabilitation will need 2 mm of additional restorative space. The clinician proposes immediate definitive bonding without a trial.
Which conclusion is most defensible?

A	New symptomatic occlusal contacts should be identified and corrected when causally linked
B	A change in OVD requires evaluation of function, speech, comfort and adaptation
C	A stable anterior guidance scheme may need protection during new posterior restoration design
D	A reproducible closure slide is a finding to document, not automatic grounds for equilibration
E	Conservative TMD care is appropriate before irreversible bite modification

See answer and justification for Q1412 →

Q1413 ADVANCED · Block I

A stressed patient has bilateral masseter tenderness, diffuse morning pain and no significant joint noise or limitation of opening. Pain is provoked by muscle palpation.
Which diagnosis or management pathway is most likely?

A	Possible inflammatory joint disease: medical/specialist evaluation when systemic features exist
B	Disc displacement without reduction: structured assessment of restricted opening
C	Disc displacement with reduction: reassurance and monitoring if minimally symptomatic
D	Non-odontogenic persistent pain: avoid further irreversible dental treatment
E	Muscle-related pain: conservative management and reassessment

See answer and justification for Q1413 →

Q1414 ADVANCED · Block I

A patient has painless reciprocal clicking on opening and closing, 45 mm of mouth opening and no interference with everyday function.
Which diagnosis or management pathway is most likely?

A	Disc displacement with reduction: reassurance and monitoring if minimally symptomatic
B	Possible inflammatory joint disease: medical/specialist evaluation when systemic features exist
C	Disc displacement without reduction: structured assessment of restricted opening
D	Non-odontogenic persistent pain: avoid further irreversible dental treatment
E	Muscle-related pain: conservative management and reassessment

See answer and justification for Q1414 →

Q1415 ADVANCED · Block I

A patient had clicking for months, then abrupt loss of the click and now opens only 24 mm with pain at the affected joint.
Which diagnosis or management pathway is most likely?

A	Possible inflammatory joint disease: medical/specialist evaluation when systemic features exist
B	Disc displacement without reduction: structured assessment of restricted opening
C	Disc displacement with reduction: reassurance and monitoring if minimally symptomatic
D	Muscle-related pain: conservative management and reassessment
E	Non-odontogenic persistent pain: avoid further irreversible dental treatment

See answer and justification for Q1415 →

Q1416 ADVANCED · Block I

A patient has bilateral painful joints, prolonged morning stiffness and swollen small hand joints with a known systemic inflammatory condition.
Which diagnosis or management pathway is most likely?

A	Disc displacement with reduction: reassurance and monitoring if minimally symptomatic
B	Disc displacement without reduction: structured assessment of restricted opening
C	Possible inflammatory joint disease: medical/specialist evaluation when systemic features exist
D	Muscle-related pain: conservative management and reassessment
E	Non-odontogenic persistent pain: avoid further irreversible dental treatment

See answer and justification for Q1416 →

Q1417 ADVANCED · Block I

A full posterior quadrant of indirect restorations is planned. The patient shows reproducible guidance and eccentric contacts that were not recorded in the original laboratory prescription.
What is the most appropriate treatment implication?

A	Verify centric and eccentric contacts before finalising an extensive posterior restoration
B	Treat a newly created high contact from a recent crown with measured correction
C	Use a protective, reversible option for nocturnal parafunctional overload where indicated
D	Assess whether a patient has enough restorative space before changing OVD
E	Avoid occlusal grinding to treat otherwise uncomplicated TMD

See answer and justification for Q1417 →

Q1418 ADVANCED · Block I

A heavy nocturnal bruxist has multiple repaired posterior onlays and wishes to reduce future damage; there is no evidence of a primary joint disorder.
What is the most appropriate treatment implication?

A	Treat a newly created high contact from a recent crown with measured correction
B	Assess whether a patient has enough restorative space before changing OVD
C	Use a protective, reversible option for nocturnal parafunctional overload where indicated
D	Avoid occlusal grinding to treat otherwise uncomplicated TMD
E	Verify centric and eccentric contacts before finalising an extensive posterior restoration

See answer and justification for Q1418 →

Q1419 ADVANCED · Block I

A patient with longstanding intermittent masseter pain and normal dentition asks for permanent adjustment of all crowns as the first treatment.
What is the most appropriate treatment implication?

A	Treat a newly created high contact from a recent crown with measured correction
B	Assess whether a patient has enough restorative space before changing OVD
C	Avoid occlusal grinding to treat otherwise uncomplicated TMD
D	Verify centric and eccentric contacts before finalising an extensive posterior restoration
E	Use a protective, reversible option for nocturnal parafunctional overload where indicated

See answer and justification for Q1419 →

Q1420 ADVANCED · Block I

An anterior tooth-wear case has a small localised deficit but stable posterior contacts and maintained facial height. Addition at current contacts is impossible without space creation.
What is the most appropriate treatment implication?

A	Verify centric and eccentric contacts before finalising an extensive posterior restoration
B	Avoid occlusal grinding to treat otherwise uncomplicated TMD
C	Use a protective, reversible option for nocturnal parafunctional overload where indicated
D	Assess whether a patient has enough restorative space before changing OVD
E	Treat a newly created high contact from a recent crown with measured correction

See answer and justification for Q1420 →

BLOCK J · Tooth wear, the failing dentition, interfaces and trauma (36)

Q1070 ADVANCED · Block J

A 37-year-old has mild, non-progressive anterior wear, no sensitivity, satisfactory aesthetics and consistent prevention. What is most appropriate?

A	Address uncontrolled acid exposure before definitive restoration
B	Investigate major occlusal instability and seek specialist planning
C	Use localised additive composite with the Dahl concept after risk control
D	Rehabilitate the entire mouth immediately at increased OVD
E	Monitor and prevent before irreversible restoration when disease is stable and asymptomatic

See answer and justification for Q1070 →

Q1071 ADVANCED · Block J

A 44-year-old has localised palatal anterior wear, dentine sensitivity despite prevention, no space for restoration and intact posterior support. The patient values repairability. What is a reasonable first restorative approach?

A	Use localised additive composite with the Dahl concept after risk control
B	Investigate major occlusal instability and seek specialist planning
C	Rehabilitate the entire mouth immediately at increased OVD
D	Monitor and prevent before irreversible restoration when disease is stable and asymptomatic
E	Address uncontrolled acid exposure before definitive restoration

See answer and justification for Q1071 →

Q1072 ADVANCED · Block J

A 39-year-old with widespread severe erosion, ongoing frequent self-induced vomiting and recurrent caries asks for full-mouth crowns next week. What is the critical first step?

A	Monitor and prevent before irreversible restoration when disease is stable and asymptomatic
B	Rehabilitate the entire mouth immediately at increased OVD
C	Address uncontrolled acid exposure before definitive restoration
D	Use localised additive composite with the Dahl concept after risk control
E	Investigate major occlusal instability and seek specialist planning

See answer and justification for Q1072 →

Q1073 ADVANCED · Block J

A 59-year-old with multiple failing restorations, local periodontal defects and variable tooth prognosis asks for definitive dentures at the first visit. Which first-stage decision is necessary?

A	Avoid periodontal assessment because the patient chiefly wants dentures
B	Preserve strategic restorable teeth for a phased treatment plan
C	Extract all teeth immediately because multiple restorations have failed
D	Create a tooth-by-tooth prognosis after controlling disease and testing restorability
E	Use provisional stabilisation and re-evaluation before committing to a final rehabilitation

See answer and justification for Q1073 →

Q1074 ADVANCED · Block J

A 64-year-old has two sound canines with favourable periodontal support but otherwise poor mandibular teeth. What preservation strategy should be discussed?

A	Use provisional stabilisation and re-evaluation before committing to a final rehabilitation
B	Create a tooth-by-tooth prognosis after controlling disease and testing restorability
C	Avoid periodontal assessment because the patient chiefly wants dentures
D	Preserve strategic restorable teeth for a phased treatment plan
E	Extract all teeth immediately because multiple restorations have failed

See answer and justification for Q1074 →

Q1075 ADVANCED · Block J

A patient has recently stabilised chronic periodontitis but shows uncertain motivation and intermittent attendance. They demand an expensive full-arch fixed restoration immediately. What is prudent?

A	Avoid periodontal assessment because the patient chiefly wants dentures
B	Use provisional stabilisation and re-evaluation before committing to a final rehabilitation
C	Preserve strategic restorable teeth for a phased treatment plan
D	Create a tooth-by-tooth prognosis after controlling disease and testing restorability
E	Extract all teeth immediately because multiple restorations have failed

See answer and justification for Q1075 →

Q1133 ADVANCED · Block J

A 34-year-old with progressive palatal erosion reports nocturnal reflux and frequent acidic regurgitation. Sensitivity is mild and no urgent restorative need is present. What is first?

A	Provide full-mouth porcelain crowns as the immediate first step
B	Address reflux or vomiting and introduce appropriate prevention before definitive coverage
C	Assume all palatal wear is bruxism and give a hard splint as the sole treatment
D	Plan a local additive restoration once causes are controlled and symptoms justify it
E	Coordinate medical assessment of a possible eating disorder sensitively

See answer and justification for Q1133 →

Q1134 ADVANCED · Block J

A 22-year-old with severe palatal erosion, enlarged parotids and calluses on fingers denies vomiting. What is the most appropriate response?

A	Coordinate medical assessment of a possible eating disorder sensitively
B	Assume all palatal wear is bruxism and give a hard splint as the sole treatment
C	Provide full-mouth porcelain crowns as the immediate first step
D	Address reflux or vomiting and introduce appropriate prevention before definitive coverage
E	Plan a local additive restoration once causes are controlled and symptoms justify it

See answer and justification for Q1134 →

Q1135 ADVANCED · Block J

A 49-year-old has stable, previously erosive anterior wear with some dentine sensitivity and no interocclusal space. Reflux is treated; posterior contacts are stable. What could be considered?

A	Address reflux or vomiting and introduce appropriate prevention before definitive coverage
B	Assume all palatal wear is bruxism and give a hard splint as the sole treatment
C	Coordinate medical assessment of a possible eating disorder sensitively
D	Provide full-mouth porcelain crowns as the immediate first step
E	Plan a local additive restoration once causes are controlled and symptoms justify it

See answer and justification for Q1135 →

Q1136 ADVANCED · Block J

A patient with multifactorial wear wants evidence that the lesions have progressed over 18 months. Which record provides a useful direct comparison?

A	Plan preventive and behavioural changes with periodic reassessment
B	Infer progression from a single BEWE score without a prior baseline
C	Use the Smith and Knight index when detailed surface-by-surface documentation is required
D	Use serial clinical photographs and appropriate casts or scans to assess progression
E	Increase OVD in every patient with BEWE score above zero

See answer and justification for Q1136 →

Q1137 ADVANCED · Block J

A 40-year-old with frequent acidic drinks has mild BEWE changes, no symptoms and no documented progression. What is a reasonable approach?

A	Infer progression from a single BEWE score without a prior baseline
B	Use the Smith and Knight index when detailed surface-by-surface documentation is required
C	Use serial clinical photographs and appropriate casts or scans to assess progression
D	Increase OVD in every patient with BEWE score above zero
E	Plan preventive and behavioural changes with periodic reassessment

See answer and justification for Q1137 →

Q1138 ADVANCED · Block J

A research audit requires recording wear severity on each tooth surface rather than only the worst sextant. What index is more suited?

A	Use serial clinical photographs and appropriate casts or scans to assess progression
B	Use the Smith and Knight index when detailed surface-by-surface documentation is required
C	Increase OVD in every patient with BEWE score above zero
D	Plan preventive and behavioural changes with periodic reassessment
E	Infer progression from a single BEWE score without a prior baseline

See answer and justification for Q1138 →

Q1421 ADVANCED · Block J

A patient has localised severe palatal anterior wear, intact posterior occlusal stops and little space for restoration. Acid exposure is controlled and ongoing sensitivity affects function.
Which restorative strategy is most appropriate?

A	Accept monitoring if wear is stable, asymptomatic and function remains adequate
B	Use a local cuspal-coverage strategy for a structurally compromised tooth
C	Use additive anterior direct composite with a Dahl-type approach after risk control
D	Arrange specialist assessment for severe generalised wear with multiple modifying factors
E	Control active acid exposure and monitor before initiating definitive work

See answer and justification for Q1421 →

Q1422 ADVANCED · Block J

A 32-year-old has active, rapidly progressing erosive wear from frequent acidic drinks, and wants veneers immediately. They have not tried risk reduction.
Which restorative strategy is most appropriate?

A	Control active acid exposure and monitor before initiating definitive work
B	Use a local cuspal-coverage strategy for a structurally compromised tooth
C	Arrange specialist assessment for severe generalised wear with multiple modifying factors
D	Accept monitoring if wear is stable, asymptomatic and function remains adequate
E	Use additive anterior direct composite with a Dahl-type approach after risk control

See answer and justification for Q1422 →

Q1423 ADVANCED · Block J

A lower first molar has a large restoration, reproducible cusp pain on release and a visible incomplete crack confined to the crown. Pulp tests are normal.
Which restorative strategy is most appropriate?

A	Control active acid exposure and monitor before initiating definitive work
B	Use additive anterior direct composite with a Dahl-type approach after risk control
C	Arrange specialist assessment for severe generalised wear with multiple modifying factors
D	Accept monitoring if wear is stable, asymptomatic and function remains adequate
E	Use a local cuspal-coverage strategy for a structurally compromised tooth

See answer and justification for Q1423 →

Q1424 ADVANCED · Block J

A patient has extensive generalised erosive and attritional wear, reduced available vertical space, unstable periodontal support and multiple missing posterior teeth.
Which restorative strategy is most appropriate?

A	Use a local cuspal-coverage strategy for a structurally compromised tooth
B	Control active acid exposure and monitor before initiating definitive work
C	Accept monitoring if wear is stable, asymptomatic and function remains adequate
D	Use additive anterior direct composite with a Dahl-type approach after risk control
E	Arrange specialist assessment for severe generalised wear with multiple modifying factors

See answer and justification for Q1424 →

Q1425 ADVANCED · Block J

A patient drinks sour beverages throughout the day; occlusal cusps are rounded and restorations stand proud of smooth surrounding enamel.
Which finding most accurately identifies the dominant wear process?

A	Unexplained rapid palatal erosion with systemic indicators warrants medical evaluation
B	Matched planar facets and hypertrophic elevators suggest attrition
C	Smooth cupped surfaces and restorations standing proud suggest erosion
D	Widespread cervical lesions with multiple mechanisms require mixed-aetiology assessment
E	Sharp notches related to an abrasive habit suggest abrasion

See answer and justification for Q1425 →

Q1426 ADVANCED · Block J

A patient reports intense nocturnal clenching; upper and lower canine facets match directly and there is masseter enlargement.
Which finding most accurately identifies the dominant wear process?

A	Matched planar facets and hypertrophic elevators suggest attrition
B	Sharp notches related to an abrasive habit suggest abrasion
C	Smooth cupped surfaces and restorations standing proud suggest erosion
D	Unexplained rapid palatal erosion with systemic indicators warrants medical evaluation
E	Widespread cervical lesions with multiple mechanisms require mixed-aetiology assessment

See answer and justification for Q1426 →

Q1427 ADVANCED · Block J

A patient brushes horizontally with an abrasive paste and develops narrow cervical V-shaped defects on exposed roots, asymmetric by brushing access.
Which finding most accurately identifies the dominant wear process?

A	Widespread cervical lesions with multiple mechanisms require mixed-aetiology assessment
B	Matched planar facets and hypertrophic elevators suggest attrition
C	Sharp notches related to an abrasive habit suggest abrasion
D	Unexplained rapid palatal erosion with systemic indicators warrants medical evaluation
E	Smooth cupped surfaces and restorations standing proud suggest erosion

See answer and justification for Q1427 →

Q1428 ADVANCED · Block J

A patient has palatal smooth erosion, opposing incisal wear facets and cervical notches, plus both acidic drinks and vigorous brushing.
Which finding most accurately identifies the dominant wear process?

A	Matched planar facets and hypertrophic elevators suggest attrition
B	Sharp notches related to an abrasive habit suggest abrasion
C	Smooth cupped surfaces and restorations standing proud suggest erosion
D	Unexplained rapid palatal erosion with systemic indicators warrants medical evaluation
E	Widespread cervical lesions with multiple mechanisms require mixed-aetiology assessment

See answer and justification for Q1428 →

Q1429 ADVANCED · Block J

An adult with a failing partially dentate arch has several teeth that might retain a future prosthesis, but many have uncertain endodontic status and deep periodontal defects.
What should dominate definitive rehabilitation planning?

A	Prioritise plaque and caries control before extensive crowns or dentures
B	Preserve strategically useful roots when suitable for an overdenture
C	Refer severe combined pathology for specialist multidisciplinary planning
D	Use an interim prosthesis when the dentition is changing rapidly
E	Establish the periodontal and endodontic prognosis of strategic teeth first

See answer and justification for Q1429 →

Q1430 ADVANCED · Block J

A 55-year-old requests complete-arch crowns but has rampant new caries, high plaque scores and progressive periodontal disease despite irregular attendance.
What should dominate definitive rehabilitation planning?

A	Preserve strategically useful roots when suitable for an overdenture
B	Use an interim prosthesis when the dentition is changing rapidly
C	Refer severe combined pathology for specialist multidisciplinary planning
D	Establish the periodontal and endodontic prognosis of strategic teeth first
E	Prioritise plaque and caries control before extensive crowns or dentures

See answer and justification for Q1430 →

Q1431 ADVANCED · Block J

Two lower canines are endodontically and periodontally stable in an otherwise edentulous arch, and the patient wants to preserve remaining ridge support.
What should dominate definitive rehabilitation planning?

A	Prioritise plaque and caries control before extensive crowns or dentures
B	Refer severe combined pathology for specialist multidisciplinary planning
C	Establish the periodontal and endodontic prognosis of strategic teeth first
D	Preserve strategically useful roots when suitable for an overdenture
E	Use an interim prosthesis when the dentition is changing rapidly

See answer and justification for Q1431 →

Q1432 ADVANCED · Block J

An elderly patient needs several staged extractions and uncertain healing. They want teeth for social functioning but cannot yet commit to a definitive rehabilitation plan.
What should dominate definitive rehabilitation planning?

A	Prioritise plaque and caries control before extensive crowns or dentures
B	Use an interim prosthesis when the dentition is changing rapidly
C	Refer severe combined pathology for specialist multidisciplinary planning
D	Preserve strategically useful roots when suitable for an overdenture
E	Establish the periodontal and endodontic prognosis of strategic teeth first

See answer and justification for Q1432 →

Q1433 ADVANCED · Block J

A root-filled crowned molar has pain on release, a persistent sinus and a narrow isolated deep pocket alongside a J-shaped radiolucency.
Which action best protects the tooth and patient?

A	Assess a suspected vertical root fracture before attempting retreatment or recrowning
B	Stabilise a traumatised vital tooth and monitor pulpal outcome as indicated
C	Control periodontal inflammation before finalising prosthetic abutments
D	Investigate a possible combined endodontic-periodontal lesion with relevant testing
E	Refer a complex luxation or avulsion injury according to current trauma guidance

See answer and justification for Q1433 →

Q1434 ADVANCED · Block J

A lower premolar has a deep isolated probing defect and a periapical lesion, but pulp testing is equivocal and there is no clear crack pattern.
Which action best protects the tooth and patient?

A	Control periodontal inflammation before finalising prosthetic abutments
B	Investigate a possible combined endodontic-periodontal lesion with relevant testing
C	Stabilise a traumatised vital tooth and monitor pulpal outcome as indicated
D	Assess a suspected vertical root fracture before attempting retreatment or recrowning
E	Refer a complex luxation or avulsion injury according to current trauma guidance

See answer and justification for Q1434 →

Q1435 ADVANCED · Block J

A teenager has a luxated, vital anterior tooth after trauma, now repositioned and stable; current tests do not establish pulpal necrosis.
Which action best protects the tooth and patient?

A	Assess a suspected vertical root fracture before attempting retreatment or recrowning
B	Investigate a possible combined endodontic-periodontal lesion with relevant testing
C	Stabilise a traumatised vital tooth and monitor pulpal outcome as indicated
D	Refer a complex luxation or avulsion injury according to current trauma guidance
E	Control periodontal inflammation before finalising prosthetic abutments

See answer and justification for Q1435 →

Q1436 ADVANCED · Block J

An adult attends after complex intrusion of a mature upper incisor with significant displacement and associated alveolar injury.
Which action best protects the tooth and patient?

A	Stabilise a traumatised vital tooth and monitor pulpal outcome as indicated
B	Control periodontal inflammation before finalising prosthetic abutments
C	Investigate a possible combined endodontic-periodontal lesion with relevant testing
D	Refer a complex luxation or avulsion injury according to current trauma guidance
E	Assess a suspected vertical root fracture before attempting retreatment or recrowning

See answer and justification for Q1436 →

Q1437 ADVANCED · Block J

A patient has widespread smooth palatal erosion and nocturnal regurgitation despite a low-acid diet; they want full-coverage crowns immediately.
Which intervention should be prioritised?

A	Monitor slow wear when there is no evidence of progression or functional harm
B	Control gastro-oesophageal reflux risk and reduce acid challenges before restoring
C	Use additive restorations where local wear is symptomatic and tissue-preserving space creation is feasible
D	Assess possible eating-disorder-related acid exposure with sensitive referral
E	Modify an occlusal overload before repeating a fractured restoration

See answer and justification for Q1437 →

Q1438 ADVANCED · Block J

A patient has sensitivity from local anterior wear, stable posterior support and no restorative space. The causal diet and grinding behaviour are now controlled.
Which intervention should be prioritised?

A	Assess possible eating-disorder-related acid exposure with sensitive referral
B	Use additive restorations where local wear is symptomatic and tissue-preserving space creation is feasible
C	Monitor slow wear when there is no evidence of progression or functional harm
D	Modify an occlusal overload before repeating a fractured restoration
E	Control gastro-oesophageal reflux risk and reduce acid challenges before restoring

See answer and justification for Q1438 →

Q1439 ADVANCED · Block J

A new posterior onlay fractured at the functional cusp. A single eccentric contact is heavy and was introduced by recent cementation.
Which intervention should be prioritised?

A	Modify an occlusal overload before repeating a fractured restoration
B	Monitor slow wear when there is no evidence of progression or functional harm
C	Use additive restorations where local wear is symptomatic and tissue-preserving space creation is feasible
D	Control gastro-oesophageal reflux risk and reduce acid challenges before restoring
E	Assess possible eating-disorder-related acid exposure with sensitive referral

See answer and justification for Q1439 →

Q1440 ADVANCED · Block J

A patient in their twenties has rapidly progressing palatal tooth loss, enlarged parotids and a history suggesting possible induced vomiting; discussion is difficult.
Which intervention should be prioritised?

A	Use additive restorations where local wear is symptomatic and tissue-preserving space creation is feasible
B	Modify an occlusal overload before repeating a fractured restoration
C	Monitor slow wear when there is no evidence of progression or functional harm
D	Assess possible eating-disorder-related acid exposure with sensitive referral
E	Control gastro-oesophageal reflux risk and reduce acid challenges before restoring

See answer and justification for Q1440 →

Q1441 ADVANCED · Block J

A vital premolar has a small chip of composite at one margin. The remaining restoration is sound, bonding is possible and there is no crack or secondary caries.
Which treatment best matches the restoration failure?

A	Indirect cuspal coverage for repeated posterior cusp fracture with adequate structure
B	Conservative repair of an isolated marginal direct-composite defect with sound tooth
C	Assess restorability and ferrule for a compromised root-treated tooth
D	Remove active disease or unstable caries risk before replacing a large restoration
E	Refer a severely compromised arch for multidisciplinary strategy

See answer and justification for Q1441 →

Q1442 ADVANCED · Block J

A large posterior MOD composite has failed twice through fracture of the remaining cusps. The pulp is vital and adhesive margins remain available.
Which treatment best matches the restoration failure?

A	Refer a severely compromised arch for multidisciplinary strategy
B	Conservative repair of an isolated marginal direct-composite defect with sound tooth
C	Indirect cuspal coverage for repeated posterior cusp fracture with adequate structure
D	Remove active disease or unstable caries risk before replacing a large restoration
E	Assess restorability and ferrule for a compromised root-treated tooth

See answer and justification for Q1442 →

Q1443 ADVANCED · Block J

A root-treated premolar has a repeatedly loose crown and inadequate coronal dentine. The last post was long and intact, but no ferrule was present.
Which treatment best matches the restoration failure?

A	Conservative repair of an isolated marginal direct-composite defect with sound tooth
B	Remove active disease or unstable caries risk before replacing a large restoration
C	Assess restorability and ferrule for a compromised root-treated tooth
D	Refer a severely compromised arch for multidisciplinary strategy
E	Indirect cuspal coverage for repeated posterior cusp fracture with adequate structure

See answer and justification for Q1443 →

Q1444 ADVANCED · Block J

Multiple crown margins show new root caries, plaque and active periodontal inflammation; the patient requests immediate replacement of every crown.
Which treatment best matches the restoration failure?

A	Conservative repair of an isolated marginal direct-composite defect with sound tooth
B	Refer a severely compromised arch for multidisciplinary strategy
C	Assess restorability and ferrule for a compromised root-treated tooth
D	Indirect cuspal coverage for repeated posterior cusp fracture with adequate structure
E	Remove active disease or unstable caries risk before replacing a large restoration

See answer and justification for Q1444 →

BLOCK K · Implants and digital prosthodontics (36)

Q1076 ADVANCED · Block K

A 6-year-old implant crown is surrounded by erythematous mucosa, bleeding on probing and increased plaque; radiographs show no progressive bone loss beyond initial remodelling. What best describes management?

A	Suspect peri-implantitis and arrange appropriate disease management or specialist referral
B	Investigate excess subgingival cement and correct the prosthetic cause
C	Assume a loose abutment screw without examining tissues
D	Remove the implant immediately because bleeding is present
E	Diagnose peri-implant mucositis and intensify plaque control with prosthesis access review

See answer and justification for Q1076 →

Q1077 ADVANCED · Block K

A molar implant has persistent bleeding and suppuration, deepening pockets and new progressive crestal bone loss relative to baseline radiographs. What is the priority?

A	Diagnose peri-implant mucositis and intensify plaque control with prosthesis access review
B	Investigate excess subgingival cement and correct the prosthetic cause
C	Remove the implant immediately because bleeding is present
D	Suspect peri-implantitis and arrange appropriate disease management or specialist referral
E	Assume a loose abutment screw without examining tissues

See answer and justification for Q1077 →

Q1078 ADVANCED · Block K

A recently cemented implant crown has localised swelling with radiopaque material near its subgingival margin; the implant is stable and baseline bone is unchanged. What cause deserves attention?

A	Assume a loose abutment screw without examining tissues
B	Investigate excess subgingival cement and correct the prosthetic cause
C	Remove the implant immediately because bleeding is present
D	Suspect peri-implantitis and arrange appropriate disease management or specialist referral
E	Diagnose peri-implant mucositis and intensify plaque control with prosthesis access review

See answer and justification for Q1078 →

Q1079 ADVANCED · Block K

An extensive distal-extension RPD requires accurate functional registration of compressible mucosa. The available intraoral scan captures the tooth surfaces but not the loaded saddle tissues. What is appropriate?

A	Choose CBCT as the routine replacement for an intraoral scan
B	Digitally design the prosthesis while explaining that biology and occlusion still govern the outcome
C	Confirm scan completeness, bite registration and laboratory verification before manufacture
D	Assume all intraoral scans reproduce deep subgingival margins without tissue control
E	Use a conventional impression or adjunct records where scanning cannot capture the needed anatomy

See answer and justification for Q1079 →

Q1080 ADVANCED · Block K

A posterior zirconia bridge is milled from a scan, but several occlusal surfaces and the bite relationship were captured incompletely. What should happen before manufacture?

A	Digitally design the prosthesis while explaining that biology and occlusion still govern the outcome
B	Choose CBCT as the routine replacement for an intraoral scan
C	Confirm scan completeness, bite registration and laboratory verification before manufacture
D	Use a conventional impression or adjunct records where scanning cannot capture the needed anatomy
E	Assume all intraoral scans reproduce deep subgingival margins without tissue control

See answer and justification for Q1080 →

Q1081 ADVANCED · Block K

A patient assumes a CAD/CAM crown means no follow-up is needed. How should the digital workflow be framed?

A	Choose CBCT as the routine replacement for an intraoral scan
B	Confirm scan completeness, bite registration and laboratory verification before manufacture
C	Digitally design the prosthesis while explaining that biology and occlusion still govern the outcome
D	Assume all intraoral scans reproduce deep subgingival margins without tissue control
E	Use a conventional impression or adjunct records where scanning cannot capture the needed anatomy

See answer and justification for Q1081 →

Q1139 ADVANCED · Block K

A patient reports movement of a screw-retained implant crown. The crown moves but the implant fixture is immobile, the mucosa is healthy and radiographic bone levels are stable. What is most likely?

A	Investigate peri-implant inflammation and cleanability before prosthetic remake
B	Assess screw-retained retrieval where recurrent cement-associated inflammation occurs
C	Check and correct a loose abutment screw and evaluate occlusion and components
D	Refer when true fixture mobility or significant progressive bone loss is detected
E	Assume every implant crown movement represents implant failure

See answer and justification for Q1139 →

Q1140 ADVANCED · Block K

A cement-retained implant crown shows recurrent bleeding, a deep margin and radiopaque subgingival material; bone levels are stable. Which option should be considered?

A	Assess screw-retained retrieval where recurrent cement-associated inflammation occurs
B	Investigate peri-implant inflammation and cleanability before prosthetic remake
C	Refer when true fixture mobility or significant progressive bone loss is detected
D	Check and correct a loose abutment screw and evaluate occlusion and components
E	Assume every implant crown movement represents implant failure

See answer and justification for Q1140 →

Q1141 ADVANCED · Block K

A patient has genuine implant fixture mobility with pain and marked new radiographic bone destruction. What is appropriate?

A	Assess screw-retained retrieval where recurrent cement-associated inflammation occurs
B	Investigate peri-implant inflammation and cleanability before prosthetic remake
C	Check and correct a loose abutment screw and evaluate occlusion and components
D	Refer when true fixture mobility or significant progressive bone loss is detected
E	Assume every implant crown movement represents implant failure

See answer and justification for Q1141 →

Q1142 ADVANCED · Block K

A partially edentulous patient is considered for implant replacement close to the mandibular canal, and conventional imaging cannot define the available buccolingual bone. What is appropriate?

A	Take CBCT for every routine crown, bridge and denture case
B	Use appropriate 3-D imaging only after identifying a clear clinical question and justified need
C	Review prosthesis-driven position, restorative space and nerve proximity before surgery
D	Recognise that metal artefacts and segmentation errors can affect digital planning
E	Proceed with drilling based solely on a printed guide without verification

See answer and justification for Q1142 →

Q1143 ADVANCED · Block K

A radiopaque metal crown produces streak artefacts in a CBCT used for planning an adjacent implant site. How should the scan be interpreted?

A	Review prosthesis-driven position, restorative space and nerve proximity before surgery
B	Use appropriate 3-D imaging only after identifying a clear clinical question and justified need
C	Proceed with drilling based solely on a printed guide without verification
D	Take CBCT for every routine crown, bridge and denture case
E	Recognise that metal artefacts and segmentation errors can affect digital planning

See answer and justification for Q1143 →

Q1144 ADVANCED · Block K

A virtual implant position is centred in bone but would emerge through the buccal face of the planned crown and has inadequate prosthetic space. What should be prioritised?

A	Review prosthesis-driven position, restorative space and nerve proximity before surgery
B	Take CBCT for every routine crown, bridge and denture case
C	Proceed with drilling based solely on a printed guide without verification
D	Recognise that metal artefacts and segmentation errors can affect digital planning
E	Use appropriate 3-D imaging only after identifying a clear clinical question and justified need

See answer and justification for Q1144 →

Q1445 ADVANCED · Block K

A six-year-old implant crown has bleeding on probing and erythema. Radiographs show no progressive supporting bone loss beyond stable early remodelling.
Which diagnosis and initial management are most appropriate?

A	Excess cement-related inflammation: investigate and remove the causative excess where indicated
B	Mechanical screw-related complication: verify prosthetic components and connection
C	Immediate implant removal is not indicated until diagnosis and prognosis are established
D	Suspected peri-implantitis: assess progressive bone loss and arrange appropriate management/referral
E	Peri-implant mucositis: address plaque and access for cleaning without assuming bone loss

See answer and justification for Q1445 →

Q1446 ADVANCED · Block K

A patient has peri-implant bleeding, suppuration, 7-mm pockets and new crater-like bone loss compared with baseline after years of previous stability.
Which diagnosis and initial management are most appropriate?

A	Excess cement-related inflammation: investigate and remove the causative excess where indicated
B	Immediate implant removal is not indicated until diagnosis and prognosis are established
C	Mechanical screw-related complication: verify prosthetic components and connection
D	Peri-implant mucositis: address plaque and access for cleaning without assuming bone loss
E	Suspected peri-implantitis: assess progressive bone loss and arrange appropriate management/referral

See answer and justification for Q1446 →

Q1447 ADVANCED · Block K

An implant crown rotates slightly under function, but mucosa is healthy, radiographs show no progressive bone loss and a faint metallic click is heard.
Which diagnosis and initial management are most appropriate?

A	Suspected peri-implantitis: assess progressive bone loss and arrange appropriate management/referral
B	Mechanical screw-related complication: verify prosthetic components and connection
C	Excess cement-related inflammation: investigate and remove the causative excess where indicated
D	Immediate implant removal is not indicated until diagnosis and prognosis are established
E	Peri-implant mucositis: address plaque and access for cleaning without assuming bone loss

See answer and justification for Q1447 →

Q1448 ADVANCED · Block K

A cement-retained implant crown develops localized inflammation shortly after placement. The margin is deep and imaging suggests residual subgingival cement.
Which diagnosis and initial management are most appropriate?

A	Excess cement-related inflammation: investigate and remove the causative excess where indicated
B	Mechanical screw-related complication: verify prosthetic components and connection
C	Suspected peri-implantitis: assess progressive bone loss and arrange appropriate management/referral
D	Immediate implant removal is not indicated until diagnosis and prognosis are established
E	Peri-implant mucositis: address plaque and access for cleaning without assuming bone loss

See answer and justification for Q1448 →

Q1449 ADVANCED · Block K

A single implant crown was placed recently and the patient is clinically stable. Baseline peri-implant crestal levels need documented comparison for maintenance.
Which planning investigation is best justified?

A	Review digital scan registration and fit rather than assume the file is correct
B	Use a small-field CBCT where a 3-D anatomical question will change implant planning
C	Reassess occlusal and prosthetic space using clinical and diagnostic records
D	Refer when anatomy or augmentation complexity exceeds the operator’s scope
E	Obtain a clinically appropriate periapical baseline for monitoring bone levels around an implant

See answer and justification for Q1449 →

Q1450 ADVANCED · Block K

Implant placement is being considered close to a mental foramen and suspected anterior loop. Conventional imaging cannot safely answer the available bone-volume question.
Which planning investigation is best justified?

A	Obtain a clinically appropriate periapical baseline for monitoring bone levels around an implant
B	Use a small-field CBCT where a 3-D anatomical question will change implant planning
C	Reassess occlusal and prosthetic space using clinical and diagnostic records
D	Review digital scan registration and fit rather than assume the file is correct
E	Refer when anatomy or augmentation complexity exceeds the operator’s scope

See answer and justification for Q1450 →

Q1451 ADVANCED · Block K

An edentulous patient wants implant-supported fixed teeth but has severe loss of restorative height and inadequate interarch space from over-erupted antagonists.
Which planning investigation is best justified?

A	Review digital scan registration and fit rather than assume the file is correct
B	Reassess occlusal and prosthetic space using clinical and diagnostic records
C	Use a small-field CBCT where a 3-D anatomical question will change implant planning
D	Refer when anatomy or augmentation complexity exceeds the operator’s scope
E	Obtain a clinically appropriate periapical baseline for monitoring bone levels around an implant

See answer and justification for Q1451 →

Q1452 ADVANCED · Block K

A patient has severe post-traumatic ridge loss, prior surgical complications and requests complex augmentation and full-arch implants in an office without suitable facilities.
Which planning investigation is best justified?

A	Refer when anatomy or augmentation complexity exceeds the operator’s scope
B	Review digital scan registration and fit rather than assume the file is correct
C	Obtain a clinically appropriate periapical baseline for monitoring bone levels around an implant
D	Use a small-field CBCT where a 3-D anatomical question will change implant planning
E	Reassess occlusal and prosthetic space using clinical and diagnostic records

See answer and justification for Q1452 →

Q1453 ADVANCED · Block K

An implant overdenture patient has stable implants but poor cleaning, heavy plaque and bleeding despite few symptoms. They have not had recall for two years.
Which maintenance priority most directly reduces risk?

A	Review parafunction and prosthetic load before replacing repeatedly fractured components
B	Refer suspected progressive peri-implant bone loss for appropriate intervention
C	Establish accessible cleaning and a patient-specific peri-implant recall pathway
D	Verify retrievability and system compatibility before servicing an implant crown
E	Correct a poorly cleansable emergence profile before recurrent inflammation worsens

See answer and justification for Q1453 →

Q1454 ADVANCED · Block K

A posterior implant crown has fractured its veneering ceramic repeatedly. Tissue health is excellent but the patient has strong parafunction and extreme lateral loading.
Which maintenance priority most directly reduces risk?

A	Review parafunction and prosthetic load before replacing repeatedly fractured components
B	Refer suspected progressive peri-implant bone loss for appropriate intervention
C	Correct a poorly cleansable emergence profile before recurrent inflammation worsens
D	Establish accessible cleaning and a patient-specific peri-implant recall pathway
E	Verify retrievability and system compatibility before servicing an implant crown

See answer and justification for Q1454 →

Q1455 ADVANCED · Block K

An implant crown has a severely convex subgingival contour that traps plaque, with recurrent bleeding despite reasonable patient effort.
Which maintenance priority most directly reduces risk?

A	Refer suspected progressive peri-implant bone loss for appropriate intervention
B	Review parafunction and prosthetic load before replacing repeatedly fractured components
C	Verify retrievability and system compatibility before servicing an implant crown
D	Correct a poorly cleansable emergence profile before recurrent inflammation worsens
E	Establish accessible cleaning and a patient-specific peri-implant recall pathway

See answer and justification for Q1455 →

Q1456 ADVANCED · Block K

A clinician proposes unscrewing a six-year-old implant crown without knowing the brand, driver or connection; the screw access has been previously filled.
Which maintenance priority most directly reduces risk?

A	Review parafunction and prosthetic load before replacing repeatedly fractured components
B	Refer suspected progressive peri-implant bone loss for appropriate intervention
C	Verify retrievability and system compatibility before servicing an implant crown
D	Correct a poorly cleansable emergence profile before recurrent inflammation worsens
E	Establish accessible cleaning and a patient-specific peri-implant recall pathway

See answer and justification for Q1456 →

Q1457 ADVANCED · Block K

An intraoral scan of a subgingival crown margin shows repeated holes and ambiguous borders due to blood and moisture; the technician asks whether to proceed.
Which digital-workflow response is most appropriate?

A	Assess digital articulation and occlusal registration before final milling
B	Use high-quality clinical verification rather than assuming CAD/CAM eliminates errors
C	Verify marginal capture and scan accuracy before accepting a digital impression
D	Use conventional or combined records where extensive edentulous mobile tissues are inadequately scanned
E	Check support, accuracy and manufacturing tolerance of a surgical guide

See answer and justification for Q1457 →

Q1458 ADVANCED · Block K

An extensively edentulous arch has mobile soft-tissue anatomy and very few stable scan landmarks. Consecutive complete-arch scans disagree.
Which digital-workflow response is most appropriate?

A	Use high-quality clinical verification rather than assuming CAD/CAM eliminates errors
B	Use conventional or combined records where extensive edentulous mobile tissues are inadequately scanned
C	Verify marginal capture and scan accuracy before accepting a digital impression
D	Assess digital articulation and occlusal registration before final milling
E	Check support, accuracy and manufacturing tolerance of a surgical guide

See answer and justification for Q1458 →

Q1459 ADVANCED · Block K

A digitally designed two-unit posterior prosthesis seats on the model but shows large clinical occlusal discrepancy. Interocclusal records were taken with unstable closure.
Which digital-workflow response is most appropriate?

A	Check support, accuracy and manufacturing tolerance of a surgical guide
B	Verify marginal capture and scan accuracy before accepting a digital impression
C	Use conventional or combined records where extensive edentulous mobile tissues are inadequately scanned
D	Use high-quality clinical verification rather than assuming CAD/CAM eliminates errors
E	Assess digital articulation and occlusal registration before final milling

See answer and justification for Q1459 →

Q1460 ADVANCED · Block K

A computer-guided implant plan is correct digitally, but the tooth-supported guide rocks on the remaining dentition and does not fully seat clinically.
Which digital-workflow response is most appropriate?

A	Use conventional or combined records where extensive edentulous mobile tissues are inadequately scanned
B	Verify marginal capture and scan accuracy before accepting a digital impression
C	Check support, accuracy and manufacturing tolerance of a surgical guide
D	Use high-quality clinical verification rather than assuming CAD/CAM eliminates errors
E	Assess digital articulation and occlusal registration before final milling

See answer and justification for Q1460 →

Q1461 ADVANCED · Block K

A healthy edentulous patient has comfortable conventional dentures and adequate chewing. They have no desire for surgery or extra prosthetic maintenance.
Which prosthetic option is most appropriate?

A	Plan retrievability and hygiene access for a proposed implant restoration
B	Consider a conventional denture when function is satisfactory and surgery is declined
C	Refer for assessment where severe ridge deficiency or medical complexity alters implant risk
D	Control peri-implant disease before elective replacement or addition
E	Assess a mandibular implant-retained overdenture for difficult conventional retention

See answer and justification for Q1461 →

Q1462 ADVANCED · Block K

A patient has excellent upper complete-denture retention but repeated lower-denture instability despite sound clinical techniques; they are medically suitable and motivated.
Which prosthetic option is most appropriate?

A	Refer for assessment where severe ridge deficiency or medical complexity alters implant risk
B	Assess a mandibular implant-retained overdenture for difficult conventional retention
C	Plan retrievability and hygiene access for a proposed implant restoration
D	Control peri-implant disease before elective replacement or addition
E	Consider a conventional denture when function is satisfactory and surgery is declined

See answer and justification for Q1462 →

Q1463 ADVANCED · Block K

A fixed posterior implant crown is being planned in a patient with previous screw loosening and difficult access for cleaning.
Which prosthetic option is most appropriate?

A	Control peri-implant disease before elective replacement or addition
B	Assess a mandibular implant-retained overdenture for difficult conventional retention
C	Consider a conventional denture when function is satisfactory and surgery is declined
D	Plan retrievability and hygiene access for a proposed implant restoration
E	Refer for assessment where severe ridge deficiency or medical complexity alters implant risk

See answer and justification for Q1463 →

Q1464 ADVANCED · Block K

A patient requires advanced grafting adjacent to important anatomical structures and has severe systemic risk modifiers; the operator has only basic implant experience.
Which prosthetic option is most appropriate?

A	Consider a conventional denture when function is satisfactory and surgery is declined
B	Assess a mandibular implant-retained overdenture for difficult conventional retention
C	Control peri-implant disease before elective replacement or addition
D	Refer for assessment where severe ridge deficiency or medical complexity alters implant risk
E	Plan retrievability and hygiene access for a proposed implant restoration

See answer and justification for Q1464 →

Q1465 ADVANCED · Block K

A complete-arch implant bar designed from an intraoral scan shows a large passive-fit discrepancy. Short scans were accurate but repeated full-arch scans accumulated drift.
Which explanation best accounts for the discrepancy?

A	Incorrect digital jaw relation despite accurately scanned teeth
B	Stitching and registration error in a long-span intraoral scan
C	Lack of adequate functional soft-tissue information in a scan of a complete denture arch
D	Manufacturing tolerance or distortion after milling/printing
E	Unexpected clinical change since records were taken, requiring repeat assessment

See answer and justification for Q1465 →

Q1466 ADVANCED · Block K

A digitally fabricated complete denture has accurate artificial tooth geometry but poor border seal. Scanning was done without any functional border record.
Which explanation best accounts for the discrepancy?

A	Unexpected clinical change since records were taken, requiring repeat assessment
B	Manufacturing tolerance or distortion after milling/printing
C	Lack of adequate functional soft-tissue information in a scan of a complete denture arch
D	Incorrect digital jaw relation despite accurately scanned teeth
E	Stitching and registration error in a long-span intraoral scan

See answer and justification for Q1466 →

Q1467 ADVANCED · Block K

A milled posterior onlay fits margins precisely but is significantly high in occlusion. The scan bite was recorded while the patient was posturing forward.
Which explanation best accounts for the discrepancy?

A	Manufacturing tolerance or distortion after milling/printing
B	Lack of adequate functional soft-tissue information in a scan of a complete denture arch
C	Stitching and registration error in a long-span intraoral scan
D	Incorrect digital jaw relation despite accurately scanned teeth
E	Unexpected clinical change since records were taken, requiring repeat assessment

See answer and justification for Q1467 →

Q1468 ADVANCED · Block K

A printed trial prosthesis fits digitally but shows a uniform systematic offset from the cast. The scan and design have been independently verified.
Which explanation best accounts for the discrepancy?

A	Incorrect digital jaw relation despite accurately scanned teeth
B	Stitching and registration error in a long-span intraoral scan
C	Unexpected clinical change since records were taken, requiring repeat assessment
D	Lack of adequate functional soft-tissue information in a scan of a complete denture arch
E	Manufacturing tolerance or distortion after milling/printing

See answer and justification for Q1468 →

BLOCK L · Complications, failures and maintenance (22)

Q1082 ADVANCED · Block L

A bridge has failed twice on the same abutment while plaque and periodontal status have deteriorated. What should precede the next bridge?

A	Repair a limited defect where prognosis and design allow
B	Assess tissue support, occlusion and hygiene before denture remake
C	Refer a recurrent high-complexity failure for specialist assessment
D	Diagnose biological and mechanical causes before replacing the restoration
E	Simply repeat the original design with stronger cement

See answer and justification for Q1082 →

Q1083 ADVANCED · Block L

A complete denture is reported loose after two years; ridge resorption, sore areas and worn occlusion are present. What should be reviewed before proposing a new set?

A	Repair a limited defect where prognosis and design allow
B	Refer a recurrent high-complexity failure for specialist assessment
C	Assess tissue support, occlusion and hygiene before denture remake
D	Simply repeat the original design with stronger cement
E	Diagnose biological and mechanical causes before replacing the restoration

See answer and justification for Q1083 →

Q1084 ADVANCED · Block L

A complex full-arch prosthesis has recurrent unexplained framework fractures despite multiple remakes and careful routine adjustments. What is prudent?

A	Assess tissue support, occlusion and hygiene before denture remake
B	Diagnose biological and mechanical causes before replacing the restoration
C	Simply repeat the original design with stronger cement
D	Repair a limited defect where prognosis and design allow
E	Refer a recurrent high-complexity failure for specialist assessment

See answer and justification for Q1084 →

Q1145 ADVANCED · Block L

A cobalt-chromium RPD major connector fractures after years of flexure and uneven support. The patient requests the same design in a thicker alloy. What is most useful?

A	Evaluate a persistent peri-prosthetic ulcer before a routine repair
B	Investigate abutment support and implant components in a recurrent fracture
C	Repair a minor removable component with a technically suitable method where function is sound
D	Provide a stronger metal without reviewing design
E	Assess fit, flexure and occlusal loading before making a replacement

See answer and justification for Q1145 →

Q1146 ADVANCED · Block L

An acrylic denture has a small accessible fracture of a non-functional flange after accidental damage, but fit and occlusion are otherwise satisfactory. What may be appropriate?

A	Investigate abutment support and implant components in a recurrent fracture
B	Repair a minor removable component with a technically suitable method where function is sound
C	Provide a stronger metal without reviewing design
D	Assess fit, flexure and occlusal loading before making a replacement
E	Evaluate a persistent peri-prosthetic ulcer before a routine repair

See answer and justification for Q1146 →

Q1147 ADVANCED · Block L

An implant-supported prosthesis has two recurrent screw fractures and a unilateral long cantilever. What must be evaluated?

A	Investigate abutment support and implant components in a recurrent fracture
B	Provide a stronger metal without reviewing design
C	Assess fit, flexure and occlusal loading before making a replacement
D	Repair a minor removable component with a technically suitable method where function is sound
E	Evaluate a persistent peri-prosthetic ulcer before a routine repair

See answer and justification for Q1147 →

Q1469 ADVANCED · Block L

A new complete denture produces a pinpoint ulcer over a bony prominence; pressure-indicating material confirms a local high spot.
Which action is most appropriate?

A	Repair a small isolated fracture when design and support are otherwise sound
B	Re-evaluate denture support and occlusion after recurrent fracture
C	Reline a stable prosthesis whose fitting surface no longer adapts
D	Relieve a confirmed pressure area after identifying local contact
E	Investigate a suspicious persistent lesion rather than adjusting repeatedly

See answer and justification for Q1469 →

Q1470 ADVANCED · Block L

An RPD acrylic saddle chips after accidental impact. The framework, rest seats and occlusion remain good, and there is no indication of systemic failure.
Which action is most appropriate?

A	Reline a stable prosthesis whose fitting surface no longer adapts
B	Relieve a confirmed pressure area after identifying local contact
C	Repair a small isolated fracture when design and support are otherwise sound
D	Investigate a suspicious persistent lesion rather than adjusting repeatedly
E	Re-evaluate denture support and occlusion after recurrent fracture

See answer and justification for Q1470 →

Q1471 ADVANCED · Block L

A maxillary complete denture has split at the midline three times. The patient has a heavy anterior bite and the base flexes over a broad torus.
Which action is most appropriate?

A	Investigate a suspicious persistent lesion rather than adjusting repeatedly
B	Re-evaluate denture support and occlusion after recurrent fracture
C	Reline a stable prosthesis whose fitting surface no longer adapts
D	Relieve a confirmed pressure area after identifying local contact
E	Repair a small isolated fracture when design and support are otherwise sound

See answer and justification for Q1471 →

Q1472 ADVANCED · Block L

A long-term denture is loose because of ridge resorption, but the patient likes the tooth arrangement and the base is structurally intact.
Which action is most appropriate?

A	Re-evaluate denture support and occlusion after recurrent fracture
B	Reline a stable prosthesis whose fitting surface no longer adapts
C	Relieve a confirmed pressure area after identifying local contact
D	Investigate a suspicious persistent lesion rather than adjusting repeatedly
E	Repair a small isolated fracture when design and support are otherwise sound

See answer and justification for Q1472 →

Q1473 ADVANCED · Block L

A patient with clasp-retained RPDs now has early root lesions beneath clasp shoulders, xerostomia and frequent sugar intake. The metal framework is mechanically sound.
Which maintenance issue should be addressed first?

A	Review periodontal support around strategic abutments before considering another bridge
B	Investigate persistent neuropathic symptoms that cannot be explained mechanically
C	Reinforce prevention and scheduled recall for a high-caries-risk RPD wearer
D	Investigate progressive peri-implant bone loss with appropriate referral
E	Manage denture-related mucosal inflammation and cleaning practices

See answer and justification for Q1473 →

Q1474 ADVANCED · Block L

An existing fixed bridge appears well fitted but one abutment has new deep pockets and progressive mobility; the patient wants just a new crown.
Which maintenance issue should be addressed first?

A	Manage denture-related mucosal inflammation and cleaning practices
B	Investigate progressive peri-implant bone loss with appropriate referral
C	Reinforce prevention and scheduled recall for a high-caries-risk RPD wearer
D	Investigate persistent neuropathic symptoms that cannot be explained mechanically
E	Review periodontal support around strategic abutments before considering another bridge

See answer and justification for Q1474 →

Q1475 ADVANCED · Block L

An implant-supported bridge has suppuration and serial evidence of progressive crestal bone loss. The patient requests routine polishing only.
Which maintenance issue should be addressed first?

A	Reinforce prevention and scheduled recall for a high-caries-risk RPD wearer
B	Investigate progressive peri-implant bone loss with appropriate referral
C	Investigate persistent neuropathic symptoms that cannot be explained mechanically
D	Review periodontal support around strategic abutments before considering another bridge
E	Manage denture-related mucosal inflammation and cleaning practices

See answer and justification for Q1475 →

Q1476 ADVANCED · Block L

An upper denture wearer has diffuse palatal erythema and sleeps in the prosthesis; heavy biofilm is present on the fitting surface.
Which maintenance issue should be addressed first?

A	Investigate persistent neuropathic symptoms that cannot be explained mechanically
B	Manage denture-related mucosal inflammation and cleaning practices
C	Reinforce prevention and scheduled recall for a high-caries-risk RPD wearer
D	Investigate progressive peri-implant bone loss with appropriate referral
E	Review periodontal support around strategic abutments before considering another bridge

See answer and justification for Q1476 →

Q1477 ADVANCED · Block L

A crown dislodges and there is soft carious dentine at the margin; the patient asks for immediate recementation before travel.
What is the most appropriate approach to a failed crown or bridge?

A	Check a resin-bonded framework for contamination and design failure before rebonding
B	Offer a removable alternative where abutment prognosis no longer supports a bridge
C	Assess pulp and root prognosis following repeated unexplained tenderness
D	Investigate occlusal load and connector dimensions after recurrent bridge fracture
E	Assess and treat active marginal caries before replacing a loose crown

See answer and justification for Q1477 →

Q1478 ADVANCED · Block L

A long-span posterior bridge has fractured at its connector twice. Abutments are healthy, but there is heavy parafunction and a thin connector.
What is the most appropriate approach to a failed crown or bridge?

A	Assess pulp and root prognosis following repeated unexplained tenderness
B	Check a resin-bonded framework for contamination and design failure before rebonding
C	Offer a removable alternative where abutment prognosis no longer supports a bridge
D	Assess and treat active marginal caries before replacing a loose crown
E	Investigate occlusal load and connector dimensions after recurrent bridge fracture

See answer and justification for Q1478 →

Q1479 ADVANCED · Block L

An anterior adhesive bridge debonds without fracture. The abutment enamel is intact, but the retainer has poor surface treatment and an unfavorable heavy contact.
What is the most appropriate approach to a failed crown or bridge?

A	Assess pulp and root prognosis following repeated unexplained tenderness
B	Assess and treat active marginal caries before replacing a loose crown
C	Investigate occlusal load and connector dimensions after recurrent bridge fracture
D	Check a resin-bonded framework for contamination and design failure before rebonding
E	Offer a removable alternative where abutment prognosis no longer supports a bridge

See answer and justification for Q1479 →

Q1480 ADVANCED · Block L

A root-treated crowned tooth is persistently tender and has an isolated deep periodontal pocket despite a well-fitting crown; imaging shows an unusual lateral radiolucency.
What is the most appropriate approach to a failed crown or bridge?

A	Assess and treat active marginal caries before replacing a loose crown
B	Investigate occlusal load and connector dimensions after recurrent bridge fracture
C	Offer a removable alternative where abutment prognosis no longer supports a bridge
D	Assess pulp and root prognosis following repeated unexplained tenderness
E	Check a resin-bonded framework for contamination and design failure before rebonding

See answer and justification for Q1480 →

Q1481 ADVANCED · Block L

A patient has a comfortable functioning complete denture, healthy mucosa and good cleaning, and asks whether no further review is needed.
Which follow-up plan best fits the clinical finding?

A	Urgent escalation for red-flag pathology detected during prosthetic review
B	Prompt specialist review for suspected implant disease or complex failure
C	Routine periodic review of a stable prosthesis and oral tissues
D	Early review after a new adjustment to confirm mucosal healing and function
E	Frequent risk-based review while active disease is being brought under control

See answer and justification for Q1481 →

Q1482 ADVANCED · Block L

A local pressure sore from a new mandibular flange has just been relieved and the patient can now close comfortably.
Which follow-up plan best fits the clinical finding?

A	Frequent risk-based review while active disease is being brought under control
B	Early review after a new adjustment to confirm mucosal healing and function
C	Prompt specialist review for suspected implant disease or complex failure
D	Urgent escalation for red-flag pathology detected during prosthetic review
E	Routine periodic review of a stable prosthesis and oral tissues

See answer and justification for Q1482 →

Q1483 ADVANCED · Block L

A patient with new root caries, poor plaque control and recently worsening xerostomia has begun a prevention programme during denture maintenance.
Which follow-up plan best fits the clinical finding?

A	Early review after a new adjustment to confirm mucosal healing and function
B	Frequent risk-based review while active disease is being brought under control
C	Prompt specialist review for suspected implant disease or complex failure
D	Urgent escalation for red-flag pathology detected during prosthetic review
E	Routine periodic review of a stable prosthesis and oral tissues

See answer and justification for Q1483 →

Q1484 ADVANCED · Block L

A patient has a persistent indurated lateral-tongue ulcer that has not healed despite elimination of flange trauma, and cervical nodes are now enlarged.
Which follow-up plan best fits the clinical finding?

A	Frequent risk-based review while active disease is being brought under control
B	Early review after a new adjustment to confirm mucosal healing and function
C	Urgent escalation for red-flag pathology detected during prosthetic review
D	Routine periodic review of a stable prosthesis and oral tissues
E	Prompt specialist review for suspected implant disease or complex failure

See answer and justification for Q1484 →

BLOCK M · Governance, law and communication (18)

Q1085 ADVANCED · Block M

A patient requests a bridge but the clinician believes an implant is more profitable. Both options have reasonable indications. What must the consultation include?

A	Present risks, alternatives including no treatment, and the patient-specific implications
B	Document the complication candidly and follow duty-of-candour processes
C	Proceed using the option the clinician prefers without exploring patient goals
D	Confirm understanding and capacity and offer supported decision-making
E	Withhold information because it may distress the patient

See answer and justification for Q1085 →

Q1086 ADVANCED · Block M

An older patient with fluctuating cognition agrees to extractions but cannot repeat basic consequences during today’s consultation. The treatment is elective. What is most appropriate?

A	Document the complication candidly and follow duty-of-candour processes
B	Withhold information because it may distress the patient
C	Present risks, alternatives including no treatment, and the patient-specific implications
D	Confirm understanding and capacity and offer supported decision-making
E	Proceed using the option the clinician prefers without exploring patient goals

See answer and justification for Q1086 →

Q1087 ADVANCED · Block M

A laboratory error leads to an ill-fitting definitive bridge already cemented; the patient has pain and a remake is required. What communication approach is appropriate?

A	Present risks, alternatives including no treatment, and the patient-specific implications
B	Document the complication candidly and follow duty-of-candour processes
C	Withhold information because it may distress the patient
D	Confirm understanding and capacity and offer supported decision-making
E	Proceed using the option the clinician prefers without exploring patient goals

See answer and justification for Q1087 →

Q1148 ADVANCED · Block M

A complex restorative plan includes extraction and bridge versus denture alternatives, with substantial differences in maintenance. What is essential in the clinical record?

A	Disclose a clinical error and arrange corrective care with candid documentation
B	Share identifiable patient photographs in a public group without consent
C	Communicate a clear written maintenance and referral plan at handover
D	Record only the selected treatment and fee
E	Document the diagnostic reasoning, alternatives, patient goals and agreed risks

See answer and justification for Q1148 →

Q1149 ADVANCED · Block M

A Level 2 practitioner has completed a prosthetic phase and returns the patient to the referring GDP for long-term periodontal maintenance. What is required?

A	Share identifiable patient photographs in a public group without consent
B	Disclose a clinical error and arrange corrective care with candid documentation
C	Communicate a clear written maintenance and referral plan at handover
D	Record only the selected treatment and fee
E	Document the diagnostic reasoning, alternatives, patient goals and agreed risks

See answer and justification for Q1149 →

Q1150 ADVANCED · Block M

A patient notices that a shade was mistakenly selected and the cemented crown needs replacement. The error is confirmed and preventable. What is professionally required?

A	Share identifiable patient photographs in a public group without consent
B	Disclose a clinical error and arrange corrective care with candid documentation
C	Record only the selected treatment and fee
D	Communicate a clear written maintenance and referral plan at handover
E	Document the diagnostic reasoning, alternatives, patient goals and agreed risks

See answer and justification for Q1150 →

Q1485 ADVANCED · Block M

An RPD is delivered with an accidental sharp flange causing a significant tissue injury. The clinician recognises the error after the patient reports pain.
What is the most appropriate professional response?

A	Respect and record an informed refusal after discussing options and consequences
B	Document a near miss and analyse the process to reduce recurrence
C	Refer beyond personal competence even if the patient strongly prefers treatment locally
D	Assess patient understanding with accessible communication and teach-back
E	Explain a complication openly and arrange appropriate follow-up in accordance with duty of candour

See answer and justification for Q1485 →

Q1486 ADVANCED · Block M

A prosthesis with incorrect patient identification is intercepted in the laboratory before being fitted. No patient was harmed, but several labels were similar.
What is the most appropriate professional response?

A	Document a near miss and analyse the process to reduce recurrence
B	Explain a complication openly and arrange appropriate follow-up in accordance with duty of candour
C	Respect and record an informed refusal after discussing options and consequences
D	Assess patient understanding with accessible communication and teach-back
E	Refer beyond personal competence even if the patient strongly prefers treatment locally

See answer and justification for Q1486 →

Q1487 ADVANCED · Block M

A capable patient declines a proposed fixed bridge and prefers no replacement after a balanced discussion of functional implications.
What is the most appropriate professional response?

A	Explain a complication openly and arrange appropriate follow-up in accordance with duty of candour
B	Respect and record an informed refusal after discussing options and consequences
C	Assess patient understanding with accessible communication and teach-back
D	Document a near miss and analyse the process to reduce recurrence
E	Refer beyond personal competence even if the patient strongly prefers treatment locally

See answer and justification for Q1487 →

Q1488 ADVANCED · Block M

A clinician is asked to undertake extensive implant augmentation beyond their training. The patient offers additional payment to avoid referral.
What is the most appropriate professional response?

A	Explain a complication openly and arrange appropriate follow-up in accordance with duty of candour
B	Refer beyond personal competence even if the patient strongly prefers treatment locally
C	Respect and record an informed refusal after discussing options and consequences
D	Assess patient understanding with accessible communication and teach-back
E	Document a near miss and analyse the process to reduce recurrence

See answer and justification for Q1488 →

Q1489 ADVANCED · Block M

A patient disputes whether risks of a resin-bonded bridge were explained. The record states only “consent obtained” and lacks options or patient concerns.
Which record or governance action is most appropriate?

A	Record a patient-specific maintenance plan and support responsibilities
B	Document baseline findings, risks, options, consent and treatment discussions
C	Ensure radiation justification and clinical evaluation are recorded
D	Use a structured clinical audit to measure and improve recall or complication outcomes
E	Provide a clear referral with urgency, clinical findings and relevant investigations

See answer and justification for Q1489 →

Q1490 ADVANCED · Block M

A department wants to reduce repeat visits from fractured dentures. Cases are recorded inconsistently and there are no baseline complication rates.
Which record or governance action is most appropriate?

A	Record a patient-specific maintenance plan and support responsibilities
B	Provide a clear referral with urgency, clinical findings and relevant investigations
C	Document baseline findings, risks, options, consent and treatment discussions
D	Ensure radiation justification and clinical evaluation are recorded
E	Use a structured clinical audit to measure and improve recall or complication outcomes

See answer and justification for Q1490 →

Q1491 ADVANCED · Block M

A patient asks why a panoramic scan was taken for a single tooth when a recent periapical already answered the question. The clinical notes contain no justification.
Which record or governance action is most appropriate?

A	Use a structured clinical audit to measure and improve recall or complication outcomes
B	Provide a clear referral with urgency, clinical findings and relevant investigations
C	Ensure radiation justification and clinical evaluation are recorded
D	Record a patient-specific maintenance plan and support responsibilities
E	Document baseline findings, risks, options, consent and treatment discussions

See answer and justification for Q1491 →

Q1492 ADVANCED · Block M

A patient has suspected progressive peri-implant bone loss and recurrent suppuration; the receiving specialist asks for the original implant system and earlier bone-level images.
Which record or governance action is most appropriate?

A	Ensure radiation justification and clinical evaluation are recorded
B	Use a structured clinical audit to measure and improve recall or complication outcomes
C	Record a patient-specific maintenance plan and support responsibilities
D	Document baseline findings, risks, options, consent and treatment discussions
E	Provide a clear referral with urgency, clinical findings and relevant investigations

See answer and justification for Q1492 →

Q1493 ADVANCED · Block M

A patient who speaks very little English is asked to consent to several elective bridge preparations; the only interpretation has been gestures and a relative who answers for them.
What is the most appropriate communication step?

A	Communicate a treatment error and immediate mitigation transparently
B	Document a significant treatment change and seek renewed informed consent
C	Check decision-specific capacity and optimise the person’s ability to decide
D	Use a qualified interpreter or reliable supported communication for complex consent
E	Discuss foreseeable long-term maintenance and alternatives in a balanced way

See answer and justification for Q1493 →

Q1494 ADVANCED · Block M

A patient with cognitive impairment becomes more engaged when options are explained slowly and illustrated. A colleague argues that the diagnosis automatically makes consent invalid.
What is the most appropriate communication step?

A	Use a qualified interpreter or reliable supported communication for complex consent
B	Communicate a treatment error and immediate mitigation transparently
C	Check decision-specific capacity and optimise the person’s ability to decide
D	Discuss foreseeable long-term maintenance and alternatives in a balanced way
E	Document a significant treatment change and seek renewed informed consent

See answer and justification for Q1494 →

Q1495 ADVANCED · Block M

A patient wants implant-supported fixed teeth because they expect never to need dental maintenance again. Their hygiene is currently poor.
What is the most appropriate communication step?

A	Document a significant treatment change and seek renewed informed consent
B	Use a qualified interpreter or reliable supported communication for complex consent
C	Check decision-specific capacity and optimise the person’s ability to decide
D	Discuss foreseeable long-term maintenance and alternatives in a balanced way
E	Communicate a treatment error and immediate mitigation transparently

See answer and justification for Q1495 →

Q1496 ADVANCED · Block M

During bridge preparation, it becomes clear that a key abutment is unrestorable and the plan must change substantially from the original agreement.
What is the most appropriate communication step?

A	Discuss foreseeable long-term maintenance and alternatives in a balanced way
B	Document a significant treatment change and seek renewed informed consent
C	Use a qualified interpreter or reliable supported communication for complex consent
D	Communicate a treatment error and immediate mitigation transparently
E	Check decision-specific capacity and optimise the person’s ability to decide

See answer and justification for Q1496 →

BLOCK N · Integrated cases (13)

Q1151 ADVANCED · Block N

A 49-year-old with high caries activity, uncontrolled reflux and progressive anterior wear asks for immediate veneers. The oral hygiene score is poor and symptoms are modest. What is the priority?

A	Stabilise disease and prepare a phased risk-based plan
B	Provide full-mouth irreversible ceramics immediately
C	Extract all remaining teeth as the default solution
D	Choose local reversible repair after risk control where appropriate
E	Arrange appropriate multidisciplinary assessment of complexity

See answer and justification for Q1151 →

Q1152 ADVANCED · Block N

A 51-year-old has extensive erosive wear, multiple missing posterior units, severe periodontal attachment loss and uncertain occlusal position. They request full-mouth reconstruction in one visit. What is prudent?

A	Extract all remaining teeth as the default solution
B	Arrange appropriate multidisciplinary assessment of complexity
C	Provide full-mouth irreversible ceramics immediately
D	Choose local reversible repair after risk control where appropriate
E	Stabilise disease and prepare a phased risk-based plan

See answer and justification for Q1152 →

Q1153 ADVANCED · Block N

A 45-year-old after successful reflux treatment has one sensitive worn upper incisor with sound enamel and stable occlusal support. The patient wants a repairable solution. What may be appropriate?

A	Choose local reversible repair after risk control where appropriate
B	Provide full-mouth irreversible ceramics immediately
C	Stabilise disease and prepare a phased risk-based plan
D	Extract all remaining teeth as the default solution
E	Arrange appropriate multidisciplinary assessment of complexity

See answer and justification for Q1153 →

Q1154 ADVANCED · Block N

A patient has one failing posterior bridge, untreated periodontitis and mobile potential abutments. They want the bridge immediately replaced with a longer span. What comes first?

A	Offer no replacement when function is good and the patient is satisfied
B	Refer severe progressive peri-implant bone loss for specialist management
C	Choose a hygienic, maintainable RPD when fixed work is high-risk and patient preference supports it
D	Complete periodontal treatment and reassess abutments before long-span bridgework
E	Investigate implant replacement after disease stabilisation, risk assessment and consent

See answer and justification for Q1154 →

Q1155 ADVANCED · Block N

A 74-year-old with stable periodontal maintenance has lost several posterior teeth, declines surgery and values an appliance they can remove for cleaning. What is reasonable?

A	Refer severe progressive peri-implant bone loss for specialist management
B	Complete periodontal treatment and reassess abutments before long-span bridgework
C	Investigate implant replacement after disease stabilisation, risk assessment and consent
D	Offer no replacement when function is good and the patient is satisfied
E	Choose a hygienic, maintainable RPD when fixed work is high-risk and patient preference supports it

See answer and justification for Q1155 →

Q1156 ADVANCED · Block N

A maintained implant bridge develops suppuration, deep pockets and documented progressive crestal bone loss despite improved hygiene. What is the appropriate next action?

A	Refer severe progressive peri-implant bone loss for specialist management
B	Investigate implant replacement after disease stabilisation, risk assessment and consent
C	Offer no replacement when function is good and the patient is satisfied
D	Complete periodontal treatment and reassess abutments before long-span bridgework
E	Choose a hygienic, maintainable RPD when fixed work is high-risk and patient preference supports it

See answer and justification for Q1156 →

Q1178 ADVANCED · Block N

A 63-year-old has widespread caries, variable periodontal prognosis, poor self-care and multiple failing crowns, but asks to start definitive implants next week. What is the strongest first step?

A	Consider a maintainable removable alternative after explaining fixed options
B	Provide irreversible full-mouth reconstruction immediately
C	Refer severe multidisciplinary complexity for specialist-led planning
D	Offer a conservative local adhesive restoration where prognosis is favourable
E	Adopt a staged plan with disease stabilisation, provisionalisation and reassessment

See answer and justification for Q1178 →

Q1179 ADVANCED · Block N

A 42-year-old has an isolated fractured occlusal cusp, excellent hygiene and stable periodontium; the tooth is restorable with sound margins and the patient wants the least invasive repair. What is appropriate?

A	Provide irreversible full-mouth reconstruction immediately
B	Adopt a staged plan with disease stabilisation, provisionalisation and reassessment
C	Consider a maintainable removable alternative after explaining fixed options
D	Offer a conservative local adhesive restoration where prognosis is favourable
E	Refer severe multidisciplinary complexity for specialist-led planning

See answer and justification for Q1179 →

Q1180 ADVANCED · Block N

A 58-year-old has severe bone loss, unstable jaw relationship, major maxillofacial defects after surgery and requests complex fixed prosthetic rehabilitation. What is appropriate?

A	Consider a maintainable removable alternative after explaining fixed options
B	Adopt a staged plan with disease stabilisation, provisionalisation and reassessment
C	Offer a conservative local adhesive restoration where prognosis is favourable
D	Refer severe multidisciplinary complexity for specialist-led planning
E	Provide irreversible full-mouth reconstruction immediately

See answer and justification for Q1180 →

Q1497 ADVANCED · Block N

A 58-year-old with active root caries, BPE concerns, xerostomia from polypharmacy and a missing upper premolar requests an expensive fixed bridge immediately.
Which overall plan is most appropriate?

A	Use a conservative adhesive replacement after confirming stable disease and favourable anatomy
B	Offer a provisional removable solution while complex diagnostic issues are addressed
C	Document informed refusal and maintain prevention when the patient declines replacement
D	Refer for integrated specialist planning of major acquired or developmental defects
E	Stabilise active disease, clarify abutment prognosis and phase conservative options

See answer and justification for Q1497 →

Q1498 ADVANCED · Block N

After an accident, a 34-year-old has lost two anterior teeth; neighbouring teeth have uncertain vitality and the patient needs a presentable smile for work within two weeks.
Which overall plan is most appropriate?

A	Document informed refusal and maintain prevention when the patient declines replacement
B	Use a conservative adhesive replacement after confirming stable disease and favourable anatomy
C	Stabilise active disease, clarify abutment prognosis and phase conservative options
D	Offer a provisional removable solution while complex diagnostic issues are addressed
E	Refer for integrated specialist planning of major acquired or developmental defects

See answer and justification for Q1498 →

Q1499 ADVANCED · Block N

A 28-year-old with a healed isolated lateral-incisor space has sound enamel-rich adjacent teeth, stable periodontal support and a favourable occlusion; they prefer a reversible minimally invasive approach.
Which overall plan is most appropriate?

A	Offer a provisional removable solution while complex diagnostic issues are addressed
B	Refer for integrated specialist planning of major acquired or developmental defects
C	Document informed refusal and maintain prevention when the patient declines replacement
D	Stabilise active disease, clarify abutment prognosis and phase conservative options
E	Use a conservative adhesive replacement after confirming stable disease and favourable anatomy

See answer and justification for Q1499 →

Q1500 ADVANCED · Block N

A patient has severe amelogenesis imperfecta, extensive hypodontia, active periodontal problems and major occlusal discrepancies requiring combined orthodontic-restorative reconstruction.
Which overall plan is most appropriate?

A	Offer a provisional removable solution while complex diagnostic issues are addressed
B	Document informed refusal and maintain prevention when the patient declines replacement
C	Use a conservative adhesive replacement after confirming stable disease and favourable anatomy
D	Refer for integrated specialist planning of major acquired or developmental defects
E	Stabilise active disease, clarify abutment prognosis and phase conservative options

See answer and justification for Q1500 →